

A NEURAL NETWORK BASED MODEL FOR BIOMASS GASIFICATION IN
FLUIDIZED BED

BY

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ABSTRACT

Biomass is a renewable energy resource and its utilization has received great attention due to its life cycle carbon-neutrality and the potential to substitute fossil fuel to produce a variety of energy-related products. Thermochemical gasification is an important route for conversion of biomass that results in a product gas mainly consisting of H₂, CO, CO₂, CH₄ and other light hydrocarbons that can be used as fuel gas to generate power, or as well as raw material to produce a variety of chemicals. Among the existing gasifiers, fluidized beds (FB) offer many advantages such as high conversion efficiency and great flexibility over types of feedstock.

Modeling and simulation of FB gasifier is a problem of considerable theoretical and computational difficulty mainly due to large number of chemical and physical processes in the reactor and the lack of proper characterization of solid fuel. The biggest challenge that the existing models face is lack of general predictivity. Here, we present a black-box model that is significantly more accurate than the existing models in predicting the carbon conversion efficiency, yield and composition of product gas under a wide range of operating conditions for different types of feedstock.

Initially in this study, the information regarding elemental composition (C, H, N, O, S), proximate analysis (Volatile matter, Fixed carbon, Ash), and structural component (Cellulose, Hemicellulose, Lignin) of biomass was consolidated using canonical correlation analysis (CCA).

Subsequently, more than 200 data sets of biomass gasification in fluidized bed were collected featuring a wide range of operating condition and fuel types. An axiom-based reasoning was used to develop a multiphase statistical pathway needed as a

precondition to effectively quantify the entanglements of different important factors in the process.

Specifically, by creating an interconnected chain of analysis based on trigonometric functions, geometric projections, and design of a statistical inference tool utilizing neural network units, multiple partial measures of associations between biomass constituents, and operating condition were effectively consolidated and embedded in a single characteristic matrix that consequently led to detection of monotonic relationships for prediction of carbon conversion efficiency and product gas yield. Overall, the black box model in comparison to three different models showed better accuracy in predicting four major components of product gas (H_2 , CH_4 , CO , and CO_2), over the largest applicable range of all the influential parameters of the process, namely, temperature, air equivalent ratio, steam to biomass ratio, and type of fuel.

In part of our methodology, we introduce a novel technique for obtaining a dynamical property value for stationary objects, based on a “specific computational time” of an “abstract mechanical operation on characteristics matrices”. The specific computational time (sct) showed excellent capability in capturing the non-equilibrium factor of the process which itself was function of several interrelated variables.

CHAPTER 1

INTRODUCTION

1.1 Climate change and renewable energy

The United Nation (UN) has estimated that in 2040 population of the world will exceed 9 billion [1]. This significant growth in population along with the energy intensity of modern life will evidently lead to a similar hike in demand for energy resources.

The International Energy Outlook 2016 (IEO16) has projected total world consumption of marketed energy to expand from 549 quadrillion Btu in 2012 to 629 quadrillion Btu in 2020 and to 815 quadrillion Btu in 2040, a nearly 50% increase from 2012 to 2040 [1]. Figure 1 shows how the consumption of different sources of energy drift.

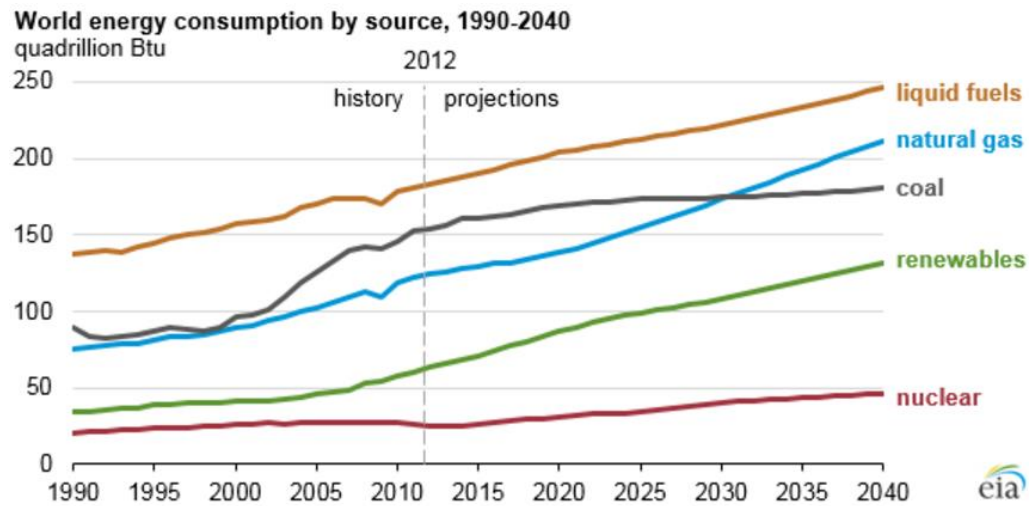


Figure 1. World consumption of energy by source. [1]

It can be readily observed that fossil fuel has been the most significant source of world energy over decades. Abundance and low prices can be viewed as most important reasons of fossil fuel dominance. Although burning fossil fuels has played a key role in

creating a rapid economic expansion for industrialized world, its utilization has resulted in release of billions of metric tons of CO₂ into the atmosphere [1]. Figure 2 displays the increasing rate of CO₂ emissions for fossil fuels.

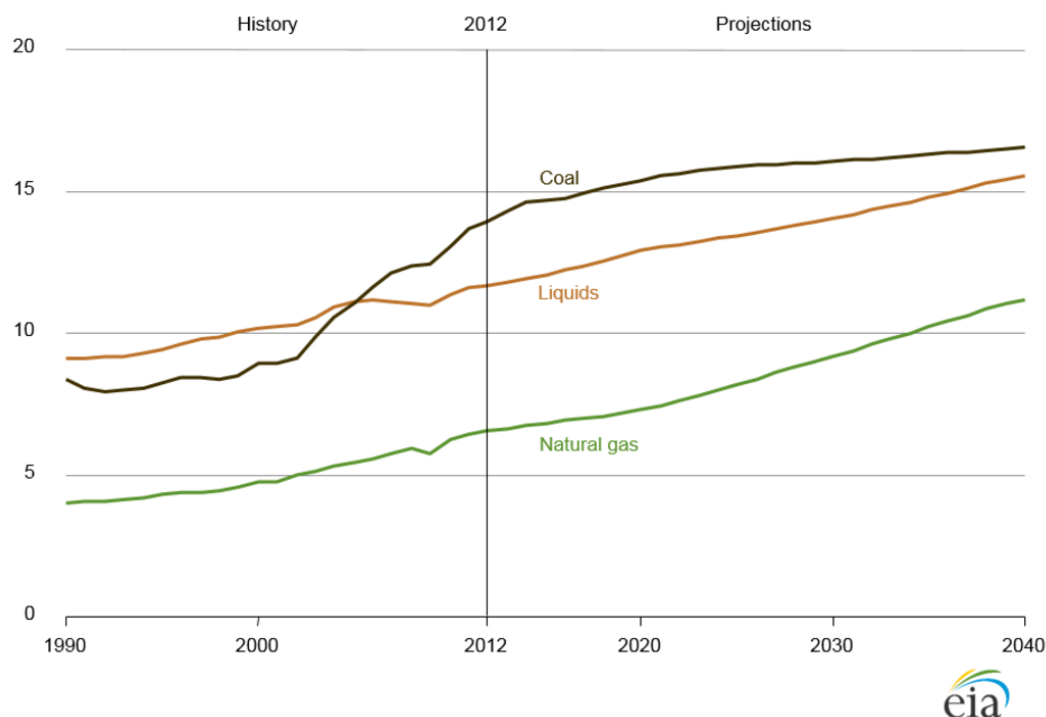


Figure 2. World energy-related carbon dioxide emissions by fuel (billion metric tons) [1]

The climate center at National Aeronautics and Space Administration reported CO₂ level in atmosphere has increased by more than 40% since the start of the industrial revolution from 280 ppm to 399 ppm as of 2015 (Fig. 3).

This spike in CO₂ level is of concern because it has been estimated that if greenhouse gas emissions continue at the present rate (Fig. 2) Earth's surface temperature could exceed historical values as early as 2047, with potentially harmful effects on ecosystems, biodiversity and the livelihoods of people worldwide [2].

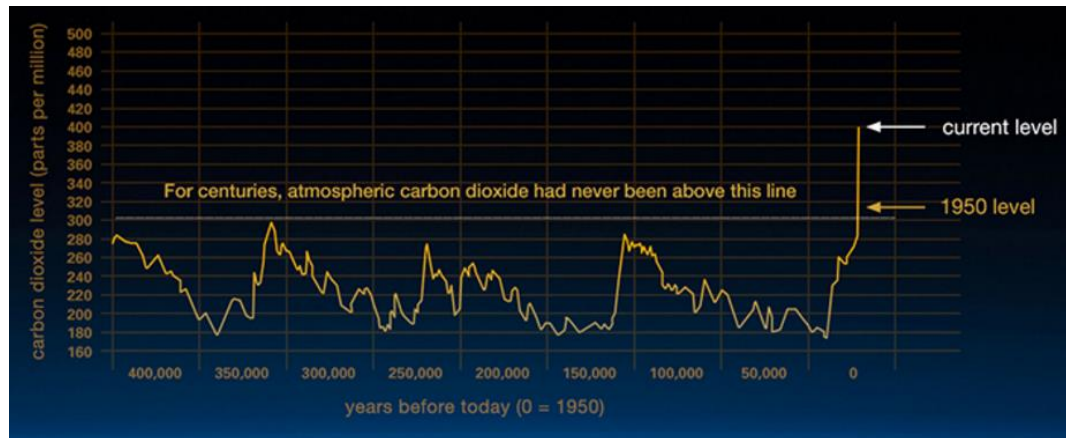


Figure 3. CO2 Level (ppm) over Time [2]

Necessity of Renewable Energy:

These concerns pushed policy makers to intensify their effort to mitigate CO2 emissions. A global agreement on combat climate change was signed by world leaders in Paris in December 2015. This climate and energy framework aims to achieve carbon neutrality in the second half of this century and to hold global warming well below 2 °C relative to pre-industrial levels [3]. The three key targets that have been set for the year 2030 are the followings:

1. At least 40 % cuts in GHG emissions (from 1990 levels)
2. At least 27 % share for renewable energy
3. At least 27 % improvement in energy efficiency

As part of efforts to cut GHG emissions, advancement of carbon capture and sequestration (CCS) technology has been pursued by numerous researchers but there seems to be several technical and economic challenges that must be overcome before cost-effective CCS technologies can be implemented. One of the largest challenges facing CCS technology is the cost. A study by Worely Parsons in 2009 determined that the cost of CCS for power generation, based on the use of commercially available

technology ranges from US\$57-107 per ton of CO₂ avoided or US\$42-90 per ton of CO₂ captured [4]. Another study done at MIT [5] estimated CCS raises the costs of electricity (compared to an uncontrolled plant) by between 30 and 80%.

The results of these studies imply “the low-price factor” that for long has been contributing to the dominance of fossil derived products in the market will be significantly weakened in a world where carbon emissions are regulated. Therefore, many efforts have been made to develop technologies for renewable resources of energy that have the potential of zero or almost zero GHG emissions such as wind energy, solar energy, biofuel and biomass.

In a study on role of innovative technologies under the global zero emissions, Koji [6] conducted global modeling over the period 2010–2150 in which energy, biomass, and materials supply and costs were minimized by linear programming. The model simulation suggested that zero emissions energy systems may be possible toward the end of the century, while allowing for global economic growth. Figure 4 shows the primary energy supply structures for zero emissions scenarios.

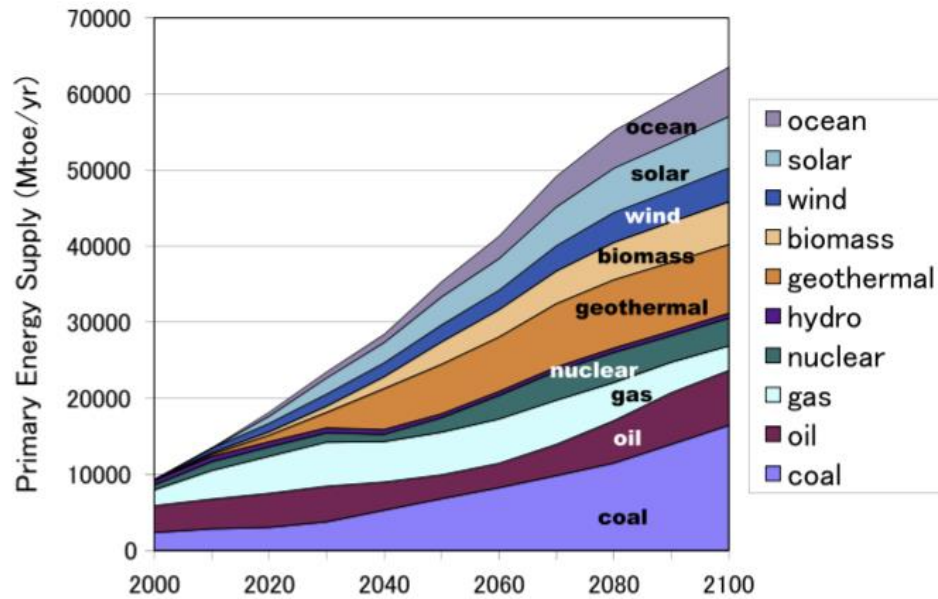


Figure 4. Global primary energy supply structure [6]

1.2 Biomass gasification in fluidized bed

“Biomass is contemporaneous (non-fossil) and complex biogenic organic–inorganic solid product generated by natural and anthropogenic (technogenic) processes, and comprises: (1) natural constituents originated from growing land and water-based vegetation via photosynthesis or generated via animal and human food digestion; and (2) technogenic products derived via processing of the above natural constituents” [7]. Biomass consists of a broad range of material, it includes woody materials like pellets, sawdust, herbaceous and agricultural residues such as straws and husk as well as aquatic materials like seaweed and microalgae. The components of biomass include cellulose, hemicelluloses, lignin, extractives, lipids, proteins, simple sugars, starches, water, HC, ash, and other compounds [7].

From environmental point of view, biomass has negligible content of sulfur, nitrogen and ash, which gives lower emissions of sulfur dioxide (SO₂), nitrogen oxides (NO_x) and soot than conventional fossil fuels. That said, the most important characteristic of biomass that makes it an attractive option for fuels, power and chemicals production is its carbon neutrality.

When an energy production activity produces no net increase in greenhouse gas (GHG) emissions on a life-cycle basis typically is classified as “carbon neutral”. Figure 5 illustrates a simplified notion of cyclic carbon neutrality [8].

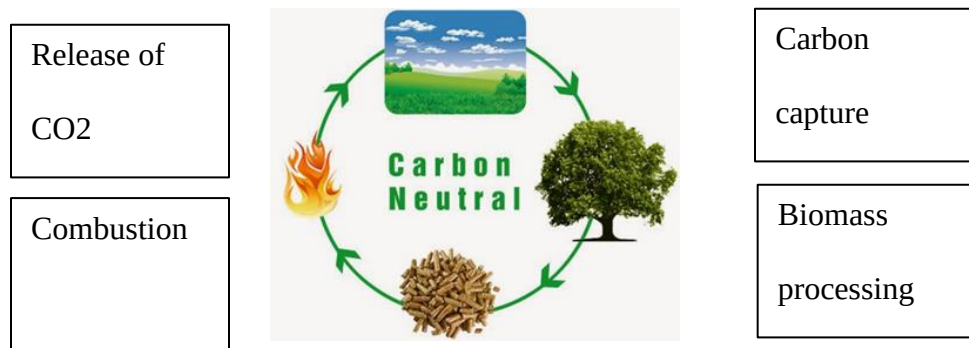


Figure 5. Biomass carbon neutrality cycle [8]

A comprehensive report from IEA Bioenergy T38 [9] shows that the CO₂ balance between growth and harvesting of biomass in sustainably managed forests, and combustion of woody biomass in comparison with fossil fuels with regard to CO₂ emissions leads to the conclusion that biomass harvested from sustainably managed forests should be regarded as “carbon neutral” as the vitality and CO₂ absorption is sustained and kept on the same level [10].

Gasification is one of the several routes for biomass-based energy production. These routes include liquefaction, pyrolysis, supercritical water extraction, gasification, air–steam gasification (Figure 6). This study is focused on biomass gasification.

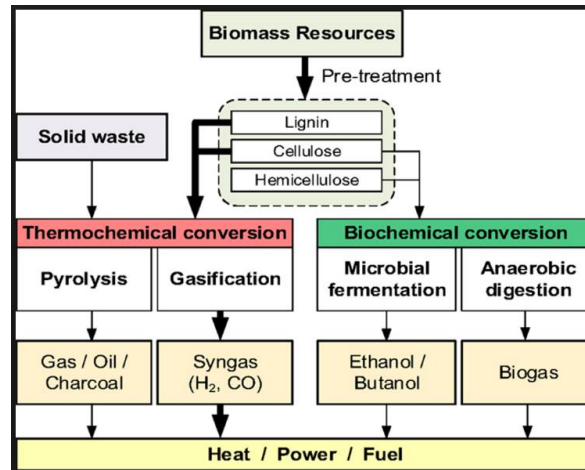


Figure 6. Main routes of biomass conversion [11]

Gasification is the result of partial oxidation of carbonaceous solid fuel with a gasifying agent, usually air, oxygen or steam. The schematic in Figure 7 depicts the major constituents of the gasification process:

1. Feed streams including solid fuel, oxidant and steam.
2. Pressure and temperature under which the gasifier operates.
3. Gasifier output which is a gaseous mixture of H_2 , CO , CO_2 ... and is interchangeably called produced gas, product gas or syngas.
4. Application of interest or final product.

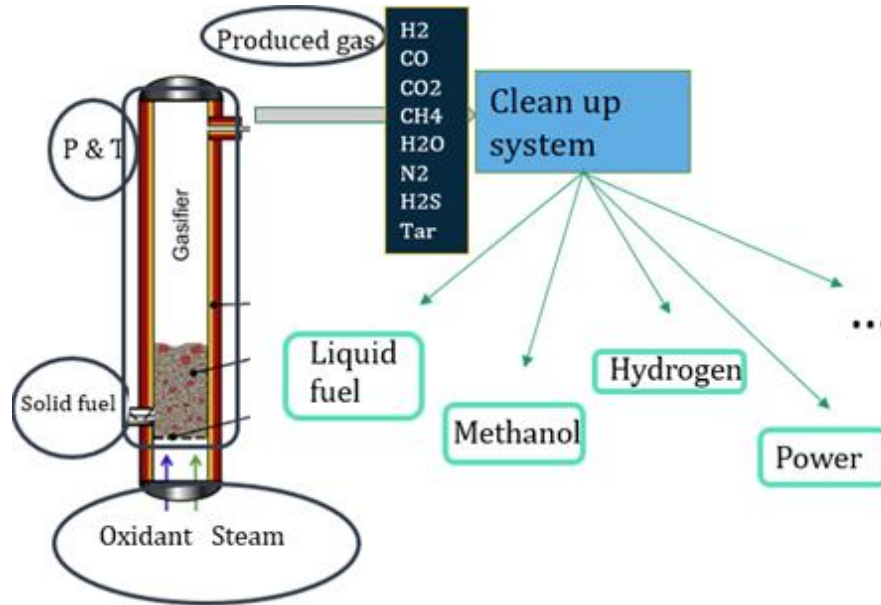
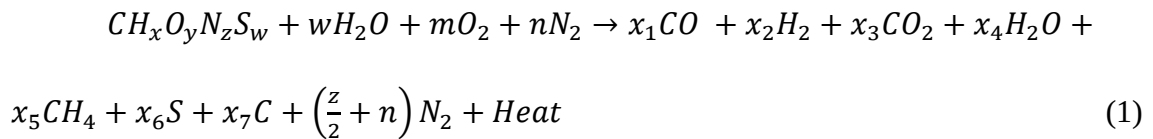


Figure 7. Schematic of gasification process

Gasification consists of many sub-processes, such as drying, devolatilization, pyrolysis and partial oxidation and comprises complicated combinations of numerous reactions which are listed in Table 1 from Gomez [12]. Drying of the feedstocks takes place until 120 °C and volatile species are generated below 500 °C. Gasification of char can begin at around 350 °C. The heat supplied can be intrinsically produced via exothermic combustion reactions or can be supplied from outside sources [13].

A simplified global biomass gasification reaction can be represented by Eq. (1):



Where w , m and n are the molar quantity of water, oxygen and nitrogen per mole of biomass fuel, respectively. Molar indices x , y , z and w can be calculated from ultimate and proximate elemental analysis of feedstock.

Table 1. Major reactions in gasification process [12],[14]

Category	Reaction	$\Delta H_{298} \left(\frac{\text{kJ}}{\text{mol}} \right)$	Name
Char Combustion	$C + O_2 \rightarrow CO_2$	-394	Complete combustion
	$C + 0.5O_2 \rightarrow CO$	-111	Partial combustion
Char Gasification	$C + 2H_2 \rightarrow CH_4$	-75	Hydrogen Gasification
	$C + CO_2 \rightarrow 2CO$	173	Boudouard
	$C + H_2O \rightarrow H_2 + CO$	131	Steam Gasification
Homogenous volatile oxidation	$CO + 0.5O_2 \rightarrow CO_2$	-283	Carbon monoxide oxidation
	$H_2 + 0.5O_2 \rightarrow H_2O$	-242	Hydrogen oxidation
	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	-283	Methane oxidation
	$CO + H_2O \rightarrow H_2 + CO_2$	-41	Water gas shift
Tar reaction	$C_nH_m + \left(\frac{n}{2}\right)O_2 \rightarrow nCO$ $+ \left(\frac{m}{2}\right)H_2$		Partial oxidation
	$C_nH_m + nCO_2 \rightarrow 2nCO$ $+ \left(\frac{m}{2}\right)H_2$	Highly endothermic	Dry reforming
	$C_nH_m + nH_2O \rightarrow nCO_2$ $+ \left(\frac{m}{2} + n\right)H_2$	+(200-300)	Steam reforming
	$C_nH_m + \left(2n - \frac{m}{2}\right)H_2 \rightarrow nCH_4$		Hydrogenation
	$C_nH_m + \left(\frac{m}{4}\right)CH_4 + \left(n - \frac{m}{4}\right)C$		Thermal cracking

A gasifier can process a broad range of organic carbon containing material such as agriculture and forestry residue, negative value feedstocks such as coke and municipal solid waste into product gas. Although the versatility feature of gasification technology is a great advantage, it also leads to complication of the process, mainly because accepting ultra-wide range of feed is a two-sided issue. On one hand, it provides an operator more options and freedom to adjust its operation with respect to changing conditions for

example the price and availability of a feedstock in region, on the other hand, it makes it difficult to predict the output of a gasifier.

For output, the quality of product gas is usually specified in the context of downstream application. These quality specs primary include key characteristics of gas composition, namely H₂:CO ratio (Fischer Tropsch applications), amount of hydrocarbon like CH₄, C₂H₆, ... and its heating value (Fuel gas), and CO₂ concentration (Choice of CCS technology).

In addition to product gas composition, the product gas yield $\left(\frac{Nm^3}{Kg\ biomass}\right)$ and carbon conversion efficiency (*CCE* %) are two influential parameters in design (vessel sizing and scale up) and operation (load adjustment).

Examples of quality specifications that are considered favorable for a specific application are listed in Table 2.

Table 2. Product gas quality specification [15]

Product	Synthetic fuels	Methanol	Hydrogen	Fuel gas	
	FT Gasoline & Diesel			Boiler	Turbine
—	0.6	~2.0	High	Unimportant	Unimportant
	Low	Low	Unimportant	Not critical	Not critical
Hydrocarbons	Low	Low	Low	High	High
	Low	Low	High	Low	
Pressure, bar	~20-30	~150(Vapor phase) ~50(Liquid phase)	~28	Low	~400
Temperature, C	200-300	100-200	100-200	250	500-600

It can be seen for example, when a product gas is intended to be used as fuel gas, maximizing the heating value is favorable whereas if the overall objective is to eventually manufacture methanol, meeting a specific ratio of hydrogen to carbon monoxide of gas (~ 2.0) is important and its heating value is unimportant.

For any application of interest, the product gas composition, yield and conversion depend on:

1. Biomass class, type and properties,
2. Operating condition like pressure, temperature and oxidant to fuel ratio

$$\left(\frac{Air}{Biomass}, \frac{H_2O}{Biomass} \right),$$

3. Type of gasifier

Major advantage of gasification is its versatility in terms of accepting a wide range of feedstocks and having many applications for the produced gas such as “Fuel gas for direct firing in thermal applications, such as kilns and boilers, cofiring in existing coal-fired boilers, gas for engines, turbines and fuel cells generating electricity” [12] as well as other fuels and chemicals.

Gasifiers can be classified by type of gasification agent (air blown, oxygen blown, ...), heat source (direct, indirect), pressure (atmospheric, pressurized) and reactor bed type. Most of the commercially available gasifiers fall into these 3 categories:

- Fixed-bed gasifiers (also referred as moving-bed gasifiers):
 - This type of reactor is simple to construct and generally has high carbon conversion.
- Entrained-flow gasifiers:

- Coal particles concurrently react at high speed in a suspension mode. Short residence time, high temperature, high pressure and large capacities are other features of this type of reactor.
- Fluidized-bed gasifiers (Circulating/Bubbling):
 - This type of reactor processes crushed/shredded solid fuels at lower temperature and it is known for high fuel flexibility, good gas solid contact, high conversion efficiency and good scale up potential. [15]

Among the existing gasifiers, the fluidized bed gasifier has many advantages, such as high flexibility regarding feedstock type and size, easy scale up, uniform temperature distribution, and high carbon conversion efficiency [12]. The flexibility of fluidized bed gasifier in accepting various types of feed allows the optimization of operation with respect to the price and availability of feedstock in a region.

1.3 Modeling

Modeling biomass gasification in fluidized bed can be undertaken with different purposes such as concept studies, preliminary design, or complex simulation of a single unit. An overall impression from current state of fluidized bed gasification modeling is lack of computational reliability. Current models are limited mainly due to lack of generality and sufficient accuracy in predictivity [12],[16][17].

The underlying reasons that give rise to these issues can be categorized as following:

- a) Complexity of fluidized bed dynamics. Gasification is a complex multiphase reactive solid/gas system that involves several chemical and physical sub processes such as drying, devolatilization, pyrolysis, etc [16].

It is not completely clear which processes are dominant at each stage of the gasification. The closest representation of the real system would require a rigorous solution of mass, energy and momentum equations for all phases and species, and an accurate knowledge of the process kinetics. However, there are multiple obstacles to obtain generalized gasification kinetics.

- b) Limitations of experimental data. Despite availability of a large number of fuel option for gasification, experimental studies in fluidized beds are only available for limited types of fuels and ranges of temperature, air and steam to biomass ratios.

For the available data in literature, often, the information regarding minor components of product gas (e.g. H_2S, NH_3), tar, soot, and slug is not reported which makes it difficult to attain mass and energy closure on the overall gasification process.

Moreover, the experimental finding regarding the relationship between a pair of design parameter/product gas characteristic (e.g. impact of moisture content of biomass on $\frac{H_2}{CO}$ ratio in product gas, or impact of gasification temperature on product gas heating value) significantly varies across different cases.

- c) Biomass characterization. One of the most influential aspects of gasification process is feedstock. Biomass is a heterogenous complex biogenic organic–inorganic solid that is described with multiple levels of composition such as elemental analysis, proximate analysis, structural

components, and ash composition to distinguish its different types (Table 3.).

Table 3. Biomass characterizations

Ultimate analysis (Elemental composition)	C, H, N, O, S, Cl
Proximate analysis	Volatile matter, Fixed carbon, Ash content
Structural component	Cellulose, Hemicellulose, Lignin
Ash composition	SiO ₂ , CaO, K ₂ O, P ₂ O ₅ , Al ₂ O ₃ , MgO, Fe ₂ O ₃ , SO ₃ , Na ₂ O, TiO ₂

Although numerous studies [18], [19], [20] have demonstrated that each of biomass characteristics specially ash profile of biomass can have significant catalytic impact on the process, based on our literature review, none of the models has attempted to consider the impact of ash composition and structural components of biomass along with other properties. Most models use elemental composition and a few specific types of biomass such as examples listed in Table 4 & 5.

- d) A large variety of biomass classes. Biomass can be classified in several groups such as woody, herbaceous, aquatic, animal waste, municipal solid waste, and etc. These groups themselves can be classified to other sub groups based on different criteria such as origin, organic component, minerals and mode of element occurrence (Figure 8) [7].

Table 4. Examples of models using specific types of biomass and characterization

Model Ref	Biomass type	Type of biomass characterization used in modeling
[21]	Sorghum Wood chips Dairy manure	Ultimate analysis (Elemental composition)
[22]	Coconut shell Charcoal	Ultimate analysis
[23]	Pine Sawdust Rice husk	Ultimate analysis
[24]	Various	Proximate analysis

Table 5. Example of models developed for specific temperature ranges and biomass type

Model Ref	Temp	Feedstock	Remarks
[25]	1073-1573	Sewage sludge- SB coal	Equilibrium based-BFB
[26]	920-1323	Biomass: beech chips	Equilibrium based-Dual bed
[27]	1273-1323	Agricultural waste	Equilibrium based-BFB 2 zones
[28]	700-900	Pine sawdust size: 0.25-0.75mm	Fluidization model-Based on modified Two-Phase Theory
[29]	Not reported	Woodchips-Almond Shell Grass Straw Cornstalk RDF pellets	Fluidization model-Based on Two-Phase Theory-Reaction only in the emulsion phase

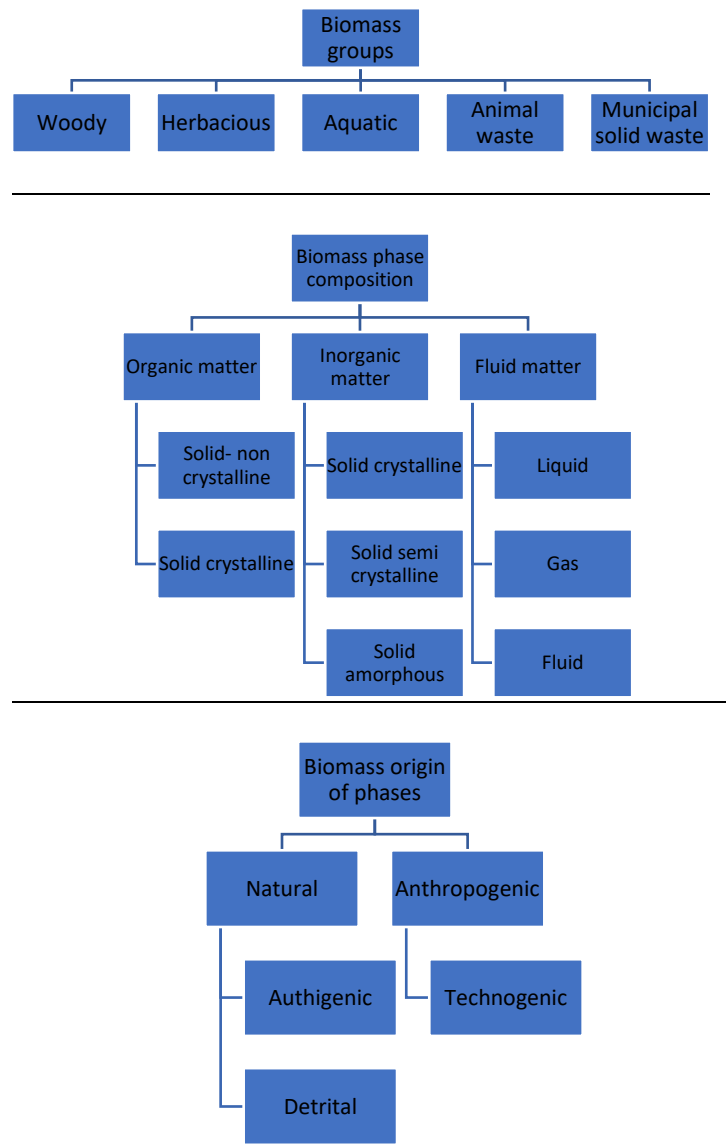


Figure 8. Biomass classification based on source and origin

1.4 Research Objective

The overall objective of this study is to develop a general model for biomass gasification in fluidized bed suitable for first stage analysis by adopting effective strategies to cope with complex dynamics of the process, limitation of experimental data, and lack of proper characterization of solid fuel. The research approach of this study is to

first develop a comprehensive and diverse data base of biomass gasification in circulating and bubbling fluidized bed. Next step is to resolve data quality issues such as missing values, irregular cardinality, and mass and energy balance closure of data sets. The progression from experimental data to insight to model starts with applying a sequence of machine learning predictive data analytics to extract features and patterns related to many properties of biomass that in previous models that have not been taken into account for, namely the structural component of biomass (cellulose, hemicellulose, lignin), the proximate analysis of biomass fuel (volatile matter, fixed carbon, ash content), and ash composition. This study aims to extend the generality of the model through a systematic quantification of collective impact of different critical factors in play to make it applicable to a wide range of biomass classes. From practical point of view, it is desirable to maintain the computational time of model reasonably short (i.e. close to the black box models). Such a model will allow one to optimize the blend of biomass fuel and operating condition to meet desirable specifications for various applications of interest including power, hydrogen, methanol and Fischer-Tropsch hydrocarbons.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

In this chapter, different components of the biomass gasification process are discussed in detail. Mainly, the influence of critical process parameters such as temperature, oxidant to fuel ratio and nature of feedstock, are illustrated by examples. This includes cases that study the impact of influential parameters on product gas composition, carbon conversion, and/or tar formation. In particular, the emphasis is on demonstrating the sporadic state of experimental data and analyses regarding the impact of process parameters across different types of biomass type and operating condition.

In the second part, various modeling strategies in biomass gasification in fluidized bed are discussed, followed by examples of generality and predictivity issues. The literature review suggests the existing models lack holistic approach to cope with irregular patterns that arise when a certain change in a process parameter has multiple possible consequences depending on the operating conditions. This major bottleneck appears to be the main factor of non-generality of models.

2.2 Biomass as fuel source

Biomass is quite a complicated system with highly variable characteristics that depend on various factors. The most efficient utilization of biomass requires a systematic characterization of its composition and properties. Vassilev et al. [7] conducted an extended overview of biomass varieties as solid fuel resources according to their biological diversity, source and origin, and reported an extremely high variation of moisture, ash yield (shortly ash) and different genetic types of inorganic matter in

biomass along with significant distinctions among the specified natural and anthropogenic biomass groups and sub-groups.

It was also found that “the major and minor elements in biomass, in decreasing order of abundance, are commonly C, O, H, N, Ca, K, Si, Mg, Al, S, Fe, P, Cl, Na, Mn, and Ti. The biomass chemical composition is significantly different from that of coal and the variations among biomass composition were also found to be greater than for coal. Natural biomass is normally: (1) highly enriched in $Mn > K > P > Cl > Ca > (Mg, Na) > O > \text{moisture} > \text{volatile matter}$; (2) slightly enriched in H; and (3) depleted in ash, Al, C, Fe, N, S, Si and Ti in comparison with coal [7].” A general classification presented in Vassilev et al. [7] is given in Table 6.

These different groups and subgroups of biomasses presented here are only one of the multiple ways to classify this complex solid fuel. Depending on the purpose of the analysis, biomass can be characterized based on any single or bundle of the characteristics listed shown Figure 9.

The type of feedstock influences the tar and char yield[30]. For example, “it was found in experiments that the maximum tar production was around 60% for paper, 35% for wood and only 30% for sawdust during the thermal degradation process [31]”. This demonstrates that the gasification processes are heavily influenced by the biomass characteristics. In addition to chemical composition, moisture and inorganic species, ash content can play an important role in defining a biomass feedstock[12], [15], [16]. This is particularly important since high ash content results in a decrease in conversion efficiency owing to ash agglomeration [32]. Therefore, a model aimed to predict the conversion efficiency of the gasification needs to take this factor into account. However, biomass ash

is a highly variable heterogeneous mixture of organic, inorganic and fluid matter that its impact on the process cannot be easily evaluated (Table 7) [33].

Table 6. A general classification of biomass [7]

Biomass group	Biomass sub-groups, varieties and species
<u>Wood and woody biomass</u>	Coniferous or deciduous; angiospermous or gymnospermous; soft or hard; stems, branches, foliage, bark, chips, lumps, pellets, briquettes, sawdust, sawmill and others from various wood species
<u>Herbaceous and agricultural biomass</u>	Grasses and flowers (alfalfa, arundo, bamboo, banana, brassica, cane, cynara, miscanthus, switchgrass, timothy, others), Straws (barley, bean, flax, corn, mint, oat, rape, rice, rye, sesame, sunflower, wheat, others), Other residues (fruits, shells, husks, hulls, pits, pips, grains, seeds, coir, stalks, cobs, kernels, bagasse, food, fodder, pulps, cakes, others)
<u>Aquatic biomass</u>	Marine or freshwater algae; macroalgae (blue, green, blue-green, brown, red) or microalgae; seaweed, kelp, lake weed, water hyacinth, others
<u>Animal and human biomass wastes</u>	Bones, meat-bone meal, chicken litter, various manures, others
<u>Contaminated and industrial biomass wastes</u>	Municipal solid waste, demolition wood, refuse-derived fuel, sewage sludge, hospital waste, paper-pulp sludge, waste papers, paperboard waste, chipboard, fibreboard, plywood, wood pallets and boxes

Table 7. Phase composition of biomass ash. [33]

Matter	State and type of constituent	Phase, mineral and practice morphology
Inorganic matter	Solid, noncrystalline (amorphous)	Spherical and spheroidal glass particles (massive, vesicular, cenospheres, plerospheres, dermaspheres, ferrospheres, ferrispheres, others)
	Solid, noncrystalline (amorphous)	Angular, irregular and other aspherical glass particles (massive, hollow, vesicular, fibres, needles, cylinders, discs, slabs, sheets, lacy, spongy, porous, elongated, rounded, agglomerates, others)
	Solid, crystalline to semi-Crystalline (mineral)	Crystals, grains, aggregates, clusters, ferrospheres, ferrispheres, skeletal spheres and spheroids, agglomerates and others of various minerals or poorly crystallized mineraloids (silicates, oxides and hydroxides, sulphates, sulphides, sulphosalts, sulphites, thiosulphates, phosphates, carbonates, chlorides, chlorites, chlorates, nitrates, native elements, pure metals, metallic alloys, others)
	Solid, crystalline (mineral)	Crystals, grains, and aggregates of organic minerals (whewellite-weddellite and other groups)
Organic matter	Solid, noncrystalline (amorphous) to semi-crystalline	Char including semi-coked, coked, soot and tar particles (angular, irregular, rounded, massive, hollow, vesicular, spongy, porous, skeleton, needles, whiskers, fibres, scaly, flaky, sheets, cluster of rosettes, cylinders, discs, slabs, spherical, spheroidal, agglomerates, others)
Fluid matter	Fluid, liquid, gas	Moisture and gas and gas-liquid inclusions associated with both inorganic and organic matter

2.3 Temperature, ER, gasifying agent, residence time

In addition to important role of feedstock, numerous experimental studies [34], [35], [36] have demonstrated the critical impact of gasification temperature on the gasifier output. Typically, a higher gasification temperature results in higher char conversion and lower tar yield, and presence of heavier hydrocarbons in the gas product [12]. Therefore, it is desirable to have a higher gasification temperature. Typical ranges of gasification temperatures for agricultural waste fuels, residue derived fuel (RDF), and woody biomass were reported between 750 - 850 °C, 800 - 900 °C, and 850 - 950 °C

respectively [37]. However, “The desire for lower ash accumulation necessitates lower temperatures which practically restrict the temperatures up to 750 °C during biomass gasification. It is found that H₂ yield at equilibrium initially increases to a limit, then decreases, with a rise in temperature. Therefore, the optimal operational gasifier temperature can only be determined after evaluating several key parameters such as desired gas composition and calorific content, tar yield and char conversion.

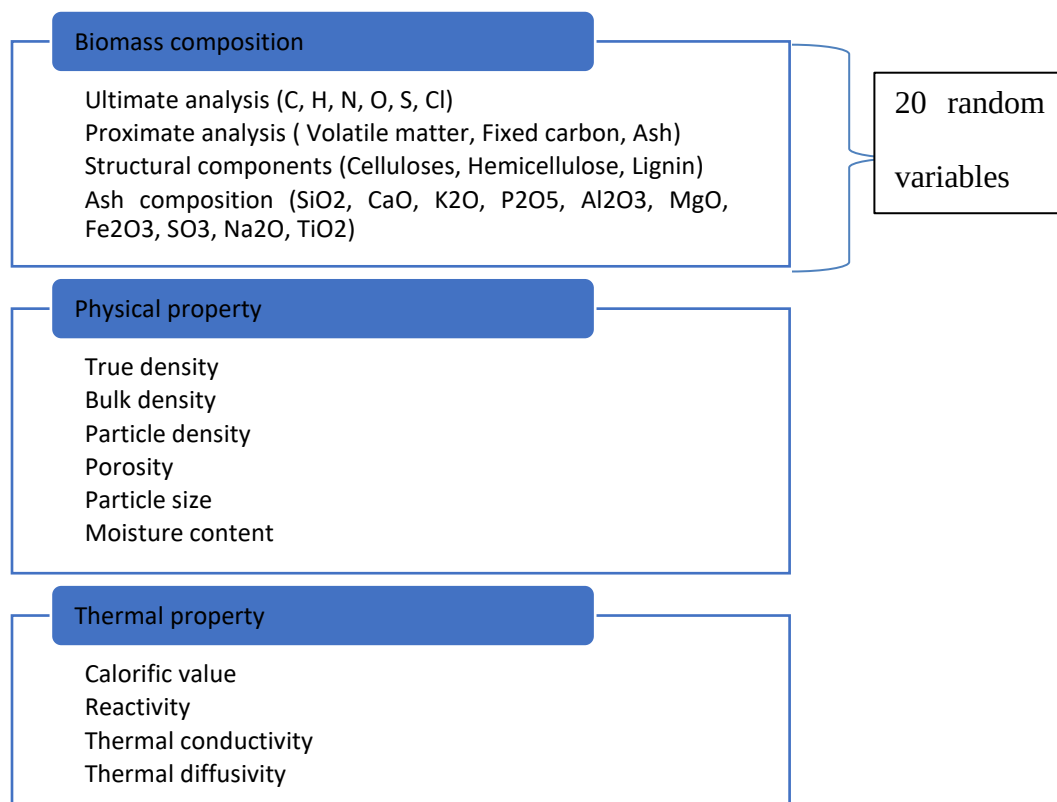


Figure 9. Various defining characteristics of biomass fuel

Pressurized gasification of biomass was studied by Knight et al. [38] and Wang et al. [39]. Higher pressures with greater ERs result in the decrease of light hydrocarbons and tar yield along with complete conversion of carbon. The quantity of polycyclic

aromatic hydrocarbons (PAH) increases with a rise in gasifier pressure, although an overall decrease in the tar was observed.” [40]

Gasifying agent: the most common mediums used in gasification process are air, steam, and carbon dioxide. The heat content of the product gas is a function of the blend of gasification medium. The product gas from air gasification gives a heating value of around 4 to 7 MJ/Nm³ since N₂ in the air dilutes syngas and results in low calorific content [41], [42], [43]. When the pure O₂ is used, the heating value of syngas can be as high as 28 MJ/Nm³ [44]. Steam or a combination of steam and O₂ produces syngas of intermediate calorific value (10 to 18 MJ/Nm³) [42]. Furthermore, the blend of air and steam yields higher quantities of H₂ than air alone, and also helps in decreasing the process energy requirements. One of the great challenges of gasification modeling is to predict the heating value of the product gas. This arises due to the fact that the process not only can be run by many possible blends of air/O₂/steam/CO₂ but also the impact of a certain blend or ratio of oxidants to fuel has different outcomes depending on the nature of feedstock that is being applied on. Gasification occurs in a restricted oxygen environment, the actual air-to-biomass ratio with respect to the stoichiometric requirement for complete conversion (ER) plays a key role in the process [45]. Studies [46], [47] showed a decrease in ER resulted in increase of H₂ and CO in syngas. However, it was also seen that when the ER is reduced below a certain threshold, the overall yield of syngas starts to decrease.

In another experimental study [48] on different residual biomass fuels for circulating fluidized bed gasification that concluded the ratio of steam to biomass (SBR)

is a key process parameter since it has a significant impact on the product gas yield and heating value, the data presented in Table 8.

Table 8. An experimental study considering multiple fuels with limited operating condition [48]. Numerical results of gasification tests using different biomass fuels. Abriviation: ar: as received, HHV: efficiency, na: not analysed, HC: hydrocarbons up to naphthalene

Table no.	0	1	2	3	4a	4b	5
Fuel A	Willow	Demolitio n wood	Park wood	Chip board material s	Verge grass	Verge grass	Demolitio n wood +paper residue sludge
Process conditions							
Fuel flow kg/h ar	69	61	56.2	60.7	87.8	102.6	82.6
Water content of fuel wt% ar	17.5	6.1	3.5	10.8	7.3	7.3	10.2
Energy input (HHV) kW	323	324	305	303	386	450	367
Temperature °C	827	847	861	843	803	815	829
Fuel gas							
CO vol% dry	9.40	11.35	11.65	9.64	9.89	10.64	9.24
H ₂ vol% dry	7.20	7.05	6.77	6.42	7.28	8.45	6.08
CO ₂ vol% dry	17.10	15.75	15.51	15.63	15.61	15.18	16.11
CH ₄ vol% dry	3.30	3.25	3.17	2.77	2.58	2.79	2.81
C ₂ H ₄ vol% dry	1.10	0.85	0.99	0.87	1.15	1.04	1.02
C ₂ H ₆ vol% dry	0.10	0.03	0.04	0.03	0.05	0.04	0.04
Benzene vol% dry	0.21	0.21	0.25	0.22	0.12	0.13	0.19
Toluene vol% dry	0.065	0.031	0.043	0.040	0.035	0.041	0.054
Xylene vol% dry	0.013	0.005	0.009	0.007	0.004	0.005	0.070
Estimated remaining HCs	0.26	0.19	0.22	0.20	0.25	0.23	0.24

While findings in such experimental studies improve understanding the mechanism of the process, the limitations of the experiments from which they are drawn, significantly complicates inference of general assessments. For example, the experimental data of the study presented in Figure 10 can be used to analyze impact of [ER, time] on [HHV, Yield], but the assessments cannot be generalized to all biomass types because the experiment only used coir pith as feedstock.

Sl. No	ER	Time, min	CO, %	CO ₂ , %	H ₂ , %	CH ₄ , %	N ₂ , %	O ₂ , %	Gas Yield, MJ Nm ³ kg ⁻¹	HHV, Nm ⁻³
1.	0.30	10	08.71	14.62	08.49	1.52	66.41	0.25	1.98	2.65
2.		20	09.26	14.88	07.89	1.36	66.38	0.23	2.02	2.58
3.		30	10.12	12.12	09.54	1.89	66.08	0.25	2.26	3.08
4.		40	10.28	11.63	10.17	1.53	66.21	0.18	2.18	3.04
5.		50	11.53	11.58	10.55	2.67	63.50	0.17	2.21	3.67
6.		60	12.57	11.05	10.61	3.82	61.78	0.17	2.32	4.23
7.	0.40	10	08.24	15.61	07.54	1.08	67.21	0.32	2.24	2.31
8.		20	09.21	15.22	08.62	1.59	65.12	0.24	2.12	2.75
9.		30	10.52	14.17	08.79	1.32	65.00	0.20	2.24	2.82
10.		40	11.12	12.83	09.53	2.54	63.77	0.21	2.68	3.44
11.		50	10.91	11.69	10.46	2.50	64.25	0.19	2.52	3.52
12.		60	12.68	11.83	10.57	3.60	61.14	0.18	2.69	4.16
13.	0.50	10	06.54	16.24	05.62	1.91	69.37	0.32	2.73	2.18
14.		20	08.94	15.48	06.87	1.21	67.29	0.21	3.14	2.36
15.		30	08.31	14.32	07.23	1.86	68.00	0.28	2.83	2.57
16.		40	10.52	13.67	09.61	1.33	64.68	0.19	3.24	2.93
17.		50	11.61	13.14	09.78	2.42	62.88	0.17	2.90	3.49
18.		60	12.47	12.68	09.64	2.91	62.19	0.11	2.84	4.14

Figure 10. Experimental data with modest ranges for ER and residence time but only for one type of fuel (i.e. Coir pith)[49]

There's another research method in which the gasification process is divided into a series of sub processes (i.e. drying, devolatilization, and pyrolysis...) to make it easier to study impact of operational variables on various fuels. This strategy essentially aims to maximize likelihood of finding less ambiguous trends in each zone by temporarily neglecting the less dominant processes.

One of the critical stages during gasification process is pyrolysis which strongly impacts the amount of tar and heavy hydrocarbons in product gas [16]. In this method, “the release of volatile gases in the pyrolysis zone needs to be predicted for the estimation of the gas produced in the reduction zone, where conversion of char, tar and

methane takes place [12].” Figure 11 illustrates the position and critical role of pyrolysis step with respect to the overall process.

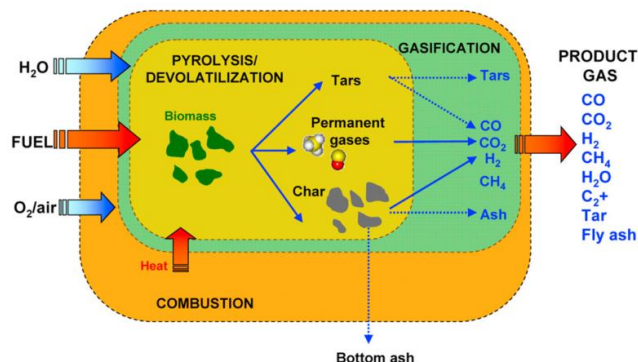


Figure 11. Pyrolysis/devolatilization of biomass fuel [12]

Wedneqin et al [50] studied impact of biomass ash content on pyrolysis yield and economic and environmental of biofuel production. The results show that high ash content of biomass leads to greater biochar (char) formation and has a negative impact on the fuel yield. “Biomass ash content of 0.3–7.7 wt% increases biochar yield from 0.13 to 0.16 kg/kg of biomass, and decreases biofuel yields from 87.3 to 40.7 gallons per tonne. Higher O/C ratio (0.88–1.12) in biomass decreases biochar yield and increases biofuel yields within the same ash content level. Higher ash content of biomass increases minimum fuel selling price while higher O/C ratio of biomass decreases price within the same ash content level. The impact of ash and O/C ratio of biomass on GHG emissions are not consistent for all feedstocks (Figure 12).”

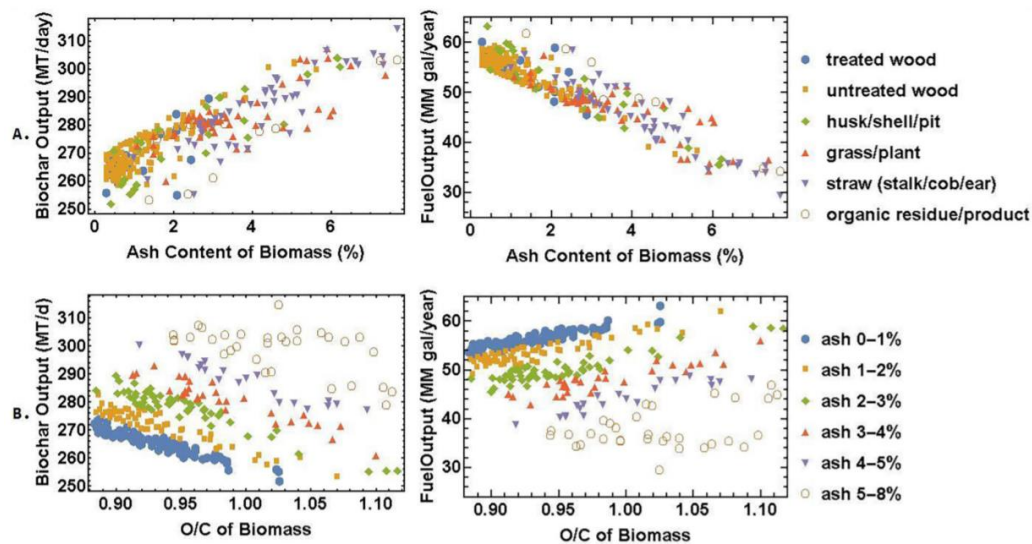


Figure 12. A. The impact of ash content of biomass on biochar and fuel output B. The impact of O/C of six biomass ash content.[50]

In another example of experimental study [51] two woody biomass fuels Agrol and Willow, and one agricultural residue Dry Distiller's Grain with Solubles (Figure 13) were tested using an atmospheric circulating fluidized bed gasifier. the effect of operational variables (i.e. SBR, ER, Temperature, and bed material) on the product gas composition and tar formation from these fuels were investigated and the experimental results show there is significant variation in the composition of the product gas produced.

Xiangmei et al [51] reported "Among all the experiments, the averaged concentration of H₂ obtained from Agrol, willow and DDGS over the temperature range from 800 to 820 C was around 24 vol.%, 28 vol.% and 20 vol.%, respectively, on a N₂ free basis. A fairly high amount of H₂S (w2300 ppmv), COS (w200 ppmv) and trace amounts of methyl mercaptan (<3 ppmv) on a N₂ free basis were obtained from DDGS. Due to a relatively high content of K and Cl in DDGS fuel, an alkali-getter (e.g. kaolin) was added to avoid agglomeration during gasification. Higher temperatures and SBR values were favorable for increasing the mole ratio of H₂ to CO and the tar

decomposition but less advantageous for the formation of CH₄.” The results for three types of biomass (Figure 13) are shown in Figures 14-17.

Types of fuel	Agrol	Willow	DDGS
Moisture (wt.% a.r.)	8	8	12
Proximate Analysis (wt.%, dry)			
Volatile matters	74.7	69.8	67.2
Fixed carbon	16.0	20.1	15.5
Ash content [%]	0.14	2.52	4.82
Ultimate Analysis (wt.%, dry)			
C	51.0	50.3	48.2
H	6.26	6.17	6.54
O	38.2	37.4	31.2
N	0.15	0.69	5.52
S	0.002	0.002	0.76
Cl	0.01	0.01	0.21
Elemental Ash Analysis (wt.% of Ash at 550 °C)			
Al ₂ O ₃	2.68	4.23	0.05
BaO	0.41	0.06	0.01
CaO	33.30	26.31	2.87
Fe ₂ O ₃	1.32	1.54	0.26
K ₂ O	16.50	13.13	38.07
MgO	7.79	2.89	8.87
MnO	3.96	0.30	0.15
Na ₂ O	1.00	1.67	10.04
SiO ₂	6.46	33.80	1.86
TiO ₂	0.06	0.18	—
C	5.24	3.61	0.33
Cl	0.07	0.11	0.02
SO ₃	2.46	2.27	10.04

Figure 13. Biomass fuel characteristics [51]

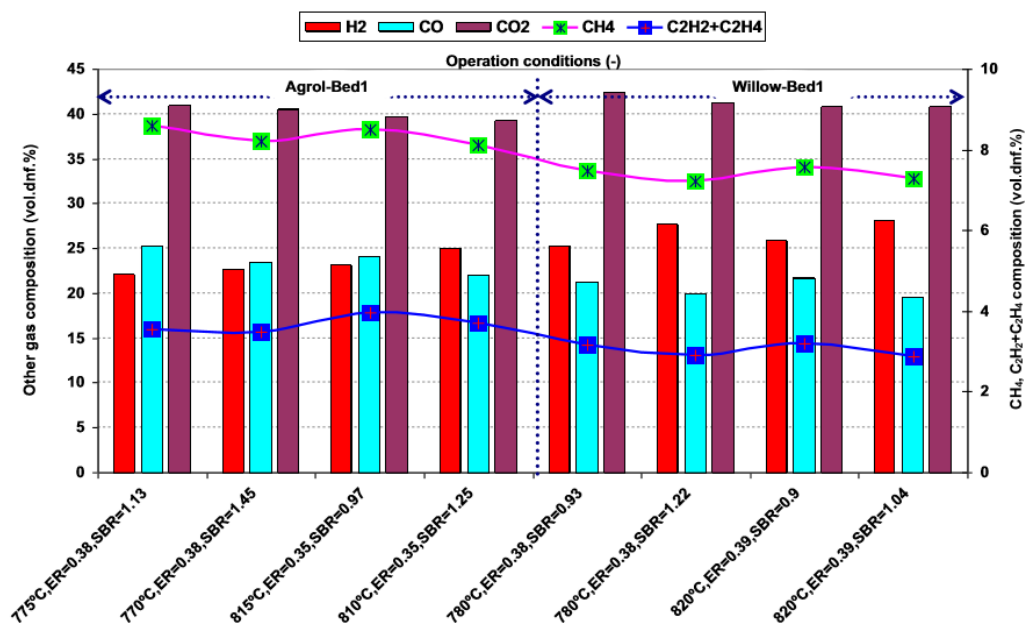


Figure 14. Effect of operational variables on Agrol and Willow using treated Austrian olive as bed material [51]

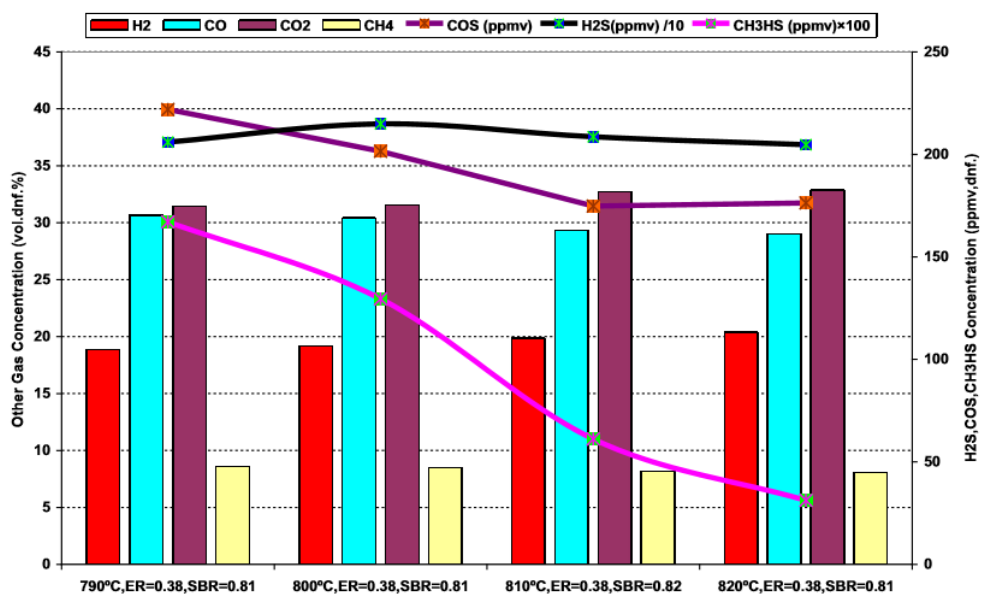


Figure 15. Effect of operational variables on DDGS using Fresh Austrian olive bed material [51]

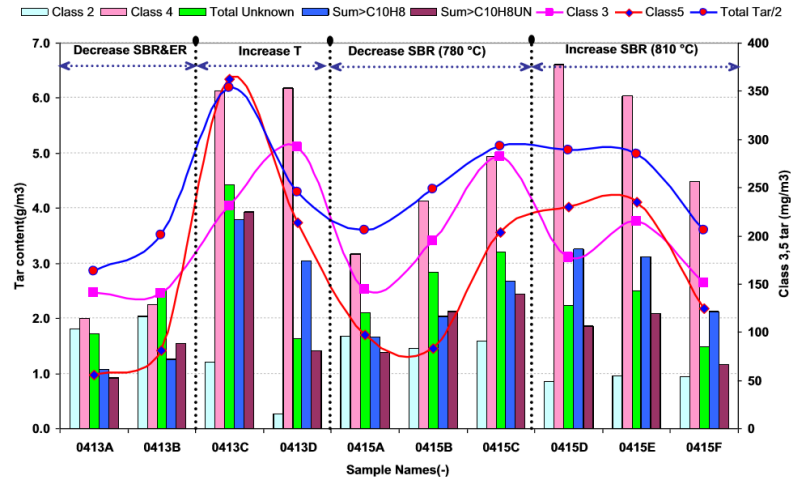


Figure 16. Tar formation under different operating conditions for Argol [51]

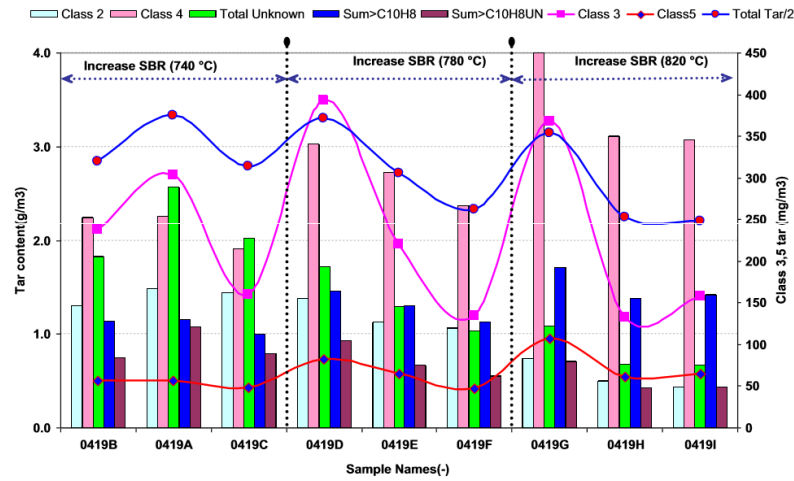


Figure 17. Tar formation under different operating conditions for Willow[51]

The effect of operational variables on formation of different classes of tar is shown in Figure 18.

Class	Name	Tar components
Class 2	Heterocyclic aromatics	phenol, o-cresol, m-cresol and p-cresol
Class 3	Aromatics (1 ring)	benzene, toluene, m/p-xylene, o-xylene
Class 4	Light PAHs (2, 3 ring)	indan, indene, naphthalene, 2-methylnaphthalene, 1-methylnaphthalene, biphenyl, acenaphthylene, acenaphthene, fluorene, phenanthrene, anthracene
Class 5	Heavy PAHs (>3-ring)	Fluoranthene, pyrene

Figure 18. Tar classes [51]

In summary, the reviewed studies clearly show the discussions on effect of temperature, ER, SBR, and biomass properties on product gas characteristics are extremely variable and dependent on different factors that constitute the experiment.

The large number of feedstock options, random variables describing them, and different interactions between these variables, raises an issue, known as curse of dimensionality. “A challenge posed by curse of dimensionality is a statistical challenge because the number of possible configurations of set of random variables is much larger than the number of available experimental data sets”.[52]

2.4 Modeling approaches

According to the reviews by Gomez [12], Mirmoshtaghi [17] and Radmanesh [53] there are two major approaches in modeling the Fluidized bed biomass gasifiers depending on the aim of the study, dynamic and black-box approach.

In dynamic modeling, kinetics and/or hydrodynamics of the bed are considered. The most rigorous form of dynamic modeling is computational fluid dynamics (CFD) which is based on applying mass, species, heat and momentum balances along with formulation of the boundary conditions and use of constitutive relationships for each phase.

A subcategory of dynamic models, fluidization models do not consider the complex gas-solid dynamics in the bed, but still maintain the fluid dynamic effects by assuming a multiphase pattern in the bed and considering different regions in the reactor.

Although this modeling approach has an important place as a design aid in describing many chemical processes, their success significantly depends on the chemical kinetics they use as source term.

The kinetics of gasification is developed only to a limited extent. Mainly because the limited amount of data has made it difficult to explain the inconsistent behavior of different feedstock with variable reactivities. But the reactivity of different biomass depends on a number of factors, in particular the following factors[16].

- The porous structure of biomass, i.e. inner structure, surface and active sites
- The crystal structure of fixed carbon
- The catalytic effects of ash content of biomass

That said, if accurate and detailed information on kinetics and geometry of the reactor is available, the CFD approach gives a better interpretation of the real system. This is usually the case when one type of fuel is being studied.

For example, Anil et al [54] developed a CFD model to predict the effect of equivalence ratio, steam to biomass ratio, steam temperature and air preheating on product gas composition in a circulating fluidized bed. It is found that when equivalence ratio is increased from 0.15 to 0.4, H₂ concentration is decreased by 72% and the variation of steam to biomass ratio from 0.5 to 1.5 enhanced the H₂ concentration by 87%.

Although useful insights are obtained, the only feedstock used in the analysis is sawdust, if the impact of several classes of feedstocks were to be evaluated, the time-consuming procedure of CFD should be repeated for each class and its associated kinetics.

Gomez [12] presented a comparative review of fluidization models for biomass gasification (Figure 19).

BFBFG	Nikoo [359]	Radmanesh et al. [69]	Sadaka et al. [147,360,361]	Fiaschi and Michelini [68]	Bilodeau et al. [212]	Jiang and Morey [67]	van den Aarsen [219]	Raman et al. [362]
A. Model characteristics								
Type of Reactor model	LM	LM	LM	LM	LM	LM	LM	LM
Dimensionality	SS 1D	SS 1D	TM 1D	SS 1D	SS 1D	SS 1D	SS 1D	TM 2D
Bottom bed model	Based on modified Two-Phase Theory with properties varying with height	- Based on KLM - CCBMM for the mixing of char in emulsion	Divided into three zones: jet, bubbling and slugging.	Based on Two-Phase Theory Reaction only in the emulsion phase	Based on Modified Two-Phase Theory	From [219]	Extended Two-Phase Theory with key parameters checked by own experiments	Based on modified Two-Phase Theory with properties varying with height
Freeboard model fluid-dynamics	Based on Eq. (27) with $a = 1.8/u$ and $\rho_w = 0$	No	No	Based on elutriation constants and Eq.(27)	Plug flow in an homogeneous emulsion	Based on Eq. (27) with measured parameter a .	No	No
Reactions in Freeboard	Char reactions assuming perfect mixing	No	No	No	The same as bottom bed with a single gas phase	Gas species determined by WGS in equilibrium	No	No
Thermal balance	T imposed	No for reactor/Yes for particle	For reactor in each zone and also for the particle	For the bottom bed by lumping bubble and emulsion	For bottom bed and freeboard (compartments)	Overall, T_b is input to determine T_b and biomass feed rate	T imposed (adjusted by airflow rate)	No
Devolatilization model	Instantaneous and equilibrium	Yes (2 models) Model 1: Tar generation model from [160,161]. Model 2: from their own tests.	Based on CHO balances with 3 adjustable parameters (values NR)	Instantaneous, NR gas distribution	Instantaneous (equilibrium assumed)	From [219]. Species distribution from own FB test with corn cob	Pyrolysis takes place in a finite region determined by fuel particle size. Species distribution from own FB tests	Relative yield of gas and char from TGA (conducted by authors) Volatile yields distribution internal report
Char particle conversion model (PCM) and reactivity	No PCM Reactivity from coal	No PCM Char- CO_2 reactivity from coal. Langmuir-Hinshelwood kinetics for char- H_2O reactivity	No	PCM from [257].	Kinetic model from [363]	From [219].	Shrinking model with kinetics and physical parameters determined in own measurements in an FB	No PCM. Taken from [302] for coal
Fragmentation and attrition	No	No	No	Simplified model. Parameters NR	No	No	No	No

Figure 19. Fluidization models for bubbling fluidized bed [12]

An alternative to CFD and Fluidization models are black box models that do not deal with the process of inside the gasifier. The essential concept behind black-box models is to have the overall heat and mass over the entire reactor balanced while giving reasonably accurate predictions of gasifier output including product gas composition, yield, and thermal efficiency. Black box models are usually easy to use and require few inputs. Due to simplicity and independency of gasifier design, it is considered a suitable option for the purposes of preliminary and system level analyses.

However, most of black box models have narrow ranges of applicability in terms of type of feedstock and operating conditions (e.g. Table 3) which eventually leads to partially optimized design and operation. Currently, the system level analyses are performed by looking for the least costly mode of operation in a limited design space (i.e. models with correlations fit to experiments in a specific plant (Fitting-data models or empirical models)).

The complexity of black box model and the type of assumptions vary widely from one to another depending of the aim: from simple heat and mass balances to predict the overall performance to the prediction of the product gas and solids composition. Some modelers assume global equilibrium over the entire process while others complement equilibria by imposing chemical reactions in selected reactors with empirical correlations obtained from experiments.

The thermodynamic equilibrium method is independent of gasifier design and relies on the assumption that thermodynamic equilibrium is attained in the outlet streams. Such models, which are based on evaluation of equilibrium constants for reactions under consideration, and non-stoichiometric models, which are based on the direct minimization of Gibbs free energy. In the non-stoichiometry method, no reaction is defined and only species are specified.

Jarungthammachote [22] used the Gibbs free energy minimization to predict the composition of six major components (i.e. CO, CO₂, H₂, H₂O, CH₄ and N₂) in the product gas. At the equilibrium state the total Gibbs free energy of system is minimized.

$$G^t = \sum_{i=1}^N n_i \underline{G}_i \quad (2)$$

Where n_i is the number of moles of species i and $\underline{G}_i$ is the Gibbs free energy of species i that can be represented by:

$$\underline{G}_i(T, P, y_i) = \underline{G}_i^\circ + RT \ln \left(\frac{\widehat{\phi}_i y_i P}{P_i^\circ} \right), \text{ where } \underline{G}_i^\circ \text{ is partial Gibbs free energy of reference state.} \quad (3)$$

Given most fluidized bed gasifiers operate at atmospheric pressure, and high temperatures: we have $\mu_i = \underline{G}_i^\circ + RT \ln (y_i)$ where y_i is mole fraction of species i in the mixture, therefore:

$$G^t = \sum_{i=1}^N n_i \underline{G}_i^\circ + \sum_{i=1}^N n_i RT \ln (y_i) \quad (4)$$

The last step is to find values of n_i which minimizes the objective function G^t . The appropriate method, which is usually performed for minimization of Gibbs free energy is Lagrange multipliers [55]. In order to calculate Gibbs free energy $\underline{G}_i^\circ$ at specific temperature T :

$$\underline{G}_i^\circ = \underline{H}_i^\circ - T \underline{S}_i^\circ \quad (5)$$

The thermodynamic properties that are function of temperature and are described by polynomial equations:

$$\frac{C_p(T)}{R} = a_1 + a_2 T + a_3 T^2 + a_4 T^3 + a_5 T^4 \quad (6)$$

$$\frac{\underline{H}^\circ(T)}{RT} = a_1 + a_2 \frac{T}{2} + a_3 \frac{T^2}{3} + a_4 \frac{T^3}{4} + a_5 \frac{T^4}{5} + \frac{b_1}{T} \quad (7)$$

$$\frac{\underline{S}^\circ(T)}{T} = a_1 \ln \ln T + a_2 T + a_3 \frac{T^2}{2} + a_4 \frac{T^3}{3} + a_5 \frac{T^4}{4} + b_2 \quad (8)$$

The coefficients can be obtained from NASA technical memorandum 4513 [56].

As for energy balance, the first law of thermodynamic is performed for the process by:

$$\sum_{reactant} n_r \underline{H}_r^\circ(T_r) = \sum_{product} n_p \underline{H}_p^\circ(T_p) + Q \quad (9)$$

Q is the heat of gasification process and plays an important role on the gasifier output. However, specifying this critical parameter is one of the greatest challenges of gasification modeling. Because gasification involves many reactions as listed in Table 1 (e.g. char combustion, char gasification, volatile oxidations and tar reactions...). The way heat produced by exothermic reactions gets consumed by endothermic reactions depend on several factors such as nature of biomass (e.g. reactivity of char) and hydrodynamics of the bed [12],[16]. For instance, simple assumptions such as $Q = \%1$ of HHV of feedstock by Altafini et al [57] only stands when applied to experimental data from a specific plant. These issues are countered by modifying the models using correlations and empirical variables, to match the measured data from gasifiers.

The most common method is coupling empirical correlations for major reactions such as water gas shift and methane steam reformation. Since according to many experimental data, these two reactions do not reach equilibrium in most cases (Chapter 3) while having the most impact on major components of product gas.

Many researchers have suggested that to achieve a higher level of accuracy in quasi-equilibrium models, the empirical parameters in the correlations for improving equilibrium models should be adjusted to the experimental data from the same plant that is modeled indicating that such models cannot be generalized.

Kersten [30] reviewed and compared different quasi-equilibrium models for biomass gasification in fluidized bed gasifiers. He studied two methods: 1. implementing empirical correlations in the Schlapfer model [31] and 2. using the quasi-equilibrium temperature (QET) in the Gumz model[58]. Kersten concluded that the Gumz model with QET yields better result for H₂ and CO predictions.

Mirmoshtaghi et al [17] compared predictivity of three modified gasification models with the following conditions (Figure 20, Figure 21).

Models name	MODEL I	MODEL II	MODEL III
Reference to models	[27]	[28]	[21]
Oxidizing agent(s)	Air	Air/Steam	Air/Steam
Temp. range (°C)	730–815	856–955	700–900
ER range	0.22–0.53	0.28–0.39	0.19–0.27
S/B range	—	0.08–0.28	0–4.04
Flowrate (kg/h)	15.5–48.5	18–58	0.445–0.512
Modification methods from EMs	Assigning temperature lower than equilibrium temperature in RGIBBS block for specific components and reactions	Using the empirical correlation for non-equilibrium components in product gas by calculator block	Including reaction kinetics and hydrodynamics of the bed as an external FORTRAN subroutine. Including an empirical nonlinear regression with specific temperatures (700, 750, 800, 850 and 900 °C) for CH ₄ concentration.

Figure 20. Modified equilibrium models simulated in ASPEN plus

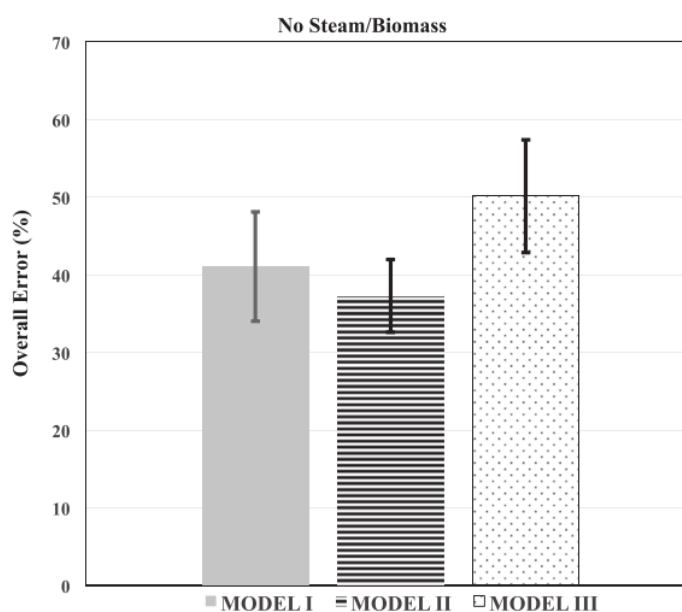


Figure 21. Comparison of models based on average overall error

They concluded that “existing empirical correlations can be further improved by considering more experimental data. The updated model features better accuracy in the prediction of product gas composition in a larger range of operating conditions. Additionally, combining the quasi equilibrium temperature method with a kinetic and

hydrodynamic approach results in a model that features less overall error than the original model based on a kinetic and hydrodynamic approach.”[17]. Hannula and Kurkela [59] developed a semi empirical model for pressurized air blown gasification of biomass using experimental data of two types of sawdust. Eight main blocks were used to model different stages of the process, as illustrated in Figure 22.

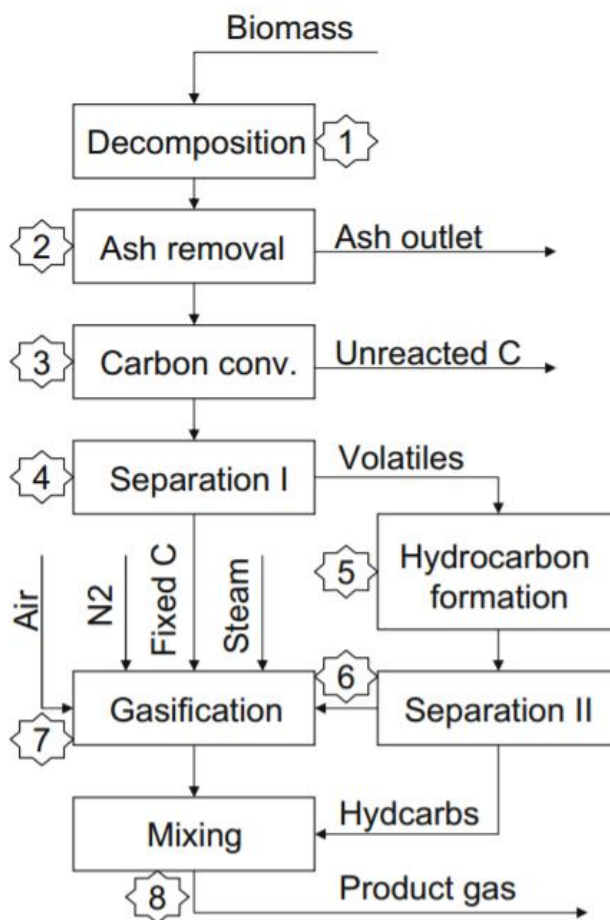


Figure 22. A Schematic illustration of Hannuka model [59]

Empirical correlations were used to overcome two main limitations of equilibrium approach, lack of prediction for carbon conversion and underprediction of CH₄, other

heavy hydrocarbons and nitrogen related compounds. These correlations which are function of equivalent ratio are listed in Figure 23.

Carbon conversion	$25.7 \times E + 88.5$
NH ₃ conversion	$0.819 - 1.154 \times E$
<i>Hydrocarbon conversions</i>	
CH ₄	$0.5166 - 0.8621 \times E$
C ₂ H ₂	0.0046
C ₂ H ₄	$0.138 - 0.311 \times E$
C ₂ H ₆	$0.02 - 0.038 \times E$

E = air ratio.

Figure 23. Empirical correlations used in Hannula model [59]

For validation purpose, six set points (Figure 24) with different feedstocks and operating conditions.

Set point	1	2	3	4	5	6
Feedstock	Pine sawdust	Pine chips	Forest residues	Pine bark	Eucalyptus chips	Wheat straw
Pressure (bar)	5	5	5	5	5	5
Air-to-fuel ratio (kg/kg-daf)	2.33	1.65	1.97	2.27	1.90	1.37
Gasifier outlet temperature (°C)	930	905	890	980	935	835
Steam-to-fuel ratio (kg/kg-daf)	0.20	0.09	0.14	0.12	0.14	0.31

daf = dry, ash free.

Figure 24. Validation set used in Hannula study [59]

Hannula reported that the model is capable of predicting the concentrations of main product gas components with an average relative error of 14%. The greatest weakness of the model pertains to the prediction capability of carbon conversion when using different fuels. The model was shown to be suitable for simulating the gasification

of pine sawdust, pine and eucalyptus chips as well as forest residues, but not for pine bark or wheat straw. The difference between measurements and model predictions are given in (Fig. 25) for 6 six validation set points.

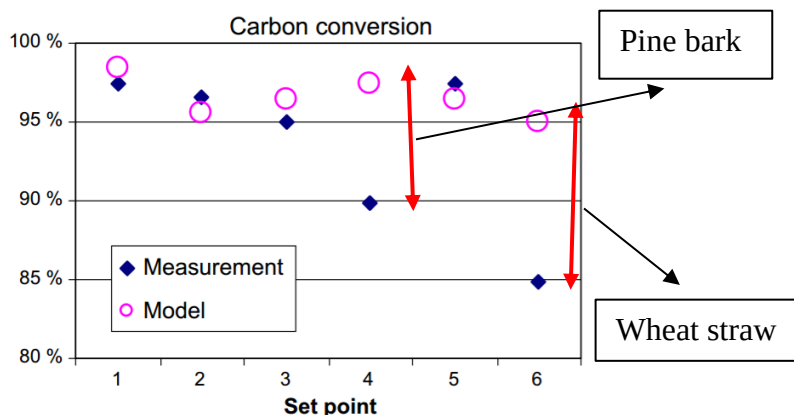


Figure 25. Poor prediction of carbon conversion for pine bark and wheat straw [59]

Authors discussed that the difference between reactivities of pine bark and wheat straw with the feedstock used in parameter fitting can be one of the possible reasons for this outcome.

In addition, a look at the properties of feedstocks (Figure 26) used to validate the model does not definitively conclude what distinct characteristics of pine bark and wheat straw can be considered as underlying reason for their different experimental outcome.

Name	Proximate analysis					Ultimate analysis					Structural component		
	VM	FC	M	A	C	O	H	N	S		Cell	Hem	Lig
Pine sawdust	70.4	14.2	15.3	0.1	51	42.9	6	0.1	0.01		45.9	26.4	27.7
Pine chips	66.9	20	7.6	5.5	52.8	40.5	6.1	0.5	0.09		48.1	23.5	28.4
Forest residue	69.6	15.5	7.3	7.6	51.7	41.5	6.3	0.4	0.13		44.8	33.4	21.8
Pine bark	70.2	23.3	4.7	1.8	53.8	39.9	5.9	0.3	0.07		48.1	23.5	28.4
Eucalyptus bark	68.7	15.1	12	4.2	48.7	45.3	5.7	0.3	0.05		52.7	15.4	31.9
Wheat straw	67.2	16.3	10.1	6.4	49.4	43.6	6.1	0.7	0.17		44.5	33.2	22.3

Figure 26. Biomass fuel properties of samples used to validate the model

Therefore, the correlations in this model were not suitable for broad analysis in terms of type of biomass fuel.

2.5 Concluding remarks

The starting point of analysis is the biomass. In summary, the characterization of biomass with respect to its application in gasification process can be divided into three categories:

- 1) Types of analyses that are well known, usually investigated in experiments, and are directly used as inputs of majority of models (i.e. elemental composition).
- 2) Types of analyses that their impacts are recognized but not investigated in experiments as extensively as type 1, and therefore their consideration in models are limited (i.e. proximate analysis and structural components).
- 3) The unknown characteristics (or extremely variable features) of biomass fuel that results in presence of an irreducible amount of uncertainty about the feedstock composition and structure (i.e. biomass ash).

Considering the limitation and specificity of experimental studies on one hand, and inconsistent behavior of gasification process under different conditions, the main challenge becomes how to infer fundamental mechanism of fluidized bed gasification despite of missing information and entangled nature of influential parameters (i.e. the interaction between a highly variable heterogenous feedstock with inevitable uncertainty, with three main design parameters of temperature, ER, and SBR).

Given that most studies only cover a subspace of possible gasification scenarios (mode of operation), and the fact existing empirical correlations can be further improved by considering more experimental data, the first step is to develop a comprehensive and

diverse database by collecting a large number of lab, pilot and commercial experimental data with different fuels and operating temperature and oxidant/biomass equivalent ratios. Subsequently, consistency and quality of data should be checked, and additional analyses be performed only when data is consolidated on one basis and unreliable data is eliminated.

Besides the statistical challenge and availability of only limited experimental data, one of the fundamental research gaps identified in this review is neglecting to consider the effect of different characteristics of fuel on the process such as the structural component of biomass (cellulose, hemicellulose, lignin), the proximate analysis of biomass fuel (volatile matter, fixed carbon, ash content), and ash composition.

Therefore, an integral part of our strategy is to detect possible partial and sporadic associations and correlations between these different aspects of fuel to enable utilization of part of info regarding biomass as precursor of the less known part.

It seems in order to effectively apply the partial impact of each parameter under a broad range of biomass gasification in fluidized bed, there is a need for a systematic quantification method capable of capturing the latent and underlying relations between different component of the process to be applicable under wide ranges of conditions, and tolerate high levels of uncertainty. A computational framework built upon this vision will be able to optimize the blend of feedstock, operating condition with respect to different applications of interest in a more robust manner.

A preliminary illustration of our approach can be expressed in the following form. Since using a single function to describe the entire process can be very inefficient, our approach is to understand the process statistically, and in terms of a hierarchy of

concepts, with each concept defined through its relation to simpler ones. More generally, suppose that we observe a quantitative response Y with q variables from an Input X with p predictors.

$$X^p = [X_1 X_2 \dots X_p] \quad Y^q = [Y_1 Y_2 \dots Y_q] \quad (10)$$

We assume there is some relationship between Y and X which in a very general form can be written as:

$$Y = F(X) + \quad (11)$$

Now the true challenge is how to quantize the expanded system of equations, both in terms of form of the function (F) and individual respective weights, for our problem:

$$Input = [C, H, O, N, S, Cell, Hem, Lig, Vm, Fc, Ash, Temp, ER, SBR]$$

$$y_{H_2} = W_{H_2}^C F^C + W_{H_2}^H F^H + W_{H_2}^O F^O + \dots + W_{H_2}^T F^T + W_{H_2}^{ER} F^{ER} + W_{H_2}^{SBR} F^{SBR} \quad (12)$$

$$y_{CO} = W_{CO}^C F^C + W_{CO}^H F^H + W_{CO}^O F^O + \dots + W_{CO}^T F^T + W_{CO}^{ER} F^{ER} + W_{CO}^{SBR} F^{SBR} \quad (13)$$

$$y_{CO_2} = W_{CO_2}^C F^C + W_{CO_2}^H F^H + W_{CO_2}^O F^O + \dots + W_{CO_2}^T F^T + W_{CO_2}^{ER} F^{ER} + W_{CO_2}^{SBR} F^{SBR} \quad (14)$$

$$y_{CH_4} = W_{CH_4}^C F^C + W_{CH_4}^H F^H + W_{CH_4}^O F^O + \dots + W_{CH_4}^T F^T + W_{CH_4}^{ER} F^{ER} + W_{CH_4}^{SBR} F^{SBR} \quad (15)$$

$$CCE_{\%} = W_{\%}^C F^C + W_{\%}^H F^H + W_{\%}^O F^O + \dots + W_{\%}^T F^T + W_{\%}^{ER} F^{ER} + W_{\%}^{SBR} F^{SBR} \quad (16)$$

$$Yield = W_{\%}^C F^C + W_{\%}^H F^H + W_{\%}^O F^O + \dots + W_{\%}^T F^T + W_{\%}^{ER} F^{ER} + W_{\%}^{SBR} F^{SBR} \quad (17)$$

CHAPTER 3

DATABASE DEVELOPMENT

3.1 Experimental data collection and organization

In first step, a total number of 238 datasets of biomass gasification in bubbling and circulating fluidized bed were collected from available experimental studies in different scales (i.e. lab, pilot, and commercial). The database contains a diverse set of biomass types and few blends of coal and biomass (Table 9).

Table 9. Types of feedstock available in the database

Feedstock	
Pine sawdust	RIL 1(natural wood)
Beech wood	RIL 2 (waste wood)
Wood pellet	Olive oil waste
Coir pitch	80%PC 20%Pine
Rice husk	80%PC 20%Bagasse
Sawdust	90%PC 10%PE
Palm kernel shell	100% German coal
Coconut shell	80%GC 20%Pine
Olive oil waste	Cypress
Agrol	Hemlock
Willow	Cedar
DDGS	Jatropha
Forest residue	Verge grass
Coffee husk	Sunflower
Vine pruning residue	Park wood
Rubber wood chip	Paper residue sludge

In this study, a dataset with the following information and format is used in analyses:

Biomass properties:

- C in feed (%wt), dry ash-free basis)
- H in feed (%wt), dry ash-free basis)

- O in feed (%wt), dry ash-free basis)
- N in feed (%wt), dry ash-free basis)
- S in feed (%wt), dry ash-free basis)
- Moisture content (%wt) (wet basis)
- Volatile matter (%wt), dry with ash basis)
- Fixed carbon (%wt), dry with ash basis)
- Ash (%wt), dry with ash basis)
- Cellulose, Hemicellulose, Lignin (%wt)
- Ash composition (%wt)

Operating condition:

Temperature (C)

Air equivalent ratio, ER

Steam to biomass ratio, SBR

The output of gasifier:

Gas Yield $\left(\frac{Nm^3}{kg \text{ biomass } daf} \right)$,

Carbon conversion efficiency (CCE %),

Product gas composition (% Vol),

Preliminary steps taken to organize and consolidate datasets:

1. Elemental composition is normalized with respect to (C, H, O, N, and S).

Therefore, Cl is neglected to maintain a uniform set of composition, this is necessary because Cl is not reported in majority of datasets. Therefore, values may slightly differ from what appears in original papers.

2. The parameters of approximate formula of biomass and its molar mass is calculated for molar basis of $CH_xO_yN_zS_w$.
3. Boie relation [60] is used to estimate higher heating value of biomass (i.e. defined as amount of heat released by the unit of mass or volume fuel combusted):

$$HHV_{biomass} = 351.70 C + 1162.49 H + 104.6 S - 62.88 \left(\frac{kJ}{kg} \right) \quad (17)$$

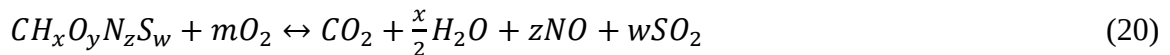
by subtracting the heat of vaporization of water, lower heating value (LHV) is obtained:

$$LHV_{biomass} = HHV - \Delta H_{Latent-steam} \left(\frac{2H}{100} + \frac{MC}{100} \right) \left(\frac{kJ}{kg} \right) \quad (18)$$

4. For data sets of BFB2, 6, and 10 the moisture content is assumed the typical value of 7 percent.
5. For data sets of BFB 8, the moisture content is calculated based on mass fractions of mixture.
6. The amount of air going into gasifier can be calculated from the dimensionless variable of Equivalent air ratio ER, defined as:

$$ER = \frac{\left(\frac{Air}{Biomass} \right)_{actual}}{\left(\frac{Air}{biomass} \right)_{stoich}} \quad (19)$$

For a complete combustion (i.e. carbon is converted to CO₂ and hydrogen is converted to H₂O):



$m = \left(1 + \frac{x}{4} - \frac{y}{2}\right)$ gives the required number of moles of oxygen for complete combustion of 1 mole $CH_xO_yN_zS_w$.

Assuming air's oxygen percent is 21%, and by calculating molar mass of $CH_xO_yN_zS_w$. The mass ratio of $\left(\frac{Air}{biomass}\right)$ from stoichiometric calculations can be determined.

$$\left(\frac{Air}{biomass}\right)_{stoich} = \frac{\frac{m}{0.21} \times 29}{MW_{biomass}} \quad (21)$$

Subsequently, when ER is multiplied, the actual amount of air per **1 kg biomass** (daf) entering the gasifier, the molar flow rate of nitrogen exiting gasifier can be estimated:

$$Air_{actual} = ER * \left(\frac{Air}{biomass}\right)_{stoich}, \quad n_{N_2} = Air_{actual} * \frac{0.79 * 1000}{29}, \quad (22)$$

The lower heating value of product gas can be calculated by knowing mass fraction of H₂, CO, and CH₄.

$$LHV_{Product-gas} = LHV_{H_2} * X_{H_2} + LHV_{CH_4} * X_{CH_4} + LHV_{CO} * X_{CO} \quad (23)$$

$$LHV_{H_2} = 11900, LHV_{CH_4} = 5000, LHV_{CO} = 10100 \left(\frac{kJ}{kg}\right) \quad (24)$$

Carbon conversion efficiency and cold gas efficiency (CGE) are defined as ratio of amount of carbon present in the product gas to amount of carbon input of gasifier, and the energy content of the fuel gas divided by total energy content of the incoming fuel.

$$CCE \% = \frac{C_{CO} + C_{CO_2} + C_{CH_4}}{C_{biomass}} * 100 \quad (25)$$

$$CGE \% = \frac{LHV_{Product-gas}}{LHV_{biomass} * m_{fuel}} \quad (26)$$

The database is divided into 7 groups based on completeness of information.

The strategy to close material balance is described in the following:

Table 10. Seven groups based on completeness of information

Group	Yield	CCE%	X_{N_2}
1	✓	✓	
2	✓	✓	✓
3	✓		✓
4	✓		
5		✓	
6			✓
7			

Group 1. The reported CCE is used to calculate a yield that is compatible with the reported composition of 4 major compounds, the adjusted yield is used to calculate X_{N_2} .

Group 2. It is inherently over specified. The reported CCE is used to calculate a yield that is compatible with the reported composition of 4 major compounds, the adjusted yield is used to calculate X_{N_2} .

Group 3. The reported yield and mole fractions of 4 major compounds are used to calculate CCE. The same test applies, if reasonable values for CCE (70-100) are obtained, it is accepted. X_{N_2} is recalculated to check its consistency with reported yield, if not the new X_{N_2} is accepted as final value.

Group 4. Using yield and mole fractions of 4 major compounds CCE is calculated, if the calculated CCE falls between 70 and 100, it is accepted and the same reported yield is used to calculate X_{N_2} . If CCE is awfully out of the interval,

CCE is assumed 90 percent and a new yield is calculated. The new yield is used to calculate X_{N_2} .

Group 5. The reported CCE is always reasonable (70-100). The yield is calculated based on CCE which guarantees 100 percent carbon closure, then the calculated yield is used to calculate X_{N_2} .

Group 6. The mole fraction of nitrogen X_{N_2} and the number of moles of nitrogen from ER calculation is used to determine the yield. The calculated yield and mole fractions of 4 major compounds are used to calculate CCE. The same test applies, if reasonable values for CCE (70-100) are obtained, it is accepted. If not however, the reported X_{N_2} is discarded, a CCE of 90 is assumed and another yield is calculated. The new calculated yield is used to calculate a new X_{N_2} .

Group 7. These datasets will not be involved in developing the yield and CCE modules of the overall model.

3.2 Complete database

The collected experimental data (from literature listed in Table 11) are presented based on biomass type and ultimate analysis proximate analysis, operating condition and product gas composition, and gas yield and CCE% in appendices E, F, G and H.

Table 11. References of collected experimental data mns

Bed Type	Ref	Bed Type	Ref	Bed Type	Ref
BFB 1	[61]	BFB 9	[62]	CFB 2	[51]
BFB 2	[53]	BFB 10	[63]	CFB 3	[48]
BFB 3	[64]	BFB 11	[45]	CFB 4	[65]
BFB 4	[49]	BFB 12	[66]	CFB 5	[32]
BFB 5	[67]	BFB 13	[68]	CFB 6	[69]
BFB 6	[70]	BFB 14	[71]	CFB 7	[72]
BFB 7	[73]	BFB 15	[74]		
BFB 8	[76]	CFB 1	[77]		

CHAPTER 4

METHODOLOGY

4.1 Canonical correlation analysis of biomass properties

As mentioned in previous chapters, our strategy is to first construct a congruent composite set representing each sample of database, and then through performing a series of functional analyses, detect predictors of composition of product gas, carbon conversion and yield. In the process, we utilize a selective set of machine learning algorithms, differential geometry/trigonometry concepts, neural networks, calculus of variations, and work with various mathematical objects, such as scalars, vectors, matrices, polygons, and pairwise relations constructed from statistical analysis of data.

We begin with development of correlations between different biomass property groups using Canonical Component Analysis (CCA):

Composition (C, H, N, O, S), proximate analysis (Volatile matter, Fixed carbon, Ash), structural component (Cellulose, Hemicellulose, Lignin) and ash composition from 38 types of biomass is collected, as given in Tables 12-14. [78]

Table 12. Ultimate and proximate analysis of 38 types of biomass (Page 1 of 2)

No	Name	VM	FC	A	C	O	H	N	S	No
1	Beech bark	73.69	18.56	7.75	51.4	41.8	6	0.7	0.11	1
2	Birch bark	78.49	19.43	2.07	57	35.7	6.7	0.5	0.1	2
3	Eucalyptus bark	78.07	17.16	4.77	48.7	45.3	5.7	0.3	0.05	3
4	Oak sawdust	86.21	13.45	0.34	50.1	43.9	5.9	0.1	0.01	4
5	Oak wood	78.07	21.39	0.53	50.6	42.9	6.1	0.3	0.1	5
6	Olive wood	79.55	17.24	3.21	49	44.9	5.4	0.7	0.03	6
7	Pine bark	73.66	24.45	1.89	53.8	39.9	5.9	0.3	0.07	7
8	Pine chips	72.40	21.65	5.95	52.8	40.5	6.1	0.5	0.09	8
9	Pine pruning	82.32	15.02	2.66	51.9	41.3	6.3	0.5	0.01	9
10	Pine sawdust	83.12	16.77	0.12	51	42.9	6	0.1	0.01	10
11	Spruce bark	73.47	23.36	3.17	53.6	40	6.2	0.1	0.1	11

Table 12. Ultimate and proximate analysis of 38 types of biomass (Page 2 of 2)

No	Name	VM	FC	A	C	O	H	N	S	No
12	Spruce wood	81.14	18.33	0.54	52.3	41.2	6.1	0.3	0.1	12
13	Wood	84.06	15.73	0.22	49.6	44.1	6.1	0.1	0.06	13
14	Wood residue	77.99	16.58	5.43	51.4	41.9	6.1	0.5	0.08	14
15	Bamboo whole	81.61	17.47	0.92	52	42.5	5.1	0.4	0.04	15
16	Reed canary grass	73.46	17.66	8.88	49.4	42.7	6.3	1.5	0.15	16
17	Sweet sorghum	77.20	18.06	4.73	49.7	43.7	6.1	0.4	0.09	17
18	Switchgrass	80.36	14.53	5.11	49.7	43.4	6.1	0.7	0.11	18
19	Barley straw	76.16	18.53	5.31	49.4	43.6	6.2	0.7	0.13	19
20	Oat straw	80.50	13.62	5.88	48.8	44.6	6	0.5	0.08	20
21	Rape straw	77.44	17.85	4.71	48.5	44.5	6.4	0.5	0.1	21
22	Rice straw	64.29	15.58	20.13	50.1	43	5.7	1	0.16	22
23	Straw	73.40	15.75	10.84	48.8	44.5	5.6	1	0.13	23
24	Wheat straw	74.75	18.13	7.12	49.4	43.6	6.1	0.7	0.17	24
25	Almond shells	74.89	21.77	3.34	50.3	42.5	6.2	1	0.05	25
26	Coconut shells	73.74	23.01	3.24	51.1	43.1	5.6	0.1	0.1	26
27	Cotton husks	78.41	18.15	3.44	50.4	39.8	8.4	1.4	0.01	27
28	Hazelnut shells	77.05	21.44	1.51	51.5	41.6	5.5	1.4	0.04	28
29	Olive husks	79.08	18.67	2.25	50	42.1	6.2	1.6	0.05	29
30	Olive pits	77.00	19.91	3.09	52.8	39.4	6.6	1.1	0.07	30
31	Olive residue	67.34	25.50	7.16	58.4	34.2	5.8	1.4	0.23	31
32	Rice husks	62.75	19.24	18.01	49.3	43.7	6.1	0.8	0.08	32
33	Sunflower husks	76.02	20.90	3.08	50.4	43	5.5	1.1	0.03	33
34	Walnut blows	80.78	16.86	2.35	54.9	36.9	6.7	1.4	0.11	34
35	Walnut hulls	79.65	17.47	2.88	55.1	36.5	6.7	1.6	0.12	35
36	Walnut shells	59.33	37.88	2.79	49.9	42.4	6.2	1.4	0.09	36
37	Wood residue	78.48	18.22	3.30	52.4	41.2	6	0.4	0.04	37
38	Straw residue	75.08	16.72	8.20	51.7	41.5	6.3	0.4	0.13	38

Table 13. Structural component of the 38 types of biomass under study. [7] (Page 1 of 2)

Ref	Name	Cellulose	Hemicellulose	Lignin
1	Beech bark	45.2	32.7	22.1
2	Birch bark	50.2	32.8	17
3	Eucalyptus bark	52.7	15.4	31.9
4	Oak sawdust	58.4	31.4	10.2
5	Oak wood	58.4	31.4	10.2
6	Olive wood	25	24.6	50.4
7	Pine bark	48.1	23.5	28.4
8	Pine chips	48.1	23.5	28.4
9	Pine pruning	48.1	23.5	28.4
10	Pine sawdust	45.9	26.4	27.7
11	Spruce bark	43.6	27.4	29

Table 13. Structural component of the 38 types of biomass under study. [7] (Page 2 of 2)

Ref	Name	Cellulose	Hemicellulose	Lignin
12	Spruce wood	47	25.3	27.7
13	Wood	42	22	36
14	Wood residue	50.6	24.7	24.7
15	Bamboo whole	43.9	26.5	29.6
16	Reed canary grass	49.2	39.1	11.7
17	Sweet sorghum grass	50.6	24.7	24.7
18	Switchgrass	48.7	38.4	12.9
19	Barley straw	48.6	29.7	21.7
20	Oat straw	44.8	33.4	21.8
21	Rape straw	54.8	23.2	22
22	Rice straw	52.3	32.8	14.9
23	Straw	44.5	33.2	22.3
24	Wheat straw	44.5	33.2	22.3
25	Almond shells	50.7	28.9	20.4
26	Coconut shells	40.3	27.8	31.9
27	Cotton husks	82.8	17.1	0.1
28	Hazelnut shells	26.6	30	43.4
29	Olive husks	25	24.6	50.4
30	Olive pits	25	24.6	50.4
31	Olive residue	25	24.6	50.4
32	Rice husks	43.8	31.6	24.6
33	Sunflower husks	56.5	28	15.5
34	Walnut blows	28.1	26.6	45.3
35	Walnut hulls and blows	28.1	26.6	45.3
36	Walnut shells	28.1	26.6	45.3
37	Wood-agricultural residue	46.8	30.4	22.8
38	Wood-straw residue	44.8	33.4	21.8

Table 14. Ash composition of the 38 types of biomass under study [78] (Page 1 of 2)

Ref	Name	SiO2	CaO	K2O	P2O5	Al2O3	MgO	Fe2O3	SO3	Na2O	TiO2
1	Beech bark	12.4	68.2	2.6	2.3	0.12	11.5	1.1	0.8	0.9	0.1
2	Birch bark	4.38	69.06	8.99	4.13	0.55	5.92	2.24	2.75	1.85	0.13
3	Eucalyptus bark	10.04	57.74	9.29	2.35	3.1	10.91	1.12	3.47	1.86	0.12
4	Oak sawdust	29.93	15.56	31.99	1.9	4.27	5.92	4.2	3.84	2	0.39
5	Oak wood	48.95	17.48	9.49	1.8	9.49	1.1	8.49	2.6	0.5	0.1
6	Olive wood	10.24	41.47	25.16	10.75	2.02	3.03	0.88	2.65	3.67	0.13
7	Pine bark	9.2	56.83	7.78	5.02	7.2	6.19	2.79	2.83	1.97	0.19
8	Pine chips	68.18	7.89	4.51	1.56	7.04	2.43	5.45	1.19	1.2	0.55
9	Pine pruning	7.76	44.1	22.32	5.73	2.75	11.33	1.25	4.18	0.42	0.17
10	Pine sawdust	9.71	48.88	14.38	6.08	2.34	13.8	2.1	2.22	0.35	0.14

Table 14. Ash composition of the 38 types of biomass under study [78] (Page 2 of 2)

Ref	Name	SiO ₂	CaO	K ₂ O	P ₂ O ₅	Al ₂ O ₃	MgO	Fe ₂ O ₃	SO ₃	Na ₂ O	TiO ₂
11	Spruce bark	6.13	72.39	7.22	2.69	0.68	4.97	1.9	1.88	2.02	0.12
12	Spruce wood	49.3	17.2	9.6	1.9	9.4	1.1	8.3	2.6	0.5	0.1
13	Wood	23.15	37.35	11.59	2.9	5.75	7.26	3.27	4.95	2.57	1.2
14	Wood residue	53.15	11.66	4.85	1.37	12.64	3.06	6.24	1.99	4.47	0.57
15	Bamboo whole	9.92	4.46	53.38	20.33	0.67	6.57	0.67	3.68	0.31	0.01
16	Reed canary	84.92	3.31	2.93	3.88	1.32	1.42	1.04	1.04	0.09	0.05
17	Sweet sorghum	66.85	10.41	9.49	3.47	0.81	3.12	0.58	3.47	1.74	0.06
18	Switchgrass	66.25	10.21	9.64	3.92	2.22	4.71	1.36	0.83	0.58	0.28
19	Barley straw	50.78	9.89	28.18	2.97	0.67	2.87	0.95	2.22	1.39	0.08
20	Oat straw	37.79	12.03	26.84	6.14	4.69	4.45	2.17	4.93	0.72	0.24
21	Rape straw	40.8	30.68	13.45	2.22	5.45	2	2	2.67	0.44	0.29
22	Rice straw	77.2	2.46	12.59	0.98	0.55	2.71	0.5	1.18	1.79	0.04
23	Straw	57.14	6.7	25.82	2.74	0.76	1.67	0.53	3.89	0.7	0.05
24	Wheat straw	50.35	8.21	24.89	3.54	1.54	2.74	0.88	4.24	3.52	0.09
25	Almond shells	16.96	11.55	53.48	4.93	2.99	4.51	2.78	0.93	1.76	0.11
26	Coconut shells	66.75	2.41	8.48	1.54	8.48	1.54	6.16	0.01	4.62	0.01
27	Cotton husks	10.93	20.95	50.2	4.05	1.32	7.59	1.92	1.72	1.31	0.01
28	Hazelnut shells	33.7	15.4	30.4	3.2	3.1	7.9	3.8	1.1	1.3	0.1
29	Olive husks	32.7	14.5	4.3	2.5	8.4	4.2	6.3	0.6	26.2	0.3
30	Olive pits	21.48	19.97	16.44	9.71	5.95	3.84	4.25	2.3	15.77	0.29
31	Olive residue	22.26	12.93	42.79	6.09	4.1	5.84	1.99	3.73	0.12	0.15
32	Rice husks	94.48	0.97	2.29	0.54	0.21	0.19	0.22	0.92	0.16	0.02
33	Sunflower husks	23.66	15.31	28.53	7.13	8.75	7.33	4.27	4.07	0.8	0.15
34	Walnut blows	6.41	27.64	34.67	10.28	2.25	14.34	1.05	2.33	0.92	0.11
35	Walnut hulls	8.29	20.03	39.65	7.52	2.92	16.21	1.37	2.71	1.19	0.11
36	Walnut shells	23.32	16.72	33.03	6.21	2.4	13.51	1.5	2.2	1	0.1
37	Wood-residue	37.18	25.7	7.76	2.22	11.07	4.77	5.77	2.03	2.57	0.93
38	Straw residue	57.83	11.51	6.67	1.08	9.77	2.66	4.97	1.88	3.11	0.52

The main goal is to incorporate the consideration of the effect of ultimate, proximate, structural and ash content in predicting the composition of product gas. In addition, this method develops relationships between different types of biomass analyses

utilizing statistically inferred features that express various ranges of irregularity in biomass. In the following sections, the step by step of this statistical process is described for Elemental composition, Proximate analysis, and structural component. The ash composition is treated separately due to more heterogeneity, the main part of methodology is presented in chapter 6 and needs to be completed in future.

“CCA is a method for exploring the relationships between two multivariate sets of variables (vectors), all measured on the same individual [79].”

$$X = [X_1 \ X_2 \ ..X_p \], Y = [Y_1 \ Y_2 \ ..Y_q \] \quad (27)$$

$$U_1 = a_{11}X_1 + a_{12}X_2 + \dots + a_{1p}X_p$$

$$U_2 = a_{21}X_1 + a_{22}X_2 + \dots + a_{2p}X_p$$

...

$$U_p = a_{p1}X_1 + a_{p2}X_2 + \dots + a_{pp}X_p$$

&

$$V_1 = b_{11}Y_1 + b_{12}Y_2 + \dots + b_{1q}Y_q$$

$$V_2 = b_{21}Y_1 + b_{22}Y_2 + \dots + b_{2q}Y_q$$

...

$$V_p = b_{p1}Y_1 + b_{p2}Y_2 + \dots + b_{pq}Y_q$$

$$(U_i, V_i)$$

(28)

In this setup, (U1, V1) is the first canonical variate pair, and (U2, V2) would be the second canonical variate pair and so on... With $p \leq q$ there are p canonical covariate pair

The objective is to find linear combinations that maximize the correlations between the members of each canonical variate pair.

“We can compute the variance of U_i variables using the following expression [79]”:

$$var(U_i) = \sum_{k=1}^p \sum_{l=1}^p a_{ik} a_{il} cov(X_k, X_l) \quad (29)$$

The coefficients a_{i1} through a_{ip} that appear in the double sum are the same coefficients that appear in the definition of U_i . The covariances between the k th and l th X-variables are multiplied by the corresponding coefficients a_{ik} and a_{il} for the variate U_i .

Similar calculations can be made for the variance of V_i as shown below [79]:

$$var(V_i) = \sum_{k=1}^p \sum_{l=1}^q b_{ik} b_{il} cov(Y_k, Y_l) \quad (30)$$

Then calculate the covariance between U_i and V_i as [79]:

$$Cov(U_i, V_i) = \sum_{k=1}^p \sum_{l=1}^q a_{ik} b_{il} cov(X_k, Y_l) \quad (31)$$

The correlation between U_i and V_i is calculated by taking the covariance between two variables and divide it by the square root of the product of the variances:

$$\frac{cov(U_i, V_j)}{\sqrt{var(U_i)var(V_j)}} \quad (32)$$

“The canonical correlation is a specific type of correlation. The canonical correlation for the i^{th} canonical variate pair is simply the correlation between U_i and V_i

$$[80]”: \rho_i^* = \frac{cov(U_i, V_j)}{\sqrt{var(U_i)var(V_j)}} \quad (33)$$

The specific type of correlation defined in equation 33 should be maximized.

For first canonical variate pair, (U1, V1):

The coefficients $a_{11}, a_{12}, \dots, a_{1p}$ and $b_{11}, b_{12}, \dots, b_{1q}$ should be selected to maximize the canonical correlation ρ_{*1} of the first canonical variate pair, with the constraint that variances of the two canonical variates in that pair are equal to one.

$$\text{var}(U_1) = \text{var}(V_1) = 1 \quad (34)$$

This is required so that unique values for the coefficients are obtained. [80]

For the second canonical variate pair: (U2, V2)

Likewise, we want the coefficients $a_{21}, a_{22}, \dots, a_{2p}$ and $b_{21}, b_{22}, \dots, b_{2q}$ that maximize the canonical correlation ρ_{*2} of the second canonical variate pair, (U2, V2). However, in addition to the constraint that variances of the two canonical variates in this pair be equal to one, we require the additional constraints that (U1, U2), and (V1, V2) have to be uncorrelated. In addition, the combinations (U1, V2) and (U2, V1) must be uncorrelated. In summary, the overall constraints become:

$$\left. \begin{aligned} \text{var}(U_2) &= \text{var}(V_2) = 1 \\ \text{Cov}(U_1, U_2) &= \text{Cov}(V_1, V_2) = 0 \\ \text{Cov}(U_1, V_2) &= \text{Cov}(U_2, V_1) = 0 \end{aligned} \right\} \quad (35)$$

Basically, we require that all of the remaining correlations equal zero. This procedure is repeated for each pair of canonical variates. By applying this dimension reduction technique, one can find linear combinations of X's (one group of properties describing biomass) that are correlated with linear combinations of Y's (another group of properties describing biomass).

Our method in using CCA analysis to amalgamate different properties of biomass into a whole is multifaceted and involves a preconditioning step. Specifically, first, an internal linkage is made between members of each group. This is done by picking the variable with least variation coefficient in each group as the base of normalization (i.e. C% for elemental, FC for proximate, and Hem for structural). This is followed by a logarithmic transformation of ratios to set up a one-to-one correspondence between the elements of the groups (i.e. a group isomorphism).

Next, the three groups of elementals, proximate and structural are linked together performing “two” separate CCA (i.e. the minimum number of edges to ensure connectedness of 3 vertices). Overall, we form some sort of association between different types of biomass that has two particular qualities: 1. It is fully chained, and 2. It has hierarchy.

Conceptually, these steps may be viewed as depicted in Figure 27.

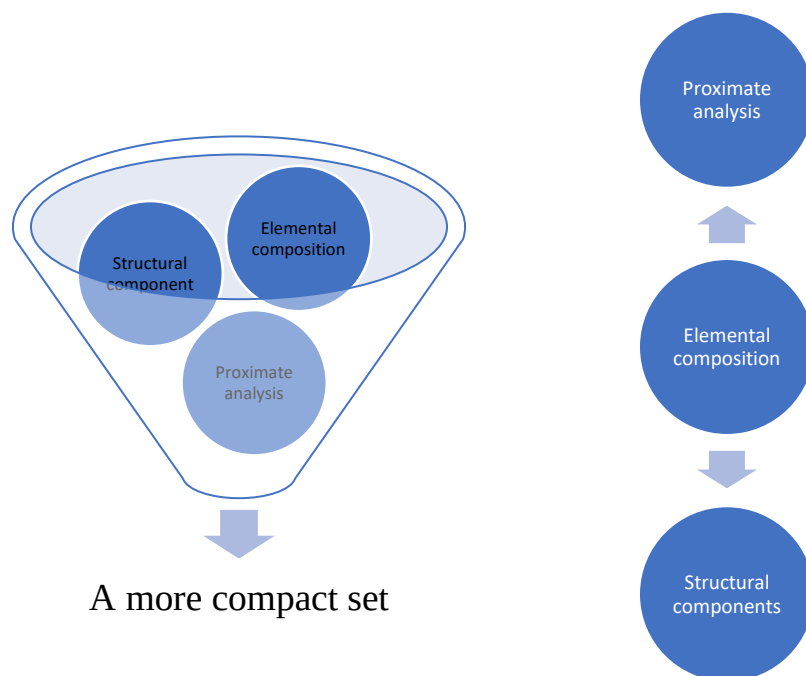


Figure 27. Biomass representative array [E1, E2, P1, S1]

Where Elemental Composition Quotient (E), Proximate Composition Quotient (P), Structural Composition Quotient (S) are indices representing their class or perspective on the feedstock. The following are the transformed composition for the elemental, proximate (dry basis) and structural components (Figures 28-29).

In our study, 38 types of biomass are considered as the sample space of the biomass system, and their properties are expressed using vectors.

	Name	Ln C/O	Ln C/H	Ln C/N	Ln C/S
1	Beech bark	0.21	2.15	4.30	6.15
2	Birch bark	0.47	2.14	4.74	6.35
3	Eucalyptus bark	0.07	2.15	5.09	6.88
4	Oak sawdust	0.13	2.14	6.22	8.52
5	Oak wood	0.17	2.12	5.13	6.23
6	Olive wood	0.09	2.21	4.25	7.40
7	Pine bark	0.30	2.21	5.19	6.64
8	Pine chips	0.27	2.16	4.66	6.37
9	Pine pruning	0.23	2.11	4.64	8.55
10	Pine sawdust	0.17	2.14	6.23	8.54
11	Spruce bark	0.29	2.16	6.28	6.28
12	Spruce wood	0.24	2.15	5.16	6.26
13	Wood	0.12	2.10	6.21	6.72
14	Wood residue	0.20	2.13	4.63	6.47
15	Bamboo whole	0.20	2.32	4.87	7.17
16	Reed canary grass	0.15	2.06	3.49	5.80
17	Sweet sorghum grass	0.13	2.10	4.82	6.31
18	Switchgrass	0.14	2.10	4.26	6.11
19	Barley straw	0.12	2.08	4.26	5.94
20	Oat straw	0.09	2.10	4.58	6.41
21	Rape straw	0.09	2.03	4.57	6.18
22	Rice straw	0.15	2.17	3.91	5.75
23	Straw	0.09	2.16	3.89	5.93
24	Wheat straw	0.12	2.09	4.26	5.67
25	Almond shells	0.17	2.09	3.92	6.91
26	Coconut shells	0.17	2.21	6.24	6.24
27	Cotton husks	0.24	1.79	3.58	8.53
28	Hazelnut shells	0.21	2.24	3.61	7.16
29	Olive husks	0.17	2.09	3.44	6.91
30	Olive pits	0.29	2.08	3.87	6.63
31	Olive residue	0.54	2.31	3.73	5.54
32	Rice husks	0.12	2.09	4.12	6.42
33	Sunflower husks	0.16	2.22	3.82	7.43
34	Walnut blows	0.40	2.10	3.67	6.21
35	Walnut hulls and blows	0.41	2.11	3.54	6.13
36	Walnut shells	0.16	2.09	3.57	6.32
37	Wood-agricultural resid	0.24	2.17	4.88	7.18
38	Wood-straw residue	0.22	2.10	4.86	5.99

Figure 28. Log transformation of elemental composition based on C

	Name	Ln(FC/VM)	Ln(FC/A)	Ln (Hem/Cell)	Ln (Hem/Lig)
1	Beech bark	-1.38	0.87	-0.32	0.39
2	Birch bark	-1.40	2.24	-0.43	0.66
3	Eucalyptus bark	-1.52	1.28	-1.23	-0.73
4	Oak sawdust	-1.86	3.68	-0.62	1.12
5	Oak wood	-1.29	3.69	-0.62	1.12
6	Olive wood	-1.53	1.68	-0.02	-0.72
7	Pine bark	-1.10	2.56	-0.72	-0.19
8	Pine chips	-1.21	1.29	-0.72	-0.19
9	Pine pruning	-1.70	1.73	-0.72	-0.19
10	Pine sawdust	-1.60	4.96	-0.55	-0.05
11	Spruce bark	-1.15	2.00	-0.46	-0.06
12	Spruce wood	-1.49	3.53	-0.62	-0.09
13	Wood	-1.68	4.28	-0.65	-0.49
14	Wood residue	-1.55	1.12	-0.72	0.00
15	Bamboo whole	-1.54	2.94	-0.50	-0.11
16	Reed canary grass	-1.43	0.69	-0.23	1.21
17	Sweet sorghum grass	-1.45	1.34	-0.72	0.00
18	Switchgrass	-1.71	1.05	-0.24	1.09
19	Barley straw	-1.41	1.25	-0.49	0.31
20	Oat straw	-1.78	0.84	-0.29	0.43
21	Rape straw	-1.47	1.33	-0.86	0.05
22	Rice straw	-1.42	-0.26	-0.47	0.79
23	Straw	-1.54	0.37	-0.29	0.40
24	Wheat straw	-1.42	0.93	-0.29	0.40
25	Almond shells	-1.24	1.87	-0.56	0.35
26	Coconut shells	-1.16	1.96	-0.37	-0.14
27	Cotton husks	-1.46	1.66	-1.58	5.14
28	Hazelnut shells	-1.28	2.65	0.12	-0.37
29	Olive husks	-1.44	2.11	-0.02	-0.72
30	Olive pits	-1.35	1.86	-0.02	-0.72
31	Olive residue	-0.97	1.27	-0.02	-0.72
32	Rice husks	-1.18	0.07	-0.33	0.25
33	Sunflower husks	-1.29	1.91	-0.70	0.59
34	Walnut blows	-1.57	1.97	-0.05	-0.53
35	Walnut hulls and blows	-1.52	1.80	-0.05	-0.53
36	Walnut shells	-0.45	2.61	-0.05	-0.53
37	Wood-agricultural resid	-1.46	1.71	-0.43	0.29
38	Wood-straw residue	-1.50	0.71	-0.29	0.43

Figure 29. Log transformation of proximate analysis based on FC, and structural component based on Hem

Here, we plug our “logarithmic variables” of Elemental and Proximate, in the set of equations (28), as X and Y, respectively.

$$U_1^{elem} = a_{11} \ln \ln \left(\frac{C}{O} \right) + a_{12} \ln \ln \left(\frac{C}{H} \right) + a_{13} \ln \ln \left(\frac{C}{N} \right) + a_{14} \ln \ln \left(\frac{C}{S} \right) \quad (36)$$

$$U_2^{elem} = a_{21} \ln \ln \left(\frac{C}{O} \right) + a_{22} \ln \ln \left(\frac{C}{H} \right) + a_{23} \ln \ln \left(\frac{C}{N} \right) + a_{24} \ln \ln \left(\frac{C}{S} \right) \quad (37)$$

...

$$V_1^{prox} = b_{11} \ln \ln \left(\frac{FC}{VM} \right) + b_{12} \ln \ln \left(\frac{FC}{A} \right) \quad (38)$$

$$V_2^{prox} = b_{21} \ln \ln \left(\frac{FC}{VM} \right) + b_{22} \ln \ln \left(\frac{FC}{A} \right) \quad (39)$$

Using basic logarithm rules, the mathematical representation of (E, P) reduces to the following:

$$U_1^{elem} = \ln \ln \left(\frac{C}{O} \right)^{a_{11}} + \ln \ln \left(\frac{C}{H} \right)^{a_{12}} + \ln \ln \left(\frac{C}{N} \right)^{a_{13}} + \ln \ln \left(\frac{C}{S} \right)^{a_{14}} =$$

$$\ln \ln \left[\left(\frac{C}{O} \right)^{a_{11}} \left(\frac{C}{H} \right)^{a_{12}} \left(\frac{C}{N} \right)^{a_{13}} \left(\frac{C}{S} \right)^{a_{14}} \right] \quad (40)$$

$$V_1^{prox} = \ln \ln \left(\frac{FC}{VM} \right)^{b_{11}} + \ln \ln \left(\frac{FC}{A} \right)^{b_{12}} = \ln \ln \left[\left(\frac{FC}{VM} \right)^{b_{11}} \left(\frac{FC}{A} \right)^{b_{12}} \right] \quad (41)$$

Where the powers of our statistical ensembles are obtained from performing canonical correlation analysis of 38 types of biomass using XLSTAT software [88].

In a similar fashion, a relationship between Elemental vs Structural components can be obtained.

The results of CCA for Elemental vs Proximate are as following (Table 15):

Where $F1 = (E_1, P_1)$

Table 15. Coefficients obtained from CCA analysis for $F1 = (E_1, P_1)$

Standardized canonical coefficients (Y1)		
	F1	
Y11		0.167
Y12		0.011
Y13		0.635
Y14		0.651
Standardized canonical coefficients (Y2)		
	F1	
Y21		-0.382
Y22		0.903

Therefore, the powers of the following relationship are obtained (Table 16), yielding the correlation depicted in Figure 30.

$$U_1^{elem} = \ln \ln \left[\left(\frac{C}{O} \right)^{a_{11}} \left(\frac{C}{H} \right)^{a_{12}} \left(\frac{C}{N} \right)^{a_{13}} \left(\frac{C}{S} \right)^{a_{14}} \right]$$

$$V_1^{prox} = \ln \ln \left[\left(\frac{FC}{VM} \right)^{b_{11}} \left(\frac{FC}{A} \right)^{b_{12}} \right]$$

Table 16. Powers of (elemental, proximate) variate

Coefficient	Value
a_{11}	0.167
a_{12}	0.011
a_{13}	0.635
a_{14}	0.651
b_{11}	-0.382
b_{12}	0.903

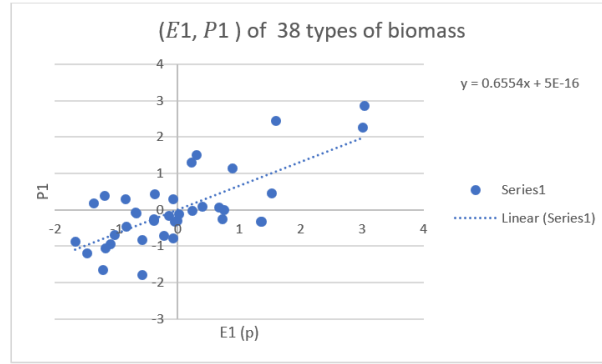


Figure 30. Elemental character vs Proximate character (E1 vs P1)

Therefore, given the elemental composition and proximate analysis (E1 vs P1) is calculated as:

$$E_1 = \frac{C^{1.45}}{O^{0.17}N^{0.63}S^{0.65}} \quad (42)$$

$$P_1 = \frac{FC^{0.52}VM^{0.38}}{A^{0.9}} \quad (43)$$

It should be noted, in case the proximate analysis of a fuel was unavailable, the parameter P_1 should be estimated based on the relation in Figure 31.

The results of CCA for Elemental vs Structural component (Table 17):

Where $F2 = (E_2, S_1)$

Table 17. Coefficients obtained from CCA analysis for $F1 = (E_1, P_1)$

Standardized canonical coefficients (Y1)		
	F2	
Y11		-0.010
Y12		-0.878
Y13		0.233
Y14		0.342
Standardized canonical coefficients (Y2)		
	F2	
Y21		-0.484
Y22		0.653

The powers of the following relationship are obtained (Table 18), yielding the correlation depicted in Figure 34.

$$U_1^{elem} = a_{11} \ln \ln \left(\frac{C}{O} \right) + a_{12} \ln \ln \left(\frac{C}{H} \right) + a_{13} \ln \ln \left(\frac{C}{N} \right) + a_{14} \ln \ln \left(\frac{C}{S} \right)$$

$$V_1 = b_{11} \ln \ln \left(\frac{Hem}{Cell} \right) + b_{12} \ln \ln \left(\frac{Hem}{Lig} \right)$$

Table 18. Powers of (elemental, structural) variate

a_{11}	-0.010
a_{12}	-0.878
a_{13}	0.233
a_{14}	0.342
b_{11}	-0.484
b_{12}	0.653

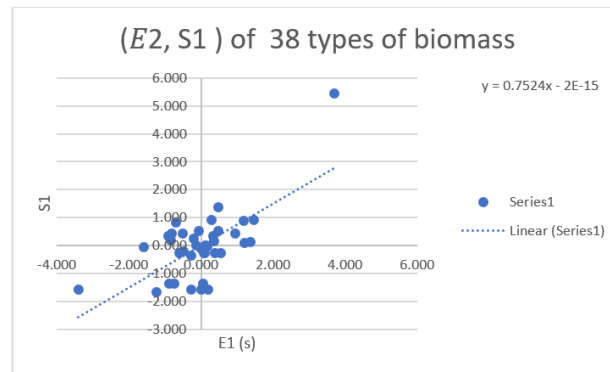


Figure 31. Elemental character vs Proximate character (E1vs P1)

Where:

$$E_2 = \ln\left(\frac{H^{0.88}}{C^{0.31}N^{0.23}S^{0.34}}\right) \quad (44)$$

$$S_1 = \ln\left(\frac{Hem^{0.17}Cell^{0.48}}{Lig^{0.65}}\right) \quad (45)$$

In this setup, the relationship between E1 and E2 is as depicted in Figure 32.

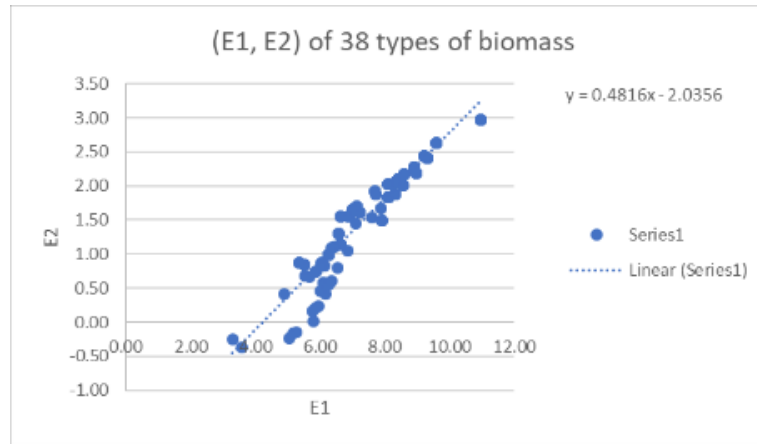


Figure 32. Elemental character (p) vs Elemental character (s) (E1vs E2)

These relationships are considered as the building blocks of our biomass characterization.

4.2 Trigonometric/ Geometric transformations

In the next step, we use trigonometric identities to develop an assortment of variables representing product gas composition.

In trigonometry, the basic relationship between the sine and the cosine is given by the Pythagorean identity.

$$\sin^2\theta + \cos^2\theta = 1 \quad (46)$$

On the other hand, sum of mole fractions of different components in product gas mixture is also one based on Daltons law.

$$\Sigma \quad y_{H_2} + y_{CH_4} + y_{CO} + y_{CO_2} = 1 \quad (47)$$

Considering the following 3 combinations:

$$\sin^2 \theta_1 = y_{H_2} + y_{CH_4} \quad \cos^2 \theta_1 = y_{CO} + y_{CO_2} \quad (48)$$

$$\sin^2 \theta_2 = y_{H_2} + y_{CO} \quad \cos^2 \theta_2 = y_{CH_4} + y_{CO_2} \quad (49)$$

$$\sin^2 \theta_3 = y_{H_2} + y_{CO_2} \quad \cos^2 \theta_3 = y_{CH_4} + y_{CO} \quad (50)$$

Knowing $\tan^2 \theta_i = \frac{\sin^2 \theta_i}{\cos^2 \theta_i}$ and using the inverse of the tangent function, we may

build the following representative vector of product gas composition for each dataset:

$$\theta_i = [\theta_1, \theta_2, \theta_3], i = 1, 2, \dots, n$$

Note that each element of this vector “single-handedly” represents the total composition of product gas. Therefore, the whole vector contains information about the “same event” from 3 different points of view. Then, by assuming each θ as a side of a triangle, we may construct the triangle representative of the product gas composition. Figure 33. illustrates progress from θ (in radian) on the unit circle to the triangle representative.

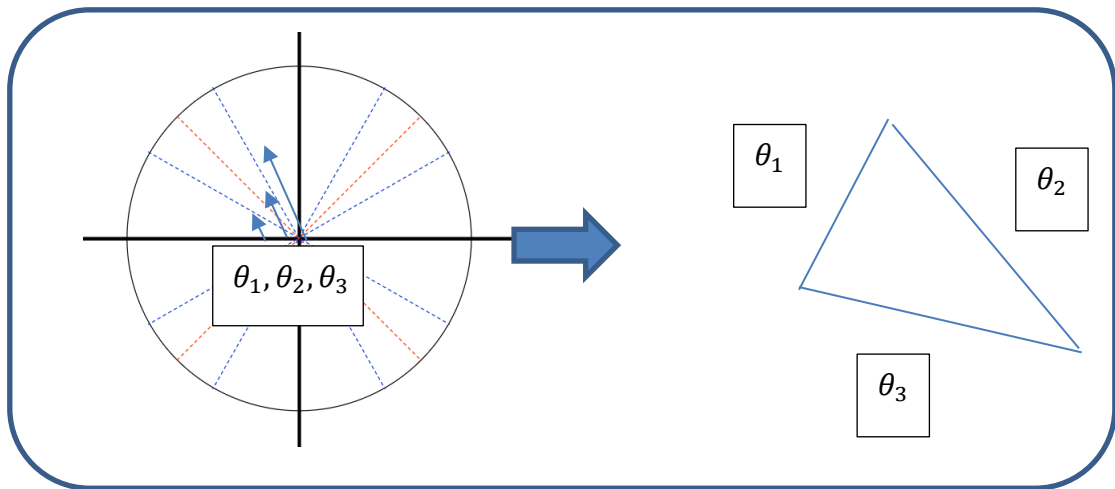


Figure 33. Illustrates progress from θ (in radian) on the unit circle to the triangle

This transformation enables us to choose from a very rich variety of trig functions in unit circle and also many well-established properties of triangles. For example, *circumcenter* is the point of concurrency of the perpendicular bisectors of a triangle; *centroid* is the point of concurrency of the medians of a triangle; *orthocenter* is the point of concurrency of the altitudes in a triangle. Although, theoretically, all of the trigonometric functions of an angle θ can be constructed geometrically (Fig. 34), we only use few of them according to our need in each analysis.

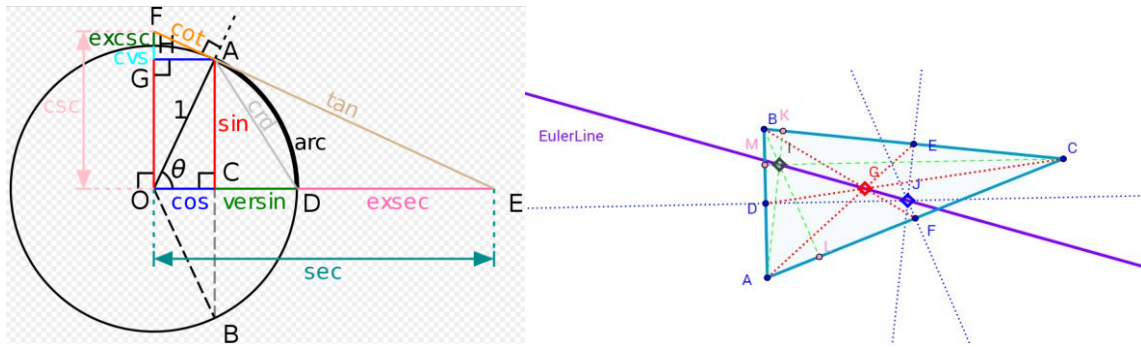


Figure 34. Graphical illustration of trigonometric functions of an angle θ and triangle centers [81]

The calculations of $\theta = [\theta_1, \theta_2, \theta_3]$ and other parameters for the first 50 datasets is given in Figures 35-36.

N° of data set										Radian			sum(theta)
	H2+CH4	CO+CO2	H2+CO	CH4+CO2	H2+CO2	CH4+CO							
1	0.38	0.62	0.64	0.36	0.41	0.59	0.60	1.79	0.71	0.66	0.93	0.70	2.29
2	0.42	0.58	0.68	0.32	0.47	0.53	0.72	2.09	0.90	0.70	0.97	0.76	2.43
3	0.44	0.56	0.70	0.30	0.50	0.50	0.80	2.33	1.02	0.73	0.99	0.79	2.51
4	0.47	0.53	0.71	0.29	0.55	0.45	0.90	2.42	1.22	0.76	1.00	0.83	2.59
5	0.48	0.52	0.73	0.27	0.59	0.41	0.91	2.70	1.41	0.76	1.02	0.87	2.66
6	0.45	0.55	0.69	0.31	0.49	0.51	0.83	2.26	0.97	0.74	0.98	0.78	2.50
7	0.44	0.56	0.70	0.30	0.48	0.52	0.80	2.32	0.91	0.73	0.99	0.76	2.48
8	0.44	0.56	0.71	0.29	0.48	0.52	0.79	2.42	0.91	0.73	1.00	0.76	2.49
9	0.43	0.57	0.70	0.30	0.48	0.52	0.77	2.34	0.94	0.72	0.99	0.77	2.48
10	0.43	0.57	0.69	0.31	0.51	0.49	0.76	2.27	1.04	0.72	0.99	0.79	2.50
11	0.41	0.59	0.77	0.23	0.49	0.51	0.70	3.31	0.98	0.70	1.07	0.78	2.54
12	0.43	0.57	0.70	0.30	0.46	0.54	0.77	2.34	0.86	0.72	0.99	0.75	2.46
13	0.44	0.56	0.70	0.30	0.48	0.52	0.78	2.35	0.92	0.72	0.99	0.76	2.48
14	0.44	0.56	0.70	0.30	0.50	0.50	0.79	2.33	1.02	0.73	0.99	0.79	2.51
15	0.44	0.56	0.70	0.30	0.51	0.49	0.78	2.28	1.04	0.72	0.99	0.80	2.51
16	0.42	0.58	0.71	0.29	0.47	0.53	0.73	2.49	0.90	0.71	1.01	0.76	2.47
17	0.43	0.57	0.72	0.28	0.48	0.52	0.76	2.53	0.94	0.72	1.01	0.77	2.49
18	0.43	0.57	0.69	0.31	0.51	0.49	0.75	2.25	1.05	0.71	0.98	0.80	2.49
19	0.42	0.58	0.69	0.31	0.53	0.47	0.71	2.21	1.12	0.70	0.98	0.81	2.49
20	0.29	0.71	0.63	0.37	0.54	0.46	0.40	1.67	1.17	0.57	0.91	0.82	2.30
21	0.32	0.68	0.65	0.35	0.50	0.50	0.47	1.88	0.99	0.60	0.94	0.78	2.32
22	0.23	0.77	0.53	0.47	0.66	0.34	0.30	1.13	1.91	0.50	0.82	0.94	2.26
23	0.35	0.65	0.76	0.24	0.42	0.58	0.53	3.15	0.73	0.63	1.06	0.71	2.39
24	0.37	0.63	0.68	0.32	0.52	0.48	0.60	2.15	1.08	0.66	0.97	0.80	2.43
25	0.32	0.68	0.67	0.33	0.51	0.49	0.46	2.02	1.03	0.60	0.96	0.79	2.35
26	0.19	0.81	0.51	0.49	0.61	0.39	0.23	1.06	1.58	0.45	0.80	0.90	2.14
27	0.38	0.62	0.59	0.41	0.63	0.37	0.62	1.42	1.71	0.67	0.87	0.92	2.46
28	0.40	0.60	0.60	0.40	0.62	0.38	0.67	1.51	1.64	0.69	0.89	0.91	2.48
29	0.40	0.60	0.60	0.40	0.61	0.39	0.67	1.50	1.54	0.69	0.89	0.89	2.46
30	0.40	0.60	0.56	0.44	0.67	0.33	0.68	1.27	2.00	0.69	0.84	0.96	2.49
31	0.36	0.64	0.61	0.39	0.61	0.39	0.57	1.55	1.56	0.65	0.89	0.90	2.44
32	0.38	0.62	0.61	0.39	0.57	0.43	0.61	1.56	1.32	0.66	0.90	0.85	2.41
33	0.36	0.64	0.60	0.40	0.62	0.38	0.57	1.51	1.65	0.65	0.89	0.91	2.44
34	0.37	0.63	0.60	0.40	0.58	0.42	0.58	1.51	1.38	0.65	0.89	0.86	2.40
35	0.30	0.70	0.62	0.38	0.52	0.48	0.44	1.61	1.09	0.58	0.90	0.81	2.29
36	0.28	0.72	0.68	0.32	0.45	0.55	0.40	2.09	0.83	0.56	0.97	0.74	2.27
37	0.27	0.73	0.64	0.36	0.52	0.48	0.37	1.75	1.08	0.55	0.92	0.80	2.27
38	0.28	0.72	0.64	0.36	0.49	0.51	0.39	1.78	0.98	0.56	0.93	0.78	2.27
39	0.30	0.70	0.66	0.34	0.53	0.47	0.42	1.90	1.11	0.58	0.94	0.81	2.33
40	0.32	0.68	0.66	0.34	0.50	0.50	0.47	1.98	1.01	0.60	0.95	0.79	2.34
41	0.27	0.73	0.65	0.35	0.51	0.49	0.37	1.84	1.04	0.55	0.94	0.80	2.28
42	0.29	0.71	0.65	0.35	0.51	0.49	0.40	1.87	1.04	0.57	0.94	0.80	2.30
43	0.34	0.66	0.65	0.35	0.58	0.42	0.51	1.83	1.37	0.62	0.93	0.86	2.42
44	0.29	0.71	0.55	0.45	0.65	0.35	0.41	1.20	1.83	0.57	0.83	0.93	2.33
45	0.21	0.79	0.45	0.55	0.72	0.28	0.27	0.80	2.55	0.48	0.73	1.01	2.22
46	0.32	0.68	0.51	0.49	0.58	0.42	0.46	1.04	1.35	0.60	0.80	0.86	2.25
47	0.23	0.77	0.37	0.63	0.70	0.30	0.30	0.60	2.30	0.50	0.66	0.99	2.15
48	0.26	0.74	0.45	0.55	0.66	0.34	0.36	0.82	1.98	0.54	0.74	0.95	2.23
49	0.27	0.73	0.56	0.44	0.55	0.45	0.36	1.27	1.21	0.54	0.85	0.83	2.22
50	0.30	0.70	0.56	0.44	0.61	0.39	0.44	1.27	1.56	0.58	0.85	0.90	2.32

Figure 35. Computation of representative vector of product gas (first 50 datasets)

s (semiperimeter)	K (area)	L (eq)	R (circumradius)	r (inradius)
1.143	0.230	0.729	0.465	0.201
1.214	0.265	0.782	0.487	0.218
1.254	0.284	0.810	0.501	0.227
1.297	0.309	0.845	0.512	0.238
1.329	0.324	0.865	0.525	0.244
1.251	0.284	0.810	0.498	0.227
1.241	0.276	0.798	0.499	0.222
1.244	0.275	0.798	0.502	0.221
1.241	0.275	0.797	0.499	0.222
1.248	0.281	0.806	0.498	0.225
1.272	0.271	0.791	0.535	0.213
1.230	0.268	0.787	0.498	0.218
1.241	0.275	0.797	0.499	0.222
1.254	0.284	0.810	0.501	0.227
1.253	0.284	0.810	0.499	0.227
1.237	0.268	0.787	0.504	0.217
1.247	0.275	0.796	0.506	0.220
1.247	0.281	0.805	0.498	0.225
1.247	0.281	0.805	0.497	0.225
1.152	0.230	0.729	0.464	0.200
1.162	0.234	0.735	0.473	0.201
1.129	0.203	0.684	0.472	0.180
1.196	0.215	0.704	0.547	0.179
1.217	0.262	0.778	0.490	0.216
1.174	0.236	0.739	0.480	0.201
1.072	0.178	0.642	0.450	0.166
1.229	0.276	0.799	0.483	0.225
1.240	0.284	0.810	0.486	0.229
1.232	0.281	0.806	0.482	0.228
1.244	0.283	0.808	0.492	0.227
1.218	0.270	0.790	0.480	0.222
1.207	0.268	0.787	0.474	0.222
1.222	0.271	0.792	0.482	0.222
1.201	0.264	0.781	0.472	0.220
1.147	0.231	0.731	0.460	0.202
1.134	0.207	0.691	0.485	0.183
1.137	0.219	0.711	0.464	0.192
1.133	0.218	0.709	0.465	0.192
1.165	0.233	0.733	0.474	0.200
1.171	0.236	0.739	0.478	0.202
1.140	0.218	0.709	0.469	0.191
1.151	0.225	0.720	0.471	0.195
1.211	0.261	0.776	0.482	0.216
1.166	0.233	0.734	0.472	0.200
1.112	0.164	0.615	0.544	0.147
1.127	0.229	0.728	0.446	0.204
1.074	0.148	0.585	0.550	0.138
1.114	0.198	0.676	0.479	0.177
1.110	0.215	0.705	0.443	0.194
1.162	0.238	0.742	0.463	0.205

Figure 36. Computation of other useful parameters of the product gas triangle

$$K_p(\text{area}) = \sqrt{s(s - \theta_1)(s - \theta_2)(s - \theta_3)} \text{ where } s \text{ is semiperimeter of triangle} \quad (51)$$

$$K_p(\text{area}) = \frac{L^2 \sqrt{3}}{4} \text{ (equivalent length of an equilateral triangle with area of } K) \quad (52)$$

$$K_p = rs = \frac{\theta_1 \theta_2 \theta_3}{R}, \text{ where } r \text{ is inradius and } R \text{ is circumradius of the triangle} \quad (53)$$

In a similar fashion, we would like to construct the vector representative of the elemental composition, proximate analysis and structural component of biomass in a triangle form, however we need to follow a slightly different procedure, since we are consolidating 3 different groups.

Based on the correlations developed in part 1, we have Figure 37.

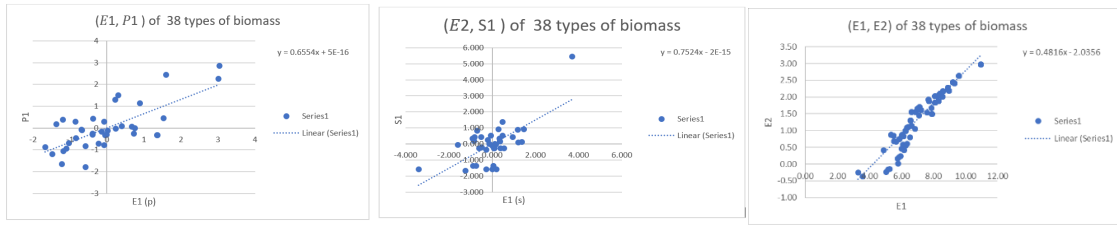


Figure 37. Correlations developed in part 1

$$E_1 = \ln\left(\frac{C^{1.45}}{O^{0.17}N^{0.63}S^{0.65}}\right), P_1 = \ln\left(\frac{FC^{0.52}VM^{0.38}}{A^{0.9}}\right),$$

$$E_2 = \ln\left(\frac{H^{0.88}}{C^{0.31}N^{0.23}S^{0.34}}\right), S_1 = \ln\left(\frac{Hem^{0.17}Cell^{0.48}}{Lig^{0.65}}\right),$$

The value of the $EPS = [E_1, E_2, S_1, P_1]$ is calculated for all the datasets. Two triangles of $\Delta E_1 E_2 S_1$ with centroid G_s and $\Delta E_1 E_2 P_1$ with centroid G_p are constructed (Fig. 38).

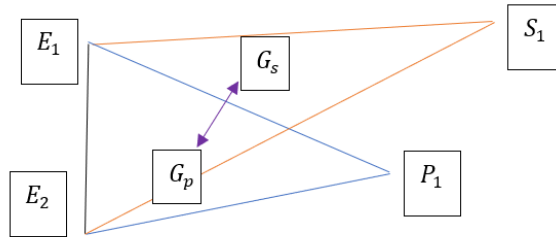


Figure 38. $\Delta E_1 E_2 S_1$ & $\Delta E_1 E_2 P_1$

We calculate the distance (d) between two centroids, and use it as a relative measure on how the structural component of biomass and its proximate analysis are

positioned with respect to its elemental composition. Therefore, G_{ps} is our first critical parameter in this regard.

The second parameter selected is the area of triangle that the base E_1E_2 makes with the middle point of $G_p - G_s$ line and denoted as (i.e. the colored area in Figure 39).

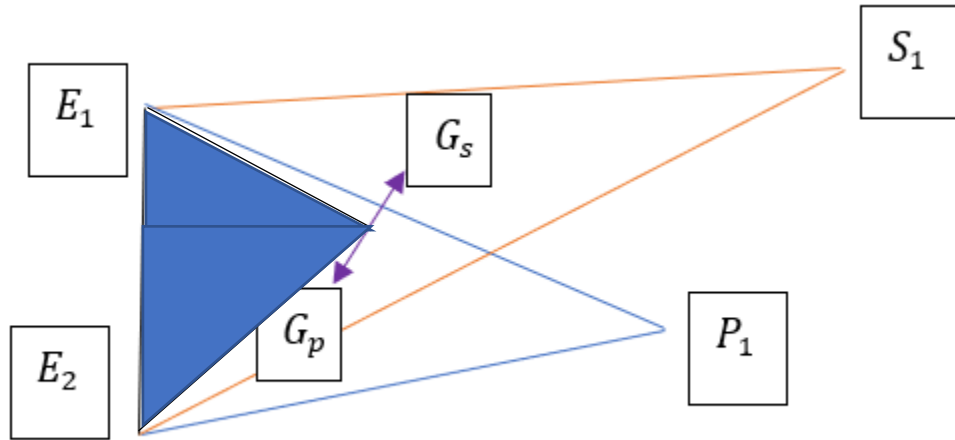


Figure 39. Illustration of the second parameter of biomass character (shaded area)

This parameter incorporates the distance between E_1E_2 in our plane which can be viewed as an enhanced packet of information regarding the elemental composition. Furthermore, the area nature of the parameter (d^2) also implies how much each data set is extended or spread overall.

Let K be the area of the triangle whose vertices are given by $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$K = \frac{1}{2} |x_1y_2 + x_2y_3 + x_3y_1 - x_2y_1 - x_3y_2 - x_1y_3| \quad (54)$$

The sample of computations is shown for the first 50 cases in Figure 40.

N° of data set	E1	E2	P1	S1	Gps	K
1	6.62	1.13	5.46	0.85	1.54	2.12
2	6.62	1.13	5.46	0.85	1.54	2.12
3	6.62	1.13	5.46	0.85	1.54	2.12
4	6.62	1.13	5.46	0.85	1.54	2.12
5	6.62	1.13	5.46	0.85	1.54	2.12
6	6.62	1.13	5.46	0.85	1.54	2.12
7	6.62	1.13	5.46	0.85	1.54	2.12
8	6.62	1.13	5.46	0.85	1.54	2.12
9	6.62	1.13	5.46	0.85	1.54	2.12
10	6.62	1.13	5.46	0.85	1.54	2.12
11	6.62	1.13	5.46	0.85	1.54	2.12
12	6.62	1.13	5.46	0.85	1.54	2.12
13	6.62	1.13	5.46	0.85	1.54	2.12
14	6.62	1.13	5.46	0.85	1.54	2.12
15	6.62	1.13	5.46	0.85	1.54	2.12
16	6.62	1.13	5.46	0.85	1.54	2.12
17	6.62	1.13	5.46	0.85	1.54	2.12
18	6.62	1.13	5.46	0.85	1.54	2.12
19	6.62	1.13	5.46	0.85	1.54	2.12
20	9.21	2.44	5.11	1.83	1.09	3.08
21	9.21	2.44	5.11	1.83	1.09	3.08
22	9.21	2.44	5.11	1.83	1.09	3.08
23	9.21	2.44	5.11	1.83	1.09	3.08
24	9.21	2.44	5.11	1.83	1.09	3.08
25	9.21	2.44	5.11	1.83	1.09	3.08
26	9.21	2.44	5.11	1.83	1.09	3.08
27	9.58	2.63	5.05	1.97	1.02	3.22
28	9.58	2.63	5.05	1.97	1.02	3.22
29	9.58	2.63	5.05	1.97	1.02	3.22
30	9.58	2.63	5.05	1.97	1.02	3.22
31	7.92	1.50	3.29	1.12	0.72	1.76
32	7.92	1.50	3.29	1.12	0.72	1.76
33	7.92	1.50	3.29	1.12	0.72	1.76
34	7.92	1.50	3.29	1.12	0.72	1.76
35	5.52	0.84	1.99	0.63	0.45	0.74
36	5.52	0.84	1.99	0.63	0.45	0.74
37	5.52	0.84	1.99	0.63	0.45	0.74
38	5.52	0.84	1.99	0.63	0.45	0.74
39	8.95	2.19	4.43	1.64	0.93	2.65
40	8.95	2.19	4.43	1.64	0.93	2.65
41	8.95	2.19	4.43	1.64	0.93	2.65
42	8.95	2.19	4.43	1.64	0.93	2.65
43	6.09	0.57	3.47	0.43	1.01	1.26
44	6.09	0.57	3.47	0.43	1.01	1.26
45	6.09	0.57	3.47	0.43	1.01	1.26
46	8.57	2.02	6.97	1.51	1.82	3.56
47	8.57	2.02	6.97	1.51	1.82	3.56
48	8.57	2.02	6.97	1.51	1.82	3.56
49	8.32	1.88	5.04	1.41	1.21	2.65
50	8.32	1.88	5.04	1.41	1.21	2.65

Figure 40. Computation of representative vector of biomass (first 50 datasets)

4.3 Construction of representative group

In the next step, a scheme with a conditional objective is implemented. The objective is to dampen high numerical variations of parameters describing events with the condition of retaining the identity of individual events. Therefore, the most important task is to hit the right level of homogeneity as a whole and independence of each case. It is necessary to establish an equal base from the starting point if we are to satisfy the homogeneity aspect. Therefore, we initiate the transformation from a circle with area equal to one ($S = 1, r = \sqrt{\frac{1}{\pi}}$) resembling a typical probability density function which is nonnegative everywhere and its integral over the entire space is equal to one. The transformative procedure for input set (i.e. Biomass properties and operating condition) is as follows:

The total area is divided into 4 sections, on the account of presence of 4 entities in the input set (i.e. biomass as a whole plus the three independent operating condition parameters, temperature (T), equivalent air ratio (ER), and steam to biomass ratio (SBR)). That said, allocation of a section to an entity is not exclusive for all, and distribution of area shares is done with following considerations:

1. The extent of involvement of a specific parameter to constitute an area is commonsurate with its significance based on the subject knowledge of the process. For example, temperature, in addition to having its own designated section, plays an auxiliary role in other sections.
2. The distribution is set up to possess the attribute of true randomness. The aim of increasing randomness intentionally, and at the earliest step of the process is to produce immunity for the model against high levels of uncertainty.

3. A lower and upper bound in calculation of each section share is considered to prevent dominance of any section. In our case, elimination of SBR section is avoided for cases where $SBR = 0$ to keep from alternating between 3 and 4 sections.
4. An internal hierarchy among sections is devised in terms of inclusiveness (Fig. 41).

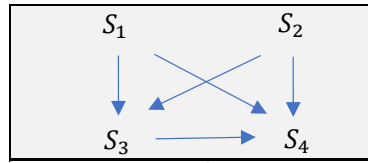


Figure 41. S_1, S_2, S_3 , and S_4 are influenced respectively by [only SBR], [T or T & ER], [Biomass, T, SBR or ALL], and [ALL]

Based on the scale of our numerical values for the input set $[d, k, T, ER, SBR]$, the following formulation and arrangement seemed to meet our conditions.

$$s_1 = \frac{3 \cdot SBR}{100} + 0.05 \quad (55)$$

$$s_2 = 0.7 * \max \left\{ \frac{T-1000}{T}, \frac{T-1000}{T} + ER \right\} \quad (56)$$

$$s_3 = \left(\frac{1+d^2}{1+d^2+\frac{10^3 k^2}{T}} \right) * (1 - (s_1 + s_2)) \quad (57)$$

$$s_4 = 1 - (s_1 + s_2 + s_3) \quad (58)$$

Transformation of a typical set of $[d, k, ER, SBR, 1000/T]$ to $[S_1, S_2, S_3, S_4]$

(Figure 42).

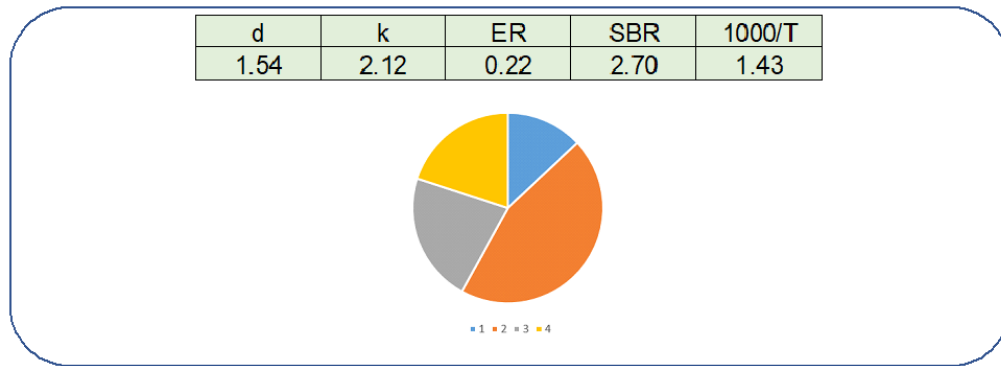


Figure 42. Illustration of the example $[s_1 = 0.13, s_2 = 0.45, s_3 = 0.22, s_4 = 0.20]$

The representation illustrated in Figure 30 has several advantages as discussed, however in following our strategy to enforce an effective system of checks and balances preventing the unfairly favoring certain subdivisions, it is decided to make use of the wellknown structure of a bicameral legislature system in which each state is represented both based on “population or size” and also “regardless of the size”. Here, we retain the property of size (i.e. area of circular sector) and for the second basis we assign “exactly one full quadrant” to each section regardless of their size (Fig. 43). Geometrically, the similarity of each partition and perpendicularity of lines separating the four areas are desirable features.

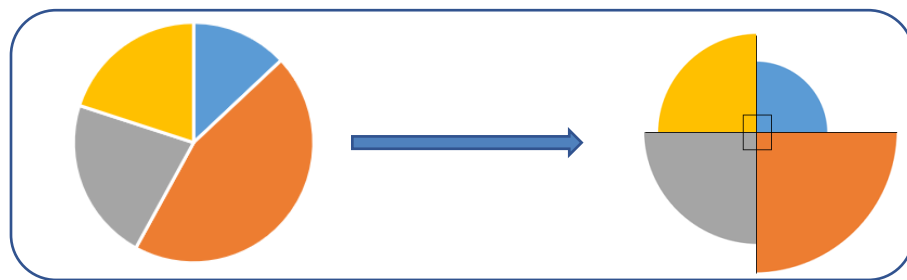
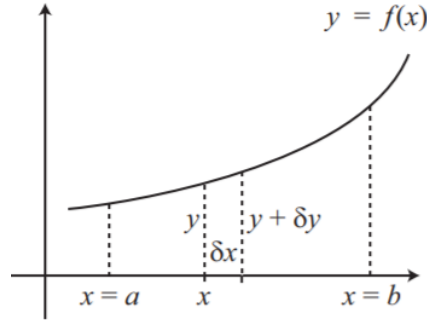


Figure 43. Assign “exactly one full quadrant”

Then, the revolutionary solid volume of quadrants is calculated to turn the 2D information to 3D to attain full geometric representation in terms of “number of dimensions”. In general we have:



$$V = \sum_{x=a}^{x=b} \partial V = \int_a^b \pi y^2 dx \quad (59)$$

Sum of the volumes for each quadrant with respect to the x and y axis yields three quarters of an sphere with the same radius.

Therefore:

for $i = 1, 2, 3, 4$ $r_i = 20 * \sqrt{\frac{S_i}{\pi}}$ (multiplied by 20 only for scaling numerical values), and

$$v_i = \pi r_i^3,$$

Another advantage of the transformation from slices with different angles to full quadrants is that quadrants can be divided into a straight segment (i.e. an isosceles right triangle) and a curvilinear segment (i.e. a cord with a fixed arc length) only by addition of one line (this is viewed as efficient binary classification).

Since the role of curvilinear segment is naturally have been addressed via the volumetric variable (v_i), we turn our attention to the straight segment which exhibited a strong relationship with its counterpart from output set (i.e. the product gas composition).

The product of ratio of altitudes ($l_i = \frac{\sqrt{2}}{2} r_i$) in right triangles (Fig. 44) showed 72%

correlation with the same variable built from output set. This strong correlation implies the efficacy of the steps taken up to this point in building highly informative characteristics.



Figure 44. Altitudes and binary areas of the kite

Additional averaging operations were done.

$$\text{Sum of the equivalent radius } R_j = \sum_1^4 r_i, \quad (60)$$

$$\text{Sum of the areas made out of binary altitudes } S_j = \frac{l_1 l_2}{2} + \frac{l_2 l_3}{2} + \frac{l_3 l_4}{2} + \frac{l_4 l_1}{2} \quad (61)$$

$$\text{A weighted sum of volumes: } V_j = \min \left(273 + \frac{v_1}{16} + \frac{v_2}{28} + \frac{v_3}{2} + \frac{v_4}{44}, 940 \right) \quad (62)$$

It can be seen that the selected weights are actually the molecular weights of the product gas composition. This type of assignment was implemented based on the assumption that we would gain more through this great leap of increasing similarity between two groups of input and output than what we may lose due to lack of an inherently one to one relation between these volumes and gas components. An analogy to this situation is where a 1v1 defending tactic is used in soccer alongside the overall strategy of defending as a group.

$$\text{Ratio of diagonals of the kite: } L_j = \frac{l_1}{l_4} \quad (63)$$

$$\text{Composite variable: } N_j = \frac{1}{2} \left(\frac{L_j * V_j}{T_j S_j * \ln(R_{j_max} - R_j)} + 12 \right) * \frac{2.3 * R}{F}, \quad (64)$$

where T_j is the temperature in °C, R =the universal gas constant, and F is the Faraday constant.

The rationale for involvement of Faraday constant is the following. To make the mathematical object created in the previous process “electrically charged” with the aim of enabling or possibly enhancing its frequency response when tested with respect to the variable of “computational time”.

In the last step of “representative” preparation we aim to recuperate its overall “uniformity” which may have been weakened due to multiplexity of steps taken after the equity circle ($S=1$).

For this purpose, the structure of Fibonacci sequence is utilized which has a prominent feature that $\frac{f_{n+1}}{f_n} = \varphi$, where φ is the golden ratio ($\frac{1+\sqrt{5}}{2}$).

The integers in Fibonacci numbers are constructed as following, starting with f_1, f_2, f_3 as seed values, and then building upon them by the convention that n th Fibonacci number is the sum of the previous two Fibonacci numbers:

$$f_1 = N_j + e^{-N_j}, f_2 = f_1 + e^{-N_j}, f_3 = e^{-N_j} * f_2, f_4 = f_3 + f_2, f_5 = f_4 + f_3, f_6 = f_5 + f_4, f_7 = f_6 + f_5, f_8 = f_7 + f_6, f_9 = f_8 + f_7, f_{10} = f_9 + f_8, f_{11} = f_{10} + f_9 \quad (65)$$

The golden ratio (approximately 1.6) starts to emerge after f_6 based on our computations and gets closer to actual value of ($\frac{1+\sqrt{5}}{2}$) moving forward. Therefore, we selected the set of $F_j = [f_6, f_7, f_8, f_9, f_{10}, f_{11}]$ as our final representatives that will go through the functional analyses. (It should be noted for convenience the subscripts 1 to 5

is used instead of 6 to 11 from now on) Figure 45 summarizes the major steps taken to obtain what we now perceive as a cohesive representation.

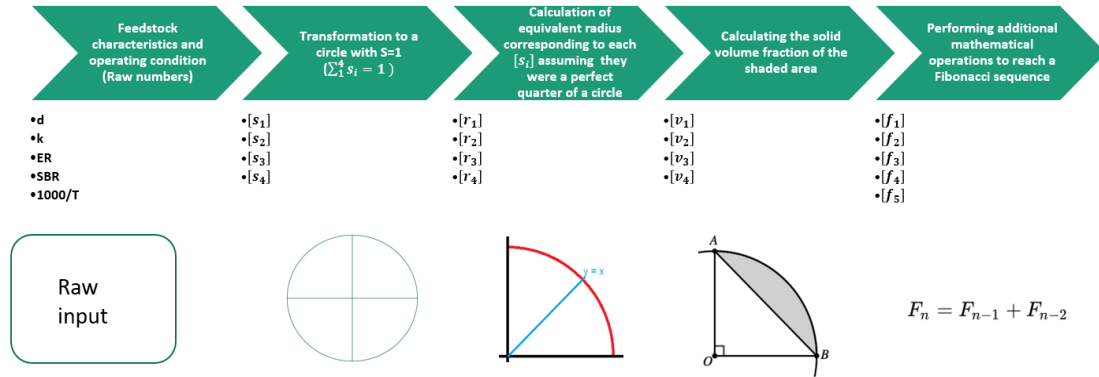


Figure 45. The progress of representative preparation

The following tables show few examples of final sets built from original data

(Figure 46):

f1	f2	f3	f4	f5
16	26	42	69	111
15	24	40	64	104
15	24	39	62	101
15	24	39	62	101
15	25	40	64	104
16	26	42	69	111
16	26	42	69	111
16	26	43	69	112
16	27	43	70	113
17	27	43	70	114
20	32	53	85	137
17	27	44	71	114
15	25	40	65	106

: Fibonacci sequence
obtained from Given
information of the
problem (Feed&
Operating condition)

Figure 46. Examples of Fibonacci sequences constructed from composite variable N_j .
(Page 1 of 2)

f1	f2	f3	f4	f5
18	29	47	76	123
18	30	48	77	125
21	35	56	90	146
23	37	59	96	155
27	43	70	114	184
21	34	55	89	144
22	36	58	93	151
24	39	63	102	164
23	37	60	96	156
22	36	58	93	151
30	49	80	129	208
23	37	59	96	155
22	35	57	91	148

: Fibonacci sequence
obtained (using
similar procedure)
from synthesis gas
composition (H₂,
CO, CO₂ and CH₄)

34	55	89	145	234
33	54	87	141	229
36	58	94	153	247
37	60	98	158	256
42	68	110	178	288
37	60	97	158	255
38	62	100	162	262
40	65	105	171	276
39	63	103	166	269
39	62	101	163	265
50	82	132	214	346
39	64	103	166	269
37	60	97	157	254

+

Figure 46. Examples of Fibonacci sequences constructed from composite variable Nj.
(Page 2 of 2)

The following plots illustrate how the distance between center of A and B differs among the 200 cases under study (First plot: a typical example, Second plot: an abnormal instance). The distance between two centers is of critical importance (Fig. 47).

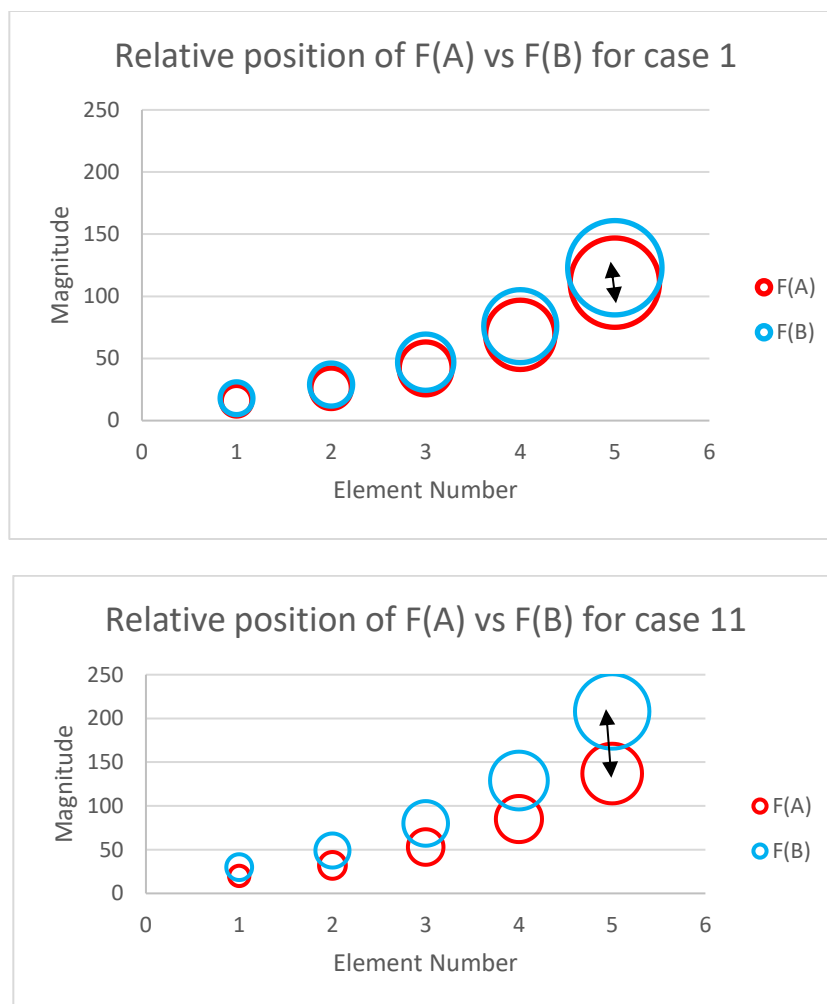


Figure 47. The centric differences of two associated Fibonacci sequences, up: typical, bottom: abnormal

At this juncture, we have successfully converted our original problem into harmonious integer sequences, representing the event (i.e. feedstock, operating condition, and product gas composition). It should be noted that carbon conversion efficiency (CCE%) and product gas yield (Nm³/kg) are the output variables that should be predicted by the model, and not included as part of the event embedded in the sequences. That will keep us two degrees of freedom away from “totality” of the process. The overall transformations practically meet the specifications of a formal mathematic “Group”,

defined in Group theory: “In mathematics, a group is an algebraic structure consisting of a set of elements equipped with an operation that combines any two elements to form a third element and that satisfies four conditions called the group axioms, namely closure, associativity, identity and invertibility. One of the most familiar examples of a group is the set of integers together with the addition operation, but the abstract formalization of the group axioms, detached as it is from the concrete nature of any particular group and its operation, applies much more widely. It allows entities with highly diverse mathematical origins in abstract algebra and beyond to be handled in a flexible way while retaining their essential structural aspects. The ubiquity of groups in numerous areas within and outside mathematics makes them a central organizing principle of contemporary mathematics.”[82]

The next step is to design a mathematical set up to serve us as a computational labrotary. Then by performing specific analyses which will be elaborated shortly, we will predict the most probable outcome of biomass gasification in fluidized bed reactor for a given input set.

The central idea of our statistical approach is “indirectness”, therefore basing the inference about the identity of an inquiry on some type of performance along an arbitrary path. The inclusiveness of the designed path and stages upon which the performance is to be monitored has direct effect on the quality and amount of information we gain. In our study, a closed, multilateral path is laid out starting from Fibbonacci sequences, followed by series of matrix operation, and ultimately reaching a fully symmetric destination (similar to the invariant point in geometric projections) that reveals accurate and deep understanding about the original problem.

4.4 Design of neural based computational lab

Here, we first give a brief and general introduction on neural networks, and then explain what role they play in our modeling.

Neural networks are conceptually quite simple computing systems, inspired by structure of biological neurons in human brain. Neural networks use computational methods to “learn” information from data without relying on a predetermined equation as a model. Deep learning is especially suited for image recognition, which is important for solving problems such as facial recognition, motion detection. “Such systems progressively learn through examples without task specific programming. Various architectures of ANNs are being used of which Multilayer Perceptron (MLP) is the most popular one. MLP consists of an input layer, a hidden layer and an output layer of neurons. While the neurons in the input layer are typically linear, the neurons in the hidden layer have nonlinear activation functions. The neurons in the output layer may be linear or nonlinear. The input layer consists of input neurons with associated weight. The hidden layer receives the signals from the input layer. There’s also a threshold parameter known as bias which is associated with each neuron in the hidden and output layers” [83]. “An elementary neuron with R inputs is shown below. Each input is weighted with an appropriate w . The sum of the weighted inputs and the bias forms the input to the transfer function f . (Fig. 48)” [84]

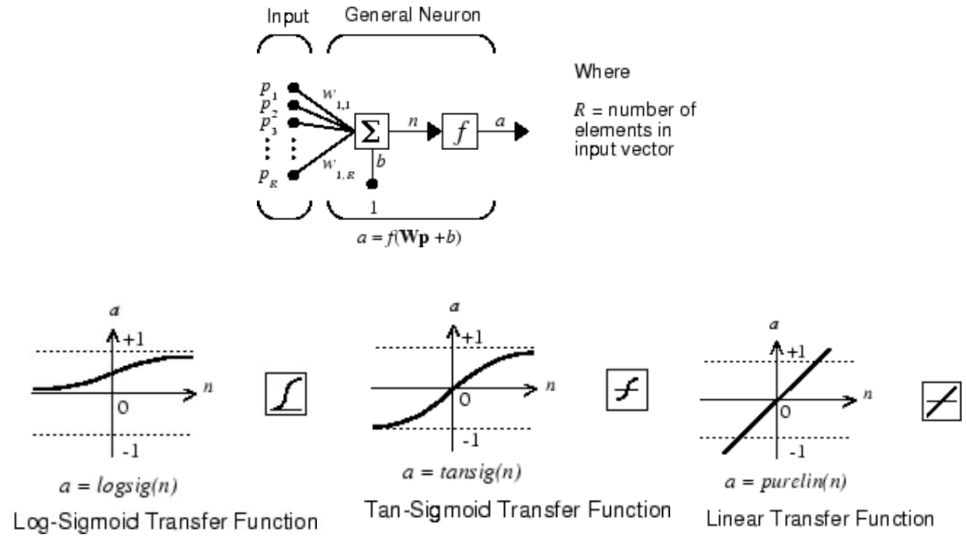
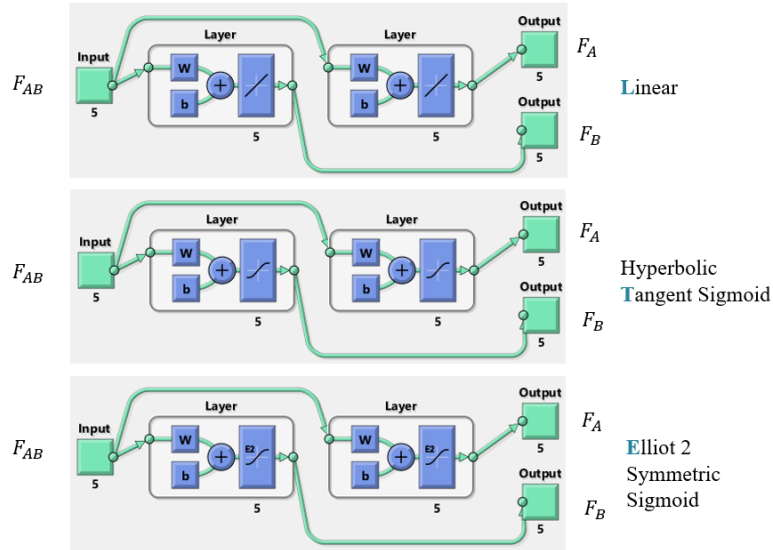


Figure 48. The structure of a general neuron as foundational unit of neural networks [84]

Development of three stable neural network unit featuring vanished gradients to serve as a computational laboratory:

Three neural network unit with three different transfer functions (i.e. linear, hyperbolic tangent sigmoid, and elliptic symmetric sigmoid) were trained by the sets of Fibonacci sequences with the configurations illustrated in Figure 49. (The ABT is chosen to exemplify the NN details).



- All NN's are composed of an input layer that receives F_{AB} , 2 hidden layers each with 5 neuron designated to predict F_A and F_B separately.
- **ABL**, **ABT**, **ABE** are 3 NN's trained based on **L**inear, **T**angential and **E**lliptical transfer function, respectively.

- There are **60** elements in ABT.
- **50** elements in terms of weight (w) (i.e. one matrix of $[]_{5 \times 5}$ for each layer).
- And, **10** elements in terms of bias (b) (i.e. one vector of $[]_{5 \times 1}$ for each layer).
- For example, the following numerical configuration, is resulted upon training the "ABT" neural network using Levenberg-Marquardt backpropagation algorithm with aim of minimizing sum of squared errors in predicting F_A and F_B .

-1.42	2.51	1.77	1.28	1.35	-0.81
-0.61	0.48	-3.08	0.94	-1.98	3.09
-2.01	-2.19	-2.27	-0.24	0.93	2.89
-2.20	-1.59	-1.98	0.28	-1.89	2.72
3.13	-1.04	1.06	-1.44	-0.93	1.55

1.43	-0.15	0.12	3.54	0.57	-4.69
2.42	-0.55	2.75	0.28	1.06	-3.94
-2.98	-0.38	2.10	-1.06	0.62	0.85
2.30	-1.04	0.77	2.35	-1.56	-0.44
2.04	0.06	-0.90	2.79	-1.47	0.67

The first layer in the figure connected to output F_B

The second layer in the figure connected to output F_A

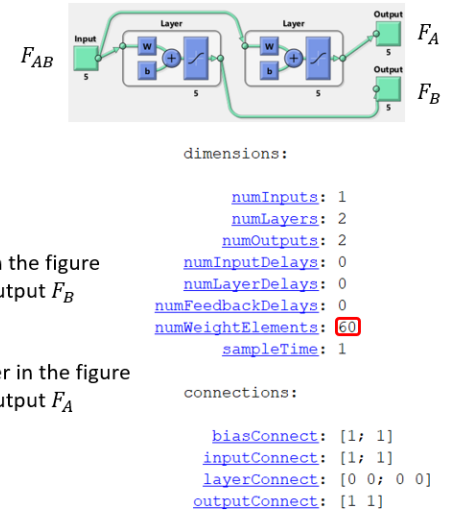


Figure 49. The detailed configuration of ABT (i.e. the neural network set up using hyperbolic tangent as transfer function)

Following the training, the neural network units shows properties of a bounded function with respect to the inputs (i.e. F_{AB}), meaning for all the input they return identical vectors of :

$$F_A = [1, 1, 1, 1, 1]; F_B = [1, 1, 1, 1, 1];$$

This seemingly short circuited configuration provides critical information on the hollistic state of data distributions. Specifically, the weights and biases calculated upon the training indirectly capture the distribution of input sets with respect to their corresponding output set. This can be viewed as an unconventional method of averaging distributions.

We will show how this particular quality can form a structure preserving map, and yield a commutative diagram across the three different neural network units.

Following our strategy to cover broad ranges of possibilities with respect to variation potentials (vertical, horizontal, and cross variations) The following scheme was arranged:

- Three different fractions of F_A (i.e. 0.4, 0.7 and 1) are considered.
- The input to ABL is in form of $[0.4 * F_A, 0.7 * F_A, F_A]$ while the input to ABT and ABE are in form of $[\frac{1}{0.4 * F_A}, \frac{1}{0.7 * F_A}, \frac{1}{F_A}]$.

Defining the quantitative measures reflecting relative deviations from:

$$F_A = [1, 1, 1, 1, 1]; F_B = [1, 1, 1, 1, 1];$$

is done with a combinatory approach using “trace” of matrices as scores. For each run of $[0.4, 0.7, 1]$ vs $[L, T, E]$ (i.e. 9 pairs) two groups of calculation is done:

1. Changes in F_B with respect to F_A , and the F_{AB}
2. Changes in F_A with respect to F_B , and F_{AB} .

It is important to notice that “changes in F_A and F_B ” exactly mean their deviations from identity vectors $F_A = [1, 1, 1, 1, 1]; F_B = [1, 1, 1, 1, 1]$.

Another important issue is that the neural network units are trained with the input set of F_{AB} where $F_{AB} = F_A + F_B$ however when we feed the

input to the neural network to monitor its changes, our input set is actually $F_{AB} = F_A$ because we do not have information about F_B . That said, the NN's behavior is unaffected and they will “act” as they are receiving some sort of F_{AB} that is constructed as $F_{AB} = F_A + F_B$. This subtle issue enables us to observe the effects of a specific F_A (i.e. an inquiry) on F_B and quantify its relation with respect to all or nearly all elements or aspects of the process.

As for the quantification method and scoring system the following matrix operation is established. All matrices have the dimension of $[]_{5 \times 5}$.

- $[B * A^T], [dB * A^T], [dB * A_{in}^T]$, (reflecting the changes of B to A) (67)
- $[A * B^T], [dA * B^T], [A_{in} * dA^T]$, (reflecting the changes of A to B) (68)

When the trace of the three matrices are summed up in each group, they yield a Figure 50 scores such as below.

		L	T	E
B to A	0.4	1.05	3.94	3.30
	0.7	1.67	3.64	3.00
	1	2.14	3.52	2.90
A to B	0.4	1.12	-1.28	3.43
	0.7	1.71	-1.12	2.94
	1	2.10	-1.04	2.70

Figure 50. Scores calculated from traces of matrices defined by Equations 67&68

Conceptually this figure summarizes the changes along 4 axioms: path, scale, AtB and BtA. Each score in table 6 is the sum of traces for three matrices ($\sum_3 []_{5 \times 5} \text{ trace}$) in aforementioned groups, it can be seen that the information in table 6 is obtained from traces of 54 matrices $[]_{5 \times 5}$ (i.e. 3 scale* 3 path* 6 measure = 54 matrices that overall consist of 1350 numerical value, each probably capturing an infinitesimal relation

between A&B) corresponding to a single inquiry. An important feature of this set up is that forms a commutative diagram between linear, hyperbolic tangent, and elliot modules as illustrated in the next Page (Fig. 51).

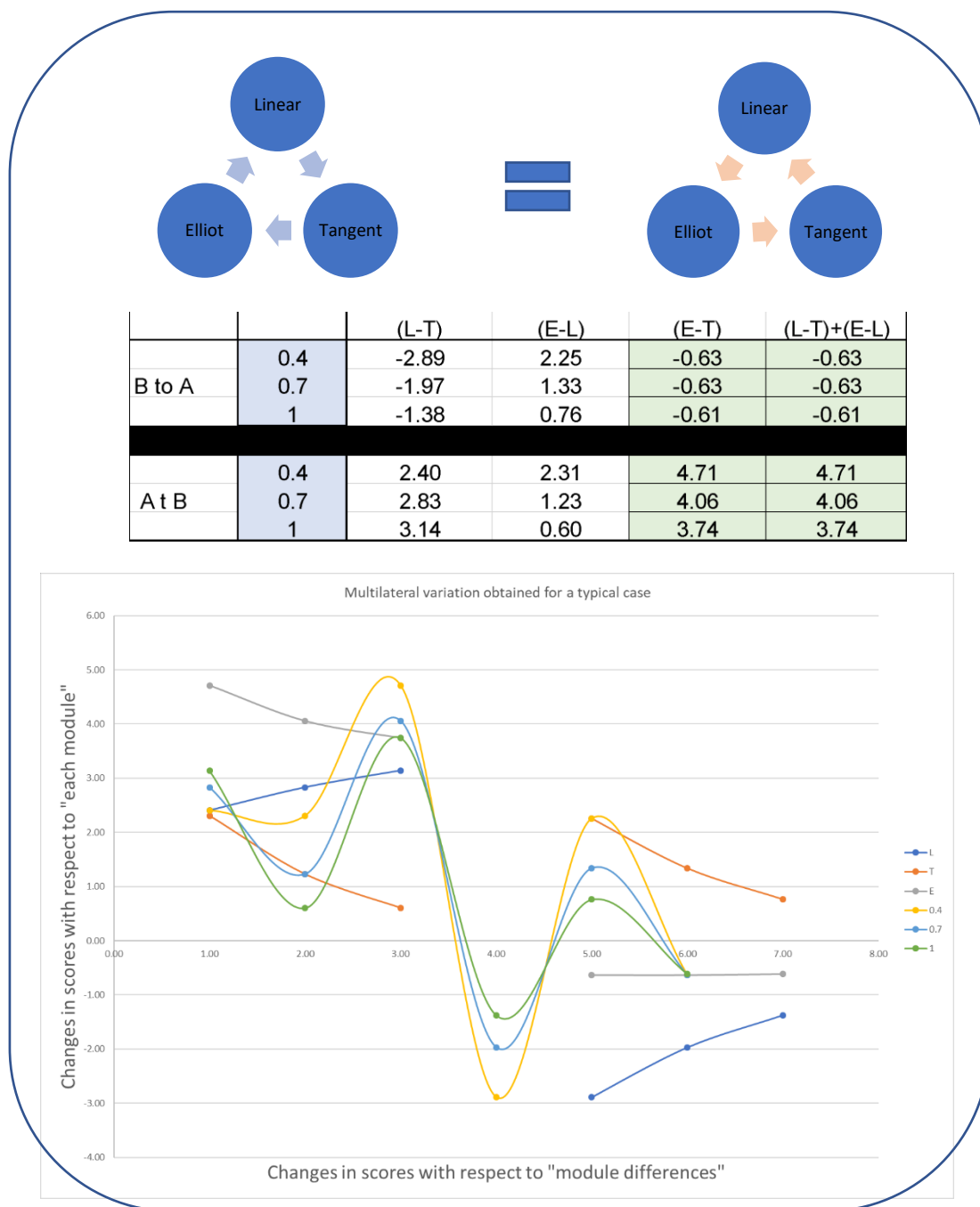


Figure 51. Elliot modules as illustrated

An important and inherent aspect of self-ruling functional analyses is to select a point where achievement of entanglement quantification can be fairly speculated. The other important consideration in these type of analyses is to decide the “form” and “dimension” (i.e single scalar, vector, involution, etc.) of the mathematical object ready for final integration. Here, since we are dealing with a convoluted object, it does not seem economical to perform rigorous structural analysis of each case, therefore, we base our judgement on the “general principle” of equilibrium for stationary objects. To be stationary all of the forces on the object must be in equilibrium, the resultant force and moment vectors must both be zero. From structural point of view, the number of equilibrium conditions required to solve different force systems can be summarized as following:

Type of force system	two dimensional	Three-dimensional
General	3	6
Coplanar	3	3
Concurrent	2	3
Parallel	2	3
Coplanar, parallel	2	2
Coplanar, concurrent	2	2
Collinear	1	1

Figure 52. Number of equilibrium conditions required to solve different force systems
(Civil engineering reference manual 41-7)

Given our aim to solve a general system, 6 equilibrium condition is needed. By considering all the 4 axioms, the summation shown in the last column in Figure 53 is chosen as the 6 equilibrium conditions.

		L	T	E	Sum
B to A	0.4	1.05	3.94	3.30	8.29
	0.7	1.67	3.64	3.00	8.31
	1	2.14	3.52	2.90	8.56
A to B	0.4	1.12	-1.28	3.43	3.26
	0.7	1.71	-1.12	2.94	3.54
	1	2.10	-1.04	2.70	3.75

Figure 53. Calculation of sum of rows to get 6 static condition

It can be seen that equilibrium conditions for changes of B with respect to A (i.e. the first 3 values) are close to the universal gas constant ($R=8.314$). This could be partially attributed to involvement of R in construction of the composite variable:

$$N_j = \frac{1}{2} \left(\frac{L_j * V_j}{T_j S_j * \ln \ln (R_{j_max} - R_j)} + 12 \right) * \frac{2.3 * R}{F},$$

where T_j was the temperature in °C, R =the universal gas constant, and F was the Faraday constant, however we are neither to claim that is necessarily the case nor think its correctness or incorrectness significantly reinforce or invalidate our course of actions. A similar argument can be applied with regard to closeness of equilibrium measures for changes of A with respect to B (i.e. the second 3 values) to the mathematical constant π .

That said, π & R can be used to quantify the measure of dispersion around “two” fixed values to normalize the “bimodal” distribution at hand. (In some sense, it is πR -reviewed!).

By normalization of the first 3 values with respect to $R=8.314$ and the second 3 scores with respect to $\pi = 3.14$, we get the numerical values with the following scale:

	1	2
1	[0.9974;0.9994;1.0290]	[1.0383;1.1259;1.1947]
2	[1.0047;0.9943;1.0162]	[1.0212;1.0961;1.1725]
3	[1.0143;1.0044;1.0274]	[1.0481;1.1511;1.2651]
4	[1.0175;1.0100;1.0354]	[1.0682;1.1863;1.3153]
5	[1.0277;1.0373;1.0773]	[1.1419;1.3093;1.4805]
6	[1.0101;1.0216;1.0608]	[1.0757;1.1913;1.2882]
7	[1.0213;1.0411;1.0887]	[1.0719;1.1846;1.2786]
8	[1.0253;1.0469;1.0928]	[1.1151;1.2655;1.4035]
9	[1.0271;1.0555;1.1081]	[1.0824;1.2032;1.3057]
10	[1.0212;1.0431;1.0875]	[1.0768;1.2099;1.3256]

Figure 54. Static vectors denoted as S_{AB} & S_{BA}

Following this transformation, dispersions around 1, now is supposed to reflect the differential information among different cases. In other words, since “1” is directly associated with 2 fundamental constants π & R , and itself reflects total equity, we may establish the connection between differentials and a fundamental equilibrium state (Equity=1). At this stage, although the non-equilibrium factor in our process has been characterized statically, there remains a fundamental gap: lack of a variable with nature of Rate & Resistance.

Basically, thermodynamics considerations do not establish the rates of chemical or physical processes because resistance is not a thermodynamic variable.

Any proposition to resolve this crucial issue must effectively combine the two seemingly paradoxical concepts of static/equilibrium and dynamics.

Here, we propose a novel technique of analysis based on “abstract physical operation on matrices” with respect to “time”. Let us exemplify the procedure by first examining 4 matrices formed by average multiplication of two static vectors for cases 1-4 as shown below:

$$S_{ci} = \frac{1}{2} (S_{ABi} * S_{BAi}^T + S_{BAi} * S_{ABi}^T), \quad (69)$$

$$S_{c1} = \begin{bmatrix} 1.035 & 1.080 & 1.130 \\ 1.080 & 1.125 & 1.176 \\ 1.130 & 1.176 & 1.229 \end{bmatrix},$$

$$S_{c2} = \begin{bmatrix} 1.025 & 1.058 & 1.107 \\ 1.058 & 1.089 & 1.139 \\ 1.107 & 1.139 & 1.191 \end{bmatrix}$$

$$S_{c3} = \begin{bmatrix} 1.063 & 1.110 & 1.180 \\ 1.110 & 1.156 & 1.226 \\ 1.180 & 1.226 & 1.299 \end{bmatrix},$$

$$S_{c4} = \begin{bmatrix} 1.086 & 1.143 & 1.222 \\ 1.143 & 1.198 & 1.278 \\ 1.222 & 1.17 & 1.361 \end{bmatrix}$$

The arrows in the following matrix indicate the direction of increase in the numerical values. This uniform pattern exists in all the 200 cases, and its emergence may be attributed to the overall harmonic tone of our analyses.

$$S_{ci} = \begin{bmatrix} S_{ci}^{11} & S_{ci}^{12} & S_{ci}^{13} \\ S_{ci}^{21} & S_{ci}^{22} & S_{ci}^{23} \\ S_{ci}^{31} & S_{ci}^{32} & S_{ci}^{33} \end{bmatrix}$$

4.5 Institution of a static/dynamic variable & integration

Now, in order to obtain a dynamical property value from a static situation, there is a need for an operation consisting some sort of change without tearing or breaking the static structure of the object. A stretching process to deform the object with condition of sorting the values from lowest to highest value seems to have this particular quality.

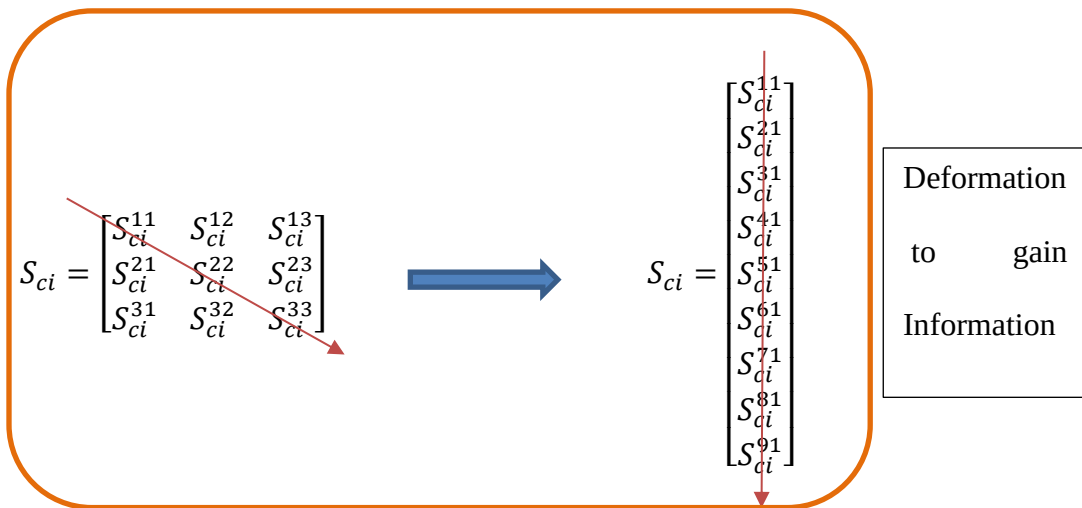


Figure 55. Deformation to gain Information

Therefore, the operation of reshaping the average product of static vectors $[]_{3 \times 3}$ to $[]_{9 \times 1}$, and arranging them from lowest to highest (as if those scores were time series or any other naturally increasing quantity such as entropy) is deemed to represent the act of “change in order”, and the duration of time from starting this specific activity till its completion is recorded. This specific computational time varied across different cases. Given the similarity of environmental condition in this test such as the device power, those differences are hypothesized to be related to stiffness of computation for different cases. This hypothesis of “relation to stiffness” is proposed based on (or inspired by) two notions:

1. The analogy between our defined event and the concepts of elasticity and strain for a physical object. (when a material is deformed under a tectonic force)
2. The theory of computational complexity that states:

“In computer science, the computational complexity, or simply complexity of an algorithm is the amount of resources required for running it. The computational complexity of a problem is the minimum of the complexities of all possible algorithms for this problem (including the unknown algorithms).

As the amount of needed resources varies with the input, the complexity is generally expressed as a function $n \rightarrow f(n)$, where n is the size of the input, and $f(n)$ is either the worst-case complexity, that is the maximum of the amount of resources that are needed for all input of size n , or the average-case complexity, that is average of the amount of resources over all input of size n . *When the nature of the resources is not explicitly given, this is usually the time needed for running the algorithm, and one talks of time complexity.* However, this depends on the computer that is used, and the time is

generally expressed as the number of needed elementary operations, which are supposed to take a constant time on a given computer, and to change by a constant factor when one changes of computer”.[85]

Specifically, we utilized an application in the MATLAB environment which allows user to record the duration time of a specific operation. The operation was done for all 200 cases and the time for each case recorded using “timeit” functionality, all the numerical values are given in appendix C.

```
For n=1:200 f = @()
tscollection(reshape(sumrows_LTE{n,3},9,1));tsc=tscollection(reshape
(sumrows_LTE{n,3},9,1));St_En(n,1)=tsc.TimeInfo.Start;
St_En(n,2)=tsc.TimeInfo.End; t1 = timeit(f); St_En(n,3)=t1; End
```

Figure 56. Function of “timeit”

An important feature that distinguishes this part of analysis from the rest, is its computational irreversibility.

“As was first argued by Rolf Landauer of IBM, in order for a computational process to be physically reversible, it must also be logically reversible. Landauer's principle is the rigorously valid observation that the oblivious erasure of n bits of known information must always incur a cost of $nkT \ln(2)$ in thermodynamic entropy. A discrete, deterministic computational process is said to be logically reversible if the transition function that maps old computational states to new ones is a one-to-one function; i.e. the output logical states uniquely determine the input logical states of the computational operation.”[86]

It is important to notice that “inverse functions” can be calculated for trigonometric, geometric transformations, the neural network formulations, etc. However, given a particular “time complexity”, the static matrix cannot be back calculated, since the relationship between them is not a bijection (Fig. 57). There is more than one configuration of that matrix with the same computational complexity.

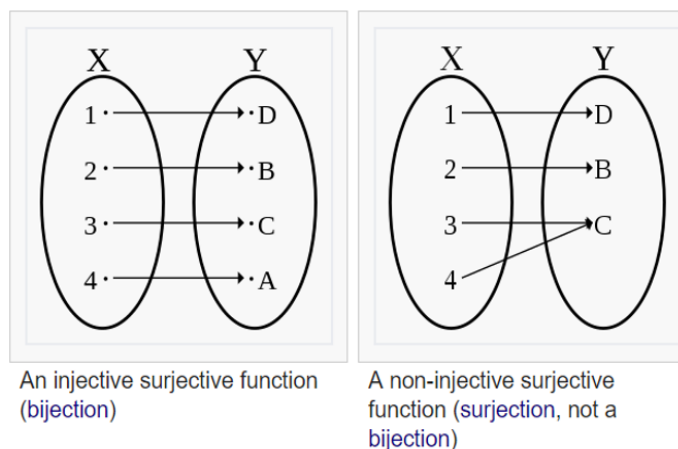


Figure 57. Surjective vs non-surjective function

Nevertheless, attainment of this dynamical/static property value proved to be extremely useful in resulting monotonic trends for important process variables such as carbon conversion efficiency and the yield of product gas (figures 14-17). The specific computational time (sct) marks the finish line of our functional analysis. In the last step, we need to perform an integration with regard to the variables we consider the most critical, and informative regarding the path traveled by our inquiry.

For the choice of integration, given the presence of two focal points in our normalization which was selected due to bimodality of the distribution, a complete “elliptic” integral of the first kind was recognized as the most compatible option. The

integration was done through a low pass filter. Specifically, the Ellipap module in MATLAB was employed. This module returns the zeros, poles, and gain of an order “n” elliptic analog lowpass filter prototype, with Rp dB of ripple in the passband, and a stopband Rs dB down from the peak value in the passband (Fig. 38) The zeros and poles are returned in length n column vectors z and p and the gain in scalar k. If n is odd, z is length n - The transfer function in factored zero-pole form returns the poles and zeros, and gain of:

$$H(S) = \frac{Z(S)}{P(S)} = k \frac{(s-z_1)(s-z_2)\dots(s-z_N)}{(s-p_1)(s-p_2)\dots(s-p_M)} \quad (70)$$

This module uses ellipke to calculate the “complete elliptic integral” of the first kind and ellipj to calculate Jacobi elliptic. It computes the Jacobi elliptic functions using the method of the arithmetic-geometric mean starting with a *triplet* of numbers. The ellipap module was fed the following three variables which were literally obtained from a *trip* along the *L*, *E*, *T* modules:

1. The specific computational time (sct)
2. The summation of 5 Fibonacci number for an inquiry (A) (i.e. starting point)
3. The absolute temperature (T)

$$a_0 = 1, b_0 = \sqrt{1-m}, c_0 = \sqrt{m}$$

$$a_i = \frac{1}{2}(a_{i-1} + b_{i-1}) \quad b_i = (a_{i-1}b_{i-1})^{\frac{1}{2}} \quad c_i = \frac{1}{2}(a_{i-1}b_{i-1})$$

$$\sin(2\phi_{n-1} - \phi_n) = \frac{c_n}{a_n} \sin(\phi_n)$$

$$\operatorname{dn}(u) = \sqrt{1-m \cdot \operatorname{sn}(u)^2} \quad \operatorname{sn}(u) = \sin \phi_0 \quad \operatorname{cn}(u) = \cos \phi_0$$

Where the Jacobi elliptic functions are

$$u = \int_0^\phi \frac{d\theta}{\sqrt{1-m \cdot \sin^2 \theta}}$$

$$\operatorname{sn}(u) = \sin \phi \quad \operatorname{cn}(u) = \cos \phi \quad \operatorname{dn}(u) = \sqrt{1-m \cdot \sin^2 \phi}$$

$$[K(m)] = \int_0^1 [(1-t^2)(1-mt^2)]^{\frac{-1}{2}} dt$$

Figure 58. The Ellipap module formulation scheme

4.6 Preliminary results

The typical results for poles, zeros and gain are shown below.

	1	2	3	4	5	6	7	8	9	10	11	12	13
1	14x1 comp...	16x1 comp...	16x1 comp...	18x1 comp...	20x1 comp...	18x1 comp...	14x1 comp...	16x1 comp...	14x1 comp...	16x1 comp...	22x1 comp...	16x1 comp...	14x1 comp...
2	15x1 comp...	16x1 comp...	17x1 comp...	18x1 comp...	20x1 comp...	18x1 comp...	15x1 comp...	16x1 comp...	15x1 comp...	16x1 comp...	23x1 comp...	16x1 comp...	15x1 comp...
3	1.4482e-47	3.9811e-53	1.9248e-52	3.9811e-58	1.2589e-60	1.2589e-55	3.1201e-52	1.2589e-55	3.0026e-52	1.2589e-55	3.6470e-53	1.2589e-55	3.4475e-52

Figure 59. Typical results for poles, zeros and gain.

The value of gains displayed in the third row are denoted with Λ . $\ln \left(\frac{\Lambda_S}{\Lambda_G} \right)$ where Λ_G is Geomean of all cases, and Λ_S the value computed for a specific case is a dimensionless number that we call the local minima of algorithm.

The difference between size of pole and zero vectors are either 1 or 0 which we call Squaricity Index (Fig. 60&61).

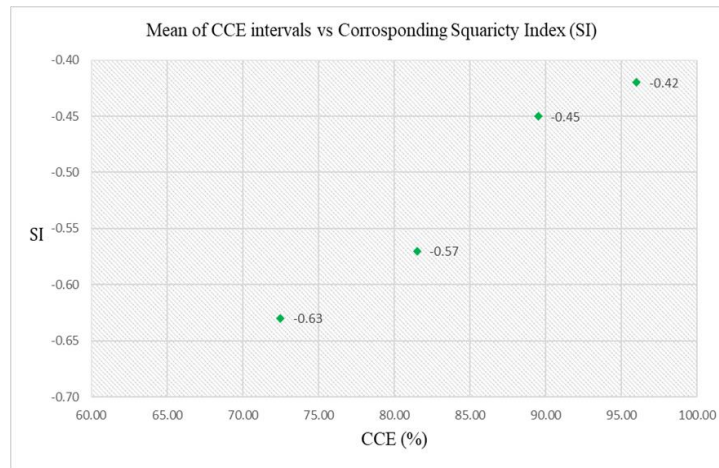


Figure 60. Carbon Conversion Efficiency vs Squaricity Index

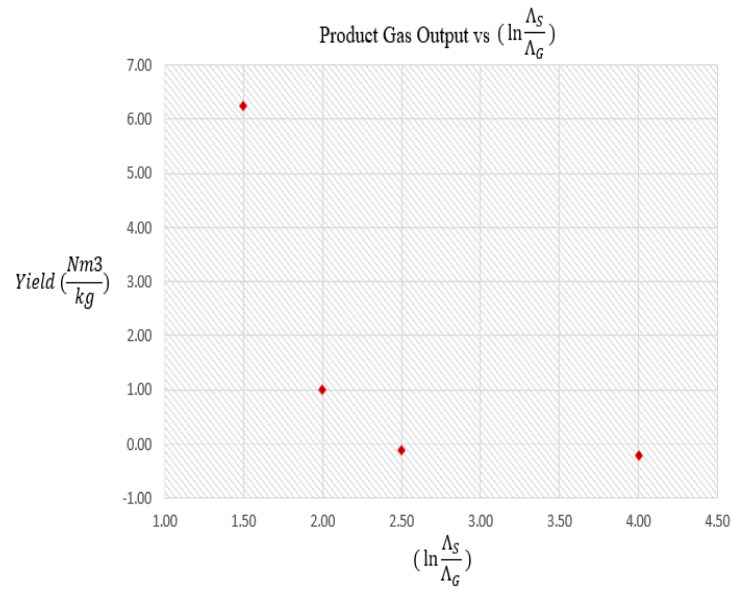
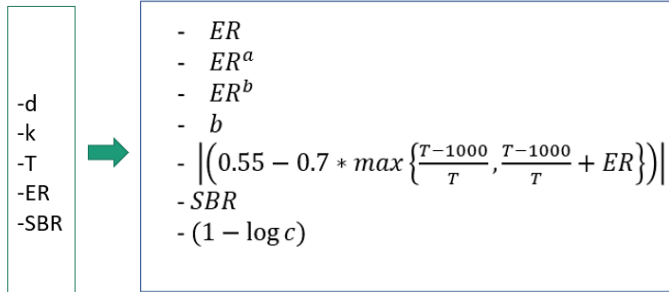


Figure 61. Syngas yield vs local minima of algorithm

For prediction of the product gas composition a neural network module was built utilizing the constructed parameters:



$$\text{Where } a = \frac{\sqrt{\left(\frac{6 \cdot SBR + 10}{\Pi}\right)}}{c}$$

$$c = \sqrt{\left(\frac{140 \cdot \max\left\{\frac{T-1000}{T}, \frac{T-1000}{T}\right\} + ER}{\Pi}\right)}$$

$$b = \frac{\Pi * \left(1 + d^2 + \frac{10^3 k^2}{T}\right) * \left(1 - 0.7 * \max\left\{\frac{T-1000}{T}, \frac{T-1000}{T}\right\} + ER\right) - \frac{3 * SBR + 5}{\Pi}}{c * 200 * (1 + d^2)}$$

$$y_{H_2} + y_{CH_4} + y_{CO} + y_{CO_2} = 1,$$

$$\cos^2 \theta_1 = y_{CO} + y_{CO_2},$$

$$\cos^2 \theta_2 = y_{CH_4} + y_{CO_2},$$

$$\cos^2 \theta_3 = y_{CH_4} + y_{CO},$$

$$\sin^2 \theta_1 = y_{H_2} + y_{CH_4},$$

$$\sin^2 \theta_2 = y_{H_2} + y_{CO},$$

$$\sin^2 \theta_3 = y_{H_2} + y_{CO_2},$$

$$y_{H_2} = 1 - \frac{\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3}{2} \quad (71)$$

$$y_{CH_4} = 1 - \frac{\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3}{2} - \sin^2 \theta_1 \quad (72)$$

$$y_{CO} = 1 - \frac{\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3}{2} - \sin^2 \theta_2 \quad (73)$$

$$y_{CO_2} = 1 - \frac{\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3}{2} - \sin^2 \theta_3 \quad (74)$$

Where the relationship between $[d, k]$, $[E_1, E_2, S_1, P_1]$, and $[C, H, O, N, S, Cell, Hem, Lig, Vm, Fc, A]$ is as follows:

$$d = \frac{P_1}{3} - \frac{S_1}{3} \quad (75)$$

$$k = \frac{1}{2} \left| \frac{P_1 + S_1}{6} * \frac{2E_1 - E_2}{3} \right| \quad (76)$$

$$E_1 = \ln\left(\frac{C^{1.45}}{O^{0.17}N^{0.63}S^{0.65}}\right), P_1 = \ln\left(\frac{FC^{0.52}VM^{0.38}}{A^{0.9}}\right), E_2 = \ln\left(\frac{H^{0.88}}{C^{0.31}N^{0.23}S^{0.34}}\right), S_1 = \ln\left(\frac{Hem^{0.17}Cell^{0.48}}{Lig^{0.65}}\right) \quad (77)$$

And $[T, ER, SBR]$ is the gasification operating condition.

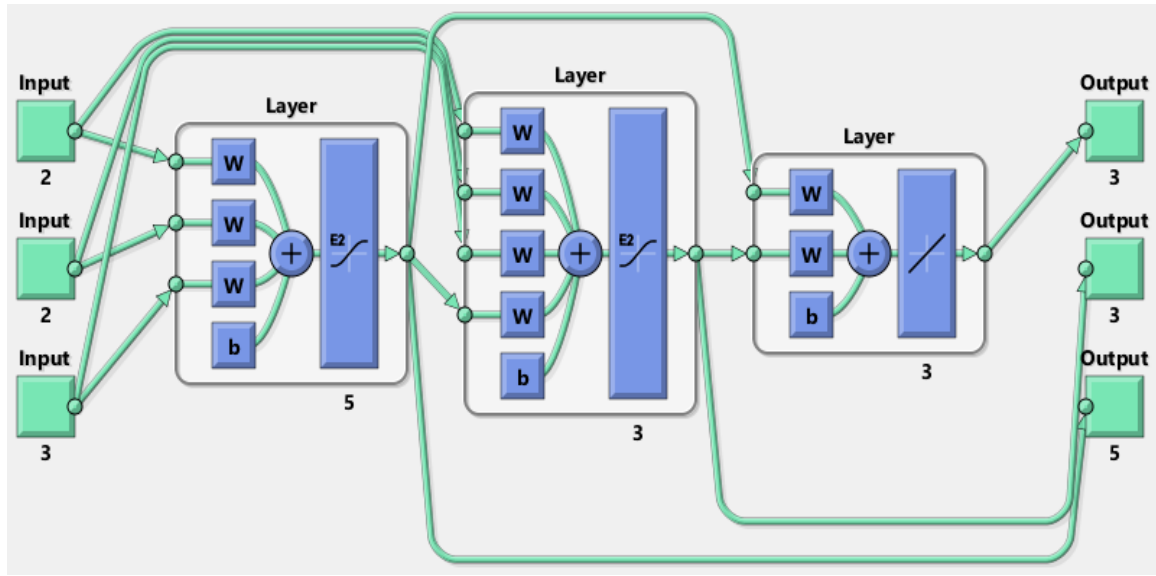


Figure 62. The module developed for prediction of product gas composition

The plots of prediction vs experimental (Overall Error=%8.6) for gas composition is presented in Figures 63-66:

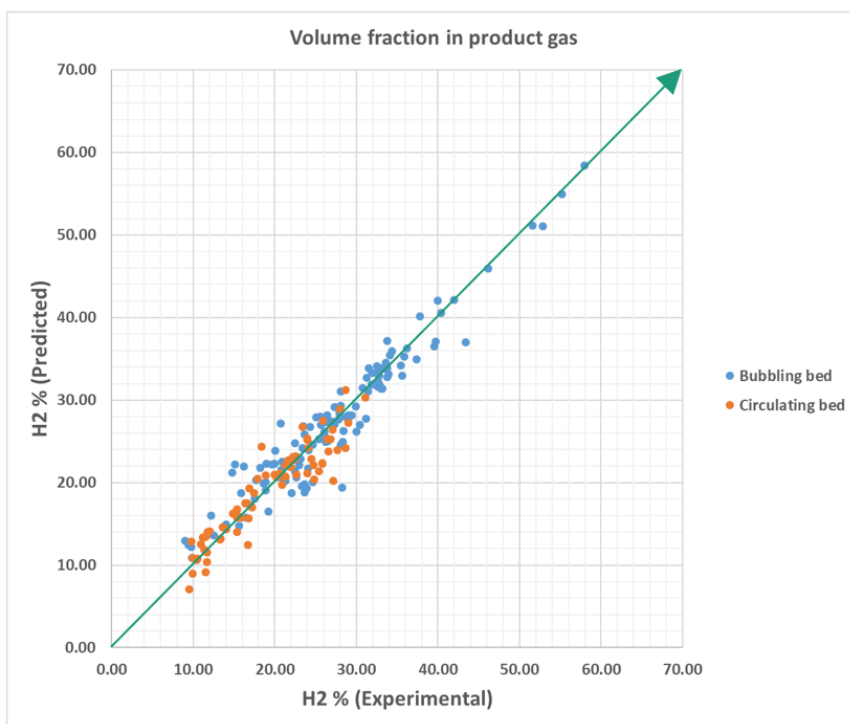


Figure 63. H₂ prediction vs experimental values

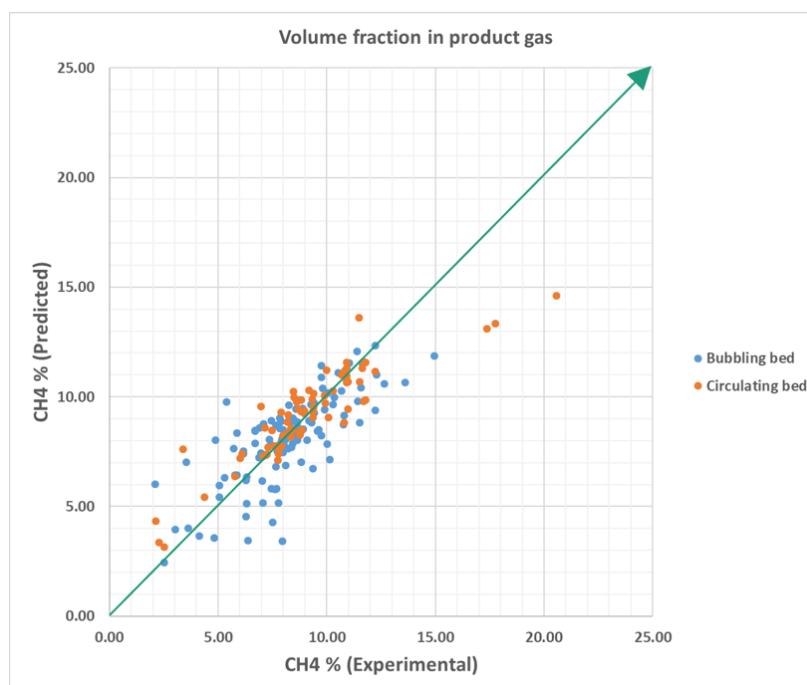


Figure 64. CH₄ prediction vs experimental values

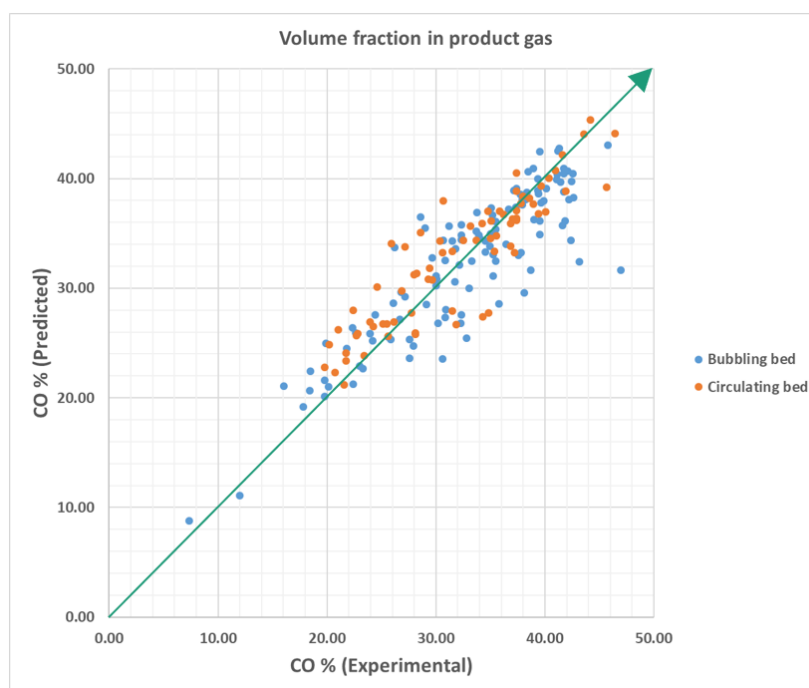


Figure 65. CO prediction vs experimental values

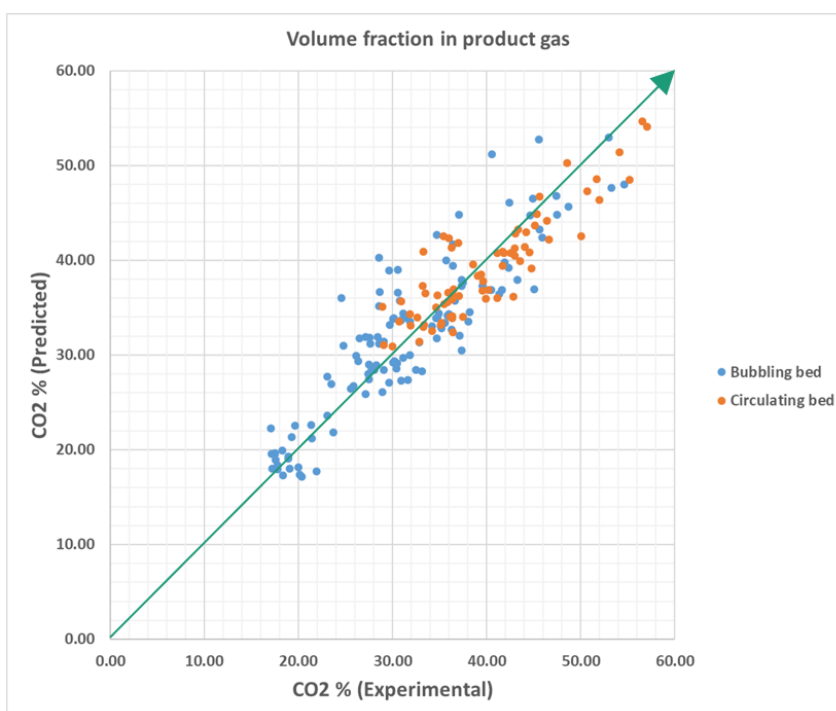


Figure 66. CO2 prediction vs experimental values

CHAPTER 5

RESULTS, DISCUSSIONS

What is presented in this chapter is essentially juxtaposition of two slightly different performance evaluation analyses, one borrowed from literature evaluating three approaches of modeling biomass gasification in fluidized bed and one done in this study in form of a sensitivity analysis on our model. The two analyses have the important common feature of evaluating models against completely new unseen datasets, a metric that is very dear to our hearts.

We consider the 114 setpoints of gas yield, 96 setpoints of carbon conversion collected from more than 15 different experiments as part of this “completely new unseen datasets” because they had not been used at any stage of our modeling effort although we had used the composition data pertinent to those gasification setpoints. It should be mentioned some modellers may have a more strict definition of completely new unseen datasets.

As for composition data, the three models evaluated by Moshtaghi [17] were tested against 63 setpoints with exception that 6 out 63 and 15 out 63 setpoints had been used in developing Model II and Model III respectively. Our model was tested against 70 new datasets.

5.1 Performance of different modeling approaches

Three approaches of modeling biomass gasification in fluidized bed “MODEL I, which restricts equilibrium to a specific temperature (QET method); MODEL II, which uses empirical correlations for carbon, CH₄, C₂H₂, C₂H₄, C₂H₆ and NH₃ conversion;

and MODEL III, which includes kinetic and hydrodynamic equationsd” was studied and compared by Moshtaghi [17].

“MODEL I, MODEL II and MODEL III (Figure 70.) are first replicated in ASPEN PLUS and verified by comparison with the model results in the original presented studies. Experimental data from different BFB and CFB gasifiers have been collected from the literature and used to evaluate the model performance, with the aim to test whether the models are also valid for experimental conditions other than those for which they were originally validated.” [17]

“The cases with input parameters in different ranges than those of the original validated experiments have been chosen for the evaluation step.” [17]

“Basically each model has been validated with exactly the same data used by the authors that developed that model. However due to further use of the input data presented in this table for evaluation of models, it is important to report all of the experimental data used for validation of the original models. Further, the experimental data that have been used for the evaluation of all models are listed in Figure 68-69. Each “setpoint” is a combination of one set of ER value, S/B ratio, temperature, pressure and load.” [17]

“To form a common basis for comparison of the different models, the same set of experimental data has been used for the evaluation of all models.” [17]

“The experimental setups chosen for evaluation of the models are listed in Figure 69. Moreover, set points **1 to 6 of CFB-EXP II** and **1 to 15 of BFB-EXP III** from Figure 68 are also used for model evaluation. These points are highlighted with italic outlined numbers in Figure 68.” [17]

	Model I	Model II	Model III	Black-box
Temperature range (span)	780-815 (35 C)	856-955 (99 C)	700-900 (200 C)	668-950 (282 C)
ER range (span)	0.22-0.53 (0.21)	0.28-0.39 (0.11)	0.19-0.27 (0.08)	0-0.84 (0.84)
SBR range (span)	-	0.08-0.28 (0.20)	0-4.04 (4.04)	0-4.7 (4.7)
Oxidizer	Air	Air/Steam	Air/Steam	Air/Steam
Method	Assuming <u>temperature lower than equilibrium</u> for specific components and reactions	Using <u>the empirical correlation for non equilibrium</u> components in product gas by calculator block	Including <u>reaction kinetics</u> and <u>hydrodynamics</u> of the bed as an external Fortran subroutine. Including an <u>empirical nonlinear regression</u> with specific temperatures for CH ₄ components	Including <u>characterization of feedstock using machine learning techniques</u> . <u>Quantifying the interconnection</u> of feedstock, operating condition, and product gas <u>using Neural network</u> framework

Figure 67. Characteristics of models evaluated by Moshtaghi [17]

The basis of comparison, the overall error (OE) is calculated according to the following. The relative error for each component at each dataset is:

$$\epsilon_i = \left| \frac{y_{ip} - y_{ie}}{y_{ie}} \right| \quad (78)$$

Where y_{ip} and y_{ie} are predicted and experimental value of product gas composition.

To evaluate the performance of the model, the total error at each dataset and the overall error (OE) is a weighted average, calculated by:

$$E_i = \sum_{i=1}^4 \epsilon_i * y_{ie} \quad (79)$$

$$\underline{OE} = \left(\frac{\sum_{i=1}^n E_i}{n} \right) \quad (80)$$

Experimental data used for validation in original paper and partly for model evaluation in this study.

Test rigs	Set point no.	ER ^a	S/B ^b	Temp (°C)
CFB-EXP I	1	0.53	0	740
	2	0.45	0	718
	3	0.40	0	766
	4	0.52	0	815
	5	0.37	0.004	772
	6	0.43	0.03	787
	7	0.34	0	718
	8	0.35	0	730
	9	0.4	0	752
	10	0.38	0	789
	11	0.22	0	701
	12	0.26	0	728
	13	0.3	0	739
	14	0.46	0	805
CFB-EXP II	1	0.28	0.074	822.5
	2	0.39	0.26	900
	3	0.43	0.21	885
	4	0.32	0.16	840
	5	0.3	0.15	855
	6	0.34	0.16	890
	7	0.31	0.156	860
	8	0.31	0.13	885
	9	0.39	0.17	900
	10	0.39	0.16	930
BFB-EXP III	1	0.19	1.56	800
	2	0.21	1.56	800
	3	0.23	1.56	800
	4	0.25	1.56	800
	5	0.27	1.56	800
	6	0.22	2.7	700
	7	0.22	2.7	750
	8	0.22	2.7	800
	9	0.22	2.7	850
	10	0.22	2.7	900
	11	0.22	0	800
	12	0.22	1.35	800
	13	0.22	2.02	800
	14	0.22	2.7	800
	15	0.22	4.04	800

^a Equivalence Ratio = Air (kg)/air stoichiometry (kg).

^b Steam (kg)/biomass (kg dry ash free).

Figure 68. Experimental data used for validation and partly evaluation

Experimental data used for model evaluation.

Test rigs	Set point no.	ER ^a	S/B ^b	Temp (°C)	Pressure (kPa ^a)	Load (Mg.m ⁻² .h ⁻¹)
CFB I	1	0.17	0	686	100	2.070
	2	0.19	0	690	100	1.951
	3	0.2	0	720	100	1.951
	4	0.22	0	919	100	1.919
	5	0.26	0	974	100	1.672
CFB II	1	0.32	0	861	100	1.790
	2	0.34	0	855	100	1.783
	3	0.34	0	822	100	1.904
	4	0.37	0	850	100	2.796
	5	0.43	0	805	100	2.548
CFB III	1	0.22	0	688	100	0.438
	2	0.25	0	726	100	0.438
	3	0.28	0	784	100	0.632
	4	0.30	0	779	100	0.438
CFB IV	1	0.38	1.13	775	100	1.942
	2	0.38	1.45	770	100	11.942
	3	0.38	0.93	780	100	2.219
	4	0.38	1.22	780	100	2.219
BFB I	1	0.26	0	800	100	0.245
	2	0.28	0	800	100	0.227
	3	0.33	0	800	100	0.193
	4	0.34	0	800	100	0.187
	5	0.44	0	800	100	0.145
BFB II	1	0	1.4	850	100	0.241
	2	0.09	1.4	850	100	0.241
	3	0.18	1.4	850	100	0.241
	4	0.28	1.4	850	100	0.241
	5	0.37	1.4	850	100	0.241
	6	0.18	1.7	750	100	0.273
	7	0.18	1.7	800	100	0.273
	8	0.18	1.7	850	100	0.273
	9	0.18	1.7	900	100	0.273
	10	0.18	1.7	950	100	0.273
	11	0	1.1	800	100	0.370
	12	0	1.4	800	100	0.289
	13	0	1.8	800	100	0.225
	14	0	2.7	800	100	0.161
	15	0	4.7	800	100	0.088
BFB III	1	0.19	0	800	100	0.271
	2	0.24	0	800	100	0.438
	3	0.27	0	800	100	0.438
	4	0.32	0	800	100	0.199

^a Equivalence Ratio = Air (kg)/air stoichiometry (kg).^b Steam (kg)/biomass (kg dry ash free).

Figure 69. Experimental data used for model evaluation

Predictions of Composition from BFB and CFB Gasifiers

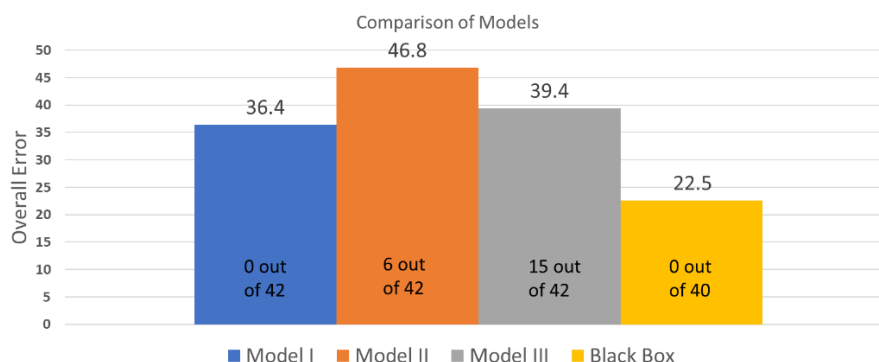


Figure 70. The overall error of models evaluated by Moshtaghi

5.2 Black box model

Black box model explains in the following sections.

5.3 Gas composition

A random sampling scheme was employed to split the database into training and test datasets. The training sets used in neural network themselves are divided into three sets of training, validation and test sets. Modeler can specify ratios of these categories.

In the following results of 3 analyses are given (i.e. 50 runs for composition module, 100 runs presenting errors of yield, CCE and composition and another 250 runs analyzing only the training errors). Information regarding the 50 and 100 runs are depicted in figure 69-70 and table 19 respectively. The 250 runs were analyzed on the training errors only to identify bad data or other meaningful patterns. Three runs out of 250 had 2 times higher errors in their composition estimates, and higher than average errors for yield and CCE estimates. The common datasets between these 3 groups mounted to 131 which is slightly more than half of the database. Therefore, elimination would make no sense and the only insight is that they collectively have a completely

different dynamics. It is recommended for future work to look into common features of these class in terms of (BFB vs CFB, oxyfuel or air blown, etc).

Runs	Database Split		Dividing the 198 data sets			Total number of test sets
	training	test	training	validation	test	
1-25	198	40	138	30	30	70
26-35	198	40	168	30	0	40
36-50	198	40	168	30	0	40

Runs	Performance Function	Training Algorithm	Training Error		Test Error	
			min	max	min	max
1-25	Sum Squared Error (SSE)	Bayesian	14.07	15.76	13.79	34.54
26-35	Mean Squared Error (MSE)	Bayesian	15.40	16.31	17.15	33.33
36-50	Mean Squared Error (MSE)	Levenberg-Marquardt backpropagation	14.33	16.25	16.04	33.14

Run	Training Error	Test Error	Run	Training Error	Test Error	Run	Training Error	Test Error
1	15.13	27.97	21	15.76	26.88	41	16.31	17.72
2	14.65	24.79	22	14.77	17.75	42	15.64	31.97
3	14.70	13.79	23	14.50	18.47	43	15.33	21.62
4	14.50	16.18	24	14.73	18.84	44	16.04	20.59
5	14.33	16.73	25	14.88	16.06	45	15.91	19.36
6	14.11	24.18	26	15.43	20.89	46	16.10	18.84
7	14.12	19.59	27	15.40	21.98	47	14.33	28.12
8	14.73	21.77	28	16.11	33.33	48	15.86	30.35
9	14.29	17.56	29	15.62	19.98	49	15.29	23.91
10	15.60	23.05	30	15.59	25.14	50	16.27	33.14
11	14.64	22.54	31	15.55	25.23			
12	14.55	20.77	32	15.52	26.04			
13	14.15	22.61	33	15.84	26.18			
14	14.28	34.54	34	16.13	17.15			
15	15.35	29.35	35	15.54	22.01			
16	14.07	22.20	36	16.25	16.04			
17	14.80	16.80	37	15.50	21.15			
18	14.37	20.34	38	15.35	29.42			
19	14.49	19.03	39	15.94	24.77			
20	14.27	18.96	40	15.65	23.04			

Figure 71. (a) Dataset ratios of 50 runs (b) Important parameters of NN development (c) Results of 50 runs

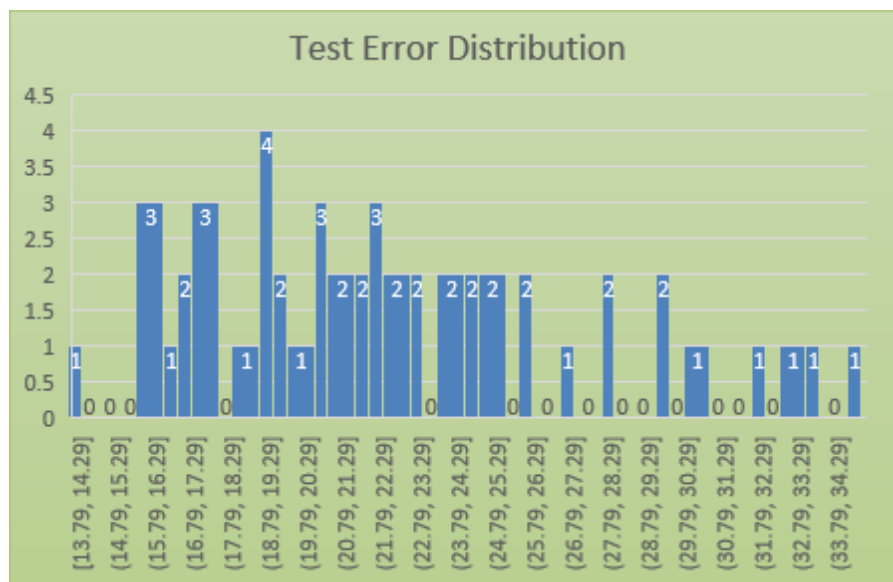


Figure 72. Test error distribution of 50 runs

Table 19. Results of additional 100 runs with modified compositions (including Yield and CCE)

Runs	Performance Function	Training Algorithm	Training Error			Test Error		
			Yield	CCE	Composition	Yield	CCE	Composition
1-20	(MSE)	Bayesian	0.35	5.76	34.22	2.57	5.89	36.69
21-30	(SSE)	Levenberg-Marquardt	0.36	5.77	24.38	0.56	5.84	26.17
31-40	(SSE)	Levenberg-Marquardt	0.35	5.74	38.19	28.21	5.97	39.66
41-100	(MSE)	Bayesian	0.36	5.79	23.67	0.69	5.77	25.12

Identification of bad data
Composition error of Runs 30, 53, 74, 107, 110, 132, 142, 153 and 213 are 2 times higher than the rest of cases (out of 250 runs). All of the aforementioned runs except run#30 also have higher than average error in their yield estimates. The CCE error of Runs 107, 110 and 153 are also higher than average.

Figure 73. Insights regarding the 250 runs (exploratory analysis)

5.4 Yield intervals

A total number of 114 datasets where gas yield was reported with following distribution (Fig 74):

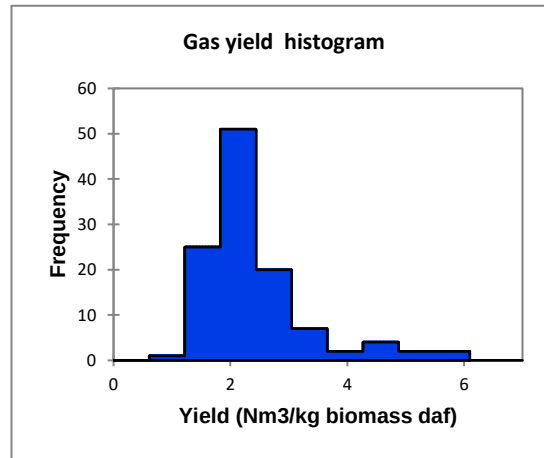


Figure 74. Product gas yields with range of (0.93-6) $\frac{\text{Nm}^3}{\text{kg biomass daf}}$

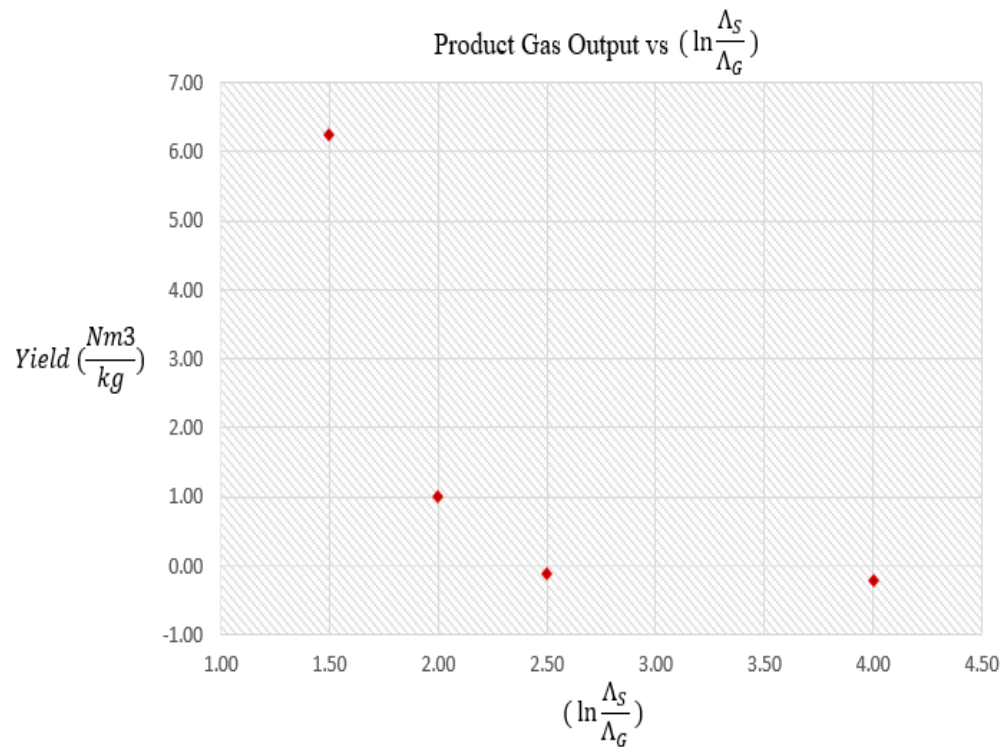


Figure 75. Syngas yield vs local minima of algorithm

5.5 Carbon Conversion Efficiency

A total number of 96 sets with CCE% were present in our database with the frequency distribution depicted in Figure 76.

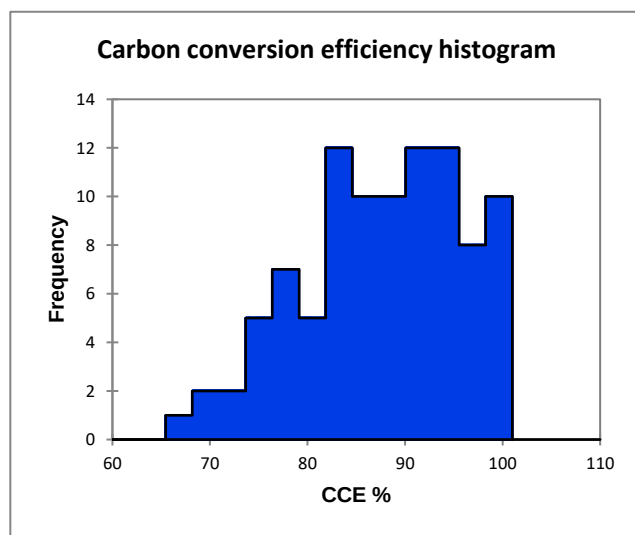


Figure 76. CCE% with range of (67.2-100)

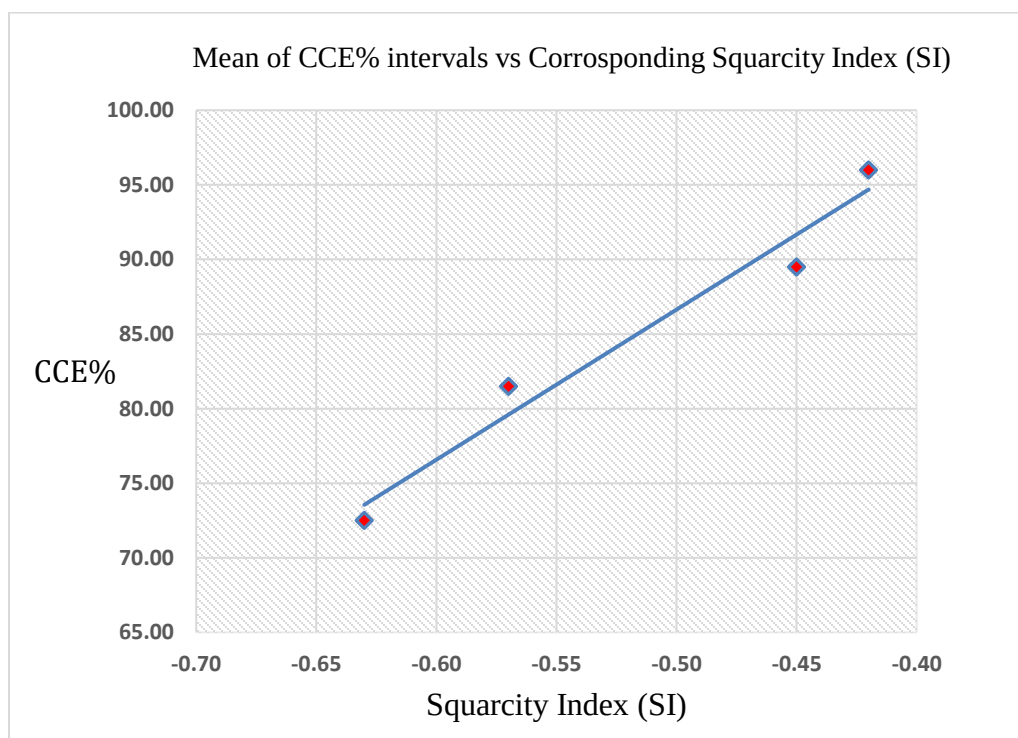


Figure 77. Carbon Conversion Efficiency vs Squarcity Index

5.6 Categorical illustration of results

An investigation was done on dispersion of errors of prediction with regard to multiple aspects of the process including those not explicitly considered in the modeling process (e.g. particle size, gasifier type, and etc). A selection of plots is shown in the following to give a categorical view of predictions for yield, CCE%, and particle size, respectively. The aspect gasifier type was already addressed in Figures 64-67.

5.7 Yield intervals

Here, dispersion of prediction errors with respect to different yield intervals is exemplified for hydrogen in Figure 78.

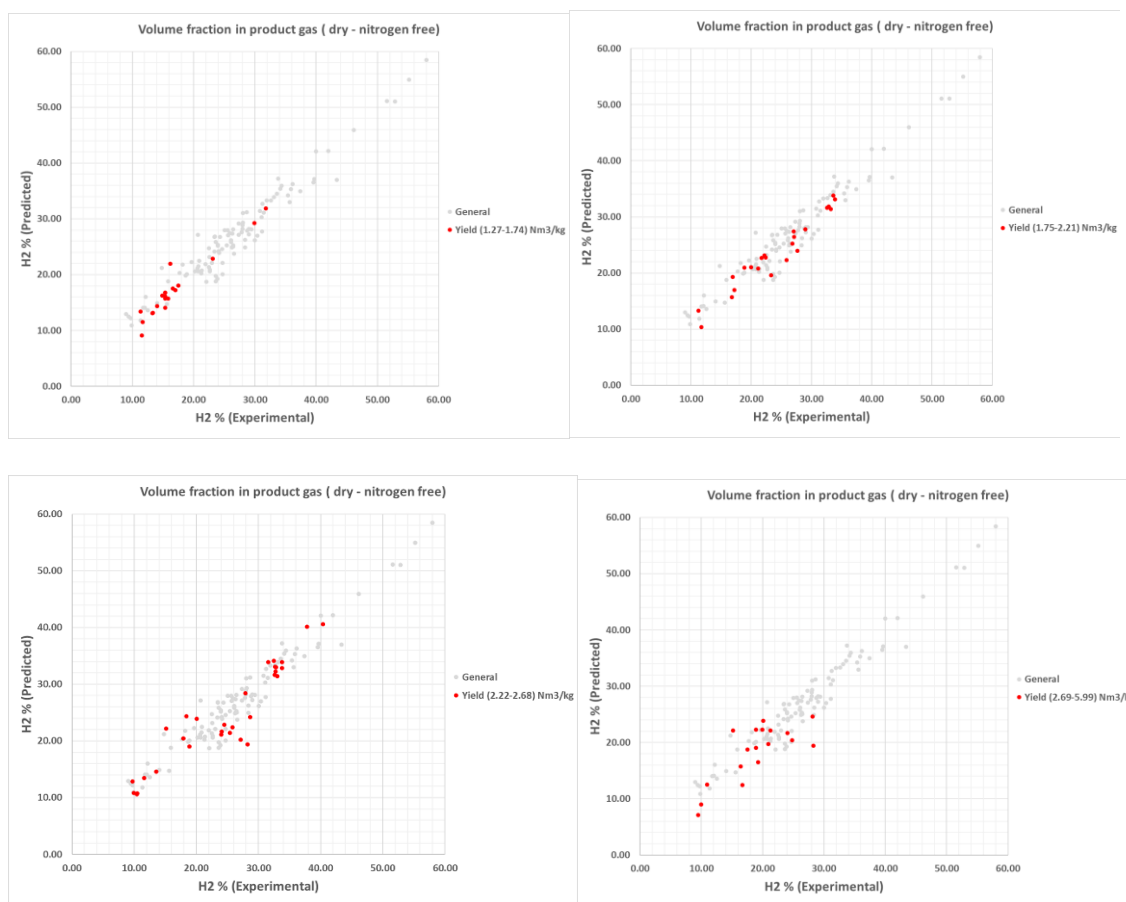


Figure 78. Results for hydrogen with focus on yield intervals

5.8 CCE% intervals

Results for 4 major syngas compounds with focus on carbon conversion efficiency depicted in Figure 79 (a, b, c, and d).

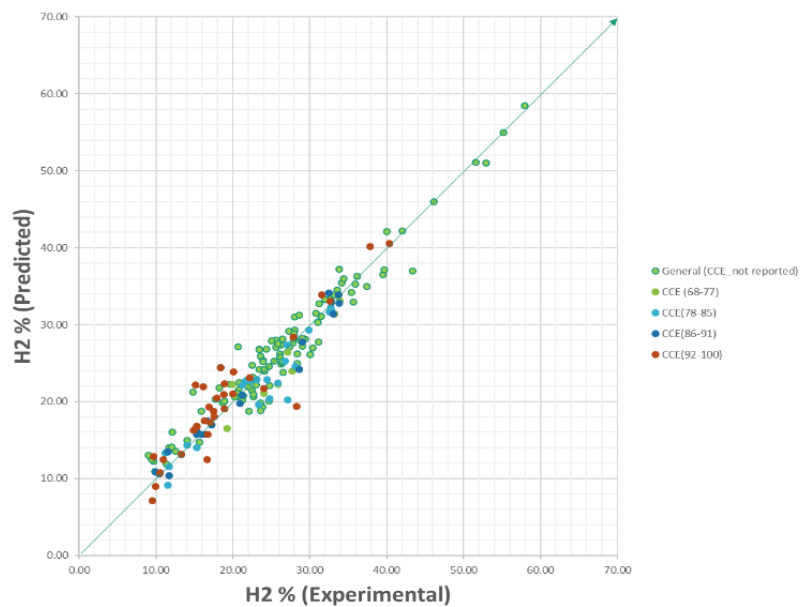


Figure 79. (a) H₂_ Dispersion of error with respect to CCE%

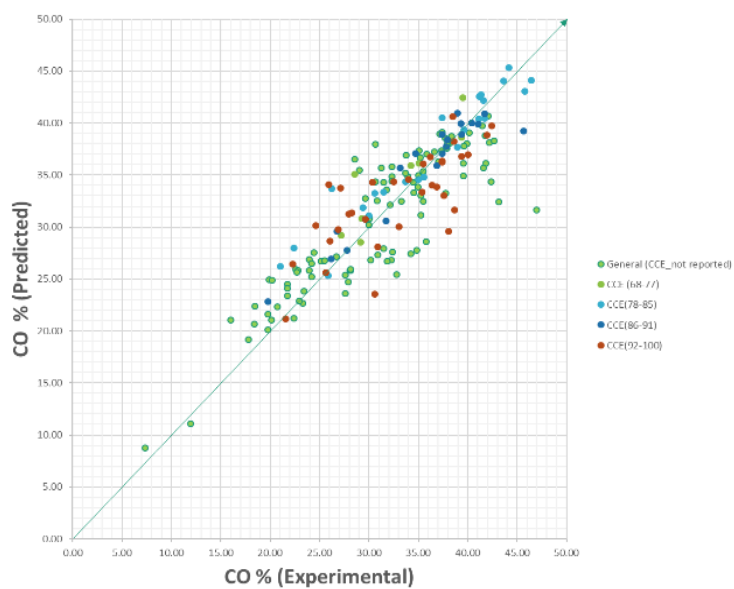


Figure 79. (b) CO_ Dispersion of error with respect to CCE%

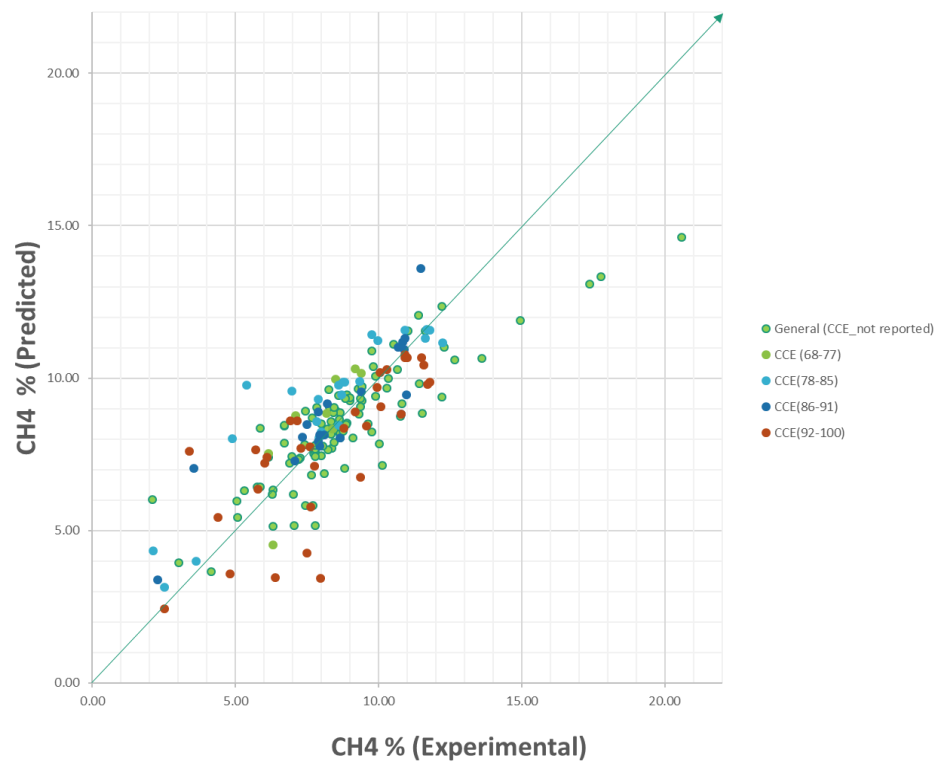


Figure 79. (c) CH₄_ Dispersion of error with respect to CCE%

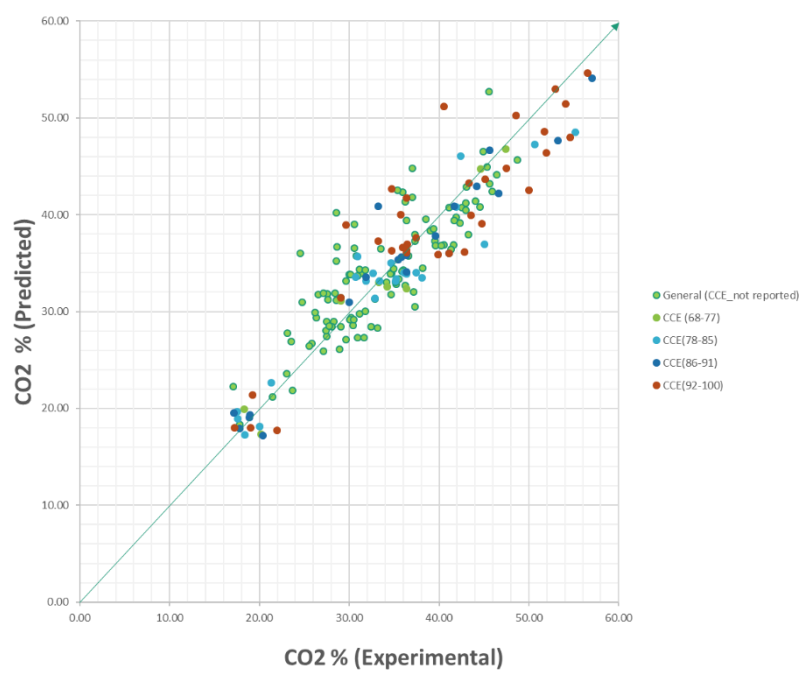


Figure 79. (d) CO₂ _ Dispersion of error with respect to CCE%

5.9 Particle size

Results for 4 major syngas compounds with focus on particle size depicted in Figure 80 (a, b, c, and d).

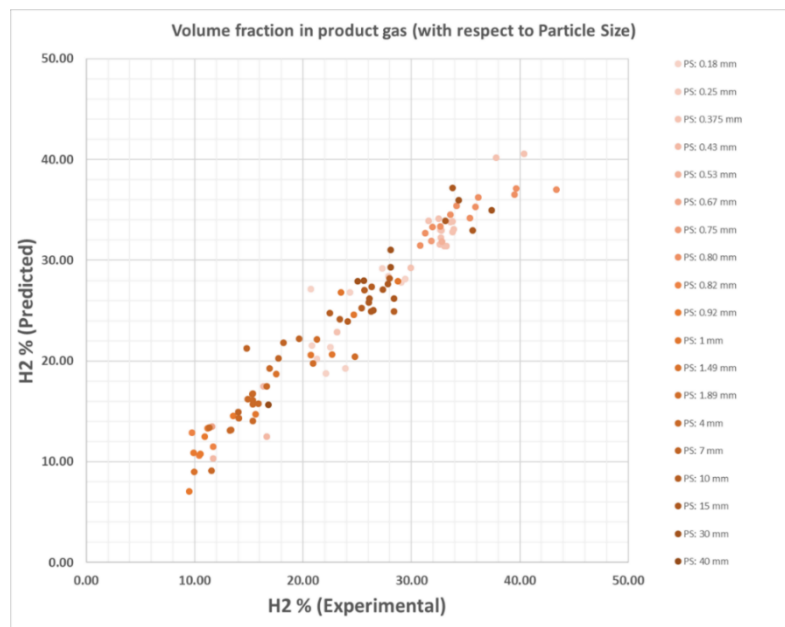


Figure 80. (a) Particle size range for hydrogen results

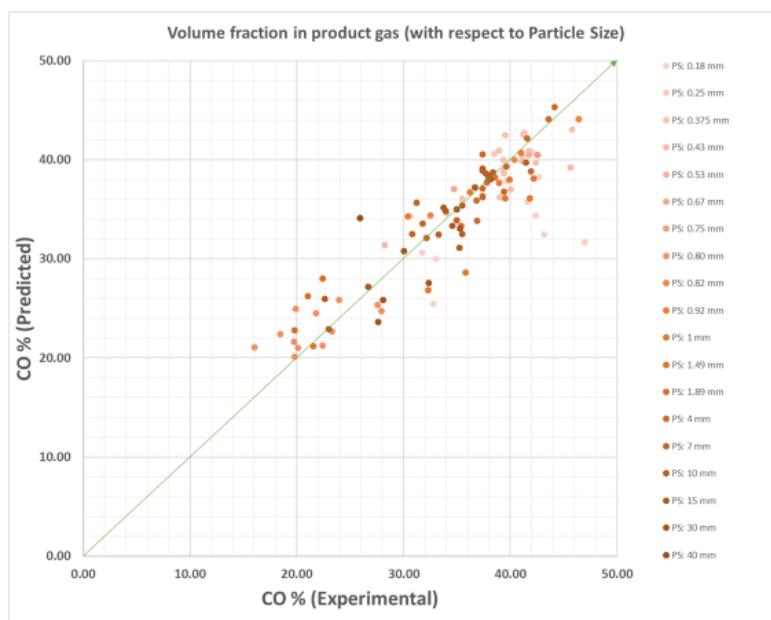


Figure 80. (b) Particle size range for carbon monoxide results

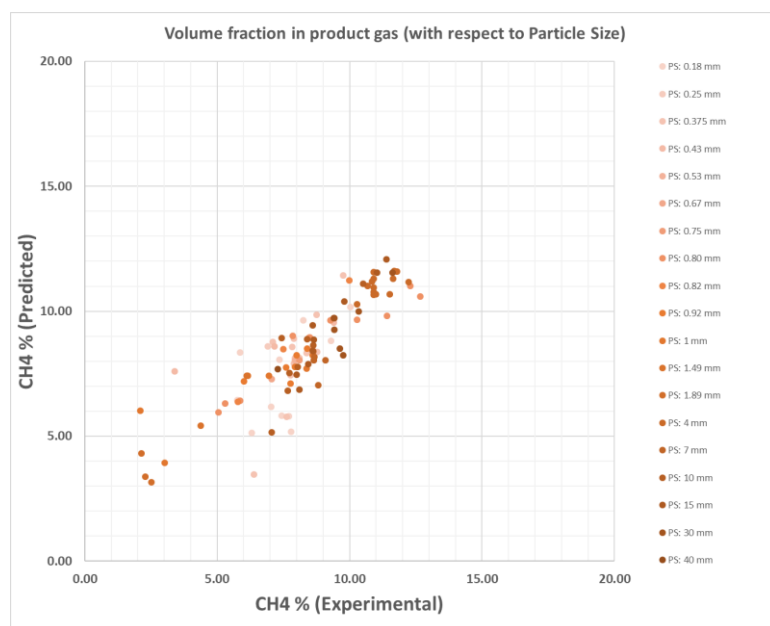


Figure 80. (c) Particle size range for methane results

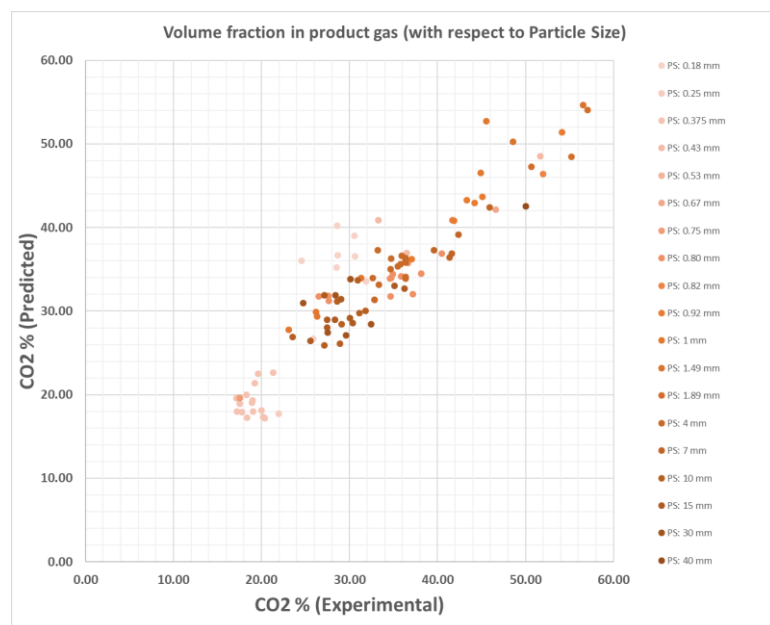


Figure 80. (d) Particle size range for carbon dioxide results

CHAPTER 6

CONCLUSIONS AND RECOMMENDATIONS

In previous chapters we showed that the black box model features better accuracy in predicting four major components of product gas (H_2 , CH_4 , CO , and CO_2) in a larger range of operating conditions and captures the physics of the phenomenon with regard to two of the most influential parameters of the process (i.e. gas yield and CCE).

- Main conclusion is that artificial neural network is a valuable and effective tool in capturing the physics of gasification process due to a more efficient way of quantifying interdependencies between various parameters.
- The ANN model developed in this study had significantly better accuracy compared to other leading models.

This was accomplished through:

- Considering more experimental data as suggested by other modelers, (building a diverse database of gasification process featuring about 40 different types of biomass and the largest range of influential parameters compared to majority of existing models)
- Developing a novel, compact characterization of biomass considering its proximate analysis and structural components using machine learning techniques
- Devising a multiphase statistical pathway needed as a precondition to effectively quantify the entanglements of different important factors in the process of biomass gasification in fluidized bed reactors

- Utilizing neural network computational framework to obtain numerical values of the model's weights and biases
- Separate results for bubbling and circulating, and then overall combined gasifiers like the example presented in the following table 20.

Table 20. Separate results for different dynamics (BFB vs. CFB vs. Overall)

Type of reactor	Sets selected for training and validation	Performance function	Training algorithm	Train yield	Train CCE	Train Composition
BFB	3x145 cell	mse	trainlm	0.30	4.44	24.55
CFB	3x79 cell	mse	trainbr	0.72	5.94	34.23
Overall 1	3x225 cell	mse	trainlm	0.39	5.81	21.96

6.1 Conclusions regarding the modeling approach

The no free lunch theorem states no machine learning algorithm is superior when averaged over the same datasets.

An important question is what subset of collected data truly represent the population under study. A neural network trained on that subset is likely to perform well against completely new dataset. In other words, it is the material you study that determines how you do on a test.

It appears that each observation is unique and that in every session of training they show resistance to all the generalizing techniques although with various vigors.

It should be added that the training error can always be lowered (e.g with increasing capacity of model) but it is the gap between training and test error that matters the most. It was decided that the best combination is when this gap is minimum. The most valuable configuration was achieved where the error on 70 previously unseen

datasets was slightly lower than the training error (13.78 vs 14.07). This particular NN was trained using Bayesian algorithm, and sum of squared error was used as performance metric. The 40 test datasets (from run 3 in Figure 72 (c) with error of 13.78) are listed in the appendix D, it should be noted the remaining 30 datasets randomly selected within the training algorithm are inaccessible.

6.2 Recommendations

One recommendation regarding the process is that it is certainly worth to investigate dynamics of bubbling and circulating bed separately, same applies for oxyfuel and air blown gasifiers. Also, if information available regarding other important variables such as catalysts used in the bed, more details about the particles (i.e. density, sphericity, ...), steam temperature, temperature of air if there is pre heating and most importantly the reactor geometry and size.

An important question arises that how successful the principle of triangle inequality is in quantifying the non-equilibrium of a process, and what was the role of centroid difference approach in predictivity of model.

“As of 2016 a rule of thumb is that a supervised deep learning algorithm will generally achieve acceptable performance with around 5000 labeled examples per category.” [52]

Working successfully with small datasets is an important research area in machine learning. In that context, applying machine learning techniques to modeling physical processes is limited by limitedness of experimental data specially for expensive experiments such as gasification.

As far as the direction of research is concerned, additional improvements become feasible by resolving the following issues and alike:

- “It is not clear now who at NETL really decides what multiphase research area to follow.” [87]
- “The Multiphase Fluid Dynamics Research Consortium (MFDRC) was established in 1998 in response to priority research needs identified in the Vision 2020 *Computational Fluid Dynamics Roadmap*.” [87]
- “That initiative is a model that should be followed if there is ever going to be an attempt to have a National Multiphase Computational Fluid Dynamics Initiative.” [87]

APPENDIX A

MATLAB CODE; RAW DATA TO FIBONACCI SEQUENCES


```

function [Tmean_struct,FibA5,FibB5] = EPS_2F5(x1)
%UNTITLED5 Summary of this function goes here: it recieves the
matrice of
%EPS:[C,H,O,N,S,VM,FC,A,Cell,Hem,Lig,ER,SBR,T(c),MC];    and
returns
%[f1,f2,f3,f4,f5] for A and B, also the target group completely
%characterized (Tmean).
% Detailed explanation goes here

epsoc=x1;
size_total=size(epsoc);
N=size_total(1,1);
elem=epsoc(:,1:5);
prox=epsoc(:,6:8);
stru=epsoc(:,9:11);
Opc=epsoc(:,12:14);%ER,SBR,T%
moist=epsoc(:,15);
mole_frac=epsoc(:,16:19);
tan_frac=zeros(N,3);
Ln_Elem=zeros(N,4);
Ln_Prox=zeros(N,2);
Ln_Stru=zeros(N,2);
e1p1e2s2dk=zeros(N,6);
%log transformation of elemntal info
for n=1:N

e1p1e2s2dk(n,1)=log(elem(n,1)^1.45/elem(n,3)^0.17/elem(n,4)^0.63/el
em(n,5)^0.65);

e1p1e2s2dk(n,3)=log(elem(n,2)^0.88/elem(n,1)^0.31/elem(n,4)^0.23/el
em(n,5)^0.34);
end
%log transformation of proximate info

for n=1:N
    cond=prox(n,1)&&prox(n,2)&&prox(n,3)>0;
    core=isAlways(cond);

```

```

if core==1
e1p1e2s2dk(n,2)=log(prox(n,2)^0.52*prox(n,1)^0.38/prox(n,3)^0.9);
else
    e1p1e2s2dk(n,2)=0.655*e1p1e2s2dk(n,1);
end
end
%log transformation of structure info

for n=1:N
    cond2=stru(n,1)&&stru(n,2)&&stru(n,3)>0;
    core2=isAlways(cond2);
    if core2==1
        e1p1e2s2dk(n,4)=log(stru(n,2)^0.17*stru(n,1)^0.48/stru(n,3)^0.65);
    else
        e1p1e2s2dk(n,4)=0.752*e1p1e2s2dk(n,3);
    end
end

    for n=1:N

        e1p1e2s2dk(n,5)=(e1p1e2s2dk(n,2)-e1p1e2s2dk(n,4))/3;

e1p1e2s2dk(n,6)=(e1p1e2s2dk(n,2)+e1p1e2s2dk(n,4))*(2*e1p1e2s2dk
(n,1)-e1p1e2s2dk(n,3))/36;
    end
    F5_Input=e1p1e2s2dk;

    S4=zeros(N,4);

    for n=1:N
        S4(n,1)=3*Opc(n,2)/100+0.05;                s21=(1000/Opc(n,3)-
1);s22=s21+Opc(n,1);maxes=max(s21,s22);
        S4(n,2)=0.7*maxes;
s31=(1+F5_Input(n,5)^2)/(1+F5_Input(n,5)^2+1000*F5_Input(n,6)/Op
c(n,3));s32=1-(S4(n,1)+S4(n,2));
        S4(n,3)=s31*s32;
        S4(n,4)=1-(S4(n,1)+S4(n,2)+S4(n,3));
    end
end

```

```

R4=zeros(N,4);
V4=zeros(N,4);
L4=zeros(N,4);
for n=1:N
    R4(n,1)=20*(abs(S4(n,1))/3.1415)^.5;
    R4(n,2)=20*(abs(S4(n,2))/3.1415)^.5;
    R4(n,3)=20*(abs(S4(n,3))/3.1415)^.5;
    R4(n,4)=20*(abs(S4(n,4))/3.1415)^.5;
end

for n=1:N
    V4(n,1)=3.1415*R4(n,1)^3;
    V4(n,2)=3.1415*R4(n,2)^3;
    V4(n,3)=3.1415*R4(n,3)^3;
    V4(n,4)=3.1415*R4(n,4)^3;
end

for n=1:N
    L4(n,1)=0.707*R4(n,1);
    L4(n,2)=0.707*R4(n,2);
    L4(n,3)=0.707*R4(n,3);
    L4(n,4)=0.707*R4(n,4);
end

RT=zeros(N,1);
VT=zeros(N,1);
ST=zeros(N,1);
LT=zeros(N,1);
NT=zeros(N,1);
EXPNT=zeros(N,1);
for n=1:N
    RT(n,1)=sum(R4(n,:),2);

VT(n,1)=min(273+V4(n,1)/16+V4(n,2)/28+V4(n,3)/2+V4(n,4)/44,940)
;

ST(n,1)=L4(n,1)*L4(n,2)/2+L4(n,2)*L4(n,3)/2+L4(n,3)*L4(n,4)/2+L4(
n,4)*L4(n,1)/2;

```

```

    LT(n,1)=L4(n,1)/L4(n,4);
    NT(n,1)=0.5*((LT(n,1)*VT(n,1)/ST(n,1)*log(23-
RT(n,1))+12)*2.3*8.3/96485*Opc(n,3));
    EXPNT(n,1)=exp(-NT(n,1));
end
Fib=zeros(N,11);
for n=1:N
    Fib(n,1)=EXPNT(n,1)+NT(n,1);Fib(n,2)=EXPNT(n,1)+Fib(n,1);
    Fib(n,3)=Fib(n,1)*Fib(n,2);
    Fib(n,4)=Fib(n,3)+Fib(n,2);
    Fib(n,5)=Fib(n,4)+Fib(n,3);
    Fib(n,6)=Fib(n,5)+Fib(n,4);
    Fib(n,7)=Fib(n,6)+Fib(n,5);
    Fib(n,8)=Fib(n,7)+Fib(n,6);
    Fib(n,9)=Fib(n,8)+Fib(n,7);
    Fib(n,10)=Fib(n,9)+Fib(n,8);
    Fib(n,11)=Fib(n,10)+Fib(n,9);

end
FibA5=Fib(:,7:11);
theta=zeros(N,3);
cauchy=zeros(N,6);
triangle=zeros(N,5);
for n=1:N

tan_frac(n,1)=(mole_frac(n,1)+mole_frac(n,2))/(mole_frac(n,3)+mole_
frac(n,4));

tan_frac(n,2)=(mole_frac(n,1)+mole_frac(n,3))/(mole_frac(n,2)+mole_
frac(n,4));

tan_frac(n,3)=(mole_frac(n,1)+mole_frac(n,4))/(mole_frac(n,2)+mole_
frac(n,3));
    theta(n,1)=atan(tan_frac(n,1)^.5);
    theta(n,2)=atan(tan_frac(n,2)^.5);
    theta(n,3)=atan(tan_frac(n,3)^.5);

```

```

end
%cauchy & triangle calculations%
for n=1:N
    EM4=theta(n,1);EN4=theta(n,2);EO4=theta(n,3);
    cauchy(n,1)=(0.5*(EM4^2-EN4^2)^2+(EM4^2-
EO4^2)^2+(EO4^2-
EM4^2)^2+6*(EM4*EN4)^2+6*(EN4*EO4)^2+6*(EO4*EM4)^2)^0.5
;
    cauchy(n,2)=(0.5*((EM4-EN4)^2+(EO4-EN4)^2+(EM4-
EO4)^2))^0.5;
    cauchy(n,3)=(EM4^2-EM4*EN4+EN4^2+3*EM4^2*EN4^2)^0.5;
    cauchy(n,4)=(EM4^2-EM4*EN4+EN4^2)^0.5;
    cauchy(n,5)=EM4*EN4*3^1/3;
    cauchy(n,6)=EM4;
    triangle(n,1)=(EM4+EN4+EO4)/2;%s
    GR4=triangle(n,1);
    triangle(n,2)=(GR4*(GR4-EM4)*(GR4-EN4)*(GR4-
EO4))^0.5;%K(area)
    GS4=triangle(n,2);
    triangle(n,3)=(4*GS4/(3)^0.5)^0.5;%(Leq)
    triangle(n,4)=EM4*EN4*EO4/4/GS4;%(R)circumradius
    triangle(n,5)=GS4/GR4;%(r)inradius
end
Tmean_struct=cell(11,N);
TL1=zeros(N,5);
TL2=zeros(N,5);
TL3=zeros(N,3);
TL4=zeros(N,15);
TL5=zeros(N,15);
TL6=zeros(N,15);
TL7=zeros(N,15);
TL8=zeros(N,15);
TL9=zeros(N,15);
TL10=zeros(N,4);
TL11=zeros(N,3);
for n=1:N

```

```

TL1(n,1)=cauchy(n,2);TL1(n,2)=cauchy(n,4);TL1(n,3)=cauchy(n,5);
    TL1(n,4)=theta(n,1);TL3(n,1)=theta(n,1);TL11(n,1)=theta(n,1);
    TL1(n,5)=triangle(n,3);TL2(n,3)=triangle(n,3);
    TL2(n,1)=triangle(n,1)-
1;TL2(n,2)=triangle(n,2);TL2(n,4)=triangle(n,4);TL2(n,5)=triangle(n,5
);
    TL3(n,1)=theta(n,1);TL3(n,2)=theta(n,2);TL3(n,3)=theta(n,3);
    TL11(n,2)=theta(n,2);TL11(n,3)=theta(n,3);

TL10(n,1)=mole_frac(n,1);TL10(n,2)=mole_frac(n,2);TL10(n,3)=mole
_frac(n,3);TL10(n,4)=mole_frac(n,4);
    end

    %Trigonometric Functions%
    HT=cell(N,2,2,3);% dim2=1 sin & dim2=2 cos% dim3=1 norm
&dim3=2 hyper
    %dim4=1,2,3 theta=1,2,3%
    for n=1:N
        t1=theta(n,1);t2=theta(n,2);t3=theta(n,3);

HT{n,1,1,1}=sin(t1);HT{n,1,1,2}=sin(t2);HT{n,1,1,3}=sin(t3);%sin

HT{n,1,2,1}=sinh(t1);HT{n,1,2,2}=sinh(t2);HT{n,1,2,3}=sinh(t3);%si
nh

HT{n,2,1,1}=cos(t1);HT{n,2,1,2}=cos(t2);HT{n,2,1,3}=cos(t3);%cos

HT{n,2,2,1}=cosh(t1);HT{n,2,2,2}=cosh(t2);HT{n,2,2,3}=cosh(t3);%c
osh
    end

    %calculating the content of TL4 to TL9%
    %TL4 (theta_1 & Norm)
    sint=HT(:,1,1,1);sintt=cell2mat(sint);
    cosint=HT(:,2,1,1);cosintt=cell2mat(cosint);
    for n=1:N

```

```
TL4(n,1)=sintt(n,1);TL4(n,2)=cosintt(n,1);TL4(n,3)=sintt(n,1)/cosintt(n,1);
```

```
    TL4(n,4)=cosintt(n,1)/sintt(n,1);TL4(n,5)=1-cosintt(n,1);TL4(n,6)=1-sintt(n,1);
```

```
TL4(n,7)=1+cosintt(n,1);TL4(n,8)=1+sintt(n,1);TL4(n,9)=TL4(n,5)/2;
```

```
TL4(n,10)=TL4(n,7)/2;TL4(n,11)=TL4(n,6)/2;TL4(n,12)=TL4(n,8)/2;
```

```
    TL4(n,13)=1/(TL4(n,2)-1);TL4(n,14)=1/(TL4(n,1)-1);TL4(n,15)=2*sin(theta(n,1)/2);
```

```
end
```

```
%TL6 (theta_2 & Norm)
```

```
sint=HT(:,1,1,2);sintt=cell2mat(sint);
```

```
cosint=HT(:,2,1,2);cosintt=cell2mat(cosint);
```

```
for n=1:N
```

```
TL6(n,1)=sintt(n,1);TL6(n,2)=cosintt(n,1);TL6(n,3)=sintt(n,1)/cosintt(n,1);
```

```
    TL6(n,4)=cosintt(n,1)/sintt(n,1);TL6(n,5)=1-cosintt(n,1);TL6(n,6)=1-sintt(n,1);
```

```
TL6(n,7)=1+cosintt(n,1);TL6(n,8)=1+sintt(n,1);TL6(n,9)=TL6(n,5)/2;
```

```
TL6(n,10)=TL6(n,7)/2;TL6(n,11)=TL6(n,6)/2;TL6(n,12)=TL6(n,8)/2;
```

```
    TL6(n,13)=1/(TL6(n,2)-1);TL6(n,14)=1/(TL6(n,1)-1);TL6(n,15)=2*sin(theta(n,2)/2);
```

```
end
```

```
%TL8 (theta_3 & Norm)
```

```
sint=HT(:,1,1,3);sintt=cell2mat(sint);
```

```
cosint=HT(:,2,1,3);cosintt=cell2mat(cosint);
```

```
for n=1:N
```

```
TL8(n,1)=sintt(n,1);TL8(n,2)=cosintt(n,1);TL8(n,3)=sintt(n,1)/cosintt(n,1);
```

```

        TL8(n,4)=cosintt(n,1)/sintt(n,1);TL8(n,5)=1-
        cosintt(n,1);TL8(n,6)=1-sintt(n,1);

        TL8(n,7)=1+cosintt(n,1);TL8(n,8)=1+sintt(n,1);TL8(n,9)=TL8(n,5)/2;

        TL8(n,10)=TL8(n,7)/2;TL8(n,11)=TL8(n,6)/2;TL8(n,12)=TL8(n,8)/2;
        TL8(n,13)=1/(TL8(n,2)-1);TL8(n,14)=1/(TL8(n,1)-
        1);TL8(n,15)=2*sin(theta(n,3)/2);
        end

        %%%% hyper

        %TL5 (theta_1 & hyper)
        sint=HT(:,1,2,1);sintt=cell2mat(sint);
        cosint=HT(:,2,2,1);cosintt=cell2mat(cosint);
        for n=1:N

        TL5(n,1)=sintt(n,1);TL5(n,2)=cosintt(n,1);TL5(n,3)=sintt(n,1)/cosintt(n
        ,1);
        TL5(n,4)=cosintt(n,1)/sintt(n,1);TL5(n,5)=1-
        cosintt(n,1);TL5(n,6)=1-sintt(n,1);

        TL5(n,7)=1+cosintt(n,1);TL5(n,8)=1+sintt(n,1);TL5(n,9)=TL5(n,5)/2;

        TL5(n,10)=TL5(n,7)/2;TL5(n,11)=TL5(n,6)/2;TL5(n,12)=TL5(n,8)/2;
        TL5(n,13)=1/(TL5(n,2)-1);TL5(n,14)=1/(TL5(n,1)-
        1);TL5(n,15)=2*sinh(theta(n,1)/2);
        end

        %TL7 (theta_2 & hyper)
        sint=HT(:,1,2,2);sintt=cell2mat(sint);
        cosint=HT(:,2,2,2);cosintt=cell2mat(cosint);
        for n=1:N

        TL7(n,1)=sintt(n,1);TL7(n,2)=cosintt(n,1);TL7(n,3)=sintt(n,1)/cosintt(n
        ,1);

```



```

        TL7(n,4)=cosintt(n,1)/sintt(n,1);TL7(n,5)=1-
        cosintt(n,1);TL7(n,6)=1-sintt(n,1);

        TL7(n,7)=1+cosintt(n,1);TL7(n,8)=1+sintt(n,1);TL7(n,9)=TL7(n,5)/2;

        TL7(n,10)=TL7(n,7)/2;TL7(n,11)=TL7(n,6)/2;TL7(n,12)=TL7(n,8)/2;
        TL7(n,13)=1/(TL7(n,2)-1);TL7(n,14)=1/(TL7(n,1)-
        1);TL7(n,15)=2*sinh(theta(n,2)/2);
        end

        %TL9 (theta_3 & hyper)
        sint=HT(:,1,2,3);sintt=cell2mat(sint);
        cosint=HT(:,2,2,3);cosintt=cell2mat(cosint);
        for n=1:N

        TL9(n,1)=sintt(n,1);TL9(n,2)=cosintt(n,1);TL9(n,3)=sintt(n,1)/cosintt(n
        ,1);

        TL9(n,4)=cosintt(n,1)/sintt(n,1);TL9(n,5)=1-
        cosintt(n,1);TL9(n,6)=1-sintt(n,1);

        TL9(n,7)=1+cosintt(n,1);TL9(n,8)=1+sintt(n,1);TL9(n,9)=TL9(n,5)/2;

        TL9(n,10)=TL9(n,7)/2;TL9(n,11)=TL9(n,6)/2;TL9(n,12)=TL9(n,8)/2;
        TL9(n,13)=1/(TL9(n,2)-1);TL9(n,14)=1/(TL9(n,1)-
        1);TL9(n,15)=2*sinh(theta(n,3)/2);
        end

        for n=1:N
            Tmean_struct{1,n}=real(TL1(n,:));
            Tmean_struct{2,n}=real(TL2(n,:));
            Tmean_struct{3,n}=real(TL3(n,:));
            Tmean_struct{4,n}=real(TL4(n,:));
            Tmean_struct{5,n}=real(TL5(n,:));
            Tmean_struct{6,n}=real(TL6(n,:));
            Tmean_struct{7,n}=real(TL7(n,:));
            Tmean_struct{8,n}=real(TL8(n,:));
            Tmean_struct{9,n}=real(TL9(n,:));

```

```

    Tmean_struct{10,n}=real(TL10(n,:));
    Tmean_struct{11,n}=real(TL11(n,:));
end

%calculation of Fibonacci for B

S4=zeros(N,4);
for n=1:N
    S4(n,1)=mole_frac(n,2);
    S4(n,2)=mole_frac(n,4);
    S4(n,3)=mole_frac(n,3);
    S4(n,4)=mole_frac(n,1);
end

R4=zeros(N,4);
V4=zeros(N,4);
L4=zeros(N,4);
for n=1:N
    R4(n,1)=2*(abs(S4(n,1))/3.1415)^.5;
    R4(n,2)=2*(abs(S4(n,2))/3.1415)^.5;
    R4(n,3)=2*(abs(S4(n,3))/3.1415)^.5;
    R4(n,4)=2*(abs(S4(n,4))/3.1415)^.5;
end

for n=1:N
    V4(n,1)=3.1415*R4(n,1)^3;
    V4(n,2)=3.1415*R4(n,2)^3;
    V4(n,3)=3.1415*R4(n,3)^3;
    V4(n,4)=3.1415*R4(n,4)^3;
end

for n=1:N
    L4(n,1)=0.707*R4(n,1);
    L4(n,2)=0.707*R4(n,2);
    L4(n,3)=0.707*R4(n,3);
    L4(n,4)=0.707*R4(n,4);
end
RT=zeros(N,1);

```

```

VT=zeros(N,1);
ST=zeros(N,1);
LT=zeros(N,1);
NT=zeros(N,1);
EXPNT=zeros(N,1);
for n=1:N
    RT(n,1)=sum(R4(n,:),2);
    VT(n,1)=273+V4(n,1)/16+V4(n,2)/44+V4(n,3)/28+V4(n,4)/2;

ST(n,1)=L4(n,1)*L4(n,2)/2+L4(n,2)*L4(n,3)/2+L4(n,3)*L4(n,4)/2+L4(
n,4)*L4(n,1)/2;
    LT(n,1)=L4(n,1)/L4(n,4)*L4(n,3)/L4(n,2);
    NT(n,1)=(LT(n,1)*VT(n,1)/ST(n,1)*log(23-
RT(n,1))+5)*2.303*8.314/96485.33*Opc(n,3);
    EXPNT(n,1)=exp(-NT(n,1));
end
Fib=zeros(N,11);
for n=1:N
    Fib(n,1)=EXPNT(n,1)+NT(n,1);Fib(n,2)=EXPNT(n,1)+Fib(n,1);
    Fib(n,3)=Fib(n,1)+Fib(n,2);
    Fib(n,4)=Fib(n,3)+Fib(n,2);
    Fib(n,5)=Fib(n,4)+Fib(n,3);
    Fib(n,6)=Fib(n,5)+Fib(n,4);
    Fib(n,7)=Fib(n,6)+Fib(n,5);
    Fib(n,8)=Fib(n,7)+Fib(n,6);
    Fib(n,9)=Fib(n,8)+Fib(n,7);
    Fib(n,10)=Fib(n,9)+Fib(n,8);
    Fib(n,11)=Fib(n,10)+Fib(n,9);

end
FibB5=Fib(:,7:11);

end

```

APPENDIX B

MATLAB CODE; THREE NEURAL NETWORKS TO PROCESS FIBONACCI
SEQUENCES

Linear:

```
CTABL_sh=network;CTABL_sh.numInputs=1;CTABL_sh.numLayers
=2;CTABL_sh.inputs{1}.size=5;CTABL_sh.Layers{1}.size=5;CTAB
L_sh.Layers{2}.size=5;
CTABL_sh.trainFcn='trainlm';CTABL_sh.performFcn='sse';CTABL_s
h.inputConnect=[1;1];CTABL_sh.outputConnect=[1
1];CTABL_sh.layerConnect=[0 0;0 0];CTABL_sh=init(CTABL_sh);
CTABL_sh.Layers{1}.transferFcn='purelin';CTABL_sh.Layers{2}.tra
nsferFcn='purelin';CTABL_sh.biasConnect=[1;1];CTABL_sh.trainFcn
='trainlm';CTABL_sh.performFcn='sse';
CTABL_sh=init(CTABL_sh)
```

Elliot:

```
CTABE_sh=network;CTABE_sh.numInputs=1;CTABE_sh.numLayers
=2;CTABE_sh.inputs{1}.size=5;CTABE_sh.Layers{1}.size=5;CTAB
E_sh.Layers{2}.size=5;
CTABE_sh.trainFcn='trainlm';CTABE_sh.performFcn='mse';CTABE_
sh.inputConnect=[1;1];CTABE_sh.outputConnect=[1
1];CTABE_sh.layerConnect=[0 0;0 0];CTABE_sh=init(CTABL_sh);
CTABE_sh.Layers{1}.transferFcn='elliotsig';CTABE_sh.Layers{2}.tra
nsferFcn='elliotsig';CTABE_sh.biasConnect=[1;1];CTABE_sh.trainFc
n='trainlm';CTABE_sh.performFcn='mse';
CTABE_sh=init(CTABE_sh);
```

Log:

```
CTABLog_sh=network;CTABLog_sh.numInputs=1;CTABLog_sh.nu
mLayers=2;CTABLog_sh.inputs{1}.size=5;CTABLog_sh.Layers{1}.s
ize=5;CTABLog_sh.Layers{2}.size=5;
CTABLog_sh.trainFcn='trainlm';CTABLog_sh.performFcn='mse';CTA
BLog_sh.inputConnect=[1;1];CTABLog_sh.outputConnect=[1
```

```

1];CTABLog_sh.layerConnect=[0
0];CTABLog_sh=init(CTABL_sh);
CTABLog_sh.Layers{1}.transferFcn='logsig';CTABLog_sh.Layers{2}
.transferFcn='logsig';CTABLog_sh.biasConnect=[1;1];CTABLog_sh.tr
ainFcn='trainlm';CTABLog_sh.performFcn='mse';
CTABLog_sh=init(CTABLog_sh);

```

APPENDIX C

RECORDED DURATION TIME OF OPERATION IN THE MATLAB

Table 21. Recorded duration time of operation in the MATLAB (Page 1 of 9)

Number of cases	Operation 1	Operation 2	Operation 3
1	1.035598	1.229445	8.92E-05
2	1.025971	1.191491	8.93E-05
3	1.063101	1.299789	8.27E-05
4	1.086903	1.361893	8.54E-05
5	1.173554	1.594856	7.75E-05
6	1.086583	1.366562	8.04E-05
7	1.094655	1.392055	9.16E-05
8	1.143324	1.533695	9.00E-05
9	1.111809	1.446874	0.000106
10	1.099653	1.441544	0.000112
11	1.408093	2.348946	9.61E-05
12	1.123238	1.509597	8.38E-05
13	1.069176	1.303737	8.40E-05
14	1.078588	1.334929	0.000101
15	1.05375	1.264177	8.44E-05
16	1.148501	1.543535	9.26E-05
17	1.150261	1.548547	8.34E-05
18	1.072625	1.339119	8.16E-05
19	1.095062	1.396718	7.73E-05
20	1.172664	1.655857	7.73E-05
21	1.121249	1.490662	8.00E-05
22	1.265904	1.902175	7.71E-05
23	1.376218	2.235809	7.74E-05

Table 21. Recorded duration time of operation in the MATLAB (Page 2 of 9)

Number of cases	Operation 1	Operation 2	Operation 3
24	1.088878	1.40314	7.71E-05
25	1.148483	1.578405	7.75E-05
26	1.615325	2.789688	7.70E-05
27	1.046384	1.285854	7.77E-05
28	1.022847	1.229678	7.70E-05
29	0.973954	1.119501	7.77E-05
30	1.022847	1.229678	7.69E-05
31	1.047935	1.288496	7.69E-05
32	0.978019	1.110376	7.71E-05
33	1.119872	1.449722	8.08E-05
34	1.028833	1.238036	7.71E-05
35	1.10813	1.458519	7.80E-05
36	1.402768	2.270394	7.72E-05
37	1.270431	1.918664	7.71E-05
38	1.348016	2.126956	8.99E-05
39	1.148483	1.578405	8.47E-05
40	1.194105	1.696033	8.26E-05
41	1.232094	1.81632	8.00E-05
42	1.265926	1.906206	7.92E-05
43	1.217662	1.754845	7.73E-05
44	1.787164	3.021899	7.68E-05
45	3.191657	6.205446	7.73E-05
46	1.022242	1.23095	7.95E-05

Table 21. Recorded duration time of operation in the MATLAB (Page 3 of 9)

Number of cases	Operation 1	Operation 2	Operation 3
47	1.502717	2.460459	7.76E-05
48	1.388689	2.117401	7.97E-05
49	1.128058	1.500715	7.73E-05
50	1.131738	1.529328	7.72E-05
51	1.166924	1.593539	8.26E-05
52	1.11058	1.465962	7.76E-05
53	1.124856	1.512422	7.75E-05
54	1.061742	1.315517	7.76E-05
55	1.19742	1.670102	8.64E-05
56	1.170503	1.59816	8.41E-05
57	1.229064	1.759034	8.51E-05
58	1.388023	2.171505	8.26E-05
59	1.349093	2.064643	8.24E-05
60	1.239058	1.803081	7.97E-05
61	1.276758	1.930488	9.00E-05
62	1.135537	1.554593	8.22E-05
63	1.168103	1.636328	8.91E-05
64	0.981831	1.117808	8.27E-05
65	1.073049	1.381849	8.25E-05
66	1.16063	1.604701	8.03E-05
67	1.301549	2.020561	8.05E-05
68	1.088182	1.423771	7.74E-05
69	1.136843	1.563978	7.74E-05

Table 21. Recorded duration time of operation in the MATLAB (Page 4 of 9)

Number of cases	Operation 1	Operation 2	Operation 3
70	1.188502	1.705514	7.70E-05
71	1.206948	1.744927	8.28E-05
72	1.084822	1.387986	7.77E-05
73	1.210146	1.765505	8.28E-05
74	1.273533	1.92779	8.35E-05
75	1.36958	2.219478	7.82E-05
76	0.984015	1.150027	7.74E-05
77	1.015041	1.205745	7.71E-05
78	1.284702	1.976998	7.82E-05
79	1.988573	3.27749	7.96E-05
80	2.301594	3.846162	7.73E-05
81	1.246014	1.713629	7.70E-05
82	1.319219	2.012439	7.69E-05
83	1.328149	2.056327	8.96E-05
84	1.306511	1.973959	8.67E-05
85	1.107316	1.443734	8.69E-05
86	1.032661	1.235659	8.19E-05
87	1.268794	1.831483	7.72E-05
88	1.208167	1.681998	7.71E-05
89	1.225372	1.734003	7.75E-05
90	1.177639	1.569758	7.71E-05
91	1.138047	1.527049	7.76E-05
92	1.11327	1.478148	7.72E-05

Table 21. Recorded duration time of operation in the MATLAB (Page 5 of 9)

Number of cases	Operation 1	Operation 2	Operation 3
93	1.096851	1.435854	7.71E-05
94	1.13453	1.521261	7.72E-05
95	1.245831	1.853459	7.97E-05
96	0.943038	1.020236	7.71E-05
97	1.00613	1.191846	7.75E-05
98	0.994519	1.148378	7.73E-05
99	1.017617	1.219951	7.70E-05
100	1.013181	1.194721	7.72E-05
101	1.028021	1.234565	7.82E-05
102	0.997987	1.146165	7.77E-05
103	0.987855	1.119445	7.73E-05
104	1.181953	1.654657	7.72E-05
105	1.069256	1.359094	7.72E-05
106	1.094465	1.40415	9.02E-05
107	1.043889	1.290624	8.61E-05
108	1.065663	1.336808	8.49E-05
109	1.034642	1.260377	7.75E-05
110	1.073379	1.345917	7.71E-05
111	1.032571	1.258852	7.73E-05
112	1.22228	1.762842	7.75E-05
113	1.184269	1.633111	7.75E-05
114	1.133875	1.509217	8.13E-05
115	1.276512	1.880006	7.77E-05

Table 21. Recorded duration time of operation in the MATLAB (Page 6 of 9)

Number of cases	Operation 1	Operation 2	Operation 3
116	1.113764	1.464396	7.70E-05
117	1.039067	1.252336	7.76E-05
118	1.025102	1.245779	7.72E-05
119	1.332155	2.070156	7.72E-05
120	1.187562	1.684583	8.17E-05
121	1.168004	1.611645	7.78E-05
122	1.097642	1.404804	7.68E-05
123	1.180741	1.672992	7.73E-05
124	1.181579	1.643578	7.98E-05
125	1.118339	1.443887	7.97E-05
126	1.257375	1.899166	7.74E-05
127	1.904771	3.338839	7.74E-05
128	1.402497	2.226571	8.37E-05
129	1.607533	2.651759	8.23E-05
130	1.076098	1.364197	8.40E-05
131	1.055983	1.340239	7.80E-05
132	1.047329	1.302328	8.81E-05
133	1.043889	1.290624	8.19E-05
134	1.095472	1.423585	7.75E-05
135	1.088974	1.406673	7.75E-05
136	1.095024	1.440295	7.83E-05
137	1.088974	1.406673	7.73E-05
138	1.019699	1.208622	7.72E-05

Table 21. Recorded duration time of operation in the MATLAB (Page 7 of 9)

Number of cases	Operation 1	Operation 2	Operation 3
139	1.008536	1.201711	7.87E-05
140	1.088974	1.406673	7.74E-05
141	1.038378	1.257871	7.73E-05
142	1.056834	1.32136	7.72E-05
143	1.133595	1.519825	7.75E-05
144	1.059122	1.334433	7.74E-05
145	1.05323	1.327203	7.76E-05
146	1.007524	1.197209	8.22E-05
147	1.050038	1.299122	7.75E-05
148	1.07805	1.365974	7.77E-05
149	1.084909	1.383545	7.74E-05
150	1.105955	1.441511	7.77E-05
151	1.141778	1.511696	9.50E-05
152	1.107539	1.41989	8.29E-05
153	2.443129	4.163961	8.35E-05
154	5.190843	8.58669	7.76E-05
155	2.515613	5.338154	7.76E-05
156	1.263398	1.873542	7.73E-05
157	1.136512	1.556459	7.76E-05
158	1.116926	1.498845	7.75E-05
159	1.078029	1.381086	7.74E-05
160	1.07717	1.387291	8.29E-05
161	1.148728	1.585094	7.77E-05

Table 21. Recorded duration time of operation in the MATLAB (Page 8 of 9)

Number of cases	Operation 1	Operation 2	Operation 3
162	1.070957	1.380664	7.77E-05
163	1.061743	1.334621	8.09E-05
164	1.065708	1.350584	7.79E-05
165	1.061749	1.315124	7.80E-05
166	1.04987	1.316589	7.77E-05
167	1.042801	1.283945	8.11E-05
168	1.067603	1.350383	7.78E-05
169	1.069601	1.355929	8.11E-05
170	1.081696	1.383114	8.22E-05
171	1.810438	3.111078	7.81E-05
172	1.497983	2.556203	7.78E-05
173	1.430556	2.398834	7.82E-05
174	1.528392	2.6812	8.99E-05
175	1.366871	2.21663	8.73E-05
176	1.371896	2.205335	8.37E-05
177	1.205432	1.734131	8.30E-05
178	1.311637	2.040704	7.74E-05
179	1.365697	2.190895	7.77E-05
180	1.365964	2.1994	7.77E-05
181	1.324558	2.090605	7.85E-05
182	1.213135	1.785435	7.77E-05
183	1.247526	1.869907	7.79E-05
184	1.16152	1.610965	7.74E-05

Table 21. Recorded duration time of operation in the MATLAB (Page 9 of 9)

Number of cases	Operation 1	Operation 2	Operation 3
185	1.289845	1.966233	8.10E-05
186	1.040261	1.259504	7.73E-05
187	1.133947	1.508934	8.13E-05
188	1.067215	1.331448	8.49E-05
189	1.088457	1.384876	9.14E-05
190	1.037617	1.259163	8.28E-05
191	1.083166	1.3729	7.77E-05
192	1.036236	1.257034	7.77E-05
193	1.073077	1.344982	7.74E-05
194	1.017066	1.2155	8.01E-05
195	1.052996	1.302301	7.78E-05
196	0.991032	1.146746	8.87E-05
197	1.071058	1.347814	9.13E-05
198	0.927333	0.987056	8.79E-05
199	0.995517	1.164245	7.95E-05
200	0.989252	1.146627	8.27E-05

APPENDIX D
EXPERIMENTAL DATA AND MODEL PREDICTION

	No of dataset	Experimental Data				Model Predictions				Error
		H2	CH4	CO	CO2	H2	CH4	CO	CO2	
1	1	23.14	9.76	45.77	21.33	18.25	9.71	45.43	26.61	10.56
2	13	32.50	8.66	41.07	17.78	34.51	8.74	33.17	23.58	15.80
3	29	30.39	9.76	29.65	30.20	22.85	9.21	36.80	31.14	16.18
4	52	23.63	6.72	32.34	37.31	24.37	8.77	33.85	33.00	8.62
5	56	17.54	9.19	37.58	35.70	16.32	8.77	36.61	38.30	5.20
6	60	19.23	6.15	27.18	47.44	20.23	7.45	30.51	41.82	11.24
7	61	16.45	3.55	26.77	53.23	21.61	7.21	29.37	41.80	22.85
8	69	46.15	8.24	24.18	21.43	44.50	6.92	28.65	19.93	8.94
9	81	30.80	12.66	16.04	40.50	38.93	9.14	23.40	28.53	30.99
10	88	33.61	8.50	23.28	34.61	35.86	9.10	26.49	28.55	12.11
11	91	14.78	8.65	34.96	41.61	21.84	7.96	31.58	38.63	14.11
12	94	17.73	9.09	30.84	42.35	20.73	7.56	32.04	39.68	8.40
13	97	25.05	10.34	34.99	29.62	26.09	8.38	31.77	33.77	10.36
14	114	26.31	8.63	37.95	27.11	27.87	9.30	33.38	29.45	9.14
15	117	25.40	10.52	35.52	28.57	38.51	9.02	24.72	27.75	26.22
16	121	23.02	7.67	35.12	34.19	24.62	8.99	35.14	31.25	5.87
17	122	29.02	8.43	34.51	28.04	27.92	9.22	33.51	29.35	4.21
18	123	28.25	7.88	15.86	48.00	21.21	8.45	34.48	35.86	38.36
19	139	23.96	9.38	28.13	38.54	30.56	8.42	29.21	31.81	15.37
20	146	21.98	9.89	31.87	36.26	24.97	8.32	31.01	35.69	5.99
21	150	17.89	11.70	26.86	43.55	19.91	8.55	35.02	36.53	20.35
22	151	19.97	10.07	27.13	42.83	15.47	9.08	37.40	38.05	20.53
23	153	17.22	10.96	26.18	45.64	23.56	9.69	28.08	38.67	16.48
24	157	11.53	11.68	44.17	32.62	5.77	10.77	43.56	39.90	14.56
25	162	13.27	11.80	41.60	33.34	12.45	9.92	40.02	37.62	8.55
26	165	16.63	10.68	36.86	35.83	19.91	9.04	35.57	35.47	6.56
27	167	15.34	10.92	37.39	36.36	13.66	9.66	38.97	37.71	5.88
28	168	15.33	10.90	37.40	36.36	16.71	9.27	37.20	36.82	3.66
29	185	16.67	3.39	28.25	51.69	19.03	7.55	32.02	41.40	20.59
30	193	22.40	8.70	33.70	35.20	22.18	8.68	34.29	34.85	1.18
31	206	26.42	6.30	31.50	35.78	18.12	8.61	35.64	37.64	16.60
32	207	18.90	7.13	34.65	39.32	15.78	8.35	36.20	39.67	6.22
33	208	17.98	9.17	31.97	40.89	24.89	8.68	33.18	33.24	16.27
34	209	25.33	7.06	31.19	36.42	21.37	8.51	34.38	35.74	9.30
35	213	20.25	12.40	35.95	31.40	19.92	9.56	38.29	32.23	6.33
36	220	17.60	8.66	31.80	41.94	23.82	8.62	33.59	33.98	16.01
37	225	39.20	8.19	19.75	32.86	41.36	9.12	21.30	28.22	9.27
38	228	35.04	7.76	15.93	41.27	25.12	8.90	34.24	31.73	38.90
39	229	39.90	7.55	17.14	35.41	33.16	9.11	29.30	28.43	27.43
40	232	46.99	8.07	20.33	24.62	45.26	6.62	21.48	26.64	6.35

Figure 81. Experimental data and model prediction

APPENDIX E
STARTING POINT

Table 22. Starting point (Page 1 of 19)

		Feedstock	C	H	O	N	S	Ash	Yield
1	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	1.43
2	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	1.51
3	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.23
4	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.45
5	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.53
6	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.13
7	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.25
8	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.37
9	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.18
10	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	1.88
11	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	1.46
12	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.39
13	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.34
14	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.23
15	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	1.95
16	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	1.53
17	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	1.93
18	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.37
19	BFB1	Pine	50.82	7.12	41.34	0.15	0.57	0.55	2.57
20	BFB 2	Beech wood	48.28	6.36	45.21	0.14	0.01	0.8	-
21	BFB 2	Beech wood	48.28	6.36	45.21	0.14	0.01	0.8	-
22	BFB 2	Beech wood	48.28	6.36	45.21	0.14	0.01	0.8	-
23	BFB 2	Beech wood	48.28	6.36	45.21	0.14	0.01	0.8	-
24	BFB 2	Beech wood	48.28	6.36	45.21	0.14	0.01	0.8	-
25	BFB 2	Beech wood	48.28	6.36	45.21	0.14	0.01	0.8	-
26	BFB 2	Beech wood	48.28	6.36	45.21	0.14	0.01	0.8	-
27	BFB 3	Wood pellet	51.02	7.16	41.73	0.09	0.01	0.89	-
28	BFB 3	Wood pellet	51.02	7.16	41.73	0.09	0.01	0.89	-
29	BFB 3	Wood pellet	51.02	7.16	41.73	0.09	0.01	0.89	-
30	BFB 3	Wood pellet	51.02	7.16	41.73	0.09	0.01	0.89	-
31	BFB 4	Coir pitch	47.18	3.6	48.2	1.01	0.01	6	2.21
32	BFB 4	Coir pitch	47.18	3.6	48.2	1.01	0.01	6	2.32
33	BFB 4	Coir pitch	47.18	3.6	48.2	1.01	0.01	6	2.52
34	BFB 4	Coir pitch	47.18	3.6	48.2	1.01	0.01	6	2.69
35	BFB 4	Rice husk	38.92	5.1	53.69	2.17	0.12	19.33	2.25
36	BFB 4	Rice husk	38.92	5.1	53.69	2.17	0.12	19.33	2.31
37	BFB 4	Rice husk	38.92	5.1	53.69	2.17	0.12	19.33	2.56
38	BFB 4	Rice husk	38.92	5.1	53.69	2.17	0.12	19.33	2.57
39	BFB 4	Sawdust	51.55	6.16	42.15	0.12	0.02	1.8	2.71
40	BFB 4	Sawdust	51.55	6.16	42.15	0.12	0.02	1.8	2.64
41	BFB 4	Sawdust	51.55	6.16	42.15	0.12	0.02	1.8	3.15
42	BFB 4	Sawdust	51.55	6.16	42.15	0.12	0.02	1.8	3.21

Table 22. Starting point (Page 2 of 19)

		Feedstock	C	H	O	N	S	Ash	Yield
43	BFB 5	Hambach	56.4	5.01	37.55	0.63	0.42	5.1	3.28
44	BFB 5	Hambach	56.4	5.01	37.55	0.63	0.42	5.1	5.15
45	BFB 5	Hambach	56.4	5.01	37.55	0.63	0.42	5.1	6.00
46	BFB 5	Labee	46.24	6.2	47.44	0.04	0.08	0.1	4.53
47	BFB 5	Labee	46.24	6.2	47.44	0.04	0.08	0.1	4.40
48	BFB 5	Labee	46.24	6.2	47.44	0.04	0.08	0.1	4.25
49	BFB 6	Forest residues	43	5	49.6	2.4	0.01	0.2	1.68
50	BFB 6	Forest residues	43	5	49.6	2.4	0.01	0.2	1.27
51	BFB 6	Forest residues	43	5	49.6	2.4	0.01	0.2	1.58
52	BFB 6	Coffe husks	40.1	5.6	49.1	5.2	0.01	2.5	3.78
53	BFB 6	Coffe husks	40.1	5.6	49.1	5.2	0.01	2.5	3.14
54	BFB 6	Coffe husks	40.1	5.6	49.1	5.2	0.01	2.5	2.82
55	BFB 6	Vine-pruning residues	41.3	5.5	50.6	2.6	0.01	3.1	4.28
56	BFB 6	Vine-pruning residues	41.3	5.5	50.6	2.6	0.01	3.1	1.96
57	BFB 6	Vine-pruning residues	41.3	5.5	50.6	2.6	0.01	3.1	1.85
58	BFB 7	Wallnut shell	48.01	6.04	45.5	0.15	0.05	0.32	0.99
59	BFB 7	Wallnut shell	48.01	6.04	45.5	0.15	0.05	0.32	1.07
60	BFB 7	Wallnut shell	48.01	6.04	45.5	0.15	0.05	0.32	1.27
61	BFB 7	Wallnut shell	48.01	6.04	45.5	0.15	0.05	0.32	1.52
62	BFB 7	Wallnut shell	48.01	6.04	45.5	0.15	0.05	0.32	1.52
63	BFB 7	Wallnut shell	48.01	6.04	45.5	0.15	0.05	0.32	0.93
64	BFB 7	Wallnut shell	48.01	6.04	45.5	0.15	0.05	0.32	0.92
65	BFB 7	Wallnut shell	48.01	6.04	45.5	0.15	0.05	0.32	1.06
66	BFB 7	Pistachio shell	48.01	6.04	45.5	0.15	0.05	0.32	1.13
67	BFB 7	Pistachio shell	48.01	6.04	45.5	0.15	0.05	0.32	1.43
68	BFB 7	Pistachio shell	48.01	6.04	45.5	0.15	0.05	0.32	1.13
69	BFB 7	Pistachio shell	48.01	6.04	45.5	0.15	0.05	0.32	1.34
70	BFB 7	Pistachio shell	48.01	6.04	45.5	0.15	0.05	0.32	1.29
71	BFB 7	Pistachio shell	48.01	6.04	45.5	0.15	0.05	0.32	1.65
72	BFB 7	Pistachio shell	48.01	6.04	45.5	0.15	0.05	0.32	1.63
73	BFB 8	80%PC 20%Pine	72.37	5.04	19.46	1.16	1.96	37.02	-
74	BFB 8	80%PC 20%Bagasse	73.27	5.06	18.18	1.54	1.95	37.91	-
75	BFB 8	90%PC 10%PE	78.59	5.66	12.33	1.26	2.16	0	-
76	BFB 8	100% German coal	67.9	5.1	25.7	0.8	0.5	3.69	-
77	BFB 8	80%GC 20%Pine	64.45	5.36	29.07	0.68	0.44	3.02	-

Table 22. Starting point (Page 3 of 19)

		Feedstock	C	H	O	N	S	Ash	Yield
78	BFB 8	80%GC 20%Bagasse	65.35	5.38	27.78	1.06	0.43	3.9	-
79	BFB 8	90%GC 10%PE	69.68	6.02	23.13	0.72	0.45		-
80	BFB 8	80%Coal 20%PE	71.46	6.94	20.56	0.64	0.4		-
81	BFB 8	100% South African coal	79.2	4.7	13.5	2	0.6	11.37	-
82	BFB 8	80%AC 20%Pine	73.49	5.04	19.3	1.64	0.52	9.16	-
83	BFB 8	80%AC 20%Bagasse	74.39	5.06	18.02	2.02	0.51	10.05	-
84	BFB 8	90%AC 10%PE	79.85	5.66	12.15	1.8	0.54		-
85	BFB 9	Rubber wood chip	46.4	5.7	47.7	0.2	0.01	1.1	1.90
86	BFB 9	Rubber wood chip	46.4	5.7	47.7	0.2	0.01	1.1	2.20
87	BFB 9	Rubber wood chip	46.4	5.7	47.7	0.2	0.01	1.1	2.25
88	BFB 9	Rubber wood chip	46.4	5.7	47.7	0.2	0.01	1.1	2.37
89	BFB 9	Rubber wood chip	46.4	5.7	47.7	0.2	0.01	1.1	2.49
90	BFB 10	Wood pellets	50.76	5.92	43.32	0.01	0.01	0.89	-
91	BFB 10	Wood pellets	50.76	5.92	43.32	0.01	0.01	0.89	-
92	BFB 10	Wood pellets	50.76	5.92	43.32	0.01	0.01	0.89	-
93	BFB 10	Wood pellets	50.76	5.92	43.32	0.01	0.01	0.89	-
94	BFB 10	Wood pellets	50.76	5.92	43.32	0.01	0.01	0.89	-
95	BFB 10	Wood pellets	50.76	5.92	43.32	0.01	0.01	0.89	-
96	BFB 11	RIL 1 (natural wood)	45.3	5.6	38.4	0.5	0		-
97	BFB 11	RIL 1 (natural wood)	45.3	5.6	38.4	0.5	0		-
98	BFB 11	RIL 1 (natural wood)	45.3	5.6	38.4	0.5	0		-
99	BFB 11	RIL 1 (natural wood)	45.3	5.6	38.4	0.5	0		-
100	BFB 11	RIL 1 (natural wood)	45.3	5.6	38.4	0.5	0		-
101	BFB 11	RIL 1 (natural wood)	45.3	5.6	38.4	0.5	0		-

Table 22. Starting point (Page 4 of 19)

		Feedstock	C	H	O	N	S	Ash	Yield
102	BFB 11	RIL 1 (natural wood)	45.3	5.6	38.4	0.5	0		-
103	BFB 11	RIL 1 (natural wood)	45.3	5.6	38.4	0.5	0		-
104	BFB 11	RIL 1 (natural wood)	45.3	5.6	38.4	0.5	0		-
105	BFB 11	RIL 1 (natural wood)	45.3	5.6	38.4	0.5	0		-
106	BFB 11	RIL 2 (wood waste)	57.4	7.3	26.4	0.7	0		-
107	BFB 11	RIL 2 (wood waste)	57.4	7.3	26.4	0.7	0		-
108	BFB 11	RIL 2 (wood waste)	57.4	7.3	26.4	0.7	0		-
109	BFB 11	RIL 2 (wood waste)	57.4	7.3	26.4	0.7	0		-
110	BFB 11	RIL 2 (wood waste)	57.4	7.3	26.4	0.7	0		-
111	BFB 11	RIL 2 (wood waste)	57.4	7.3	26.4	0.7	0		-
112	BFB 11	RIL 2 (wood waste)	57.4	7.3	26.4	0.7	0		-
113	BFB 11	RIL 2 (wood waste)	57.4	7.3	26.4	0.7	0		-
114	BFB 11	RIL 3 (wood waste)	50.2	7	25.9	2.3	0		-
115	BFB 11	RIL 3 (wood waste)	50.2	7	25.9	2.3	0		-
116	BFB 11	RIL 3 (wood waste)	50.2	7	25.9	2.3	0		-
117	CFB 1	Olive oil waste	52.87	7.22	38.23	1.61	0.07	8.5	2.90
118	CFB 1	Olive oil waste	52.87	7.22	38.23	1.61	0.07	8.5	5.60
119	CFB 1	Olive oil waste	52.87	7.22	38.23	1.61	0.07	8.5	4.70
120	CFB 1	Olive oil waste	52.87	7.22	38.23	1.61	0.07	8.5	5.20
121	CFB 2	Agrol	53.34	6.55	39.95	0.16	0.01	0.2	-
122	CFB 2	Agrol	53.34	6.55	39.95	0.16	0.01	0.2	-
123	CFB 2	Agrol	53.34	6.55	39.95	0.16	0.01	0.2	-
124	CFB 2	Agrol	53.34	6.55	39.95	0.16	0.01	0.2	-
125	CFB 2	Willow	53.19	6.52	39.55	0.73	0.01	2.9	-
126	CFB 2	Willow	53.19	6.52	39.55	0.73	0.01	2.9	-
127	CFB 2	Willow	53.19	6.52	39.55	0.73	0.01	2.9	-
128	CFB 2	Willow	53.19	6.52	39.55	0.73	0.01	2.9	-
129	CFB 2	Agrol	53.34	6.55	39.95	0.16	0.01	0.2	-
130	CFB 2	Agrol	53.34	6.55	39.95	0.16	0.01	0.2	-
131	CFB 2	Agrol	53.34	6.55	39.95	0.16	0.01	0.2	-

Table 22. Starting point (Page 5 of 19)

		Feedstock	C	H	O	N	S	Ash	Yield
132	CFB 2	Willow	53.19	6.52	39.55	0.73	0.01	2.9	-
133	CFB 2	Willow	53.19	6.52	39.55	0.73	0.01	2.9	-
134	CFB 2	Willow	53.19	6.52	39.55	0.73	0.01	2.9	-
135	CFB 2	DDGS	52.7	7.15	34.11	6.04	0.01	5.6	-
136	CFB 2	DDGS	52.7	7.15	34.11	6.04	0.01	5.6	-
137	CFB 2	DDGS	52.7	7.15	34.11	6.04	0.01	5.6	-
138	CFB 2	DDGS	52.7	7.15	34.11	6.04	0.01	5.6	-
139	CFB 3	Willow	48.68	5.91	44.48	0.88	0.04	2.13	2.19
140	CFB 3	Demolition wood	51.02	5.55	42.92	0.46	0.04	2.08	2.28
141	CFB 3	Chip board material	50.3	5.87	42.3	1.44	0.08	2.9	2.32
142	CFB 3	Verge grass a	50.33	5.73	41.26	2.49	0.19	17.6	1.95
143	CFB 3	Verge grass b	50.33	5.73	41.26	2.49	0.19	17.6	2.01
144	CFB 3	Paper residue sludge	43.5	5.58	48.9	1.37	0.65	47.3	1.94
145	CFB 3	Sewage Sludge	52.28	7.28	30.75	7.07	2.61	36	1.95
146	CFB 3	Woody excess fraction of ODW	52.28	7.28	30.75	7.07	2.61	36	1.74
147	CFB 3	Park wood (bio-dried)	51.55	5.71	41.34	1.16	0.25	18.3	2.00
148	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.39
149	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.56
150	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.65
151	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.77
152	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.31
153	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.40
154	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.47
155	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.52
156	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.60
157	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.42
158	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.45
159	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.50
160	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.59
161	CFB 4	Rice husk	46.56	5.82	46.85	0.54	0.23	14.57	1.55
162	CFB 5	Cypress	51.96	6.24	40.68	0.65	0.46		3.30
163	CFB 5	Cypress	51.96	6.24	40.68	0.65	0.46		2.92
164	CFB 5	Spruce/pine/fir	50.8	6.3	41.93	0.62	0.35		2.48
165	CFB 5	Hemlock	52.02	6.23	40.77	0.6	0.38		3.10
166	CFB 5	Hemlock	52.02	6.23	40.77	0.6	0.38		2.59
167	CFB 5	Hemlock	52.02	6.23	40.77	0.6	0.38		2.75

Table 22. Starting point (Page 6 of 19)

		Feedstock	C	H	O	N	S	Ash	Yield
168	CFB 5	Hemlock	52.02	6.23	40.77	0.6	0.38		2.34
169	CFB 5	Spruce pine fir	50.8	6.3	41.93	0.62	0.35		2.27
170	CFB 5	Hemlock	52.02	6.23	40.77	0.6	0.38		2.46
171	CFB 5	Cedar/Hemclok	52.71	6.16	40.21	0.52	0.4		2.51
172	CFB 5	Hemlock	52.02	6.23	40.77	0.6	0.38		2.13
173	CFB 5	Pine bark/Spruce	50.88	7.51	40.84	0.26	0.52		1.72
174	CFB 5	Mix	50.27	8.07	41.37	0.22	0.07		2.06
175	CFB 5	Mix	50.27	8.07	41.37	0.22	0.07		2.35
176	CFB 5	Mix	50.27	8.07	41.37	0.22	0.07		3.24
177	CFB 6	Jatropha	52.08	7.78	33.78	6.24	0.12	8.79	2.15
178	CFB 6	Jatropha	52.08	7.78	33.78	6.24	0.12	8.79	2.34
179	CFB 6	Jatropha	52.08	7.78	33.78	6.24	0.12	8.79	2.24
180	CFB 6	Jatropha	52.08	7.78	33.78	6.24	0.12	8.79	2.31
181	CFB 6	Willow	49.82	6.8	40.28	3.08	0.02	2.39	2.32
182	CFB 6	Willow	49.82	6.8	40.28	3.08	0.02	2.39	2.41
183	CFB 6	Willow	49.82	6.8	40.28	3.08	0.02	2.39	2.07
184	CFB 6	Willow	49.82	6.8	40.28	3.08	0.02	2.39	2.08
185	CFB 6	Sunflower	51.01	6.65	41.37	0.96	0.01	9.63	2.18
186	CFB 6	Sunflower	51.01	6.65	41.37	0.96	0.01	9.63	2.29
187	CFB 6	Sunflower	51.01	6.65	41.37	0.96	0.01	9.63	2.03
188	CFB 6	Sunflower	51.01	6.65	41.37	0.96	0.01	9.63	2.10
189	CFB 7	Gum wood	47.2	6.04	46.5	0.21	0.05	2.5	1.66
190	CFB 7	Gum wood	47.2	6.04	46.5	0.21	0.05	2.5	1.58
191	CFB 7	Gum wood	47.2	6.04	46.5	0.21	0.05	2.5	1.57
192	BFB 13	Peach stone	41	5.7	48.4	4.9	0.01	1.2	1.80
193	BFB 13	Peach stone	41	5.7	48.4	4.9	0.01	1.2	1.50
194	BFB 13	Peach stone	41	5.7	48.4	4.9	0.01	1.2	1.70
195	BFB 13	Peach stone	41	5.7	48.4	4.9	0.01	1.2	2.00
196	BFB 13	Peach stone	41	5.7	48.4	4.9	0.01	1.2	1.50
197	BFB 13	Peach stone	41	5.7	48.4	4.9	0.01	1.2	1.90
198	BFB 13	Peach stone	41	5.7	48.4	4.9	0.01	1.2	2.20
199	BFB 13	Peach stone	41	5.7	48.4	4.9	0.01	1.2	1.60
200	BFB 13	Peach stone	41	5.7	48.4	4.9	0.01	1.2	1.90

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		Feedstock	C	H	O	N	S	Ash	Yield
201	BFB 15	Miscantus	44.5	5.2	45	5.3	0.01	2.5	-
202	BFB 15	Miscantus	44.5	5.2	45	5.3	0.01	2.5	-
203	BFB 15	Miscantus	44.5	5.2	45	5.3	0.01	2.5	-
204	BFB 15	Peach stone	41	5.7	48.4	4.9	0.01	1.2	-
205	BFB 15	Peach stone	41	5.7	48.4	4.9	0.01	1.2	-
206	BFB 15	Peach stone	41	5.7	48.4	4.9	0.01	1.2	-
207	CFB 3	Park and public wood	51.26	5.89	42.55	0.26	0.05	3.23	-
208	CFB 3	Railroad ties	54.33	5.77	39.35	0.27	0.27	2.6	-
209	CFB 3	Cacao shells	51.39	5.66	36.39	3.28	3.28	10.5	-
210	BFB 12	Sawdust	54.9	5.5	39.1	0.4	0.1	1.95	-
211	BFB 12	Sawdust	54.9	5.5	39.1	0.4	0.1	1.95	-
212	BFB 12	Sawdust	54.9	5.5	39.1	0.4	0.1	1.95	-
213	BFB 12	Sawdust	54.9	5.5	39.1	0.4	0.1	1.95	-
214	BFB 12	Sawdust	54.9	5.5	39.1	0.4	0.1	1.95	-
215	BFB 12	Sawdust	54.9	5.5	39.1	0.4	0.1	1.95	-
216	BFB 12	Sawdust	54.9	5.5	39.1	0.4	0.1	1.95	-
217	BFB 12	Sawdust	54.9	5.5	39.1	0.4	0.1	1.95	-
218	BFB 12	Sawdust	50.06	5.91	43.5	0.42	0.11	0.27	-
219	BFB 12	Sawdust	50.06	5.91	43.5	0.42	0.11	0.27	-
220	BFB 12	Sawdust	50.06	5.91	43.5	0.42	0.11	0.27	-
221	BFB 12	Sawdust	50.06	5.91	43.5	0.42	0.11	0.27	-
222	BFB 12	Sawdust	50.06	5.91	43.5	0.42	0.11	0.27	-
223	BFB 12	Sawdust	50.06	5.91	43.5	0.42	0.11	0.27	-
224	BFB 12	Sawdust	50.06	5.91	43.5	0.42	0.11	0.27	-
225	CFB 7	Gum wood	47.2	6.04	46.5	0.21	0.05	2.5	1.76
226	CFB 7	Gum wood	47.2	6.04	46.5	0.21	0.05	2.5	1.83

Table 22. Starting point (Page 8 of 19)

	CCE %	Total	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
1	78.17	100.55	82.29	17.16	100	8	0.22	2.7	700
2	80.66	100.55	82.29	17.16	100	8	0.22	2.7	750
3	85.90	100.55	82.29	17.16	100	8	0.22	2.7	800
4	92.35	100.55	82.29	17.16	100	8	0.22	2.7	850
5	92.59	100.55	82.29	17.16	100	8	0.22	2.7	900
6	76.26	100.55	82.29	17.16	100	8	0.19	1.56	800
7	84.49	100.55	82.29	17.16	100	8	0.21	1.56	800
8	90.60	100.55	82.29	17.16	100	8	0.23	1.56	800
9	84.00	100.55	82.29	17.16	100	8	0.25	1.56	800
10	70.60	100.55	82.29	17.16	100	8	0.27	1.56	800
11	68.67	100.55	82.29	17.16	100	8	0.22	0	800
12	92.09	100.55	82.29	17.16	100	8	0.22	1.35	800
13	88.70	100.55	82.29	17.16	100	8	0.22	2.02	800
14	85.90	100.55	82.29	17.16	100	8	0.22	2.7	800
15	75.10	100.55	82.29	17.16	100	8	0.22	4.04	800
16	77.62	100.55	82.29	17.16	100	8	0.23	1.56	800
17	84.40	100.55	82.29	17.16	100	8	0.23	1.56	800
18	90.60	100.55	82.29	17.16	100	8	0.23	1.56	800
19	95.10	100.55	82.29	17.16	100	8	0.23	1.56	800
20	-	100.8	82	17.2	100	7	0.32	0	805
21	-	100.8	82	17.2	100	7	0.23	0	800
22	-	100.8	82	17.2	100	7	0.48	0	805
23	-	100.8	82	17.2	100	7	0.13	0	800
24	-	100.8	82	17.2	100	7	0.22	0	815
25	-	100.8	82	17.2	100	7	0.24	0	800
26	-	100.8	82	17.2	100	7	0.66	0	800
27	-	100.9	80.6	18.51	100	9.8	0.27	0	809
28	-	100.9	80.6	18.51	100	9.8	0.24	0	806
29	-	100.9	80.6	18.51	100	9.8	0.19	0	781
30	-	100.9	80.6	18.51	100	9.8	0.32	0	725
31	88.40	106	73.4	20.6	100	13.63	0.3	0	674.46
32	92.80	106	73.4	20.6	100	13.63	0.3	0	747.43
33	-	106	73.4	20.6	100	13.63	0.4	0	683.58
34	-	106	73.4	20.6	100	13.63	0.4	0	772.31
35	-	119.33	63.8	16.87	100	8.1	0.3	0	754
36	-	119.33	63.8	16.87	100	8.1	0.3	0	874
37	-	119.33	63.8	16.87	100	8.1	0.4	0	764
38	-	119.33	63.8	16.87	100	8.1	0.4	0	887
39	-	101.8	77.76	20.44	100	9.3	0.3	0	746.33
40	-	101.8	77.76	20.44	100	9.3	0.3	0	856.74
41	-	101.8	77.76	20.44	100	9.3	0.4	0	756.32

Table 22. Starting point (Page 9 of 19)

	CCE %	Total	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
42	-	101.8	77.76	20.44	100	9.3	0.4	0	876.72
43	94.00	105.11	52.25	42.65	100	15.6	0.4	0	791
44	98.00	105.11	52.25	42.65	100	15.6	0.68	0	802
45	99.00	105.11	52.25	42.65	100	15.6	0.84	0	858
46	100.00	100.1	84.4	15.5	100	9.2	0.36	0	782
47	100.00	100.1	84.4	15.5	100	9.2	0.62	0	824
48	100.00	100.1	84.4	15.5	100	9.2	0.6	0	792
49	97.90	100.21	79.8	20	100	11.3	0.44	0	815
50	92.20	100.21	79.8	20	100	11.3	0.39	0	815
51	92.90	100.21	79.8	20	100	11.3	0.47	0	790
52	78.00	102.51	83.2	14.3	100	25.3	0.6	0	815
53	72.20	102.51	83.2	14.3	100	25.3	0.58	0	790
54	74.10	102.51	83.2	14.3	100	25.3	0.45	0	790
55	89.30	103.11	83.6	13.3	100	13.3	0.45	0	790
56	83.90	103.11	83.6	13.3	100	13.3	0.15	0	790
57	84.30	103.11	83.6	13.3	100	13.3	0.15	0	815
58	-	100.07	76.78	18.7	95.8	0	0.18	1.4	750
59	-	100.07	76.78	18.7	95.8	0	0.18	1.4	800
60	-	100.07	76.78	18.7	95.8	0	0.18	1.4	850
61	-	100.07	76.78	18.7	95.8	0	0.18	1.4	900
62	-	100.07	76.78	18.7	95.8	0	0.18	1.4	950
63	-	100.07	76.78	18.7	95.8	0	0.37	1.7	850
64	-	100.07	76.78	18.7	95.8	0	0.28	1.7	850
65	-	100.07	76.78	18.7	95.8	0	0.18	1.7	850
66	-	100.07	76.78	18.7	95.8	0	0.09	1.7	850
67	-	100.07	76.78	18.7	95.8	0	0	1.7	850
68	-	100.07	76.78	18.7	95.8	0	0	1.1	800
69	-	100.07	76.78	18.7	95.8	0	0	1.4	800
70	-	100.07	76.78	18.7	95.8	0	0	1.8	800
71	-	100.07	76.78	18.7	95.8	0	0	2.7	800
72	-	100.07	76.78	18.7	95.8	0	0	4.7	800
73	-	137.01	34.77	28.21	100	8.72	0.2	0	850
74	-	137.91	32.86	29.23	100	8.34	0.2	0	850
75	-	100	0	0	0	7.2	0.2	0	850
76	-	103.69	51.29	45.02	100	18.7	0.2	0	850
77	-	103.02	57.89	39.09	100	17.28	0.2	0	850
78	-	103.9	55.98	40.11	99.99	16.9	0.2	0	850
79	-	100			0	16.83	0.2	0	850
80	-	100			0	10	0.2	0	850
81	-	111.37	27.74	60.89	100	5.9	0.2	0	850
82	-	109.15	39.04	51.79	99.99	5	0.2	0	850

Table 22. Starting point (Page 10 of 19)

	CCE %	Total	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
83	-	110.05	37.14	52.81	100	6	0.2	0	850
84	-	100			0	5.5	0.2	0	850
85	-	101.11	88.9	10	100	8.5	0.32	0	746.49
86	-	101.11	88.9	10	100	8.5	0.36	0	776.1
87	-	101.11	88.9	10	100	8.5	0.38	0	790.35
88	-	101.11	88.9	10	100	8.5	0.41	0	810.09
89	-	101.11	88.9	10	100	8.5	0.43	0	836.4
90	87.00	100.91	80.6	18.51	100	7	0.19	0	780
91	88.00	100.91	80.6	18.51	100	7	0.27	0	805
92	90.00	100.91	80.6	18.51	100	7	0.27	0.23	786
93	94.00	100.91	80.6	18.51	100	7	0.36	0.32	808
94	95.00	100.91	80.6	18.51	100	7	0.34	0.56	795
95	96.00	100.91	80.6	18.51	100	7	0.33	0.57	806
96	-	89.8			0	9	0.21	0	856
97	-	89.8			0	9	0.17	0	843
98	-	89.8			0	9	0.26	0	911
99	-	89.8			0	9	0.29	0	860
100	-	89.8			0	9	0.2	0	850
101	-	89.8			0	9	0.28	0.9	832
102	-	89.8			0	9	0.26	0.63	808
103	-	89.8			0	9	0.2	0.63	847
104	-	89.8			0	9	0.25	0	864
105	-	89.8			0	9	0.25	0	855
106	-	91.8			0	7	0.3	0	925
107	-	91.8			0	7	0.26	0	907
108	-	91.8			0	7	0.22	0	885
109	-	91.8			0	7	0.31	0	945
110	-	91.8			0	7	0.26	0	865
111	-	91.8			0	7	0.16	0	854
112	-	91.8			0	7	0.13	0.62	768
113	-	91.8			0	7	0.28	0	923
114	-	85.4			0	11.3	0.25	0	881
115	-	85.4			0	11.3	0.31	0	907
116	-	85.4			0	11.3	0.21	0	856
117	81.00	108.5	74.4	17.1	100	8.9	0.41	0	780
118	86.90	108.5	74.4	17.1	100	8.9	0.73	0	800
119	83.50	108.5	74.4	17.1	100	8.9	0.59	0	800
120	84.70	108.5	74.4	17.1	100	8.9	0.67	0	800
121	-	100.21	82.2	17.6	100	8	0.38	1.13	775
122	-	100.21	82.2	17.6	100	8	0.38	1.45	770
123	-	100.21	82.2	17.6	100	8	0.35	0.97	815

Table 22. Starting point (Page 11 of 19)

	CCE %	Total	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
124	-	100.21	82.2	17.6	100	8	0.35	1.25	810
125	-	102.9	75.5	21.6	100	8	0.38	0.93	780
126	-	102.9	75.5	21.6	100	8	0.38	1.22	780
127	-	102.9	75.5	21.6	100	8	0.39	0.9	820
128	-	102.9	75.5	21.6	100	8	0.39	1.04	820
129	-	100.21	82.2	17.6	100	8	0.33	0.98	800
130	-	100.21	82.2	17.6	100	8	0.33	1.16	800
131	-	100.21	82.2	17.6	100	8	0.4	1.15	800
132	-	102.9	75.5	21.6	100	8	0.34	0.91	800
133	-	102.9	75.5	21.6	100	8	0.34	1.14	800
134	-	102.9	75.5	21.6	100	8	0.43	1.07	815
135	-	105.61	76.7	17.7	100	12	0.38	0.81	790
136	-	105.61	76.7	17.7	100	12	0.38	0.81	800
137	-	105.61	76.7	17.7	100	12	0.38	0.81	810
138	-	105.61	76.7	17.7	100	12	0.38	0.81	820
139	97.00	102.12	81.72	16.15	100	17.5	0.37	0	827
140	91.60	102.07	78.34	19.58	100	6.1	0.36	0	847
141	92.20	102.89	76.9	20.2	100	10.8	0.38	0	843
142	94.70	117.6	65.1	17.3	100	7.3	0.37	0	803
143	92.70	117.6	65.1	17.3	100	7.3	0.36	0	815
144	88.20	147.3	43.32	9.38	100	10.2	0.37	0	829
145	85.20	135.99	55.17	8.83	100	10.2	0.32	0	827
146	98.40	135.99	55.17	8.83	100	12.8	0.6	0	827
147	96.50	118.31	60.87	20.83	100	14.8	0.43	0	805
148	83.60	114.57	70.36	15.07	100	14.43	0.22	0	668
149	93.30	114.57	70.36	15.07	100	14.43	0.25	0	711
150	94.80	114.57	70.36	15.07	100	14.43	0.27	0	745
151	96.50	114.57	70.36	15.07	100	14.43	0.3	0	770
152	81.70	114.57	70.36	15.07	100	14.43	0.21	0	672
153	84.10	114.57	70.36	15.07	100	14.43	0.22	0	700
154	84.90	114.57	70.36	15.07	100	14.43	0.24	0	714
155	85.50	114.57	70.36	15.07	100	14.43	0.25	0	750
156	88.20	114.57	70.36	15.07	100	14.43	0.27	0	764
157	82.70	114.57	70.36	15.07	100	14.43	0.23	0	704
158	87.30	114.57	70.36	15.07	100	14.43	0.24	0	710
159	91.90	114.57	70.36	15.07	100	14.43	0.26	0	736
160	96.90	114.57	70.36	15.07	100	14.43	0.27	0	748
161	98.90	114.57	70.36	15.07	100	14.43	0.28	0	766
162	97.70	99.99			0	22	0.63	0	740
163	102.00	99.99			0	9.7	0.46	0	718
164	92.90	100			0	10.5	0.42	0	766

Table 22. Starting point (Page 12 of 19)

	CCE %	Total	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
165	99.20	100			0	10	0.47	0	815
166	98.60	100			0	8.8	0.35	0	772
167	93.00	100			0	9.2	0.39	0	787
168	91.90	100			0	11.7	0.3	0	718
169	85.80	100			0	11.3	0.31	0	730
170	91.10	100			0	15	0.34	0	752
171	87.60	100			0	12.6	0.35	0	815
172	81.90	100			0	14.7	0.29	0	789
173	81.60	100.01			0	10.1	0.2	0	701
174	89.80	100			0	6.6	0.25	0	728
175	94.90	100			0	6.7	0.29	0	739
176	98.30	100			0	4.2	0.45	0	805
177	75.00	108.79	78.95	12.26	100	6.1	0.3	0	750
178	78.10	108.79	78.95	12.26	100	6.1	0.3	0	800
179	75.00	108.79	78.95	12.26	100	6.1	0.3	0	750
180	77.50	108.79	78.95	12.26	100	6.1	0.3	0	800
181	78.00	102.39	83.31	14.3	100	8.2	0.3	0	750
182	83.00	102.39	83.31	14.3	100	8.2	0.3	0	800
183	78.00	102.39	83.31	14.3	100	8.2	0.3	0	750
184	84.00	102.39	83.31	14.3	100	8.2	0.3	0	800
185	80.10	109.63	78.95	11.42	100	8.1	0.3	0	750
186	88.50	109.63	78.95	11.42	100	8.1	0.3	0	800
187	67.20	109.63	78.95	11.42	100	8.1	0.3	0	750
188	72.30	109.63	78.95	11.42	100	8.1	0.3	0	800
189	57.50	102.5	83	14.5	100	5	0.17	0	686
190	67.20	102.5	83	14.5	100	5	0.19	0	690
191	70.70	102.5	83	14.5	100	5	0.2	0	720
192	-	101.21	67.7	31.1	100	7	0.36	0	750
193	-	101.21	67.7	31.1	100	7	0.29	0	750
194	-	101.21	67.7	31.1	100	7	0.35	0	750
195	-	101.21	67.7	31.1	100	7	0.39	0	800
196	-	101.21	67.7	31.1	100	7	0.3	0	800
197	-	101.21	67.7	31.1	100	7	0.36	0	800
198	-	101.21	67.7	31.1	100	7	0.45	0	850
199	-	101.21	67.7	31.1	100	7	0.32	0	850
200	-	101.21	67.7	31.1	100	7	0.37	0	850
201	-	102.51	72.5	25	100	11.4	0.18	0	750
202	-	102.51	72.5	25	100	11.4	0.26	0	800
203	-	102.51	72.5	25	100	11.4	0.47	0	850
204	-	101.21	67.7	31.1	100	7	0.29	0	750
205	-	101.21	67.7	31.1	100	7	0.3	0	800

Table 22. Starting point (Page 13 of 19)

	CCE %	Total	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
206	-	101.21	67.7	31.1	100	7	0.32	0	850
207	-	103.24	77.61	19.16	100	3.5	0.38	0	861
208	-	102.59	79.19	18.21	100	10.6	0.34	0	855
209	-	110.5	67.84	21.66	100	13.4	0.34	0	822
210	-	101.95	74.23	23.82	100	7.5	0.19	0	770
211	-	101.95	74.23	23.82	100	7.5	0.24	0	770
212	-	101.95	74.23	23.82	100	7.5	0.3	0	770
213	-	101.95	74.23	23.82	100	7.5	0.37	0	770
214	-	101.95	74.23	23.82	100	7.5	0	0.41	770
215	-	101.95	74.23	23.82	100	7.5	0	0.63	770
216	-	101.95	74.23	23.82	100	7.5	0	0.97	770
217	-	101.95	74.23	23.82	100	7.5	0	1.1	770
218	-	100.27	83.17	16.56	100	7.5	0.19	0	770
219	-	100.27	83.17	16.56	100	10.2	0.24	0	770
220	-	100.27	83.17	16.56	100	7.5	0.3	0	770
221	-	100.27	83.17	16.56	100	7.5	0.37	0	770
222	-	100.27	83.17	16.56	100	7.5	0	0.41	770
223	-	100.27	83.17	16.56	100	7.5	0	0.63	770
224	-	100.27	83.17	16.56	100	7.5	0	0.97	770
225	80.50	102.5	83	14.5	100	5	0.22	0	919
226	94.50	102.5	83	14.5	100	5	0.26	0	974

Table 22. Starting point (Page 14 of 19)

	H2	CH4	CO	CO2	C2H6	C2H4	C2H2	N2	H2O	O2	SUM
1	21.53	9.08	42.59	19.85	0.25	4.5	0.1	-	-	-	97.9
2	28.4	8.31	39.23	18.98	0.25	3.5	0.1	-	-	-	98.77
3	32.33	7.6	37.63	18.08	0.25	3	0.1	-	-	-	98.99
4	36.52	7.37	34.28	18.39	0.25	2.5	0.1	-	-	-	99.41
5	39.62	6.26	33.34	18.89	0.25	1	0.1	-	-	-	99.46
6	32.05	7.92	37.24	17.3	0.25	3.25	0.1	-	-	-	98.11
7	30.97	8.14	38.93	16.62	0.25	3.25	0.1	-	-	-	98.26
8	31.29	7.67	39.51	16.24	0.25	3	0.1	-	-	-	98.06
9	30.97	7.44	39.13	17.44	0.25	3	0.1	-	-	-	98.33
10	31.72	6.77	37.73	19.24	0.25	3	0.1	-	-	-	98.81
11	33.67	6.38	43.13	15.73	0.25	1.25	0.1	-	-	-	100.51
12	29.89	8.32	40.18	16.33	0.25	3	0.1	-	-	-	98.07
13	30.97	8.25	39.14	16.94	0.25	3	0.1	-	-	-	98.65
14	32.32	7.56	37.66	18.14	0.25	3	0.1	-	-	-	99.03
15	32.19	7.44	37.35	18.81	0.25	3	0.1	-	-	-	99.14
16	30.54	7.78	40.84	16.82	0.25	3	0.1	-	-	-	99.33
17	31.55	7.62	40.08	16.82	0.25	3	0.1	-	-	-	99.42
18	31.63	7.62	37.57	19.67	0.25	3	0.1	-	-	-	99.84
19	31.63	6.7	37.24	21.26	0.25	3	0.1	-	-	-	100.17
20	22.66	6.16	39.9	31.28	-	-	-	-	-	-	100
21	23.47	8.37	41.84	26.33	-	-	-	-	-	-	100
22	20.7	2.11	32.28	44.91	-	-	-	-	-	-	100
23	26.29	8.27	49.63	15.81	-	-	-	-	-	-	100
24	28.77	8.61	39.53	23.09	-	-	-	-	-	-	100
25	24.68	6.96	42.19	26.16	-	-	-	-	-	-	100
26	15.62	3.03	35.8	45.56	-	-	-	-	-	-	100
27	30.02	8.28	28.57	33.13	-	-	-	-	-	-	100
28	31.14	8.9	29.01	30.95	-	-	-	-	-	-	100
29	30.39	9.76	29.65	30.2	-	-	-	-	-	-	100
30	31.47	8.9	24.43	35.2	-	-	-	-	-	-	100
31	10.55	2.67	11.53	11.58	-	-	-	63.5	-	-	99.83
32	10.61	3.82	12.57	11.05	-	-	-	61.8	-	-	99.83
33	10.46	2.5	10.91	11.69	-	-	-	64.3	-	-	99.81
34	10.57	3.6	12.68	11.83	-	-	-	61.1	-	-	99.82
35	8.59	3.21	15.42	11.68	-	-	-	61	-	-	99.9
36	8.62	3.24	19.55	10.21	-	-	-	58.3	-	-	99.95
37	7.42	2.01	14.78	10.66	-	-	-	65.1	-	-	99.93
38	7.77	2.78	16.13	10.68	-	-	-	62.6	-	-	99.94
39	9.98	2.45	17.38	11.96	-	-	-	58.1	-	-	99.82
40	10.14	3.21	17.54	10.78	-	-	-	58.3	-	-	99.96
41	8.12	2.14	16.32	11.16	-	-	-	62.1	-	-	99.88
42	9.35	2.62	17.73	11.87	-	-	-	58.3	-	-	99.86

Table 22. Starting point (Page 15 of 19)

	H2	CH4	CO	CO2	C2H6	C2H4	C2H2	N2	H2O	O2	SUM
43	10.4	2.1	13.4	10.9	-	0.1	-	57.3	5.2	-	99.4
44	7	1.4	8.9	11.8	-	0.1	-	64.8	5.3	-	99.3
45	4.5	0.6	6.1	12.6	-	0.1	-	69	6.4	-	99.3
46	5.2	3	8	9.7	-	0.5	-	65.1	8.2	-	99.7
47	3.8	2	5.6	13.7	-	0.3	-	62.7	11.4	-	99.5
48	5.3	2.1	7.3	13.3	-	0.7	-	60.2	10.6	-	99.5
49	8.2	4.6	18.6	16.7	-	-	-	48	-	-	96.1
50	8.4	4.4	18	17.1	-	-	-	48.2	-	-	96.1
51	7.6	4.4	17.9	17.1	-	-	-	49.2	-	-	96.2
52	12.4	1.6	11.4	18.7	-	-	-	52.3	-	-	96.4
53	7.6	2.4	11.1	17	-	-	-	54.2	-	-	92.3
54	7.5	2.4	10.6	18.5	-	-	-	55.2	-	-	94.2
55	5.1	1.1	8.3	16.5	-	-	-	56.4	-	-	87.4
56	10.4	2.4	11.7	20.1	-	-	-	51.2	-	-	95.8
57	12.7	2.3	14.1	17.9	-	-	-	49.1	-	-	96.1
58	31.4	8.4	20.6	35.3	1.6	0.7	2	-	-	-	100
59	37.2	7.6	17.4	35.3	0	0.6	1.9	-	-	-	100
60	38.8	8.3	18.6	32.2	0	0.2	2	-	-	-	100.1
61	38.3	8	19.3	32.7	0	0.3	1.4	-	-	-	100
62	44.8	7.1	19	28.4	0	0	1.6	-	-	-	100.9
63	27.6	7.7	15.5	46.9	0	0.1	2.3	-	-	-	100.1
64	34.3	7.6	15.6	40.4	0	0.1	2.1	-	-	-	100.1
65	39.1	7.4	16.8	34.7	0	0.1	1.9	-	-	-	100
66	43.2	7.4	18.5	28.5	0	0.4	1.9	-	-	-	99.9
67	46.8	8.3	20.2	22.3	0	0.6	1.8	-	-	-	100
68	46	7.9	19.9	24.1	0.2	0.3	1.6	-	-	-	100
69	48.5	6.9	18.5	24.6	0	0.2	1.4	-	-	-	100.1
70	51.3	6.1	15.9	25.1	0	0.4	1.3	-	-	-	100.1
71	54.4	5	11.8	27.4	0	0.3	1.1	-	-	-	100
72	57.4	4.1	7.3	30.2	0	0.1	1	-	-	-	100.1
73	35.33	8.34	19.46	35.32	-	-	-	-	-	-	98.44
74	35.54	9.12	19.46	34.14	-	-	-	-	-	-	98.25
75	28.98	11.91	15.09	38.11	-	-	-	-	-	-	94.09
76	38.9	5.22	27.14	27.14	-	-	-	-	-	-	98.41
77	42.58	4.97	23.53	27.11	-	-	-	-	-	-	98.18
78	38.9	5.76	27.39	26.01	-	-	-	-	-	-	98.05
79	31.56	9.93	19.24	35.92	-	-	-	-	-	-	96.65
80	28.78	11.31	20.05	31.95	-	-	-	-	-	-	92.09
81	34.93	7.74	19.84	36.13	-	-	-	-	-	-	98.64
82	32.95	8.34	22.83	33.94	-	-	-	-	-	-	98.05
83	33.55	8.34	22.03	34.33	-	-	-	-	-	-	98.25
84	30.56	10.92	17.67	36.52	-	-	-	-	-	-	95.66

Table 22. Starting point (Page 16 of 19)

	H2	CH4	CO	CO2	C2H6	C2H4	C2H2	N2	H2O	O2	SUM
85	13.99	8.19	33.09	39.39	-	-	-	-	-	-	94.67
86	17.17	8.17	29.97	38.97	-	-	-	-	-	-	94.27
87	18.55	8.15	30.35	37.35	-	-	-	-	-	-	94.41
88	16.63	8.53	28.93	39.73	-	-	-	-	-	-	93.82
89	12.91	8.11	28.71	42.21	-	-	-	-	-	-	91.95
90	25.58	11.63	35.27	27.52	-	-	-	-	-	-	100
91	25.05	10.34	34.99	29.62	-	-	-	-	-	-	100
92	28.06	9.42	30.06	32.46	-	-	-	-	-	-	100
93	28.08	9.42	32.36	30.14	-	-	-	-	-	-	100
94	34.38	9.62	27.6	28.39	-	-	-	-	-	-	100
95	37.41	9.75	28.09	24.75	-	-	-	-	-	-	100
96	14.2	6	18.8	15.4	-	-	-	-	-	-	54.4
97	15.2	6.6	20.4	15.7	-	-	-	-	-	-	57.9
98	14.5	4.4	19.9	13.3	-	-	-	-	-	-	52.1
99	13.3	4.2	16.5	14.6	-	-	-	-	-	-	48.6
100	15.2	4.6	19	14.7	-	-	-	-	-	-	53.5
101	17.5	4.2	11.9	18.2	-	-	-	-	-	-	51.8
102	17	4.1	11.6	18.6	-	-	-	-	-	-	51.3
103	21.1	5.1	15.8	17.2	-	-	-	-	-	-	59.2
104	13.8	4.9	19.1	14.3	-	-	-	-	-	-	52.1
105	13.3	5	17.2	15.5	-	-	-	-	-	-	51
106	10.3	3.4	16.6	13.7	-	-	-	-	-	-	44
107	11.8	3.7	17.2	13.3	-	-	-	-	-	-	46
108	13.1	4.3	18.9	13.5	-	-	-	-	-	-	49.8
109	9.4	3.2	15.9	13.3	-	-	-	-	-	-	41.8
110	11.2	3.9	17.8	13.5	-	-	-	-	-	-	46.4
111	12.8	5.3	17.9	14.4	-	-	-	-	-	-	50.4
112	14.5	3.8	17	15.8	-	-	-	-	-	-	51.1
113	13.1	3.3	19.4	11	-	-	-	-	-	-	46.8
114	12.3	3.2	16.3	14.7	-	-	-	-	-	-	46.5
115	9.9	3.3	15.1	14.7	-	-	-	-	-	-	43
116	14.8	4.3	17.6	14.3	-	-	-	-	-	-	51
117	5.4	1.6	8.6	21.7	0.3	1.6	-	59.5	-	-	98.66
118	7.3	0.8	6.9	19.9	0.1	0.8	-	63.2	-	-	98.96
119	9.3	0.8	8.4	19	0.2	0.8	-	60.6	-	-	99.05
120	7.6	0.9	7.5	19.7	0.2	0.9	-	62.5	-	-	99.26
121	22	8.5	25	42	-	-	-	-	-	-	97.5
122	23.5	8	24.5	41.5	-	-	-	-	-	-	97.5
123	23	8.5	24.5	40	-	-	-	-	-	-	96
124	25.5	8	23	39.5	-	-	-	-	-	-	96
125	25	7.5	21	43	-	-	-	-	-	-	96.5
126	27	7	20	42.5	-	-	-	-	-	-	96.5

Table 22. Starting point (Page 17 of 19)

	H2	CH4	CO	CO2	C2H6	C2H4	C2H2	N2	H2O	O2	SUM
127	25.5	7.5	22	41.5	-	-	-	-	-	-	96.5
128	28	7.5	19.5	41.5	-	-	-	-	-	-	96.5
129	22.5	9	27	37.5	-	-	-	-	-	-	96
130	23	9	27	37	-	-	-	-	-	-	96
131	20	8.5	23.5	45	-	-	-	-	-	-	97
132	27.5	8	22.5	38	-	-	-	-	-	-	96
133	30	7.5	21	38	-	-	-	-	-	-	96.5
134	24	7	22	44	-	-	-	-	-	-	97
135	18	8	31	32	-	-	-	-	-	-	89
136	19	8	30.5	31.5	-	-	-	-	-	-	89
137	20	9	29	33	-	-	-	-	-	-	91
138	20.5	8	28.5	33.5	-	-	-	-	-	-	90.5
139	7.2	3.3	9.4	17.1	0.1	1.1	-	60.5	-	-	98.67
140	7.05	3.25	11.35	15.75	0.03	0.85	-	60.4	-	-	98.7
141	6.42	2.77	9.64	15.63	0.03	0.87	-	62.9	-	-	98.28
142	7.28	2.58	9.89	15.61	0.05	1.15	-	61	-	-	97.57
143	8.45	2.79	10.64	15.18	0.04	1.04	-	59.5	-	-	97.61
144	6.08	2.81	9.24	16.11	0.04	1.02	-	63.2	-	-	98.52
145	8.02	3.19	10.53	15.02	0.04	1.12	-	60.5	-	-	98.38
146	1.8	1.24	5.34	16.09	0.02	0.4	-	73.7	-	-	98.57
147	5.38	1.72	8.31	16.04	0.02	0.6	-	66.6	-	-	98.65
148	5.33	5.4	20.42	15.08	-	3.15	-	48	-	-	97.36
149	6.16	5.32	19.38	15.36	-	3.42	-	48.2	-	-	97.83
150	6.77	5	17.92	15.81	-	3.22	-	49.1	-	-	97.84
151	7.57	4.61	16.52	16.1	-	2.69	-	50.2	-	-	97.67
152	5.36	5.78	20.61	15.52	-	3.46	-	47.3	-	-	98.05
153	6.22	5.53	19.5	15.63	-	3.18	-	47.8	-	-	97.85
154	6.48	5.36	18.25	15.96	-	2.67	-	48.7	-	-	97.38
155	7.25	4.95	17.26	16.2	-	2.71	-	49.6	-	-	97.92
156	7.46	4.79	16.53	16.07	-	2.82	-	50.6	-	-	98.24
157	6.95	4.94	16.93	16.47	-	3.99	-	48.7	-	-	98
158	7.18	5.11	17.5	17.02	-	4.13	-	50.4	-	-	101.29
159	7.27	5.17	17.73	17.24	-	4.17	-	51	-	-	102.59
160	7.27	5.17	17.72	17.23	-	4.17	-	51	-	-	102.53
161	7.76	5.52	18.92	18.41	-	4.44	-	54.5	-	-	109.5
162	5.6	1.4	6.9	18.1	-	-	-	68	-	-	100
163	3.1	1.9	11	15.1	-	-	-	68.1	-	-	99.2
164	3.2	1.9	10.7	17.1	-	-	-	67.1	-	-	100
165	3	1.9	9.6	17.1	-	-	-	68.4	-	-	100
166	4	2.9	14.7	16.5	-	-	-	61.8	-	-	99.9
167	3.8	2.7	12.6	15.7	-	-	-	65.2	-	-	100
168	5.5	3.4	16.6	15	-	-	-	59.5	-	-	100

Table 22. Starting point (Page 18 of 19)

	H2	CH4	CO	CO2	C2H6	C2H4	C2H2	N2	H2O	O2	SUM
169	3.9	2.8	15.1	15.6	-	-	-	62.5	-	-	99.9
170	3.5	2.8	13.4	15.6	-	-	-	64.8	-	-	100.1
171	4.1	2.5	12.3	16.5	-	-	-	64.6	-	-	100
172	4.2	3	14.6	15.7	-	-	-	62.6	-	-	100.1
173	5.4	4.6	21.4	14.7	-	-	-	53.9	-	-	100
174	5.1	4.1	19.9	14.5	-	-	-	56.3	-	-	99.9
175	7.3	3.2	17.9	16.3	-	-	-	55.4	-	-	100.1
176	5.9	1.2	10	18.3	-	-	-	64.6	-	-	100
177	22.8	8.1	25.2	32.1	-	-	-	-	-	-	88.2
178	26.5	6.8	34.2	30.2	-	-	-	-	-	-	97.7
179	22.6	8	32.2	31.3	-	-	-	-	-	-	94.1
180	25.2	8.4	36.9	28.7	-	-	-	-	-	-	99.2
181	23.6	8.5	29.5	34.8	-	-	-	-	-	-	96.4
182	25.5	7.8	35.1	30.3	-	-	-	-	-	-	98.7
183	21.8	9.4	31.7	37.7	-	-	-	-	-	-	100.6
184	22.4	8.7	33.7	35.2	-	-	-	-	-	-	100
185	26.8	8.6	29.4	35.1	-	-	-	-	-	-	99.9
186	28.6	8.2	33.1	29.9	-	-	-	-	-	-	99.8
187	27.1	9.4	29.3	34.2	-	-	-	-	-	-	100
188	28.4	8.4	36	29.8	-	-	-	-	-	-	102.6
189	5.5	9.64	16.8	14.9	-	1.3	-	47.7	-	1.1	96.95
190	5.34	8.35	14.4	18.9	-	2.1	-	50	-	0.9	100.01
191	5.79	8.35	17.8	16.1	-	1.2	-	50.2	-	1.1	100.56
192	8.3	3.7	18.2	13.1	0.1	1.1	0.2	51.4	-	3.9	100
193	8.5	4.1	15.7	15.1	0.2	1.1	0.1	51.1	-	4.1	100
194	10.3	3.2	12.4	16.7	0	0.5	0.1	52.9	-	3.9	100
195	8.2	3.6	16.5	14.7	0.1	1	0.1	52	-	3.8	100
196	7.4	4.2	14.9	15.2	0.1	1.1	0.1	52.2	-	4.8	100
197	11.5	2.8	14	15.9	0	0.4	0.1	51.7	-	3.6	100
198	8.1	2.9	14.1	16	0	0.3	0.1	53.2	-	3.1	97.8
199	7.7	3.7	12.9	16.5	0	0.9	0.2	53.5	-	4.6	100
200	11	3.1	13.7	16	0	0.3	0.1	51.9	-	4.1	100.2
201	9.8	6	17.4	15.2	0.1	1.5	0.1	47.6	-	-	97.7
202	11	4	13.3	15.8	0	0.6	0.1	49.7	-	-	94.5
203	9.7	3.5	12	16.8	0	0.3	0.1	52.5	-	-	94.9
204	8.5	4.1	15.7	15.1	0.2	1.1	0.1	51.1	-	-	95.9
205	7.4	4.2	14.9	15.2	0.1	1.1	0.1	52.2	-	-	95.2
206	7.7	3.7	12.9	16.5	0	0.9	0.2	53.5	-	-	95.4
207	5.38	1.72	8.31	16.04	-	-	-	-	-	-	31.45
208	5.85	2.88	10.57	13.94	-	-	-	-	-	-	33.24
209	9.02	2.34	8	16.02	-	-	-	-	-	-	35.38
210	8.5	5.5	17	14	-	-	-	-	-	-	45

Table 22. Starting point (Page 19 of 19)

	H2	CH4	CO	CO2	C2H6	C2H4	C2H2	N2	H2O	O2	SUM
211	6.7	4.5	16	15	-	-	-	-	-	-	42.2
212	4.5	4	12.5	16	-	-	-	-	-	-	37
213	3.4	3.5	11	17	-	-	-	-	-	-	34.9
214	32	8.2	23	13	-	-	-	-	-	-	76.2
215	42	7.5	22	19.5	-	-	-	-	-	-	91
216	49	6	17.5	22.5	-	-	-	-	-	-	95
217	50.5	6	17	22	-	-	-	-	-	-	95.5
218	5.7	6.8	19	14	-	-	-	-	-	-	45.5
219	4.2	5.8	17.1	15.5	-	-	-	-	-	-	42.6
220	3.6	4.4	14.2	16	-	-	-	-	-	-	38.2
221	3.2	3.6	12.5	16.2	-	-	-	-	-	-	35.5
222	20	11.6	29	9.9	-	-	-	-	-	-	70.5
223	36	11	27	16	-	-	-	-	-	-	90
224	54	5.6	14	21.5	-	-	-	-	-	-	95.1
225	13.24	7.32	16.8	15.8	-	1.2	-	44.8	-	1.1	100.26
226	16.32	6.75	16.7	15.6	-	1	-	43.3	-	0.7	100.32

APPENDIX F
NORMALIZED DATA

Table 23. Normalized data (Page 1 of 60)

		Feedstock	C	H	O	N	S
1.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
2.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
3.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
4.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
5.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
6.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
7.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
8.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
9.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
10.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
11.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
12.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
13.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
14.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
15.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
16.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
17.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
18.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
19.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
20.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01

Table 23. Normalized data (Page 2 of 60)

		Feedstock	C	H	O	N	S
21.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
22.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
23.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
24.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
25.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
26.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
27.00	BFB 3	Wood pellet	50.57	7.10	41.36	0.09	0.01
28.00	BFB 3	Wood pellet	50.57	7.10	41.36	0.09	0.01
29.00	BFB 3	Wood pellet	50.57	7.10	41.36	0.09	0.01
30.00	BFB 3	Wood pellet	50.57	7.10	41.36	0.09	0.01
31.00	BFB 4	Coir pitch	44.35	3.38	45.31	0.95	0.01
32.00	BFB 4	Coir pitch	44.35	3.38	45.31	0.95	0.01
33.00	BFB 4	Coir pitch	44.35	3.38	45.31	0.95	0.01
34.00	BFB 4	Coir pitch	44.35	3.38	45.31	0.95	0.01
35.00	BFB 4	Rice husk	31.40	4.11	43.31	1.75	0.10
36.00	BFB 4	Rice husk	31.40	4.11	43.31	1.75	0.10
37.00	BFB 4	Rice husk	31.40	4.11	43.31	1.75	0.10
38.00	BFB 4	Rice husk	31.40	4.11	43.31	1.75	0.10
39.00	BFB 4	Sawdust	50.62	6.05	41.39	0.12	0.02
40.00	BFB 4	Sawdust	50.62	6.05	41.39	0.12	0.02

Table 23. Normalized data (Page 3 of 60)

		Feedstock	C	H	O	N	S
41.00	BFB 4	Sawdust	50.62	6.05	41.39	0.12	0.02
42.00	BFB 4	Sawdust	50.62	6.05	41.39	0.12	0.02
43.00	BFB 5	Hambach	53.52	4.75	35.63	0.59	0.40
44.00	BFB 5	Hambach	53.52	4.75	35.63	0.59	0.40
45.00	BFB 5	Hambach	53.52	4.75	35.63	0.59	0.40
46.00	BFB 5	Labee	46.19	6.20	47.39	0.04	0.08
47.00	BFB 5	Labee	46.19	6.20	47.39	0.04	0.08
48.00	BFB 5	Labee	46.19	6.20	47.39	0.04	0.08
49.00	BFB 6	Forest residues	42.91	4.99	49.50	2.40	0.01
50.00	BFB 6	Forest residues	42.91	4.99	49.50	2.40	0.01
51.00	BFB 6	Forest residues	42.91	4.99	49.50	2.40	0.01
52.00	BFB 6	Coffe husks	39.10	5.46	47.87	5.07	0.01
53.00	BFB 6	Coffe husks	39.10	5.46	47.87	5.07	0.01
54.00	BFB 6	Coffe husks	39.10	5.46	47.87	5.07	0.01
55.00	BFB 6	Vine-pruning residues	40.02	5.33	49.03	2.52	0.01
56.00	BFB 6	Vine-pruning residues	40.02	5.33	49.03	2.52	0.01
57.00	BFB 6	Vine-pruning residues	40.02	5.33	49.03	2.52	0.01
58.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
59.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
60.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05

Table 23. Normalized data (Page 4 of 60)

		Feedstock	C	H	O	N	S
61.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
62.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
63.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
64.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
65.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
66.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
67.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
68.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
69.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
70.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
71.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
72.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
73.00	BFB 8	80%PC 20%Pine	45.58	3.17	12.26	0.73	1.23
74.00	BFB 8	80%PC 20%Bagasse	45.49	3.14	11.29	0.96	1.21
75.00	BFB 8	90%PC 10%PE	78.59	5.66	12.33	1.26	2.16
76.00	BFB 8	100% German coal	65.39	4.91	24.75	0.77	0.48
77.00	BFB 8	80%GC 20%Pine	62.51	5.20	28.19	0.66	0.43
78.00	BFB 8	80%GC 20%Bagasse	62.80	5.17	26.70	1.02	0.41
79.00	BFB 8	90%GC 10%PE	69.68	6.02	23.13	0.72	0.45
80.00	BFB 8	80%Coal 20%PE	71.46	6.94	20.56	0.64	0.40

Table 23. (a) Normalized data (Page 5 of 60)

		Feedstock	C	H	O	N	S
81.00	BFB 8	100% South African coal	70.19	4.17	11.96	1.77	0.53
82.00	BFB 8	80%AC 20%Pine	66.76	4.58	17.53	1.49	0.47
83.00	BFB 8	80%AC 20%Bagasse	66.91	4.55	16.21	1.82	0.46
84.00	BFB 8	90%AC 10%PE	79.85	5.66	12.15	1.80	0.54
85.00	BFB 9	Rubber wood chip	45.89	5.64	47.18	0.20	0.01
86.00	BFB 9	Rubber wood chip	45.89	5.64	47.18	0.20	0.01
87.00	BFB 9	Rubber wood chip	45.89	5.64	47.18	0.20	0.01
88.00	BFB 9	Rubber wood chip	45.89	5.64	47.18	0.20	0.01
89.00	BFB 9	Rubber wood chip	45.89	5.64	47.18	0.20	0.01
90.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
91.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
92.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
93.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
94.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
95.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
96.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
97.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
98.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
99.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
100.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00

Table 23. Normalized data (Page 6 of 60)

		Feedstock	C	H	O	N	S
101.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
102.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
103.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
104.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
105.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
106.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
107.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
108.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
109.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
110.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
111.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
112.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
113.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
114.00	BFB 11	RIL 3 (wood waste)	58.78	8.20	30.33	2.69	0.00
115.00	BFB 11	RIL 3 (wood waste)	58.78	8.20	30.33	2.69	0.00
116.00	BFB 11	RIL 3 (wood waste)	58.78	8.20	30.33	2.69	0.00
117.00	CFB 1	Olive oil waste	48.38	6.61	34.98	1.47	0.06
118.00	CFB 1	Olive oil waste	48.38	6.61	34.98	1.47	0.06
119.00	CFB 1	Olive oil waste	48.38	6.61	34.98	1.47	0.06
120.00	CFB 1	Olive oil waste	48.38	6.61	34.98	1.47	0.06

Table 23. Normalized data (Page 7 of 60)

		Feedstock	C	H	O	N	S
121.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
122.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
123.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
124.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
125.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
126.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
127.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
128.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
129.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
130.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
131.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
132.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
133.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
134.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
135.00	CFB 2	DDGS	49.75	6.75	32.20	5.70	0.01
136.00	CFB 2	DDGS	49.75	6.75	32.20	5.70	0.01
137.00	CFB 2	DDGS	49.75	6.75	32.20	5.70	0.01
138.00	CFB 2	DDGS	49.75	6.75	32.20	5.70	0.01
139.00	CFB 3	Willow	47.65	5.78	43.54	0.86	0.04
140.00	CFB 3	Demolition wood	49.96	5.44	42.03	0.45	0.04

Table 23. Normalized data (Page 8 of 60)

		Feedstock	C	H	O	N	S
141.00	CFB 3	Chip board material	48.85	5.70	41.08	1.40	0.08
142.00	CFB 3	Verge grass a	41.48	4.72	33.99	2.05	0.16
143.00	CFB 3	Verge grass b	41.48	4.72	33.99	2.05	0.16
144.00	CFB 3	Paper residue sludge	22.92	2.94	25.77	0.72	0.34
145.00	CFB 3	Sewage Sludge	33.46	4.66	19.68	4.53	1.67
146.00	CFB 3	Woody excess fraction of ODW	33.46	4.66	19.68	4.53	1.67
147.00	CFB 3	Park wood (bio-dried)	42.11	4.66	33.77	0.95	0.20
148.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
149.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
150.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
151.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
152.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
153.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
154.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
155.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
156.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
157.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
158.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
159.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
160.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20

Table 23. Normalized data (Page 9 of 60)

		Feedstock	C	H	O	N	S
161.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
162.00	CFB 5	Cypress	51.96	6.24	40.68	0.65	0.46
163.00	CFB 5	Cypress	51.96	6.24	40.68	0.65	0.46
164.00	CFB 5	Spruce/pine/fir	50.80	6.30	41.93	0.62	0.35
165.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38
166.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38
167.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38
168.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38
169.00	CFB 5	Spruce pine fir	50.80	6.30	41.93	0.62	0.35
170.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38
171.00	CFB 5	Cedar/Hemclok	52.71	6.16	40.21	0.52	0.40
172.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38
173.00	CFB 5	Pine bark/Spruce	50.88	7.51	40.84	0.26	0.52
174.00	CFB 5	Mix	50.27	8.07	41.37	0.22	0.07
175.00	CFB 5	Mix	50.27	8.07	41.37	0.22	0.07
176.00	CFB 5	Mix	50.27	8.07	41.37	0.22	0.07
177.00	CFB 6	Jatropha	47.50	7.10	30.81	5.69	0.11
178.00	CFB 6	Jatropha	47.50	7.10	30.81	5.69	0.11
179.00	CFB 6	Jatropha	47.50	7.10	30.81	5.69	0.11
180.00	CFB 6	Jatropha	47.50	7.10	30.81	5.69	0.11

Table 23. Normalized data (Page 10 of 60)

		Feedstock	C	H	O	N	S
181.00	CFB 6	Willow	48.63	6.63	39.32	3.00	0.02
182.00	CFB 6	Willow	48.63	6.63	39.32	3.00	0.02
183.00	CFB 6	Willow	48.63	6.63	39.32	3.00	0.02
184.00	CFB 6	Willow	48.63	6.63	39.32	3.00	0.02
185.00	CFB 6	Sunflower	46.10	6.01	37.39	0.87	0.01
186.00	CFB 6	Sunflower	46.10	6.01	37.39	0.87	0.01
187.00	CFB 6	Sunflower	46.10	6.01	37.39	0.87	0.01
188.00	CFB 6	Sunflower	46.10	6.01	37.39	0.87	0.01
189.00	CFB 7	Gum wood	46.02	5.89	45.34	0.20	0.05
190.00	CFB 7	Gum wood	46.02	5.89	45.34	0.20	0.05
191.00	CFB 7	Gum wood	46.02	5.89	45.34	0.20	0.05
192.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
193.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
194.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
195.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
196.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
197.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
198.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
199.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
200.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01

Table 23. Normalized data (Page 11 of 60)

		Feedstock	C	H	O	N	S
201.00	BFB 15	Miscanthus	43.39	5.07	43.88	5.17	0.01
202.00	BFB 15	Miscanthus	43.39	5.07	43.88	5.17	0.01
203.00	BFB 15	Miscanthus	43.39	5.07	43.88	5.17	0.01
204.00	BFB 15	Peach stone	40.51	5.63	47.82	4.84	0.01
205.00	BFB 15	Peach stone	40.51	5.63	47.82	4.84	0.01
206.00	BFB 15	Peach stone	40.51	5.63	47.82	4.84	0.01
207.00	CFB 3	Park and public wood	49.60	5.70	41.17	0.25	0.05
208.00	CFB 3	Railroad ties	52.92	5.62	38.33	0.26	0.26
209.00	CFB 3	Cacao shells	45.99	5.07	32.57	2.94	2.94
210.00	BFB 12	Walnut shell	53.83	5.39	38.34	0.39	0.10
211.00	BFB 12	Walnut shell	53.83	5.39	38.34	0.39	0.10
212.00	BFB 12	Walnut shell	53.83	5.39	38.34	0.39	0.10
213.00	BFB 12	Walnut shell	53.83	5.39	38.34	0.39	0.10
214.00	BFB 12	Walnut shell	53.83	5.39	38.34	0.39	0.10
215.00	BFB 12	Walnut shell	53.83	5.39	38.34	0.39	0.10
216.00	BFB 12	Walnut shell	53.83	5.39	38.34	0.39	0.10
217.00	BFB 12	Walnut shell	53.83	5.39	38.34	0.39	0.10
218.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
219.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
220.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11

Table 23. Normalized data (Page 12 of 60)

		Feedstock	C	H	O	N	S
221.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
222.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
223.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
224.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
225.00	CFB 7	Gum wood	46.02	5.89	45.34	0.20	0.05
226.00	CFB 7	Gum wood	46.02	5.89	45.34	0.20	0.05
			226.00	226.00	226.00	226.00	226.00

Table 23. Normalized data (Page 13 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
1.00	0.55	0.22	8.00	2.70	-	0.89	78.17	100
2.00	0.55	0.22	8.00	2.70	-	1.01	80.66	100
3.00	0.55	0.22	8.00	2.70	-	1.15	85.90	100
4.00	0.55	0.22	8.00	2.70	-	1.32	92.35	100
5.00	0.55	0.22	8.00	2.70	-	1.42	92.59	100
6.00	0.55	0.19	8.00	1.56	-	1.01	76.26	100
7.00	0.55	0.21	8.00	1.56	-	1.10	84.49	100
8.00	0.55	0.23	8.00	1.56	-	1.20	90.60	100
9.00	0.55	0.25	8.00	1.56	-	1.10	84.00	100
10.00	0.55	0.27	8.00	1.56	-	0.93	70.60	100
11.00	0.55	0.22	8.00	0.00	-	0.95	68.67	100
12.00	0.55	0.22	8.00	1.35	-	1.19	92.09	100
13.00	0.55	0.22	8.00	2.02	-	1.16	88.70	100
14.00	0.55	0.22	8.00	2.70	-	1.15	85.90	100
15.00	0.55	0.22	8.00	4.04	-	1.00	75.10	100
16.00	0.55	0.23	8.00	1.56	-	1.01	77.62	100
17.00	0.55	0.23	8.00	1.56	-	1.11	84.40	100
18.00	0.55	0.23	8.00	1.56	-	1.19	90.60	100
19.00	0.55	0.23	8.00	1.56	-	1.25	95.10	100
20.00	0.80	0.32	7.00	0.00	-	1.16	-	100

Table 23. Normalized data (Page 14 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
21.00	0.80	0.23	7.00	0.00	-	1.17	-	100
22.00	0.80	0.48	7.00	0.00	-	1.13	-	100
23.00	0.80	0.13	7.00	0.00	-	1.21	-	100
24.00	0.80	0.22	7.00	0.00	-	1.26	-	100
25.00	0.80	0.24	7.00	0.00	-	1.19	-	100
26.00	0.80	0.66	7.00	0.00	-	1.06	-	100
27.00	0.89	0.27	9.80	0.00	-	1.35	-	100
28.00	0.89	0.24	9.80	0.00	-	1.37	-	100
29.00	0.89	0.19	9.80	0.00	-	1.36	-	100
30.00	0.89	0.32	9.80	0.00	-	1.38	-	100
31.00	6.00	0.30	13.63	0.00	63.50	2.83	88.40	100
32.00	6.00	0.30	13.63	0.00	61.78	2.80	92.80	100
33.00	6.00	0.40	13.63	0.00	64.25	3.29	-	100
34.00	6.00	0.40	13.63	0.00	61.14	2.94	-	100
35.00	19.33	0.30	8.10	0.00	61.00	1.93	-	100
36.00	19.33	0.30	8.10	0.00	58.33	1.78	-	100
37.00	19.33	0.40	8.10	0.00	65.06	2.13	-	100
38.00	19.33	0.40	8.10	0.00	62.58	1.98	-	100
39.00	1.80	0.30	9.30	0.00	58.05	2.97	-	100
40.00	1.80	0.30	9.30	0.00	58.29	3.00	-	100

Table 23. Normalized data (Page 15 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
41.00	1.80	0.40	9.30	0.00	62.14	3.19	-	100
42.00	1.80	0.40	9.30	0.00	58.29	2.93	-	100
43.00	5.10	0.40	15.60	0.00	57.30	3.51	94.00	100
44.00	5.10	0.68	15.60	0.00	64.80	4.36	98.00	100
45.00	5.10	0.84	15.60	0.00	69.00	5.04	99.00	100
46.00	0.10	0.36	9.20	0.00	65.10	3.96	100.00	100
47.00	0.10	0.62	9.20	0.00	62.70	3.92	100.00	100
48.00	0.10	0.60	9.20	0.00	60.20	3.56	100.00	100
49.00	0.20	0.44	11.30	0.00	48.00	1.89	97.90	100
50.00	0.20	0.39	11.30	0.00	48.20	1.80	92.20	100
51.00	0.20	0.47	11.30	0.00	49.20	1.82	92.90	100
52.00	2.50	0.60	25.30	0.00	52.30	1.73	78.00	100
53.00	2.50	0.58	25.30	0.00	54.20	1.59	72.20	100
54.00	2.50	0.45	25.30	0.00	55.20	1.62	74.10	100
55.00	3.10	0.45	13.30	0.00	56.40	2.58	89.30	100
56.00	3.10	0.15	13.30	0.00	51.20	1.76	83.90	100
57.00	3.10	0.15	13.30	0.00	49.10	1.76	84.30	100
58.00	0.32	0.18	0.00	1.40	-	1.23	-	100
59.00	0.32	0.18	0.00	1.40	-	1.37	-	100
60.00	0.32	0.18	0.00	1.40	-	1.41	-	100

Table 23. Normalized data (Page 16 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
61.00	0.32	0.18	0.00	1.40	-	1.41	-	100
62.00	0.32	0.18	0.00	1.40	-	1.56	-	100
63.00	0.32	0.37	0.00	1.70	-	1.19	-	100
64.00	0.32	0.28	0.00	1.70	-	1.32	-	100
65.00	0.32	0.18	0.00	1.70	-	1.42	-	100
66.00	0.32	0.09	0.00	1.70	-	1.51	-	100
67.00	0.32	0.00	0.00	1.70	-	1.61	-	100
68.00	0.32	0.00	0.00	1.10	-	1.59	-	100
69.00	0.32	0.00	0.00	1.40	-	1.68	-	100
70.00	0.32	0.00	0.00	1.80	-	1.77	-	100
71.00	0.32	0.00	0.00	2.70	-	1.90	-	100
72.00	0.32	0.00	0.00	4.70	-	2.04	-	100
73.00	37.02	0.20	8.72	0.00	-	1.33	-	100
74.00	37.91	0.20	8.34	0.00	-	1.33	-	100
75.00	0.00	0.20	7.20	0.00	-	2.12	-	100
76.00	3.69	0.20	18.70	0.00	-	2.02	-	100
77.00	3.02	0.20	17.28	0.00	-	2.06	-	100
78.00	3.90	0.20	16.90	0.00	-	1.94	-	100
79.00	-	0.20	16.83	0.00	-	1.93	-	100
80.00	-	0.20	10.00	0.00	-	1.94	-	100

Table 23. Normalized data (Page 17 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
81.00	11.37	0.20	5.90	0.00	-	2.03	-	100
82.00	9.16	0.20	5.00	0.00	-	1.88	-	100
83.00	10.05	0.20	6.00	0.00	-	1.90	-	100
84.00	-	0.20	5.50	0.00	-	2.19	-	100
85.00	1.10	0.32	8.50	0.00	-	1.01	-	100
86.00	1.10	0.36	8.50	0.00	-	1.05	-	100
87.00	1.10	0.38	8.50	0.00	-	1.07	-	100
88.00	1.10	0.41	8.50	0.00	-	1.04	-	100
89.00	1.10	0.43	8.50	0.00	-	1.00	-	100
90.00	0.89	0.19	7.00	0.00	-	1.10	87.00	100
91.00	0.89	0.27	7.00	0.00	-	1.10	88.00	100
92.00	0.89	0.27	7.00	0.23	-	1.17	90.00	100
93.00	0.89	0.36	7.00	0.32	-	1.23	94.00	100
94.00	0.89	0.34	7.00	0.56	-	1.36	95.00	100
95.00	0.89	0.33	7.00	0.57	-	1.44	96.00	100
96.00	-	0.21	9.00	0.00	-	2.34	-	100
97.00	-	0.17	9.00	0.00	-	2.21	-	100
98.00	-	0.26	9.00	0.00	-	2.50	-	100
99.00	-	0.29	9.00	0.00	-	2.67	-	100
100.00	-	0.20	9.00	0.00	-	2.46	-	100

Table 23. Normalized data (Page 18 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
101.00	-	0.28	9.00	0.90	-	2.75	-	100
102.00	-	0.26	9.00	0.63	-	2.75	-	100
103.00	-	0.20	9.00	0.63	-	2.47	-	100
104.00	-	0.25	9.00	0.00	-	2.46	-	100
105.00	-	0.25	9.00	0.00	-	2.50	-	100
106.00	-	0.30	7.00	0.00	-	3.46	-	100
107.00	-	0.26	7.00	0.00	-	3.41	-	100
108.00	-	0.22	7.00	0.00	-	3.18	-	100
109.00	-	0.31	7.00	0.00	-	3.60	-	100
110.00	-	0.26	7.00	0.00	-	3.32	-	100
111.00	-	0.16	7.00	0.00	-	3.10	-	100
112.00	-	0.13	7.00	0.62	-	3.19	-	100
113.00	-	0.28	7.00	0.00	-	3.46	-	100
114.00	-	0.25	11.30	0.00	-	3.21	-	100
115.00	-	0.31	11.30	0.00	-	3.31	-	100
116.00	-	0.21	11.30	0.00	-	3.03	-	100
117.00	8.50	0.41	8.90	0.00	59.46	2.02	81.00	100
118.00	8.50	0.73	8.90	0.00	63.16	2.64	86.90	100
119.00	8.50	0.59	8.90	0.00	60.55	2.47	83.50	100
120.00	8.50	0.67	8.90	0.00	62.46	2.51	84.70	100

Table 23. Normalized data (Page 19 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
121.00	0.20	0.38	8.00	1.13	-	1.28	-	100
122.00	0.20	0.38	8.00	1.45	-	1.31	-	100
123.00	0.20	0.35	8.00	0.97	-	1.31	-	100
124.00	0.20	0.35	8.00	1.25	-	1.35	-	100
125.00	2.90	0.38	8.00	0.93	-	1.30	-	100
126.00	2.90	0.38	8.00	1.22	-	1.34	-	100
127.00	2.90	0.39	8.00	0.90	-	1.31	-	100
128.00	2.90	0.39	8.00	1.04	-	1.36	-	100
129.00	0.20	0.33	8.00	0.98	-	1.30	-	100
130.00	0.20	0.33	8.00	1.16	-	1.31	-	100
131.00	0.20	0.40	8.00	1.15	-	1.25	-	100
132.00	2.90	0.34	8.00	0.91	-	1.35	-	100
133.00	2.90	0.34	8.00	1.14	-	1.40	-	100
134.00	2.90	0.43	8.00	1.07	-	1.28	-	100
135.00	5.60	0.38	12.00	0.81	-	1.31	-	100
136.00	5.60	0.38	12.00	0.81	-	1.33	-	100
137.00	5.60	0.38	12.00	0.81	-	1.19	-	100
138.00	5.60	0.38	12.00	0.81	-	1.20	-	100
139.00	2.13	0.37	17.50	0.00	60.47	2.64	97.00	100
140.00	2.08	0.36	6.10	0.00	60.42	2.63	91.60	100

Table 23. Normalized data (Page 20 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
141.00	2.90	0.38	10.80	0.00	62.92	2.77	92.20	100
142.00	17.60	0.37	7.30	0.00	61.01	2.35	94.70	100
143.00	17.60	0.36	7.30	0.00	59.47	2.28	92.70	100
144.00	47.30	0.37	10.20	0.00	63.22	1.23	88.20	100
145.00	36.00	0.32	10.20	0.00	60.46	1.69	85.20	100
146.00	36.00	0.60	12.80	0.00	73.68	2.58	98.40	100
147.00	18.30	0.43	14.80	0.00	66.58	2.74	96.50	100
148.00	14.57	0.22	14.43	0.00	47.98	1.28	83.60	100
149.00	14.57	0.25	14.43	0.00	48.19	1.45	93.30	100
150.00	14.57	0.27	14.43	0.00	49.12	1.52	94.80	100
151.00	14.57	0.30	14.43	0.00	50.18	1.64	96.50	100
152.00	14.57	0.21	14.43	0.00	47.32	1.22	81.70	100
153.00	14.57	0.22	14.43	0.00	47.79	1.30	84.10	100
154.00	14.57	0.24	14.43	0.00	48.66	1.37	84.90	100
155.00	14.57	0.25	14.43	0.00	49.55	1.42	85.50	100
156.00	14.57	0.27	14.43	0.00	50.57	1.50	88.20	100
157.00	14.57	0.23	14.43	0.00	48.72	1.30	82.70	100
158.00	14.57	0.24	14.43	0.00	50.35	1.37	87.30	100
159.00	14.57	0.26	14.43	0.00	51.01	1.44	91.90	100
160.00	14.57	0.27	14.43	0.00	50.97	1.52	96.90	100

Table 23. Normalized data (Page 21 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
161.00	14.57	0.28	14.43	0.00	54.45	1.55	98.90	100
162.00	-	0.63	22.00	0.00	68.00	3.59	97.70	100
163.00	-	0.46	9.70	0.00	68.10	3.51	102.00	100
164.00	-	0.42	10.50	0.00	67.10	2.97	92.90	100
165.00	-	0.47	10.00	0.00	68.40	3.37	99.20	100
166.00	-	0.35	8.80	0.00	61.80	2.80	98.60	100
167.00	-	0.39	9.20	0.00	65.20	2.91	93.00	100
168.00	-	0.30	11.70	0.00	59.50	2.55	91.90	100
169.00	-	0.31	11.30	0.00	62.50	2.43	85.80	100
170.00	-	0.34	15.00	0.00	64.80	2.78	91.10	100
171.00	-	0.35	12.60	0.00	64.60	2.75	87.60	100
172.00	-	0.29	14.70	0.00	62.60	2.39	81.90	100
173.00	-	0.20	10.10	0.00	53.90	1.90	81.60	100
174.00	-	0.25	6.60	0.00	56.30	2.19	89.80	100
175.00	-	0.29	6.70	0.00	55.40	2.38	94.90	100
176.00	-	0.45	4.20	0.00	64.60	3.13	98.30	100
177.00	8.79	0.30	6.10	0.00	-	1.02	75.00	100
178.00	8.79	0.30	6.10	0.00	-	0.95	78.10	100
179.00	8.79	0.30	6.10	0.00	-	0.88	75.00	100
180.00	8.79	0.30	6.10	0.00	-	0.92	77.50	100

Table 23. Normalized data (Page 22 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
181.00	2.39	0.30	8.20	0.00	-	0.94	78.00	100
182.00	2.39	0.30	8.20	0.00	-	1.02	83.00	100
183.00	2.39	0.30	8.20	0.00	-	0.90	78.00	100
184.00	2.39	0.30	8.20	0.00	-	0.98	84.00	100
185.00	9.63	0.30	8.10	0.00	-	0.94	80.10	100
186.00	9.63	0.30	8.10	0.00	-	1.07	88.50	100
187.00	9.63	0.30	8.10	0.00	-	0.79	67.20	100
188.00	9.63	0.30	8.10	0.00	-	0.86	72.30	100
189.00	2.50	0.17	5.00	0.00	47.71	1.09	57.50	100
190.00	2.50	0.19	5.00	0.00	50.02	1.26	67.20	100
191.00	2.50	0.20	5.00	0.00	50.22	1.37	70.70	100
192.00	1.20	0.36	7.00	0.00	51.40	2.00	-	100
193.00	1.20	0.29	7.00	0.00	51.10	2.01	-	100
194.00	1.20	0.35	7.00	0.00	52.90	2.26	-	100
195.00	1.20	0.39	7.00	0.00	52.00	2.03	-	100
196.00	1.20	0.30	7.00	0.00	52.20	2.05	-	100
197.00	1.20	0.36	7.00	0.00	51.70	2.24	-	100
198.00	1.20	0.45	7.00	0.00	53.20	2.19	-	100
199.00	1.20	0.32	7.00	0.00	53.50	2.14	-	100
200.00	1.20	0.37	7.00	0.00	51.90	2.26	-	100

Table 23. Normalized data (Page 23 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
201.00	2.50	0.18	11.40	0.00	47.60	1.88	-	100
202.00	2.50	0.26	11.40	0.00	49.70	2.22	-	100
203.00	2.50	0.47	11.40	0.00	52.50	2.32	-	100
204.00	1.20	0.29	7.00	0.00	51.10	1.92	-	100
205.00	1.20	0.30	7.00	0.00	52.20	1.95	-	100
206.00	1.20	0.32	7.00	0.00	53.50	2.04	-	100
207.00	3.23	0.38	3.50	0.00	-	3.55	-	100
208.00	2.60	0.34	10.60	0.00	-	3.61	-	100
209.00	10.50	0.34	13.40	0.00	-	3.26	-	100
210.00	1.95	0.19	7.50	0.00	-	2.75	-	100
211.00	1.95	0.24	7.50	0.00	-	2.83	-	100
212.00	1.95	0.30	7.50	0.00	-	3.09	-	100
213.00	1.95	0.37	7.50	0.00	-	3.19	-	100
214.00	1.95	0.00	7.50	0.41	-	2.27	-	100
215.00	1.95	0.00	7.50	0.63	-	1.87	-	100
216.00	1.95	0.00	7.50	0.97	-	2.08	-	100
217.00	1.95	0.00	7.50	1.10	-	2.13	-	100
218.00	0.27	0.19	7.50	0.00	-	2.34	-	100
219.00	0.27	0.24	10.20	0.00	-	2.43	-	100
220.00	0.27	0.30	7.50	0.00	-	2.69	-	100

Table 23. Normalized data (Page 24 of 60)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
221.00	0.27	0.37	7.50	0.00	-	2.88	-	100
222.00	0.27	0.00	7.50	0.41	-	1.85	-	100
223.00	0.27	0.00	7.50	0.63	-	1.73	-	100
224.00	0.27	0.00	7.50	0.97	-	2.16	-	100
225.00	2.50	0.22	5.00	0.00	44.80	1.64	80.50	100
226.00	2.50	0.26	5.00	0.00	43.25	1.98	94.50	100
	187.00							

Table 23. Normalized data (Page 25 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
1.00	82.29	17.16	100	8.00	0.22	2.70	700
2.00	82.29	17.16	100	8.00	0.22	2.70	750
3.00	82.29	17.16	100	8.00	0.22	2.70	800
4.00	82.29	17.16	100	8.00	0.22	2.70	850
5.00	82.29	17.16	100	8.00	0.22	2.70	900
6.00	82.29	17.16	100	8.00	0.19	1.56	800
7.00	82.29	17.16	100	8.00	0.21	1.56	800
8.00	82.29	17.16	100	8.00	0.23	1.56	800
9.00	82.29	17.16	100	8.00	0.25	1.56	800
10.00	82.29	17.16	100	8.00	0.27	1.56	800
11.00	82.29	17.16	100	8.00	0.22	0.00	800
12.00	82.29	17.16	100	8.00	0.22	1.35	800
13.00	82.29	17.16	100	8.00	0.22	2.02	800
14.00	82.29	17.16	100	8.00	0.22	2.70	800
15.00	82.29	17.16	100	8.00	0.22	4.04	800
16.00	82.29	17.16	100	8.00	0.23	1.56	800
17.00	82.29	17.16	100	8.00	0.23	1.56	800
18.00	82.29	17.16	100	8.00	0.23	1.56	800
19.00	82.29	17.16	100	8.00	0.23	1.56	800
20.00	82.00	17.20	100	7.00	0.32	0.00	805

Table 23. Normalized data (Page 26 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
21.00	82.00	17.20	100	7.00	0.23	0.00	800
22.00	82.00	17.20	100	7.00	0.48	0.00	805
23.00	82.00	17.20	100	7.00	0.13	0.00	800
24.00	82.00	17.20	100	7.00	0.22	0.00	815
25.00	82.00	17.20	100	7.00	0.24	0.00	800
26.00	82.00	17.20	100	7.00	0.66	0.00	800
27.00	80.60	18.51	100	9.80	0.27	0.00	809
28.00	80.60	18.51	100	9.80	0.24	0.00	806
29.00	80.60	18.51	100	9.80	0.19	0.00	781
30.00	80.60	18.51	100	9.80	0.32	0.00	725
31.00	73.40	20.60	100	13.63	0.30	0.00	674
32.00	73.40	20.60	100	13.63	0.30	0.00	747
33.00	73.40	20.60	100	13.63	0.40	0.00	684
34.00	73.40	20.60	100	13.63	0.40	0.00	772
35.00	63.80	16.87	100	8.10	0.30	0.00	754
36.00	63.80	16.87	100	8.10	0.30	0.00	874
37.00	63.80	16.87	100	8.10	0.40	0.00	764
38.00	63.80	16.87	100	8.10	0.40	0.00	887
39.00	77.76	20.44	100	9.30	0.30	0.00	746
40.00	77.76	20.44	100	9.30	0.30	0.00	857

Table 23. Normalized data (Page 27 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
41.00	77.76	20.44	100	9.30	0.40	0.00	756
42.00	77.76	20.44	100	9.30	0.40	0.00	877
43.00	52.25	42.65	100	15.60	0.40	0.00	791
44.00	52.25	42.65	100	15.60	0.68	0.00	802
45.00	52.25	42.65	100	15.60	0.84	0.00	858
46.00	84.40	15.50	100	9.20	0.36	0.00	782
47.00	84.40	15.50	100	9.20	0.62	0.00	824
48.00	84.40	15.50	100	9.20	0.60	0.00	792
49.00	79.80	20.00	100	11.30	0.44	0.00	815
50.00	79.80	20.00	100	11.30	0.39	0.00	815
51.00	79.80	20.00	100	11.30	0.47	0.00	790
52.00	83.20	14.30	100	25.30	0.60	0.00	815
53.00	83.20	14.30	100	25.30	0.58	0.00	790
54.00	83.20	14.30	100	25.30	0.45	0.00	790
55.00	83.60	13.30	100	13.30	0.45	0.00	790
56.00	83.60	13.30	100	13.30	0.15	0.00	790
57.00	83.60	13.30	100	13.30	0.15	0.00	815
58.00	80.15	19.52	100	0.00	0.18	1.40	750
59.00	80.15	19.52	100	0.00	0.18	1.40	800
60.00	80.15	19.52	100	0.00	0.18	1.40	850

Table 23. Normalized data (Page 28 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
61.00	80.15	19.52	100	0.00	0.18	1.40	900
62.00	80.15	19.52	100	0.00	0.18	1.40	950
63.00	80.15	19.52	100	0.00	0.37	1.70	850
64.00	80.15	19.52	100	0.00	0.28	1.70	850
65.00	80.15	19.52	100	0.00	0.18	1.70	850
66.00	80.15	19.52	100	0.00	0.09	1.70	850
67.00	80.15	19.52	100	0.00	0.00	1.70	850
68.00	80.15	19.52	100	0.00	0.00	1.10	800
69.00	80.15	19.52	100	0.00	0.00	1.40	800
70.00	80.15	19.52	100	0.00	0.00	1.80	800
71.00	80.15	19.52	100	0.00	0.00	2.70	800
72.00	80.15	19.52	100	0.00	0.00	4.70	800
73.00	34.77	28.21	100	8.72	0.20	0.00	850
74.00	32.86	29.23	100	8.34	0.20	0.00	850
75.00	0.00	0.00	0	7.20	0.20	0.00	850
76.00	51.29	45.02	100	18.70	0.20	0.00	850
77.00	57.89	39.09	100	17.28	0.20	0.00	850
78.00	55.98	40.11	100	16.90	0.20	0.00	850
79.00	-	-	0	16.83	0.20	0.00	850
80.00	-	-	0	10.00	0.20	0.00	850

Table 23. Normalized data (Page 29 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
81.00	27.74	60.89	100	5.90	0.20	0.00	850
82.00	39.04	51.79	100	5.00	0.20	0.00	850
83.00	37.14	52.81	100	6.00	0.20	0.00	850
84.00	-	-	0	5.50	0.20	0.00	850
85.00	88.90	10.00	100	8.50	0.32	0.00	746
86.00	88.90	10.00	100	8.50	0.36	0.00	776
87.00	88.90	10.00	100	8.50	0.38	0.00	790
88.00	88.90	10.00	100	8.50	0.41	0.00	810
89.00	88.90	10.00	100	8.50	0.43	0.00	836
90.00	80.60	18.51	100	7.00	0.19	0.00	780
91.00	80.60	18.51	100	7.00	0.27	0.00	805
92.00	80.60	18.51	100	7.00	0.27	0.23	786
93.00	80.60	18.51	100	7.00	0.36	0.32	808
94.00	80.60	18.51	100	7.00	0.34	0.56	795
95.00	80.60	18.51	100	7.00	0.33	0.57	806
96.00	-	-	0	9.00	0.21	0.00	856
97.00	-	-	0	9.00	0.17	0.00	843
98.00	-	-	0	9.00	0.26	0.00	911
99.00	-	-	0	9.00	0.29	0.00	860
100.00	-	-	0	9.00	0.20	0.00	850

Table 23. Normalized data (Page 30 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
101.00	-	-	0	9.00	0.28	0.90	832
102.00	-	-	0	9.00	0.26	0.63	808
103.00	-	-	0	9.00	0.20	0.63	847
104.00	-	-	0	9.00	0.25	0.00	864
105.00	-	-	0	9.00	0.25	0.00	855
106.00	-	-	0	7.00	0.30	0.00	925
107.00	-	-	0	7.00	0.26	0.00	907
108.00	-	-	0	7.00	0.22	0.00	885
109.00	-	-	0	7.00	0.31	0.00	945
110.00	-	-	0	7.00	0.26	0.00	865
111.00	-	-	0	7.00	0.16	0.00	854
112.00	-	-	0	7.00	0.13	0.62	768
113.00	-	-	0	7.00	0.28	0.00	923
114.00	-	-	0	11.30	0.25	0.00	881
115.00	-	-	0	11.30	0.31	0.00	907
116.00	-	-	0	11.30	0.21	0.00	856
117.00	74.40	17.10	100	8.90	0.41	0.00	780
118.00	74.40	17.10	100	8.90	0.73	0.00	800
119.00	74.40	17.10	100	8.90	0.59	0.00	800
120.00	74.40	17.10	100	8.90	0.67	0.00	800

Table 23. Normalized data (Page 31 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
121.00	82.20	17.60	100	8.00	0.38	1.13	775
122.00	82.20	17.60	100	8.00	0.38	1.45	770
123.00	82.20	17.60	100	8.00	0.35	0.97	815
124.00	82.20	17.60	100	8.00	0.35	1.25	810
125.00	75.50	21.60	100	8.00	0.38	0.93	780
126.00	75.50	21.60	100	8.00	0.38	1.22	780
127.00	75.50	21.60	100	8.00	0.39	0.90	820
128.00	75.50	21.60	100	8.00	0.39	1.04	820
129.00	82.20	17.60	100	8.00	0.33	0.98	800
130.00	82.20	17.60	100	8.00	0.33	1.16	800
131.00	82.20	17.60	100	8.00	0.40	1.15	800
132.00	75.50	21.60	100	8.00	0.34	0.91	800
133.00	75.50	21.60	100	8.00	0.34	1.14	800
134.00	75.50	21.60	100	8.00	0.43	1.07	815
135.00	76.70	17.70	100	12.00	0.38	0.81	790
136.00	76.70	17.70	100	12.00	0.38	0.81	800
137.00	76.70	17.70	100	12.00	0.38	0.81	810
138.00	76.70	17.70	100	12.00	0.38	0.81	820
139.00	81.72	16.15	100	17.50	0.37	0.00	827
140.00	78.34	19.58	100	6.10	0.36	0.00	847

Table 23. Normalized data (Page 32 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
141.00	76.90	20.20	100	10.80	0.38	0.00	843
142.00	65.10	17.30	100	7.30	0.37	0.00	803
143.00	65.10	17.30	100	7.30	0.36	0.00	815
144.00	43.32	9.38	100	10.20	0.37	0.00	829
145.00	55.17	8.83	100	10.20	0.32	0.00	827
146.00	55.17	8.83	100	12.80	0.60	0.00	827
147.00	60.87	20.83	100	14.80	0.43	0.00	805
148.00	70.36	15.07	100	14.43	0.22	0.00	668
149.00	70.36	15.07	100	14.43	0.25	0.00	711
150.00	70.36	15.07	100	14.43	0.27	0.00	745
151.00	70.36	15.07	100	14.43	0.30	0.00	770
152.00	70.36	15.07	100	14.43	0.21	0.00	672
153.00	70.36	15.07	100	14.43	0.22	0.00	700
154.00	70.36	15.07	100	14.43	0.24	0.00	714
155.00	70.36	15.07	100	14.43	0.25	0.00	750
156.00	70.36	15.07	100	14.43	0.27	0.00	764
157.00	70.36	15.07	100	14.43	0.23	0.00	704
158.00	70.36	15.07	100	14.43	0.24	0.00	710
159.00	70.36	15.07	100	14.43	0.26	0.00	736
160.00	70.36	15.07	100	14.43	0.27	0.00	748

Table 23. Normalized data (Page 33 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
161.00	70.36	15.07	100	14.43	0.28	0.00	766
162.00	-	-	0	22.00	0.63	0.00	740
163.00	-	-	0	9.70	0.46	0.00	718
164.00	-	-	0	10.50	0.42	0.00	766
165.00	-	-	0	10.00	0.47	0.00	815
166.00	-	-	0	8.80	0.35	0.00	772
167.00	-	-	0	9.20	0.39	0.00	787
168.00	-	-	0	11.70	0.30	0.00	718
169.00	-	-	0	11.30	0.31	0.00	730
170.00	-	-	0	15.00	0.34	0.00	752
171.00	-	-	0	12.60	0.35	0.00	815
172.00	-	-	0	14.70	0.29	0.00	789
173.00	-	-	0	10.10	0.20	0.00	701
174.00	-	-	0	6.60	0.25	0.00	728
175.00	-	-	0	6.70	0.29	0.00	739
176.00	-	-	0	4.20	0.45	0.00	805
177.00	78.95	12.26	100	6.10	0.30	0.00	750
178.00	78.95	12.26	100	6.10	0.30	0.00	800
179.00	78.95	12.26	100	6.10	0.30	0.00	750
180.00	78.95	12.26	100	6.10	0.30	0.00	800

Table 23. Normalized data (Page 34 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
181.00	83.31	14.30	100	8.20	0.30	0.00	750
182.00	83.31	14.30	100	8.20	0.30	0.00	800
183.00	83.31	14.30	100	8.20	0.30	0.00	750
184.00	83.31	14.30	100	8.20	0.30	0.00	800
185.00	78.95	11.42	100	8.10	0.30	0.00	750
186.00	78.95	11.42	100	8.10	0.30	0.00	800
187.00	78.95	11.42	100	8.10	0.30	0.00	750
188.00	78.95	11.42	100	8.10	0.30	0.00	800
189.00	83.00	14.50	100	5.00	0.17	0.00	686
190.00	83.00	14.50	100	5.00	0.19	0.00	690
191.00	83.00	14.50	100	5.00	0.20	0.00	720
192.00	67.70	31.10	100	7.00	0.36	0.00	750
193.00	67.70	31.10	100	7.00	0.29	0.00	750
194.00	67.70	31.10	100	7.00	0.35	0.00	750
195.00	67.70	31.10	100	7.00	0.39	0.00	800
196.00	67.70	31.10	100	7.00	0.30	0.00	800
197.00	67.70	31.10	100	7.00	0.36	0.00	800
198.00	67.70	31.10	100	7.00	0.45	0.00	850
199.00	67.70	31.10	100	7.00	0.32	0.00	850
200.00	67.70	31.10	100	7.00	0.37	0.00	850

Table 23. Normalized data (Page 35 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
201.00	72.50	25.00	100	11.40	0.18	0.00	750
202.00	72.50	25.00	100	11.40	0.26	0.00	800
203.00	72.50	25.00	100	11.40	0.47	0.00	850
204.00	67.70	31.10	100	7.00	0.29	0.00	750
205.00	67.70	31.10	100	7.00	0.30	0.00	800
206.00	67.70	31.10	100	7.00	0.32	0.00	850
207.00	77.61	19.16	100	3.50	0.38	0.00	861
208.00	79.19	18.21	100	10.60	0.34	0.00	855
209.00	67.84	21.66	100	13.40	0.34	0.00	822
210.00	74.23	23.82	100	7.50	0.19	0.00	770
211.00	74.23	23.82	100	7.50	0.24	0.00	770
212.00	74.23	23.82	100	7.50	0.30	0.00	770
213.00	74.23	23.82	100	7.50	0.37	0.00	770
214.00	74.23	23.82	100	7.50	0.00	0.41	770
215.00	74.23	23.82	100	7.50	0.00	0.63	770
216.00	74.23	23.82	100	7.50	0.00	0.97	770
217.00	74.23	23.82	100	7.50	0.00	1.10	770
218.00	83.17	16.56	100	7.50	0.19	0.00	770
219.00	83.17	16.56	100	10.20	0.24	0.00	770
220.00	83.17	16.56	100	7.50	0.30	0.00	770

Table 23. Normalized data (Page 36 of 60)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)
221.00	83.17	16.56	100	7.50	0.37	0.00	770
222.00	83.17	16.56	100	7.50	0.00	0.41	770
223.00	83.17	16.56	100	7.50	0.00	0.63	770
224.00	83.17	16.56	100	7.50	0.00	0.97	770
225.00	83.00	14.50	100	5.00	0.22	0.00	919
226.00	83.00	14.50	100	5.00	0.26	0.00	974
	187.00	187.00		226.00	226.00	226.00	226.00

Table 23. Normalized data (Page 37 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
1.00	21.99	9.27	43.50	20.28	4.95	0.00
2.00	28.75	8.41	39.72	19.22	3.90	0.00
3.00	32.66	7.68	38.01	18.26	3.38	0.00
4.00	36.74	7.41	34.48	18.50	2.87	0.00
5.00	39.84	6.29	33.52	18.99	1.36	0.00
6.00	32.67	8.07	37.96	17.63	3.67	0.00
7.00	31.52	8.28	39.62	16.91	3.66	0.00
8.00	31.91	7.82	40.29	16.56	3.42	0.00
9.00	31.50	7.57	39.79	17.74	3.41	0.00
10.00	32.10	6.85	38.18	19.47	3.39	0.00
11.00	33.50	6.35	42.91	15.65	1.59	0.00
12.00	30.48	8.48	40.97	16.65	3.42	0.00
13.00	31.39	8.36	39.68	17.17	3.40	0.00
14.00	32.64	7.63	38.03	18.32	3.38	0.00
15.00	32.47	7.50	37.67	18.97	3.38	0.00
16.00	30.75	7.83	41.11	16.93	3.37	0.00
17.00	31.73	7.66	40.32	16.92	3.37	0.00
18.00	31.68	7.63	37.64	19.70	3.36	0.00
19.00	31.58	6.68	37.17	21.22	3.34	0.00
20.00	22.66	6.16	39.90	31.28	0.00	0.00

Table 23. Normalized data (Page 38 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
21.00	23.47	8.37	41.84	26.33	0.00	0.00
22.00	20.70	2.11	32.28	44.91	0.00	0.00
23.00	26.29	8.27	49.63	15.81	0.00	0.00
24.00	28.77	8.61	39.53	23.09	0.00	0.00
25.00	24.68	6.96	42.19	26.16	0.00	0.00
26.00	15.62	3.03	35.80	45.56	0.00	0.00
27.00	30.02	8.28	28.57	33.13	0.00	0.00
28.00	31.14	8.90	29.01	30.95	0.00	0.00
29.00	30.39	9.76	29.65	30.20	0.00	0.00
30.00	31.47	8.90	24.43	35.20	0.00	0.00
31.00	10.57	2.67	11.55	11.60	0.00	0.00
32.00	10.63	3.83	12.59	11.07	0.00	0.00
33.00	10.48	2.50	10.93	11.71	0.00	0.00
34.00	10.59	3.61	12.70	11.85	0.00	0.00
35.00	8.60	3.21	15.44	11.69	0.00	0.00
36.00	8.62	3.24	19.56	10.22	0.00	0.00
37.00	7.43	2.01	14.79	10.67	0.00	0.00
38.00	7.77	2.78	16.14	10.69	0.00	0.00
39.00	10.00	2.45	17.41	11.98	0.00	0.00
40.00	10.14	3.21	17.55	10.78	0.00	0.00

Table 23. Normalized data (Page 39 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
41.00	8.13	2.14	16.34	11.17	0.00	0.00
42.00	9.36	2.62	17.75	11.89	0.00	0.00
43.00	10.46	2.11	13.48	10.97	0.10	0.00
44.00	7.05	1.41	8.96	11.88	0.10	0.00
45.00	4.53	0.60	6.14	12.69	0.10	0.00
46.00	5.22	3.01	8.02	9.73	0.50	0.00
47.00	3.82	2.01	5.63	13.77	0.30	0.00
48.00	5.33	2.11	7.34	13.37	0.70	0.00
49.00	8.53	4.79	19.35	17.38	0.00	0.00
50.00	8.74	4.58	18.73	17.79	0.00	0.00
51.00	7.90	4.57	18.61	17.78	0.00	0.00
52.00	12.86	1.66	11.83	19.40	0.00	0.00
53.00	8.23	2.60	12.03	18.42	0.00	0.00
54.00	7.96	2.55	11.25	19.64	0.00	0.00
55.00	5.10	1.10	8.30	16.50	0.00	0.00
56.00	10.86	2.51	12.21	20.98	0.00	0.00
57.00	13.22	2.39	14.67	18.63	0.00	0.00
58.00	31.40	8.40	20.60	35.30	4.30	0.00
59.00	37.20	7.60	17.40	35.30	2.50	0.00
60.00	38.76	8.29	18.58	32.17	2.20	0.00

Table 23. Normalized data (Page 40 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
61.00	38.30	8.00	19.30	32.70	1.70	0.00
62.00	44.40	7.04	18.83	28.15	1.59	0.00
63.00	27.57	7.69	15.48	46.85	2.40	0.00
64.00	34.27	7.59	15.58	40.36	2.20	0.00
65.00	39.10	7.40	16.80	34.70	2.00	0.00
66.00	43.24	7.41	18.52	28.53	2.30	0.00
67.00	46.80	8.30	20.20	22.30	2.40	0.00
68.00	46.00	7.90	19.90	24.10	2.10	0.00
69.00	48.45	6.89	18.48	24.58	1.60	0.00
70.00	51.25	6.09	15.88	25.07	1.70	0.00
71.00	54.40	5.00	11.80	27.40	1.40	0.00
72.00	57.34	4.10	7.29	30.17	1.10	0.00
73.00	35.89	8.47	19.76	35.88	0.00	0.00
74.00	36.17	9.28	19.80	34.75	0.00	0.00
75.00	30.80	12.66	16.04	40.50	0.00	0.00
76.00	39.53	5.30	27.58	27.58	0.00	0.00
77.00	43.37	5.06	23.96	27.61	0.00	0.00
78.00	39.67	5.87	27.93	26.52	0.00	0.00
79.00	32.65	10.28	19.91	37.16	0.00	0.00
80.00	31.25	12.28	21.78	34.69	0.00	0.00

Table 23. Normalized data (Page 41 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
81.00	35.41	7.84	20.12	36.63	0.00	0.00
82.00	33.61	8.50	23.28	34.61	0.00	0.00
83.00	34.15	8.48	22.43	34.94	0.00	0.00
84.00	31.95	11.41	18.47	38.17	0.00	0.00
85.00	14.78	8.65	34.96	41.61	0.00	0.00
86.00	18.21	8.66	31.79	41.34	0.00	0.00
87.00	19.65	8.64	32.15	39.56	0.00	0.00
88.00	17.73	9.09	30.84	42.35	0.00	0.00
89.00	14.04	8.82	31.23	45.91	0.00	0.00
90.00	25.58	11.63	35.27	27.52	0.00	0.00
91.00	25.05	10.34	34.99	29.62	0.00	0.00
92.00	28.06	9.42	30.06	32.46	0.00	0.00
93.00	28.08	9.42	32.36	30.14	0.00	0.00
94.00	34.38	9.62	27.60	28.39	0.00	0.00
95.00	37.41	9.75	28.09	24.75	0.00	0.00
96.00	14.20	6.00	18.80	15.40	0.00	0.00
97.00	15.20	6.60	20.40	15.70	0.00	0.00
98.00	14.50	4.40	19.90	13.30	0.00	0.00
99.00	13.30	4.20	16.50	14.60	0.00	0.00
100.00	15.20	4.60	19.00	14.70	0.00	0.00

Table 23. Normalized data (Page 42 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
101.00	17.50	4.20	11.90	18.20	0.00	0.00
102.00	17.00	4.10	11.60	18.60	0.00	0.00
103.00	21.10	5.10	15.80	17.20	0.00	0.00
104.00	13.80	4.90	19.10	14.30	0.00	0.00
105.00	13.30	5.00	17.20	15.50	0.00	0.00
106.00	10.30	3.40	16.60	13.70	0.00	0.00
107.00	11.80	3.70	17.20	13.30	0.00	0.00
108.00	13.10	4.30	18.90	13.50	0.00	0.00
109.00	9.40	3.20	15.90	13.30	0.00	0.00
110.00	11.20	3.90	17.80	13.50	0.00	0.00
111.00	12.80	5.30	17.90	14.40	0.00	0.00
112.00	14.50	3.80	17.00	15.80	0.00	0.00
113.00	13.10	3.30	19.40	11.00	0.00	0.00
114.00	12.30	3.20	16.30	14.70	0.00	0.00
115.00	9.90	3.30	15.10	14.70	0.00	0.00
116.00	14.80	4.30	17.60	14.30	0.00	0.00
117.00	5.47	1.62	8.72	21.99	1.93	0.00
118.00	7.38	0.81	6.97	20.11	0.91	0.00
119.00	9.39	0.81	8.48	19.18	1.01	0.00
120.00	7.66	0.91	7.56	19.85	1.11	0.00

Table 23. Normalized data (Page 43 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
121.00	22.56	8.72	25.64	43.08	0.00	0.00
122.00	24.10	8.21	25.13	42.56	0.00	0.00
123.00	23.96	8.85	25.52	41.67	0.00	0.00
124.00	26.56	8.33	23.96	41.15	0.00	0.00
125.00	25.91	7.77	21.76	44.56	0.00	0.00
126.00	27.98	7.25	20.73	44.04	0.00	0.00
127.00	26.42	7.77	22.80	43.01	0.00	0.00
128.00	29.02	7.77	20.21	43.01	0.00	0.00
129.00	23.44	9.38	28.13	39.06	0.00	0.00
130.00	23.96	9.38	28.13	38.54	0.00	0.00
131.00	20.62	8.76	24.23	46.39	0.00	0.00
132.00	28.65	8.33	23.44	39.58	0.00	0.00
133.00	31.09	7.77	21.76	39.38	0.00	0.00
134.00	24.74	7.22	22.68	45.36	0.00	0.00
135.00	18.00	8.00	31.00	32.00	0.00	0.00
136.00	19.00	8.00	30.50	31.50	0.00	0.00
137.00	21.98	9.89	31.87	36.26	0.00	0.00
138.00	22.65	8.84	31.49	37.02	0.00	0.00
139.00	7.30	3.34	9.53	17.33	1.22	0.00
140.00	7.14	3.29	11.50	15.96	0.89	0.00

Table 23. Normalized data (Page 44 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
141.00	6.53	2.82	9.81	15.90	0.92	0.00
142.00	7.46	2.64	10.14	16.00	1.23	0.00
143.00	8.66	2.86	10.90	15.55	1.11	0.00
144.00	6.17	2.85	9.38	16.35	1.08	0.00
145.00	8.15	3.24	10.70	15.27	1.18	0.00
146.00	1.83	1.26	5.42	16.32	0.43	0.00
147.00	5.45	1.74	8.42	16.26	0.63	0.00
148.00	5.47	5.55	20.97	15.49	3.24	0.00
149.00	6.30	5.44	19.81	15.70	3.50	0.00
150.00	6.92	5.11	18.32	16.16	3.29	0.00
151.00	7.75	4.72	16.91	16.48	2.75	0.00
152.00	5.47	5.89	21.02	15.83	3.53	0.00
153.00	6.36	5.65	19.93	15.97	3.25	0.00
154.00	6.65	5.50	18.74	16.39	2.74	0.00
155.00	7.40	5.06	17.63	16.54	2.77	0.00
156.00	7.59	4.88	16.83	16.36	2.87	0.00
157.00	7.09	5.04	17.28	16.81	4.07	0.00
158.00	7.09	5.04	17.28	16.80	4.08	0.00
159.00	7.09	5.04	17.28	16.80	4.06	0.00
160.00	7.09	5.04	17.28	16.80	4.07	0.00

Table 23. Normalized data (Page 45 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
161.00	7.09	5.04	17.28	16.81	4.05	0.00
162.00	5.60	1.40	6.90	18.10	0.00	0.00
163.00	3.13	1.92	11.09	15.22	0.00	0.00
164.00	3.20	1.90	10.70	17.10	0.00	0.00
165.00	3.00	1.90	9.60	17.10	0.00	0.00
166.00	4.00	2.90	14.71	16.52	0.00	0.00
167.00	3.80	2.70	12.60	15.70	0.00	0.00
168.00	5.50	3.40	16.60	15.00	0.00	0.00
169.00	3.90	2.80	15.12	15.62	0.00	0.00
170.00	3.50	2.80	13.39	15.58	0.00	0.00
171.00	4.10	2.50	12.30	16.50	0.00	0.00
172.00	4.20	3.00	14.59	15.68	0.00	0.00
173.00	5.40	4.60	21.40	14.70	0.00	0.00
174.00	5.11	4.10	19.92	14.51	0.00	0.00
175.00	7.29	3.20	17.88	16.28	0.00	0.00
176.00	5.90	1.20	10.00	18.30	0.00	0.00
177.00	22.80	8.10	25.20	32.10	0.00	0.00
178.00	27.12	6.96	35.01	30.91	0.00	0.00
179.00	24.02	8.50	34.22	33.26	0.00	0.00
180.00	25.40	8.47	37.20	28.93	0.00	0.00

Table 23. Normalized data (Page 46 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
181.00	24.48	8.82	30.60	36.10	0.00	0.00
182.00	25.84	7.90	35.56	30.70	0.00	0.00
183.00	21.67	9.34	31.51	37.48	0.00	0.00
184.00	22.40	8.70	33.70	35.20	0.00	0.00
185.00	26.83	8.61	29.43	35.14	0.00	0.00
186.00	28.66	8.22	33.17	29.96	0.00	0.00
187.00	27.10	9.40	29.30	34.20	0.00	0.00
188.00	27.68	8.19	35.09	29.04	0.00	0.00
189.00	5.67	9.94	17.33	15.37	1.34	0.00
190.00	5.34	8.35	14.40	18.90	2.10	0.00
191.00	5.76	8.30	17.70	16.01	1.19	0.00
192.00	8.30	3.70	18.20	13.10	1.40	0.00
193.00	8.50	4.10	15.70	15.10	1.40	0.00
194.00	10.30	3.20	12.40	16.70	0.60	0.00
195.00	8.20	3.60	16.50	14.70	1.20	0.00
196.00	7.40	4.20	14.90	15.20	1.30	0.00
197.00	11.50	2.80	14.00	15.90	0.50	0.00
198.00	8.28	2.97	14.42	16.36	0.41	0.00
199.00	7.70	3.70	12.90	16.50	1.10	0.00
200.00	10.98	3.09	13.67	15.97	0.40	0.00

Table 23. Normalized data (Page 47 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
201.00	10.03	6.14	17.81	15.56	1.74	0.00
202.00	11.64	4.23	14.07	16.72	0.74	0.00
203.00	10.22	3.69	12.64	17.70	0.42	0.00
204.00	8.86	4.28	16.37	15.75	1.46	0.00
205.00	7.77	4.41	15.65	15.97	1.37	0.00
206.00	8.07	3.88	13.52	17.30	1.15	0.00
207.00	5.38	1.72	8.31	16.04	0.00	0.00
208.00	5.85	2.88	10.57	13.94	0.00	0.00
209.00	9.02	2.34	8.00	16.02	0.00	0.00
210.00	8.50	5.50	17.00	14.00	0.00	0.00
211.00	6.70	4.50	16.00	15.00	0.00	0.00
212.00	4.50	4.00	12.50	16.00	0.00	0.00
213.00	3.40	3.50	11.00	17.00	0.00	0.00
214.00	32.00	8.20	23.00	13.00	0.00	0.00
215.00	46.15	8.24	24.18	21.43	0.00	0.00
216.00	51.58	6.32	18.42	23.68	0.00	0.00
217.00	52.88	6.28	17.80	23.04	0.00	0.00
218.00	5.70	6.80	19.00	14.00	0.00	0.00
219.00	4.20	5.80	17.10	15.50	0.00	0.00
220.00	3.60	4.40	14.20	16.00	0.00	0.00

Table 23. Normalized data (Page 48 of 60)

	H2	CH4	CO	CO2	C2H4	C2H4
221.00	3.20	3.60	12.50	16.20	0.00	0.00
222.00	20.00	11.60	29.00	9.90	0.00	0.00
223.00	36.00	11.00	27.00	16.00	0.00	0.00
224.00	56.78	5.89	14.72	22.61	0.00	0.00
225.00	13.21	7.30	16.76	15.76	1.20	0.00
226.00	16.27	6.73	16.65	15.55	1.00	0.00
	226.00	226.00	226.00	226.00	226.00	226.00

Table 23. Normalized data (Page 49 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
1.00	0.00	0.00	0.00	0.00	0.00	100.00
2.00	0.00	0.00	0.00	0.00	0.00	100.00
3.00	0.00	0.00	0.00	0.00	0.00	100.00
4.00	0.00	0.00	0.00	0.00	0.00	100.00
5.00	0.00	0.00	0.00	0.00	0.00	100.00
6.00	0.00	0.00	0.00	0.00	0.00	100.00
7.00	0.00	0.00	0.00	0.00	0.00	100.00
8.00	0.00	0.00	0.00	0.00	0.00	100.00
9.00	0.00	0.00	0.00	0.00	0.00	100.00
10.00	0.00	0.00	0.00	0.00	0.00	100.00
11.00	0.00	0.00	0.00	0.00	0.00	100.00
12.00	0.00	0.00	0.00	0.00	0.00	100.00
13.00	0.00	0.00	0.00	0.00	0.00	100.00
14.00	0.00	0.00	0.00	0.00	0.00	100.00
15.00	0.00	0.00	0.00	0.00	0.00	100.00
16.00	0.00	0.00	0.00	0.00	0.00	100.00
17.00	0.00	0.00	0.00	0.00	0.00	100.00
18.00	0.00	0.00	0.00	0.00	0.00	100.00
19.00	0.00	0.00	0.00	0.00	0.00	100.00
20.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 23. Normalized data (Page 50 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
21.00	0.00	0.00	0.00	0.00	0.00	100.00
22.00	0.00	0.00	0.00	0.00	0.00	100.00
23.00	0.00	0.00	0.00	0.00	0.00	100.00
24.00	0.00	0.00	0.00	0.00	0.00	100.00
25.00	0.00	0.00	0.00	0.00	0.00	100.00
26.00	0.00	0.00	0.00	0.00	0.00	100.00
27.00	0.00	0.00	0.00	0.00	0.00	100.00
28.00	0.00	0.00	0.00	0.00	0.00	100.00
29.00	0.00	0.00	0.00	0.00	0.00	100.00
30.00	0.00	0.00	0.00	0.00	0.00	100.00
31.00	0.00	0.00	63.61	0.00	0.00	100.00
32.00	0.00	0.00	61.89	0.00	0.00	100.00
33.00	0.00	0.00	64.37	0.00	0.00	100.00
34.00	0.00	0.00	61.25	0.00	0.00	100.00
35.00	0.00	0.00	61.06	0.00	0.00	100.00
36.00	0.00	0.00	58.36	0.00	0.00	100.00
37.00	0.00	0.00	65.11	0.00	0.00	100.00
38.00	0.00	0.00	62.62	0.00	0.00	100.00
39.00	0.00	0.00	58.15	0.00	0.00	100.00
40.00	0.00	0.00	58.31	0.00	0.00	100.00

Table 23. Normalized data (Page 51 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
41.00	0.00	0.00	62.21	0.00	0.00	100.00
42.00	0.00	0.00	58.37	0.00	0.00	100.00
43.00	0.00	0.00	57.65	5.23	0.00	100.00
44.00	0.00	0.00	65.26	5.34	0.00	100.00
45.00	0.00	0.00	69.49	6.45	0.00	100.00
46.00	0.00	0.00	65.30	8.22	0.00	100.00
47.00	0.00	0.00	63.02	11.46	0.00	100.00
48.00	0.00	0.00	60.50	10.65	0.00	100.00
49.00	0.00	0.00	49.95	0.00	0.00	100.00
50.00	0.00	0.00	50.16	0.00	0.00	100.00
51.00	0.00	0.00	51.14	0.00	0.00	100.00
52.00	0.00	0.00	54.25	0.00	0.00	100.00
53.00	0.00	0.00	58.72	0.00	0.00	100.00
54.00	0.00	0.00	58.60	0.00	0.00	100.00
55.00	0.00	0.00	56.40	0.00	0.00	87.40
56.00	0.00	0.00	53.44	0.00	0.00	100.00
57.00	0.00	0.00	51.09	0.00	0.00	100.00
58.00	0.00	0.00	0.00	0.00	0.00	100.00
59.00	0.00	0.00	0.00	0.00	0.00	100.00
60.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 23. Normalized data (Page 52 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
61.00	0.00	0.00	0.00	0.00	0.00	100.00
62.00	0.00	0.00	0.00	0.00	0.00	100.00
63.00	0.00	0.00	0.00	0.00	0.00	100.00
64.00	0.00	0.00	0.00	0.00	0.00	100.00
65.00	0.00	0.00	0.00	0.00	0.00	100.00
66.00	0.00	0.00	0.00	0.00	0.00	100.00
67.00	0.00	0.00	0.00	0.00	0.00	100.00
68.00	0.00	0.00	0.00	0.00	0.00	100.00
69.00	0.00	0.00	0.00	0.00	0.00	100.00
70.00	0.00	0.00	0.00	0.00	0.00	100.00
71.00	0.00	0.00	0.00	0.00	0.00	100.00
72.00	0.00	0.00	0.00	0.00	0.00	100.00
73.00	0.00	0.00	0.00	0.00	0.00	100.00
74.00	0.00	0.00	0.00	0.00	0.00	100.00
75.00	0.00	0.00	0.00	0.00	0.00	100.00
76.00	0.00	0.00	0.00	0.00	0.00	100.00
77.00	0.00	0.00	0.00	0.00	0.00	100.00
78.00	0.00	0.00	0.00	0.00	0.00	100.00
79.00	0.00	0.00	0.00	0.00	0.00	100.00
80.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 23. Normalized data (Page 53 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
81.00	0.00	0.00	0.00	0.00	0.00	100.00
82.00	0.00	0.00	0.00	0.00	0.00	100.00
83.00	0.00	0.00	0.00	0.00	0.00	100.00
84.00	0.00	0.00	0.00	0.00	0.00	100.00
85.00	0.00	0.00	0.00	0.00	0.00	100.00
86.00	0.00	0.00	0.00	0.00	0.00	100.00
87.00	0.00	0.00	0.00	0.00	0.00	100.00
88.00	0.00	0.00	0.00	0.00	0.00	100.00
89.00	0.00	0.00	0.00	0.00	0.00	100.00
90.00	0.00	0.00	0.00	0.00	0.00	100.00
91.00	0.00	0.00	0.00	0.00	0.00	100.00
92.00	0.00	0.00	0.00	0.00	0.00	100.00
93.00	0.00	0.00	0.00	0.00	0.00	100.00
94.00	0.00	0.00	0.00	0.00	0.00	100.00
95.00	0.00	0.00	0.00	0.00	0.00	100.00
96.00	0.00	0.00	45.60	0.00	0.00	100.00
97.00	0.00	0.00	42.10	0.00	0.00	100.00
98.00	0.00	0.00	47.90	0.00	0.00	100.00
99.00	0.00	0.00	51.40	0.00	0.00	100.00
100.00	0.00	0.00	46.50	0.00	0.00	100.00

Table 23. Normalized data (Page 54 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
101.00	0.00	0.00	48.20	0.00	0.00	100.00
102.00	0.00	0.00	48.70	0.00	0.00	100.00
103.00	0.00	0.00	40.80	0.00	0.00	100.00
104.00	0.00	0.00	47.90	0.00	0.00	100.00
105.00	0.00	0.00	49.00	0.00	0.00	100.00
106.00	0.00	0.00	56.00	0.00	0.00	100.00
107.00	0.00	0.00	54.00	0.00	0.00	100.00
108.00	0.00	0.00	50.20	0.00	0.00	100.00
109.00	0.00	0.00	58.20	0.00	0.00	100.00
110.00	0.00	0.00	53.60	0.00	0.00	100.00
111.00	0.00	0.00	49.60	0.00	0.00	100.00
112.00	0.00	0.00	48.90	0.00	0.00	100.00
113.00	0.00	0.00	53.20	0.00	0.00	100.00
114.00	0.00	0.00	53.50	0.00	0.00	100.00
115.00	0.00	0.00	57.00	0.00	0.00	100.00
116.00	0.00	0.00	49.00	0.00	0.00	100.00
117.00	0.00	0.00	60.27	0.00	0.00	100.00
118.00	0.00	0.00	63.82	0.00	0.00	100.00
119.00	0.00	0.00	61.13	0.00	0.00	100.00
120.00	0.00	0.00	62.93	0.00	0.00	100.00

Table 23. Normalized data (Page 55 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
121.00	0.00	0.00	-	0.00	0.00	100.00
122.00	0.00	0.00	-	0.00	0.00	100.00
123.00	0.00	0.00	-	0.00	0.00	100.00
124.00	0.00	0.00	-	0.00	0.00	100.00
125.00	0.00	0.00	-	0.00	0.00	100.00
126.00	0.00	0.00	-	0.00	0.00	100.00
127.00	0.00	0.00	-	0.00	0.00	100.00
128.00	0.00	0.00	-	0.00	0.00	100.00
129.00	0.00	0.00	-	0.00	0.00	100.00
130.00	0.00	0.00	-	0.00	0.00	100.00
131.00	0.00	0.00	-	0.00	0.00	100.00
132.00	0.00	0.00	-	0.00	0.00	100.00
133.00	0.00	0.00	-	0.00	0.00	100.00
134.00	0.00	0.00	-	0.00	0.00	100.00
135.00	0.00	0.00	-	0.00	0.00	89.00
136.00	0.00	0.00	-	0.00	0.00	89.00
137.00	0.00	0.00	-	0.00	0.00	100.00
138.00	0.00	0.00	-	0.00	0.00	100.00
139.00	0.00	0.00	61.29	0.00	0.00	100.00
140.00	0.00	0.00	61.22	0.00	0.00	100.00

Table 23. Normalized data (Page 56 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
141.00	0.00	0.00	64.02	0.00	0.00	100.00
142.00	0.00	0.00	62.53	0.00	0.00	100.00
143.00	0.00	0.00	60.93	0.00	0.00	100.00
144.00	0.00	0.00	64.17	0.00	0.00	100.00
145.00	0.00	0.00	61.46	0.00	0.00	100.00
146.00	0.00	0.00	74.75	0.00	0.00	100.00
147.00	0.00	0.00	67.49	0.00	0.00	100.00
148.00	0.00	0.00	49.28	0.00	0.00	100.00
149.00	0.00	0.00	49.26	0.00	0.00	100.00
150.00	0.00	0.00	50.20	0.00	0.00	100.00
151.00	0.00	0.00	51.38	0.00	0.00	100.00
152.00	0.00	0.00	48.26	0.00	0.00	100.00
153.00	0.00	0.00	48.84	0.00	0.00	100.00
154.00	0.00	0.00	49.97	0.00	0.00	100.00
155.00	0.00	0.00	50.60	0.00	0.00	100.00
156.00	0.00	0.00	51.48	0.00	0.00	100.00
157.00	0.00	0.00	49.71	0.00	0.00	100.00
158.00	0.00	0.00	49.71	0.00	0.00	100.00
159.00	0.00	0.00	49.72	0.00	0.00	100.00
160.00	0.00	0.00	49.71	0.00	0.00	100.00

Table 23. Normalized data (Page 57 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
161.00	0.00	0.00	49.73	0.00	0.00	100.00
162.00	0.00	0.00	68.00	0.00	0.00	100.00
163.00	0.00	0.00	68.65	0.00	0.00	100.00
164.00	0.00	0.00	67.10	0.00	0.00	100.00
165.00	0.00	0.00	68.40	0.00	0.00	100.00
166.00	0.00	0.00	61.86	0.00	0.00	100.00
167.00	0.00	0.00	65.20	0.00	0.00	100.00
168.00	0.00	0.00	59.50	0.00	0.00	100.00
169.00	0.00	0.00	62.56	0.00	0.00	100.00
170.00	0.00	0.00	64.74	0.00	0.00	100.00
171.00	0.00	0.00	64.60	0.00	0.00	100.00
172.00	0.00	0.00	62.54	0.00	0.00	100.00
173.00	0.00	0.00	53.90	0.00	0.00	100.00
174.00	0.00	0.00	56.36	0.00	0.00	100.00
175.00	0.00	0.00	55.34	0.00	0.00	100.00
176.00	0.00	0.00	64.60	0.00	0.00	100.00
177.00	0.00	0.00	-	0.00	0.00	88.20
178.00	0.00	0.00	-	0.00	0.00	100.00
179.00	0.00	0.00	-	0.00	0.00	100.00
180.00	0.00	0.00	-	0.00	0.00	100.00

Table 23. Normalized data (Page 58 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
181.00	0.00	0.00	-	0.00	0.00	100.00
182.00	0.00	0.00	-	0.00	0.00	100.00
183.00	0.00	0.00	-	0.00	0.00	100.00
184.00	0.00	0.00	-	0.00	0.00	100.00
185.00	0.00	0.00	-	0.00	0.00	100.00
186.00	0.00	0.00	-	0.00	0.00	100.00
187.00	0.00	0.00	-	0.00	0.00	100.00
188.00	0.00	0.00	-	0.00	0.00	100.00
189.00	0.00	0.00	49.21	0.00	1.13	100.00
190.00	0.00	0.00	50.01	0.00	0.90	100.00
191.00	0.00	0.00	49.94	0.00	1.09	100.00
192.00	0.00	0.00	51.40	0.00	3.90	100.00
193.00	0.00	0.00	51.10	0.00	4.10	100.00
194.00	0.00	0.00	52.90	0.00	3.90	100.00
195.00	0.00	0.00	52.00	0.00	3.80	100.00
196.00	0.00	0.00	52.20	0.00	4.80	100.00
197.00	0.00	0.00	51.70	0.00	3.60	100.00
198.00	0.00	0.00	54.40	0.00	3.17	100.00
199.00	0.00	0.00	53.50	0.00	4.60	100.00
200.00	0.00	0.00	51.80	0.00	4.09	100.00

Table 23. Normalized data (Page 59 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
201.00	0.00	0.00	48.72	0.00	0.00	100.00
202.00	0.00	0.00	52.59	0.00	0.00	100.00
203.00	0.00	0.00	55.32	0.00	0.00	100.00
204.00	0.00	0.00	53.28	0.00	0.00	100.00
205.00	0.00	0.00	54.83	0.00	0.00	100.00
206.00	0.00	0.00	56.08	0.00	0.00	100.00
207.00	0.00	0.00	68.55	0.00	0.00	100.00
208.00	0.00	0.00	66.76	0.00	0.00	100.00
209.00	0.00	0.00	64.62	0.00	0.00	100.00
210.00	0.00	0.00	55.00	0.00	0.00	100.00
211.00	0.00	0.00	57.80	0.00	0.00	100.00
212.00	0.00	0.00	63.00	0.00	0.00	100.00
213.00	0.00	0.00	65.10	0.00	0.00	100.00
214.00	0.00	0.00	23.80	0.00	0.00	100.00
215.00	0.00	0.00	-	0.00	0.00	100.00
216.00	0.00	0.00	-	0.00	0.00	100.00
217.00	0.00	0.00	-	0.00	0.00	100.00
218.00	0.00	0.00	54.50	0.00	0.00	100.00
219.00	0.00	0.00	57.40	0.00	0.00	100.00
220.00	0.00	0.00	61.80	0.00	0.00	100.00

Table 23. Normalized data (Page 60 of 60)

	C2H2	NH3	N2	H2O	O2	SUM
221.00	0.00	0.00	64.50	0.00	0.00	100.00
222.00	0.00	0.00	29.50	0.00	0.00	100.00
223.00	0.00	0.00	-	0.00	0.00	90.00
224.00	0.00	0.00	-	0.00	0.00	100.00
225.00	0.00	0.00	44.68	0.00	1.10	100.00
226.00	0.00	0.00	43.11	0.00	0.70	100.00
	226.00	226.00	191.00	226.00	226.00	226.00

APPENDIX G
FINALIZED DATA

Table 24. Finalized data (Page 1 of 36)

		Feedstock	C	H	O	N	S
1.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
2.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
3.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
4.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
5.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
6.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
7.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
8.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
9.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
10.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
11.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
12.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
13.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
14.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
15.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
16.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
17.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
18.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
19.00	BFB1	Pine sawdust	50.54	7.08	41.11	0.15	0.57
20.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
21.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
22.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
23.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
24.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
25.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
26.00	BFB 2	Beech wood	47.90	6.31	44.85	0.14	0.01
27.00	BFB 3	Wood pellet	50.57	7.10	41.36	0.09	0.01
28.00	BFB 3	Wood pellet	50.57	7.10	41.36	0.09	0.01
29.00	BFB 3	Wood pellet	50.57	7.10	41.36	0.09	0.01
30.00	BFB 3	Wood pellet	50.57	7.10	41.36	0.09	0.01
31.00	BFB 4	Coir pitch	44.35	3.38	45.31	0.95	0.01
32.00	BFB 4	Coir pitch	44.35	3.38	45.31	0.95	0.01
33.00	BFB 4	Coir pitch	44.35	3.38	45.31	0.95	0.01
34.00	BFB 4	Coir pitch	44.35	3.38	45.31	0.95	0.01
35.00	BFB 4	Rice husk	31.40	4.11	43.31	1.75	0.10
36.00	BFB 4	Rice husk	31.40	4.11	43.31	1.75	0.10
37.00	BFB 4	Rice husk	31.40	4.11	43.31	1.75	0.10
38.00	BFB 4	Rice husk	31.40	4.11	43.31	1.75	0.10
39.00	BFB 4	Sawdust	50.62	6.05	41.39	0.12	0.02
40.00	BFB 4	Sawdust	50.62	6.05	41.39	0.12	0.02
41.00	BFB 4	Sawdust	50.62	6.05	41.39	0.12	0.02
42.00	BFB 4	Sawdust	50.62	6.05	41.39	0.12	0.02

Table 24. Finalized data (Page 2 of 36)

		Feedstock	C	H	O	N	S
43.00	BFB 5	Hambach	53.52	4.75	35.63	0.59	0.40
44.00	BFB 5	Hambach	53.52	4.75	35.63	0.59	0.40
45.00	BFB 5	Hambach	53.52	4.75	35.63	0.59	0.40
46.00	BFB 5	Labee	46.19	6.20	47.39	0.04	0.08
47.00	BFB 5	Labee	46.19	6.20	47.39	0.04	0.08
48.00	BFB 5	Labee	46.19	6.20	47.39	0.04	0.08
49.00	BFB 6	Forest residues	42.91	4.99	49.50	2.40	0.01
50.00	BFB 6	Forest residues	42.91	4.99	49.50	2.40	0.01
51.00	BFB 6	Forest residues	42.91	4.99	49.50	2.40	0.01
52.00	BFB 6	Coffe husks	39.10	5.46	47.87	5.07	0.01
53.00	BFB 6	Coffe husks	39.10	5.46	47.87	5.07	0.01
54.00	BFB 6	Coffe husks	39.10	5.46	47.87	5.07	0.01
55.00	BFB 6	Vine0pruning residues	40.02	5.33	49.03	2.52	0.01
56.00	BFB 6	Vine0pruning residues	40.02	5.33	49.03	2.52	0.01
57.00	BFB 6	Vine0pruning residues	40.02	5.33	49.03	2.52	0.01
58.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
59.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
60.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
61.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
62.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
63.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
64.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
65.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
66.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
67.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
68.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
69.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
70.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
71.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
72.00	BFB 7	Sawdust	47.86	6.02	45.35	0.15	0.05
73.00	BFB 8	80%PC 20%Pine	45.58	3.17	12.26	0.73	1.23
74.00	BFB 8	80%PC 20%Bagasse	45.49	3.14	11.29	0.96	1.21
75.00	BFB 8	90%PC 10%PE	78.59	5.66	12.33	1.26	2.16
76.00	BFB 8	100% German coal	65.39	4.91	24.75	0.77	0.48
77.00	BFB 8	80%GC 20%Pine	62.51	5.20	28.19	0.66	0.43
78.00	BFB 8	80%GC 20%Bagasse	62.80	5.17	26.70	1.02	0.41
79.00	BFB 8	90%GC 10%PE	69.68	6.02	23.13	0.72	0.45
80.00	BFB 8	80%Coal 20%PE	71.46	6.94	20.56	0.64	0.40
81.00	BFB 8	100% South African coal	70.19	4.17	11.96	1.77	0.53
82.00	BFB 8	80%AC 20%Pine	66.76	4.58	17.53	1.49	0.47
83.00	BFB 8	80%AC 20%Bagasse	66.91	4.55	16.21	1.82	0.46
84.00	BFB 8	90%AC 10%PE	79.85	5.66	12.15	1.80	0.54

Table 24. Finalized data (Page 3 of 36)

		Feedstock	C	H	O	N	S
85.00	BFB 9	Rubber wood chip	45.89	5.64	47.18	0.20	0.01
86.00	BFB 9	Rubber wood chip	45.89	5.64	47.18	0.20	0.01
87.00	BFB 9	Rubber wood chip	45.89	5.64	47.18	0.20	0.01
88.00	BFB 9	Rubber wood chip	45.89	5.64	47.18	0.20	0.01
89.00	BFB 9	Rubber wood chip	45.89	5.64	47.18	0.20	0.01
90.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
91.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
92.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
93.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
94.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
95.00	BFB 10	Wood pellets	50.31	5.87	42.93	0.01	0.01
96.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
97.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
98.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
99.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
100.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
101.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
102.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
103.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
104.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
105.00	BFB 11	RIL 1 (natural wood)	50.45	6.24	42.76	0.56	0.00
106.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
107.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
108.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
109.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
110.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
111.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
112.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
113.00	BFB 11	RIL 2 (wood waste)	62.53	7.95	28.76	0.76	0.00
114.00	BFB 11	RIL 3 (wood waste)	58.78	8.20	30.33	2.69	0.00
115.00	BFB 11	RIL 3 (wood waste)	58.78	8.20	30.33	2.69	0.00
116.00	BFB 11	RIL 3 (wood waste)	58.78	8.20	30.33	2.69	0.00
117.00	CFB 1	Olive oil waste	48.38	6.61	34.98	1.47	0.06
118.00	CFB 1	Olive oil waste	48.38	6.61	34.98	1.47	0.06
119.00	CFB 1	Olive oil waste	48.38	6.61	34.98	1.47	0.06
120.00	CFB 1	Olive oil waste	48.38	6.61	34.98	1.47	0.06
121.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
122.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
123.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
124.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
125.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
126.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01

Table 24. Finalized data (Page 4 of 36)

		Feedstock	C	H	O	N	S
127.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
128.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
129.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
130.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
131.00	CFB 2	Agrol	53.24	6.53	39.87	0.16	0.01
132.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
133.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
134.00	CFB 2	Willow	51.65	6.34	38.40	0.71	0.01
135.00	CFB 2	DDGS	49.75	6.75	32.20	5.70	0.01
136.00	CFB 2	DDGS	49.75	6.75	32.20	5.70	0.01
137.00	CFB 2	DDGS	49.75	6.75	32.20	5.70	0.01
138.00	CFB 2	DDGS	49.75	6.75	32.20	5.70	0.01
139.00	CFB 3	Willow	47.65	5.78	43.54	0.86	0.04
140.00	CFB 3	Demolition wood	49.96	5.44	42.03	0.45	0.04
141.00	CFB 3	Chip board material	48.85	5.70	41.08	1.40	0.08
142.00	CFB 3	Verge grass a	41.48	4.72	33.99	2.05	0.16
143.00	CFB 3	Verge grass b	41.48	4.72	33.99	2.05	0.16
144.00	CFB 3	Paper residue sludge	22.92	2.94	25.77	0.72	0.34
145.00	CFB 3	Sewage Sludge	33.46	4.66	19.68	4.53	1.67
146.00	CFB 3	Woody excess fraction of ODW	33.46	4.66	19.68	4.53	1.67
147.00	CFB 3	Park wood (bio0dried)	42.11	4.66	33.77	0.95	0.20
148.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
149.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
150.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
151.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
152.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
153.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
154.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
155.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
156.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
157.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
158.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
159.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
160.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
161.00	CFB 4	Rice husk	39.78	4.97	40.02	0.46	0.20
162.00	CFB 5	Cypress	51.96	6.24	40.68	0.65	0.46
163.00	CFB 5	Cypress	51.96	6.24	40.68	0.65	0.46
164.00	CFB 5	Spruce/pine/fir	50.80	6.30	41.93	0.62	0.35
165.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38

Table 24. Finalized data (Page 5 of 36)

		Feedstock	C	H	O	N	S
166.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38
167.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38
168.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38
169.00	CFB 5	Spruce pine fir	50.80	6.30	41.93	0.62	0.35
170.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38
171.00	CFB 5	Cedar/Hemlock	52.71	6.16	40.21	0.52	0.40
172.00	CFB 5	Hemlock	52.02	6.23	40.77	0.60	0.38
173.00	CFB 5	Pine bark/Spruce	50.88	7.51	40.84	0.26	0.52
174.00	CFB 5	Mix	50.27	8.07	41.37	0.22	0.07
175.00	CFB 5	Mix	50.27	8.07	41.37	0.22	0.07
176.00	CFB 5	Mix	50.27	8.07	41.37	0.22	0.07
177.00	CFB 6	Jatropha	47.50	7.10	30.81	5.69	0.11
178.00	CFB 6	Jatropha	47.50	7.10	30.81	5.69	0.11
179.00	CFB 6	Jatropha	47.50	7.10	30.81	5.69	0.11
180.00	CFB 6	Jatropha	47.50	7.10	30.81	5.69	0.11
181.00	CFB 6	Willow	48.63	6.63	39.32	3.00	0.02
182.00	CFB 6	Willow	48.63	6.63	39.32	3.00	0.02
183.00	CFB 6	Willow	48.63	6.63	39.32	3.00	0.02
184.00	CFB 6	Willow	48.63	6.63	39.32	3.00	0.02
185.00	CFB 6	Sunflower	46.10	6.01	37.39	0.87	0.01
186.00	CFB 6	Sunflower	46.10	6.01	37.39	0.87	0.01
187.00	CFB 6	Sunflower	46.10	6.01	37.39	0.87	0.01
188.00	CFB 6	Sunflower	46.10	6.01	37.39	0.87	0.01
189.00	CFB 7	Gum wood	46.02	5.89	45.34	0.20	0.05
190.00	CFB 7	Gum wood	46.02	5.89	45.34	0.20	0.05
191.00	CFB 7	Gum wood	46.02	5.89	45.34	0.20	0.05
192.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
193.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
194.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
195.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
196.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
197.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
198.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
199.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
200.00	BFB 13	Peach stone	40.51	5.63	47.82	4.84	0.01
201.00	BFB 15	Miscanthus	43.39	5.07	43.88	5.17	0.01
202.00	BFB 15	Miscanthus	43.39	5.07	43.88	5.17	0.01
203.00	BFB 15	Miscanthus	43.39	5.07	43.88	5.17	0.01
204.00	BFB 15	Peach stone	40.51	5.63	47.82	4.84	0.01
205.00	BFB 15	Peach stone	40.51	5.63	47.82	4.84	0.01
206.00	BFB 15	Peach stone	40.51	5.63	47.82	4.84	0.01
207.00	CFB 3	Park and public wood	49.60	5.70	41.17	0.25	0.05

Table 24. Finalized data (Page 6 of 36)

		Feedstock	C	H	O	N	S
208.00	CFB 3	Railroad ties	52.92	5.62	38.33	0.26	0.26
209.00	CFB 3	Cacao shells	45.99	5.07	32.57	2.94	2.94
210.00	BFB 12	Wallnut shell	53.83	5.39	38.34	0.39	0.10
211.00	BFB 12	Wallnut shell	53.83	5.39	38.34	0.39	0.10
212.00	BFB 12	Wallnut shell	53.83	5.39	38.34	0.39	0.10
213.00	BFB 12	Wallnut shell	53.83	5.39	38.34	0.39	0.10
214.00	BFB 12	Wallnut shell	53.83	5.39	38.34	0.39	0.10
215.00	BFB 12	Wallnut shell	53.83	5.39	38.34	0.39	0.10
216.00	BFB 12	Wallnut shell	53.83	5.39	38.34	0.39	0.10
217.00	BFB 12	Wallnut shell	53.83	5.39	38.34	0.39	0.10
218.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
219.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
220.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
221.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
222.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
223.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
224.00	BFB 12	Pistachio shell	49.92	5.89	43.38	0.42	0.11
225.00	CFB 7	Gum wood	46.02	5.89	45.34	0.20	0.05
226.00	CFB 7	Gum wood	46.02	5.89	45.34	0.20	0.05
			226.00	226.00	226.00	226.00	226.00

Table 24. Finalized data (Page 7 of 36)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
1.00	0.55	0.22	8.00	2.70	0.00	0.89	78.17	100
2.00	0.55	0.22	8.00	2.70	0.00	1.01	80.66	100
3.00	0.55	0.22	8.00	2.70	0.00	1.15	85.90	100
4.00	0.55	0.22	8.00	2.70	0.00	1.32	92.35	100
5.00	0.55	0.22	8.00	2.70	0.00	1.42	92.59	100
6.00	0.55	0.19	8.00	1.56	0.00	1.01	76.26	100
7.00	0.55	0.21	8.00	1.56	0.00	1.10	84.49	100
8.00	0.55	0.23	8.00	1.56	0.00	1.20	90.60	100
9.00	0.55	0.25	8.00	1.56	0.00	1.10	84.00	100
10.00	0.55	0.27	8.00	1.56	0.00	0.93	70.60	100
11.00	0.55	0.22	8.00	0.00	0.00	0.95	68.67	100
12.00	0.55	0.22	8.00	1.35	0.00	1.19	92.09	100
13.00	0.55	0.22	8.00	2.02	0.00	1.16	88.70	100
14.00	0.55	0.22	8.00	2.70	0.00	1.15	85.90	100
15.00	0.55	0.22	8.00	4.04	0.00	1.00	75.10	100
16.00	0.55	0.23	8.00	1.56	0.00	1.01	77.62	100
17.00	0.55	0.23	8.00	1.56	0.00	1.11	84.40	100
18.00	0.55	0.23	8.00	1.56	0.00	1.19	90.60	100
19.00	0.55	0.23	8.00	1.56	0.00	1.25	95.10	100
20.00	0.80	0.32	7.00	0.00	0.00	1.02	100.00	100
21.00	0.80	0.23	7.00	0.00	0.00	0.95	100.00	100
22.00	0.80	0.48	7.00	0.00	0.00	0.88	100.00	100
23.00	0.80	0.13	7.00	0.00	0.00	0.92	100.00	100
24.00	0.80	0.22	7.00	0.00	0.00	0.94	100.00	100
25.00	0.80	0.24	7.00	0.00	0.00	1.02	100.00	100
26.00	0.80	0.66	7.00	0.00	0.00	0.90	100.00	100
27.00	0.89	0.27	9.80	0.00	0.00	0.98	100.00	100
28.00	0.89	0.24	9.80	0.00	0.00	0.94	100.00	100
29.00	0.89	0.19	9.80	0.00	0.00	1.07	100.00	100
30.00	0.89	0.32	9.80	0.00	0.00	0.79	100.00	100
31.00	6.00	0.30	13.63	0.00	0.00	0.86	88.40	100
32.00	6.00	0.30	13.63	0.00	0.00	2.80	92.80	100
33.00	6.00	0.40	13.63	0.00	0.00	3.29	100.00	100
34.00	6.00	0.40	13.63	0.00	0.00	2.94	100.00	100
35.00	19.33	0.30	8.10	0.00	0.00	1.93	100.00	100
36.00	19.33	0.30	8.10	0.00	0.00	1.78	100.00	100
37.00	19.33	0.40	8.10	0.00	0.00	2.13	100.00	100
38.00	19.33	0.40	8.10	0.00	0.00	1.98	100.00	100
39.00	1.80	0.30	9.30	0.00	0.00	2.97	100.00	100
40.00	1.80	0.30	9.30	0.00	0.00	3.00	100.00	100
41.00	1.80	0.40	9.30	0.00	0.00	3.19	100.00	100
42.00	1.80	0.40	9.30	0.00	0.00	2.93	100.00	100

Table 24. Finalized data (Page 8 of 36)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
43.00	5.10	0.40	15.60	0.00	0.00	3.51	94.00	100
44.00	5.10	0.68	15.60	0.00	0.00	4.36	98.00	100
45.00	5.10	0.84	15.60	0.00	0.00	5.04	99.00	100
46.00	0.10	0.36	9.20	0.00	0.00	3.96	100.00	100
47.00	0.10	0.62	9.20	0.00	0.00	3.92	100.00	100
48.00	0.10	0.60	9.20	0.00	0.00	3.56	100.00	100
49.00	0.20	0.44	11.30	0.00	0.00	1.89	97.90	100
50.00	0.20	0.39	11.30	0.00	0.00	1.80	92.20	100
51.00	0.20	0.47	11.30	0.00	0.00	1.82	92.90	100
52.00	2.50	0.60	25.30	0.00	0.00	1.73	78.00	100
53.00	2.50	0.58	25.30	0.00	0.00	1.59	72.20	100
54.00	2.50	0.45	25.30	0.00	0.00	1.62	74.10	100
55.00	3.10	0.45	13.30	0.00	0.00	2.58	89.30	100
56.00	3.10	0.15	13.30	0.00	0.00	1.76	83.90	100
57.00	3.10	0.15	13.30	0.00	0.00	1.76	84.30	100
58.00	0.32	0.18	0.00	1.40	0.00	1.23	100.00	100
59.00	0.32	0.18	0.00	1.40	0.00	1.37	100.00	100
60.00	0.32	0.18	0.00	1.40	0.00	1.41	100.00	100
61.00	0.32	0.18	0.00	1.40	0.00	1.41	100.00	100
62.00	0.32	0.18	0.00	1.40	0.00	1.56	100.00	100
63.00	0.32	0.37	0.00	1.70	0.00	1.19	100.00	100
64.00	0.32	0.28	0.00	1.70	0.00	1.32	100.00	100
65.00	0.32	0.18	0.00	1.70	0.00	1.42	100.00	100
66.00	0.32	0.09	0.00	1.70	0.00	1.51	100.00	100
67.00	0.32	0.00	0.00	1.70	0.00	1.61	100.00	100
68.00	0.32	0.00	0.00	1.10	0.00	1.59	100.00	100
69.00	0.32	0.00	0.00	1.40	0.00	1.68	100.00	100
70.00	0.32	0.00	0.00	1.80	0.00	1.77	100.00	100
71.00	0.32	0.00	0.00	2.70	0.00	1.90	100.00	100
72.00	0.32	0.00	0.00	4.70	0.00	2.04	100.00	100
73.00	37.02	0.20	8.72	0.00	0.00	1.33	100.00	100
74.00	37.91	0.20	8.34	0.00	0.00	1.33	100.00	100
75.00	0.00	0.20	7.20	0.00	0.00	2.12	100.00	100
76.00	3.69	0.20	18.70	0.00	0.00	2.02	100.00	100
77.00	3.02	0.20	17.28	0.00	0.00	2.06	100.00	100
78.00	3.90	0.20	16.90	0.00	0.00	1.94	100.00	100
79.00	0.00	0.20	16.83	0.00	0.00	1.93	100.00	100
80.00	0.00	0.20	10.00	0.00	0.00	1.94	100.00	100
81.00	11.37	0.20	5.90	0.00	0.00	2.03	100.00	100
82.00	9.16	0.20	5.00	0.00	0.00	1.88	100.00	100
83.00	10.05	0.20	6.00	0.00	0.00	1.90	100.00	100
84.00	0.00	0.20	5.50	0.00	0.00	2.19	100.00	100

Table 24. Finalized data (Page 9 of 36)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
85.00	1.10	0.32	8.50	0.00	0.00	1.01	100.00	100
86.00	1.10	0.36	8.50	0.00	0.00	1.05	100.00	100
87.00	1.10	0.38	8.50	0.00	0.00	1.07	100.00	100
88.00	1.10	0.41	8.50	0.00	0.00	1.04	100.00	100
89.00	1.10	0.43	8.50	0.00	0.00	1.00	100.00	100
90.00	0.89	0.19	7.00	0.00	0.00	1.10	87.00	100
91.00	0.89	0.27	7.00	0.00	0.00	1.10	88.00	100
92.00	0.89	0.27	7.00	0.23	0.00	1.17	90.00	100
93.00	0.89	0.36	7.00	0.32	0.00	1.23	94.00	100
94.00	0.89	0.34	7.00	0.56	0.00	1.36	95.00	100
95.00	0.89	0.33	7.00	0.57	0.00	1.44	96.00	100
96.00	0.00	0.21	9.00	0.00	0.00	2.34	100.00	100
97.00	0.00	0.17	9.00	0.00	0.00	2.21	100.00	100
98.00	0.00	0.26	9.00	0.00	0.00	2.50	100.00	100
99.00	0.00	0.29	9.00	0.00	0.00	2.67	100.00	100
100.00	0.00	0.20	9.00	0.00	0.00	2.46	100.00	100
101.00	0.00	0.28	9.00	0.90	0.00	2.75	100.00	100
102.00	0.00	0.26	9.00	0.63	0.00	2.75	100.00	100
103.00	0.00	0.20	9.00	0.63	0.00	2.47	100.00	100
104.00	0.00	0.25	9.00	0.00	0.00	2.46	100.00	100
105.00	0.00	0.25	9.00	0.00	0.00	2.50	100.00	100
106.00	0.00	0.30	7.00	0.00	0.00	3.46	100.00	100
107.00	0.00	0.26	7.00	0.00	0.00	3.41	100.00	100
108.00	0.00	0.22	7.00	0.00	0.00	3.18	100.00	100
109.00	0.00	0.31	7.00	0.00	0.00	3.60	100.00	100
110.00	0.00	0.26	7.00	0.00	0.00	3.32	100.00	100
111.00	0.00	0.16	7.00	0.00	0.00	3.10	100.00	100
112.00	0.00	0.13	7.00	0.62	0.00	3.19	100.00	100
113.00	0.00	0.28	7.00	0.00	0.00	3.46	100.00	100
114.00	0.00	0.25	11.30	0.00	0.00	3.21	100.00	100
115.00	0.00	0.31	11.30	0.00	0.00	3.31	100.00	100
116.00	0.00	0.21	11.30	0.00	0.00	3.03	100.00	100
117.00	8.50	0.41	8.90	0.00	0.00	2.02	81.00	100
118.00	8.50	0.73	8.90	0.00	0.00	2.64	86.90	100
119.00	8.50	0.59	8.90	0.00	0.00	2.47	83.50	100
120.00	8.50	0.67	8.90	0.00	0.00	2.51	84.70	100
121.00	0.20	0.38	8.00	1.13	0.00	1.28	100.00	100
122.00	0.20	0.38	8.00	1.45	0.00	1.31	100.00	100
123.00	0.20	0.35	8.00	0.97	0.00	1.31	100.00	100
124.00	0.20	0.35	8.00	1.25	0.00	1.35	100.00	100
125.00	2.90	0.38	8.00	0.93	0.00	1.30	100.00	100
126.00	2.90	0.38	8.00	1.22	0.00	1.34	100.00	100

Table 24. Finalized data (Page 10 of 36)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
127.00	2.90	0.39	8.00	0.90	0.00	1.31	100.00	100
128.00	2.90	0.39	8.00	1.04	0.00	1.36	100.00	100
129.00	0.20	0.33	8.00	0.98	0.00	1.30	100.00	100
130.00	0.20	0.33	8.00	1.16	0.00	1.31	100.00	100
131.00	0.20	0.40	8.00	1.15	0.00	1.25	100.00	100
132.00	2.90	0.34	8.00	0.91	0.00	1.35	100.00	100
133.00	2.90	0.34	8.00	1.14	0.00	1.40	100.00	100
134.00	2.90	0.43	8.00	1.07	0.00	1.28	100.00	100
135.00	5.60	0.38	12.00	0.81	0.00	1.31	100.00	100
136.00	5.60	0.38	12.00	0.81	0.00	1.33	100.00	100
137.00	5.60	0.38	12.00	0.81	0.00	1.19	100.00	100
138.00	5.60	0.38	12.00	0.81	0.00	1.20	100.00	100
139.00	2.13	0.37	17.50	0.00	0.00	2.64	97.00	100
140.00	2.08	0.36	6.10	0.00	0.00	2.63	91.60	100
141.00	2.90	0.38	10.80	0.00	0.00	2.77	92.20	100
142.00	17.60	0.37	7.30	0.00	0.00	2.35	94.70	100
143.00	17.60	0.36	7.30	0.00	0.00	2.28	92.70	100
144.00	47.30	0.37	10.20	0.00	0.00	1.23	88.20	100
145.00	36.00	0.32	10.20	0.00	0.00	1.69	85.20	100
146.00	36.00	0.60	12.80	0.00	0.00	2.58	98.40	100
147.00	18.30	0.43	14.80	0.00	0.00	2.74	96.50	100
148.00	14.57	0.22	14.43	0.00	0.00	1.28	83.60	100
149.00	14.57	0.25	14.43	0.00	0.00	1.45	93.30	100
150.00	14.57	0.27	14.43	0.00	0.00	1.52	94.80	100
151.00	14.57	0.30	14.43	0.00	0.00	1.64	96.50	100
152.00	14.57	0.21	14.43	0.00	0.00	1.22	81.70	100
153.00	14.57	0.22	14.43	0.00	0.00	1.30	84.10	100
154.00	14.57	0.24	14.43	0.00	0.00	1.37	84.90	100
155.00	14.57	0.25	14.43	0.00	0.00	1.42	85.50	100
156.00	14.57	0.27	14.43	0.00	0.00	1.50	88.20	100
157.00	14.57	0.23	14.43	0.00	0.00	1.30	82.70	100
158.00	14.57	0.24	14.43	0.00	0.00	1.37	87.30	100
159.00	14.57	0.26	14.43	0.00	0.00	1.44	91.90	100
160.00	14.57	0.27	14.43	0.00	0.00	1.52	96.90	100
161.00	14.57	0.28	14.43	0.00	0.00	1.55	98.90	100
162.00	0.00	0.63	22.00	0.00	0.00	3.59	97.70	100
163.00	0.00	0.46	9.70	0.00	0.00	3.51	102.00	100
164.00	0.00	0.42	10.50	0.00	0.00	2.97	92.90	100
165.00	0.00	0.47	10.00	0.00	0.00	3.37	99.20	100
166.00	0.00	0.35	8.80	0.00	0.00	2.80	98.60	100
167.00	0.00	0.39	9.20	0.00	0.00	2.91	93.00	100
168.00	0.00	0.30	11.70	0.00	0.00	2.55	91.90	100

Table 24. Finalized data (Page 11 of 36)

	Ash	ER	MC	SBR	N2	Yield	CCE %	Total
169.00	0.00	0.31	11.30	0.00	0.00	2.43	85.80	100
170.00	0.00	0.34	15.00	0.00	0.00	2.78	91.10	100
171.00	0.00	0.35	12.60	0.00	0.00	2.75	87.60	100
172.00	0.00	0.29	14.70	0.00	0.00	2.39	81.90	100
173.00	0.00	0.20	10.10	0.00	0.00	1.90	81.60	100
174.00	0.00	0.25	6.60	0.00	0.00	2.19	89.80	100
175.00	0.00	0.29	6.70	0.00	0.00	2.38	94.90	100
176.00	0.00	0.45	4.20	0.00	0.00	3.13	98.30	100
177.00	8.79	0.30	6.10	0.00	0.00	1.02	75.00	100
178.00	8.79	0.30	6.10	0.00	0.00	0.95	78.10	100
179.00	8.79	0.30	6.10	0.00	0.00	0.88	75.00	100
180.00	8.79	0.30	6.10	0.00	0.00	0.92	77.50	100
181.00	2.39	0.30	8.20	0.00	0.00	0.94	78.00	100
182.00	2.39	0.30	8.20	0.00	0.00	1.02	83.00	100
183.00	2.39	0.30	8.20	0.00	0.00	0.90	78.00	100
184.00	2.39	0.30	8.20	0.00	0.00	0.98	84.00	100
185.00	9.63	0.30	8.10	0.00	0.00	0.94	80.10	100
186.00	9.63	0.30	8.10	0.00	0.00	1.07	88.50	100
187.00	9.63	0.30	8.10	0.00	0.00	0.79	67.20	100
188.00	9.63	0.30	8.10	0.00	0.00	0.86	72.30	100
189.00	2.50	0.17	5.00	0.00	0.00	1.09	57.50	100
190.00	2.50	0.19	5.00	0.00	0.00	1.26	67.20	100
191.00	2.50	0.20	5.00	0.00	0.00	1.37	70.70	100
192.00	1.20	0.36	7.00	0.00	0.00	2.00	100.00	100
193.00	1.20	0.29	7.00	0.00	0.00	2.01	100.00	100
194.00	1.20	0.35	7.00	0.00	0.00	2.26	100.00	100
195.00	1.20	0.39	7.00	0.00	0.00	2.03	100.00	100
196.00	1.20	0.30	7.00	0.00	0.00	2.05	100.00	100
197.00	1.20	0.36	7.00	0.00	0.00	2.24	100.00	100
198.00	1.20	0.45	7.00	0.00	0.00	2.19	100.00	100
199.00	1.20	0.32	7.00	0.00	0.00	2.14	100.00	100
200.00	1.20	0.37	7.00	0.00	0.00	2.26	100.00	100
201.00	2.50	0.18	11.40	0.00	0.00	1.88	100.00	100
202.00	2.50	0.26	11.40	0.00	0.00	2.22	100.00	100
203.00	2.50	0.47	11.40	0.00	0.00	2.32	100.00	100
204.00	1.20	0.29	7.00	0.00	0.00	1.92	100.00	100
205.00	1.20	0.30	7.00	0.00	0.00	1.95	100.00	100
206.00	1.20	0.32	7.00	0.00	0.00	2.04	100.00	100
207.00	3.23	0.38	3.50	0.00	0.00	3.55	100.00	100
208.00	2.60	0.34	10.60	0.00	0.00	3.61	100.00	100
209.00	10.50	0.34	13.40	0.00	0.00	3.26	100.00	100
210.00	1.95	0.19	7.50	0.00	0.00	2.75	100.00	100

Table 24. Finalized data (Page 13 of 36)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)	H2	CH4	CO
1.00	82.29	17.16	100	8.00	0.22	2.70	700	21.99	9.27	43.50
2.00	82.29	17.16	100	8.00	0.22	2.70	750	28.75	8.41	39.72
3.00	82.29	17.16	100	8.00	0.22	2.70	800	32.66	7.68	38.01
4.00	82.29	17.16	100	8.00	0.22	2.70	850	36.74	7.41	34.48
5.00	82.29	17.16	100	8.00	0.22	2.70	900	39.84	6.29	33.52
6.00	82.29	17.16	100	8.00	0.19	1.56	800	32.67	8.07	37.96
7.00	82.29	17.16	100	8.00	0.21	1.56	800	31.52	8.28	39.62
8.00	82.29	17.16	100	8.00	0.23	1.56	800	31.91	7.82	40.29
9.00	82.29	17.16	100	8.00	0.25	1.56	800	31.50	7.57	39.79
10.00	82.29	17.16	100	8.00	0.27	1.56	800	32.10	6.85	38.18
11.00	82.29	17.16	100	8.00	0.22	0.00	800	33.50	6.35	42.91
12.00	82.29	17.16	100	8.00	0.22	1.35	800	30.48	8.48	40.97
13.00	82.29	17.16	100	8.00	0.22	2.02	800	31.39	8.36	39.68
14.00	82.29	17.16	100	8.00	0.22	2.70	800	32.64	7.63	38.03
15.00	82.29	17.16	100	8.00	0.22	4.04	800	32.47	7.50	37.67
16.00	82.29	17.16	100	8.00	0.23	1.56	800	30.75	7.83	41.11
17.00	82.29	17.16	100	8.00	0.23	1.56	800	31.73	7.66	40.32
18.00	82.29	17.16	100	8.00	0.23	1.56	800	31.68	7.63	37.64
19.00	82.29	17.16	100	8.00	0.23	1.56	800	31.58	6.68	37.17
20.00	82.00	17.20	100	7.00	0.32	0.00	805	22.66	6.16	39.90
21.00	82.00	17.20	100	7.00	0.23	0.00	800	23.47	8.37	41.84
22.00	82.00	17.20	100	7.00	0.48	0.00	805	20.70	2.11	32.28
23.00	82.00	17.20	100	7.00	0.13	0.00	800	26.29	8.27	49.63
24.00	82.00	17.20	100	7.00	0.22	0.00	815	28.77	8.61	39.53
25.00	82.00	17.20	100	7.00	0.24	0.00	800	24.68	6.96	42.19
26.00	82.00	17.20	100	7.00	0.66	0.00	800	15.62	3.03	35.80
27.00	80.60	18.51	100	9.80	0.27	0.00	809	30.02	8.28	28.57
28.00	80.60	18.51	100	9.80	0.24	0.00	806	31.14	8.90	29.01
29.00	80.60	18.51	100	9.80	0.19	0.00	781	30.39	9.76	29.65
30.00	80.60	18.51	100	9.80	0.32	0.00	725	31.47	8.90	24.43
31.00	73.40	20.60	100	13.63	0.30	0.00	674	10.57	2.67	11.55
32.00	73.40	20.60	100	13.63	0.30	0.00	747	10.63	3.83	12.59
33.00	73.40	20.60	100	13.63	0.40	0.00	684	10.48	2.50	10.93
34.00	73.40	20.60	100	13.63	0.40	0.00	772	10.59	3.61	12.70
35.00	63.80	16.87	100	8.10	0.30	0.00	754	8.60	3.21	15.44
36.00	63.80	16.87	100	8.10	0.30	0.00	874	8.62	3.24	19.56
37.00	63.80	16.87	100	8.10	0.40	0.00	764	7.43	2.01	14.79
38.00	63.80	16.87	100	8.10	0.40	0.00	887	7.77	2.78	16.14
39.00	77.76	20.44	100	9.30	0.30	0.00	746	10.00	2.45	17.41
40.00	77.76	20.44	100	9.30	0.30	0.00	857	10.14	3.21	17.55

Table 24. Finalized data (Page 14 of 36)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)	H2	CH4	CO
41.00	77.76	20.44	100	9.30	0.40	0.00	756	8.13	2.14	16.34
42.00	77.76	20.44	100	9.30	0.40	0.00	877	9.36	2.62	17.75
43.00	52.25	42.65	100	15.60	0.40	0.00	791	10.46	2.11	13.48
44.00	52.25	42.65	100	15.60	0.68	0.00	802	7.05	1.41	8.96
45.00	52.25	42.65	100	15.60	0.84	0.00	858	4.53	0.60	6.14
46.00	84.40	15.50	100	9.20	0.36	0.00	782	5.22	3.01	8.02
47.00	84.40	15.50	100	9.20	0.62	0.00	824	3.82	2.01	5.63
48.00	84.40	15.50	100	9.20	0.60	0.00	792	5.33	2.11	7.34
49.00	79.80	20.00	100	11.30	0.44	0.00	815	8.53	4.79	19.35
50.00	79.80	20.00	100	11.30	0.39	0.00	815	8.74	4.58	18.73
51.00	79.80	20.00	100	11.30	0.47	0.00	790	7.90	4.57	18.61
52.00	83.20	14.30	100	25.30	0.60	0.00	815	12.86	1.66	11.83
53.00	83.20	14.30	100	25.30	0.58	0.00	790	8.23	2.60	12.03
54.00	83.20	14.30	100	25.30	0.45	0.00	790	7.96	2.55	11.25
55.00	83.60	13.30	100	13.30	0.45	0.00	790	5.10	1.10	8.30
56.00	83.60	13.30	100	13.30	0.15	0.00	790	10.86	2.51	12.21
57.00	83.60	13.30	100	13.30	0.15	0.00	815	13.22	2.39	14.67
58.00	80.15	19.52	100	0.00	0.18	1.40	750	31.40	8.40	20.60
59.00	80.15	19.52	100	0.00	0.18	1.40	800	37.20	7.60	17.40
60.00	80.15	19.52	100	0.00	0.18	1.40	850	38.76	8.29	18.58
61.00	80.15	19.52	100	0.00	0.18	1.40	900	38.30	8.00	19.30
62.00	80.15	19.52	100	0.00	0.18	1.40	950	44.40	7.04	18.83
63.00	80.15	19.52	100	0.00	0.37	1.70	850	27.57	7.69	15.48
64.00	80.15	19.52	100	0.00	0.28	1.70	850	34.27	7.59	15.58
65.00	80.15	19.52	100	0.00	0.18	1.70	850	39.10	7.40	16.80
66.00	80.15	19.52	100	0.00	0.09	1.70	850	43.24	7.41	18.52
67.00	80.15	19.52	100	0.00	0.00	1.70	850	46.80	8.30	20.20
68.00	80.15	19.52	100	0.00	0.00	1.10	800	46.00	7.90	19.90
69.00	80.15	19.52	100	0.00	0.00	1.40	800	48.45	6.89	18.48
70.00	80.15	19.52	100	0.00	0.00	1.80	800	51.25	6.09	15.88
71.00	80.15	19.52	100	0.00	0.00	2.70	800	54.40	5.00	11.80
72.00	80.15	19.52	100	0.00	0.00	4.70	800	57.34	4.10	7.29
73.00	34.77	28.21	100	8.72	0.20	0.00	850	35.89	8.47	19.76
74.00	32.86	29.23	100	8.34	0.20	0.00	850	36.17	9.28	19.80
75.00	0.00	0.00	0	7.20	0.20	0.00	850	30.80	12.66	16.04
76.00	51.29	45.02	100	18.70	0.20	0.00	850	39.53	5.30	27.58
77.00	57.89	39.09	100	17.28	0.20	0.00	850	43.37	5.06	23.96
78.00	55.98	40.11	100	16.90	0.20	0.00	850	39.67	5.87	27.93
79.00	0.00	0.00	0	16.83	0.20	0.00	850	32.65	10.28	19.91
80.00	0.00	0.00	0	10.00	0.20	0.00	850	31.25	12.28	21.78

Table 24. Finalized data (Page 15 of 36)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)	H2	CH4	CO
81.00	27.74	60.89	100	5.90	0.20	0.00	850	35.41	7.84	20.12
82.00	39.04	51.79	100	5.00	0.20	0.00	850	33.61	8.50	23.28
83.00	37.14	52.81	100	6.00	0.20	0.00	850	34.15	8.48	22.43
84.00	0.00	0.00	0	5.50	0.20	0.00	850	31.95	11.41	18.47
85.00	88.90	10.00	100	8.50	0.32	0.00	746	14.78	8.65	34.96
86.00	88.90	10.00	100	8.50	0.36	0.00	776	18.21	8.66	31.79
87.00	88.90	10.00	100	8.50	0.38	0.00	790	19.65	8.64	32.15
88.00	88.90	10.00	100	8.50	0.41	0.00	810	17.73	9.09	30.84
89.00	88.90	10.00	100	8.50	0.43	0.00	836	14.04	8.82	31.23
90.00	80.60	18.51	100	7.00	0.19	0.00	780	25.58	11.63	35.27
91.00	80.60	18.51	100	7.00	0.27	0.00	805	25.05	10.34	34.99
92.00	80.60	18.51	100	7.00	0.27	0.23	786	28.06	9.42	30.06
93.00	80.60	18.51	100	7.00	0.36	0.32	808	28.08	9.42	32.36
94.00	80.60	18.51	100	7.00	0.34	0.56	795	34.38	9.62	27.60
95.00	80.60	18.51	100	7.00	0.33	0.57	806	37.41	9.75	28.09
96.00	0.00	0.00	0	9.00	0.21	0.00	856	14.20	6.00	18.80
97.00	0.00	0.00	0	9.00	0.17	0.00	843	15.20	6.60	20.40
98.00	0.00	0.00	0	9.00	0.26	0.00	911	14.50	4.40	19.90
99.00	0.00	0.00	0	9.00	0.29	0.00	860	13.30	4.20	16.50
100.00	0.00	0.00	0	9.00	0.20	0.00	850	15.20	4.60	19.00
101.00	0.00	0.00	0	9.00	0.28	0.90	832	17.50	4.20	11.90
102.00	0.00	0.00	0	9.00	0.26	0.63	808	17.00	4.10	11.60
103.00	0.00	0.00	0	9.00	0.20	0.63	847	21.10	5.10	15.80
104.00	0.00	0.00	0	9.00	0.25	0.00	864	13.80	4.90	19.10
105.00	0.00	0.00	0	9.00	0.25	0.00	855	13.30	5.00	17.20
106.00	0.00	0.00	0	7.00	0.30	0.00	925	10.30	3.40	16.60
107.00	0.00	0.00	0	7.00	0.26	0.00	907	11.80	3.70	17.20
108.00	0.00	0.00	0	7.00	0.22	0.00	885	13.10	4.30	18.90
109.00	0.00	0.00	0	7.00	0.31	0.00	945	9.40	3.20	15.90
110.00	0.00	0.00	0	7.00	0.26	0.00	865	11.20	3.90	17.80
111.00	0.00	0.00	0	7.00	0.16	0.00	854	12.80	5.30	17.90
112.00	0.00	0.00	0	7.00	0.13	0.62	768	14.50	3.80	17.00
113.00	0.00	0.00	0	7.00	0.28	0.00	923	13.10	3.30	19.40
114.00	0.00	0.00	0	11.30	0.25	0.00	881	12.30	3.20	16.30
115.00	0.00	0.00	0	11.30	0.31	0.00	907	9.90	3.30	15.10
116.00	0.00	0.00	0	11.30	0.21	0.00	856	14.80	4.30	17.60
117.00	74.40	17.10	100	8.90	0.41	0.00	780	5.47	1.62	8.72
118.00	74.40	17.10	100	8.90	0.73	0.00	800	7.38	0.81	6.97
119.00	74.40	17.10	100	8.90	0.59	0.00	800	9.39	0.81	8.48
120.00	74.40	17.10	100	8.90	0.67	0.00	800	7.66	0.91	7.56
121.00	82.20	17.60	100	8.00	0.38	1.13	775	22.56	8.72	25.64

Table 24. Finalized data (Page 16 of 36)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)	H2	CH4	CO
122.00	82.20	17.60	100	8.00	0.38	1.45	770	24.10	8.21	25.13
123.00	82.20	17.60	100	8.00	0.35	0.97	815	23.96	8.85	25.52
124.00	82.20	17.60	100	8.00	0.35	1.25	810	26.56	8.33	23.96
125.00	75.50	21.60	100	8.00	0.38	0.93	780	25.91	7.77	21.76
126.00	75.50	21.60	100	8.00	0.38	1.22	780	27.98	7.25	20.73
127.00	75.50	21.60	100	8.00	0.39	0.90	820	26.42	7.77	22.80
128.00	75.50	21.60	100	8.00	0.39	1.04	820	29.02	7.77	20.21
129.00	82.20	17.60	100	8.00	0.33	0.98	800	23.44	9.38	28.13
130.00	82.20	17.60	100	8.00	0.33	1.16	800	23.96	9.38	28.13
131.00	82.20	17.60	100	8.00	0.40	1.15	800	20.62	8.76	24.23
132.00	75.50	21.60	100	8.00	0.34	0.91	800	28.65	8.33	23.44
133.00	75.50	21.60	100	8.00	0.34	1.14	800	31.09	7.77	21.76
134.00	75.50	21.60	100	8.00	0.43	1.07	815	24.74	7.22	22.68
135.00	76.70	17.70	100	12.00	0.38	0.81	790	18.00	8.00	31.00
136.00	76.70	17.70	100	12.00	0.38	0.81	800	19.00	8.00	30.50
137.00	76.70	17.70	100	12.00	0.38	0.81	810	21.98	9.89	31.87
138.00	76.70	17.70	100	12.00	0.38	0.81	820	22.65	8.84	31.49
139.00	81.72	16.15	100	17.50	0.37	0.00	827	7.30	3.34	9.53
140.00	78.34	19.58	100	6.10	0.36	0.00	847	7.14	3.29	11.50
141.00	76.90	20.20	100	10.80	0.38	0.00	843	6.53	2.82	9.81
142.00	65.10	17.30	100	7.30	0.37	0.00	803	7.46	2.64	10.14
143.00	65.10	17.30	100	7.30	0.36	0.00	815	8.66	2.86	10.90
144.00	43.32	9.38	100	10.20	0.37	0.00	829	6.17	2.85	9.38
145.00	55.17	8.83	100	10.20	0.32	0.00	827	8.15	3.24	10.70
146.00	55.17	8.83	100	12.80	0.60	0.00	827	1.83	1.26	5.42
147.00	60.87	20.83	100	14.80	0.43	0.00	805	5.45	1.74	8.42
148.00	70.36	15.07	100	14.43	0.22	0.00	668	5.47	5.55	20.97
149.00	70.36	15.07	100	14.43	0.25	0.00	711	6.30	5.44	19.81
150.00	70.36	15.07	100	14.43	0.27	0.00	745	6.92	5.11	18.32
151.00	70.36	15.07	100	14.43	0.30	0.00	770	7.75	4.72	16.91
152.00	70.36	15.07	100	14.43	0.21	0.00	672	5.47	5.89	21.02
153.00	70.36	15.07	100	14.43	0.22	0.00	700	6.36	5.65	19.93
154.00	70.36	15.07	100	14.43	0.24	0.00	714	6.65	5.50	18.74
155.00	70.36	15.07	100	14.43	0.25	0.00	750	7.40	5.06	17.63
156.00	70.36	15.07	100	14.43	0.27	0.00	764	7.59	4.88	16.83
157.00	70.36	15.07	100	14.43	0.23	0.00	704	7.09	5.04	17.28
158.00	70.36	15.07	100	14.43	0.24	0.00	710	7.09	5.04	17.28
159.00	70.36	15.07	100	14.43	0.26	0.00	736	7.09	5.04	17.28
160.00	70.36	15.07	100	14.43	0.27	0.00	748	7.09	5.04	17.28
161.00	70.36	15.07	100	14.43	0.28	0.00	766	7.09	5.04	17.28
162.00	0.00	0.00	0	22.00	0.63	0.00	740	5.60	1.40	6.90

Table 24. Finalized data (Page 17 of 36)

	Volatile matter	Fixed carbon	Total	MC	ER	SBR	Temp (°C)	H2	CH4	CO
163.00	0.00	0.00	0	9.70	0.46	0.00	718	3.13	1.92	11.09
164.00	0.00	0.00	0	10.50	0.42	0.00	766	3.20	1.90	10.70
165.00	0.00	0.00	0	10.00	0.47	0.00	815	3.00	1.90	9.60
166.00	0.00	0.00	0	8.80	0.35	0.00	772	4.00	2.90	14.71
167.00	0.00	0.00	0	9.20	0.39	0.00	787	3.80	2.70	12.60
168.00	0.00	0.00	0	11.70	0.30	0.00	718	5.50	3.40	16.60
169.00	0.00	0.00	0	11.30	0.31	0.00	730	3.90	2.80	15.12
170.00	0.00	0.00	0	15.00	0.34	0.00	752	3.50	2.80	13.39
171.00	0.00	0.00	0	12.60	0.35	0.00	815	4.10	2.50	12.30
172.00	0.00	0.00	0	14.70	0.29	0.00	789	4.20	3.00	14.59
173.00	0.00	0.00	0	10.10	0.20	0.00	701	5.40	4.60	21.40
174.00	0.00	0.00	0	6.60	0.25	0.00	728	5.11	4.10	19.92
175.00	0.00	0.00	0	6.70	0.29	0.00	739	7.29	3.20	17.88
176.00	0.00	0.00	0	4.20	0.45	0.00	805	5.90	1.20	10.00
177.00	78.95	12.26	100	6.10	0.30	0.00	750	22.80	8.10	25.20
178.00	78.95	12.26	100	6.10	0.30	0.00	800	27.12	6.96	35.01
179.00	78.95	12.26	100	6.10	0.30	0.00	750	24.02	8.50	34.22
180.00	78.95	12.26	100	6.10	0.30	0.00	800	25.40	8.47	37.20
181.00	83.31	14.30	100	8.20	0.30	0.00	750	24.48	8.82	30.60
182.00	83.31	14.30	100	8.20	0.30	0.00	800	25.84	7.90	35.56
183.00	83.31	14.30	100	8.20	0.30	0.00	750	21.67	9.34	31.51
184.00	83.31	14.30	100	8.20	0.30	0.00	800	22.40	8.70	33.70
185.00	78.95	11.42	100	8.10	0.30	0.00	750	26.83	8.61	29.43
186.00	78.95	11.42	100	8.10	0.30	0.00	800	28.66	8.22	33.17
187.00	78.95	11.42	100	8.10	0.30	0.00	750	27.10	9.40	29.30
188.00	78.95	11.42	100	8.10	0.30	0.00	800	27.68	8.19	35.09
189.00	83.00	14.50	100	5.00	0.17	0.00	686	5.67	9.94	17.33
190.00	83.00	14.50	100	5.00	0.19	0.00	690	5.34	8.35	14.40
191.00	83.00	14.50	100	5.00	0.20	0.00	720	5.76	8.30	17.70
192.00	67.70	31.10	100	7.00	0.36	0.00	750	8.30	3.70	18.20
193.00	67.70	31.10	100	7.00	0.29	0.00	750	8.50	4.10	15.70
194.00	67.70	31.10	100	7.00	0.35	0.00	750	10.30	3.20	12.40
195.00	67.70	31.10	100	7.00	0.39	0.00	800	8.20	3.60	16.50
196.00	67.70	31.10	100	7.00	0.30	0.00	800	7.40	4.20	14.90
197.00	67.70	31.10	100	7.00	0.36	0.00	800	11.50	2.80	14.00
198.00	67.70	31.10	100	7.00	0.45	0.00	850	8.28	2.97	14.42
199.00	67.70	31.10	100	7.00	0.32	0.00	850	7.70	3.70	12.90
200.00	67.70	31.10	100	7.00	0.37	0.00	850	10.98	3.09	13.67
201.00	72.50	25.00	100	11.40	0.18	0.00	750	10.03	6.14	17.81
202.00	72.50	25.00	100	11.40	0.26	0.00	800	11.64	4.23	14.07
203.00	72.50	25.00	100	11.40	0.47	0.00	850	10.22	3.69	12.64

Table 24. Finalized data (Page 18 of 36)

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Table 24. Finalized data (Page 19 of 36)

	CO2	C2H4	C2H4	C2H2	NH3	N2	H2O	O2	SUM
1.00	20.28	4.95	0.00	0.00	0.00	0.00	0.00	0.00	100.00
2.00	19.22	3.90	0.00	0.00	0.00	0.00	0.00	0.00	100.00
3.00	18.26	3.38	0.00	0.00	0.00	0.00	0.00	0.00	100.00
4.00	18.50	2.87	0.00	0.00	0.00	0.00	0.00	0.00	100.00
5.00	18.99	1.36	0.00	0.00	0.00	0.00	0.00	0.00	100.00
6.00	17.63	3.67	0.00	0.00	0.00	0.00	0.00	0.00	100.00
7.00	16.91	3.66	0.00	0.00	0.00	0.00	0.00	0.00	100.00
8.00	16.56	3.42	0.00	0.00	0.00	0.00	0.00	0.00	100.00
9.00	17.74	3.41	0.00	0.00	0.00	0.00	0.00	0.00	100.00
10.00	19.47	3.39	0.00	0.00	0.00	0.00	0.00	0.00	100.00
11.00	15.65	1.59	0.00	0.00	0.00	0.00	0.00	0.00	100.00
12.00	16.65	3.42	0.00	0.00	0.00	0.00	0.00	0.00	100.00
13.00	17.17	3.40	0.00	0.00	0.00	0.00	0.00	0.00	100.00
14.00	18.32	3.38	0.00	0.00	0.00	0.00	0.00	0.00	100.00
15.00	18.97	3.38	0.00	0.00	0.00	0.00	0.00	0.00	100.00
16.00	16.93	3.37	0.00	0.00	0.00	0.00	0.00	0.00	100.00
17.00	16.92	3.37	0.00	0.00	0.00	0.00	0.00	0.00	100.00
18.00	19.70	3.36	0.00	0.00	0.00	0.00	0.00	0.00	100.00
19.00	21.22	3.34	0.00	0.00	0.00	0.00	0.00	0.00	100.00
20.00	31.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
21.00	26.33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
22.00	44.91	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
23.00	15.81	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
24.00	23.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
25.00	26.16	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
26.00	45.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
27.00	33.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
28.00	30.95	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
29.00	30.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
30.00	35.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
31.00	11.60	0.00	0.00	0.00	0.00	63.61	0.00	0.00	100.00
32.00	11.07	0.00	0.00	0.00	0.00	61.89	0.00	0.00	100.00
33.00	11.71	0.00	0.00	0.00	0.00	64.37	0.00	0.00	100.00
34.00	11.85	0.00	0.00	0.00	0.00	61.25	0.00	0.00	100.00
35.00	11.69	0.00	0.00	0.00	0.00	61.06	0.00	0.00	100.00
36.00	10.22	0.00	0.00	0.00	0.00	58.36	0.00	0.00	100.00
37.00	10.67	0.00	0.00	0.00	0.00	65.11	0.00	0.00	100.00
38.00	10.69	0.00	0.00	0.00	0.00	62.62	0.00	0.00	100.00
39.00	11.98	0.00	0.00	0.00	0.00	58.15	0.00	0.00	100.00
40.00	10.78	0.00	0.00	0.00	0.00	58.31	0.00	0.00	100.00
41.00	11.17	0.00	0.00	0.00	0.00	62.21	0.00	0.00	100.00
42.00	11.89	0.00	0.00	0.00	0.00	58.37	0.00	0.00	100.00

Table 24. Finalized data (Page 21 of 36)

	CO2	C2H4	C2H4	C2H2	NH3	N2	H2O	O2	SUM
85.00	41.61	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
86.00	41.34	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
87.00	39.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
88.00	42.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
89.00	45.91	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
90.00	27.52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
91.00	29.62	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
92.00	32.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
93.00	30.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
94.00	28.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
95.00	24.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
96.00	15.40	0.00	0.00	0.00	0.00	45.60	0.00	0.00	100.00
97.00	15.70	0.00	0.00	0.00	0.00	42.10	0.00	0.00	100.00
98.00	13.30	0.00	0.00	0.00	0.00	47.90	0.00	0.00	100.00
99.00	14.60	0.00	0.00	0.00	0.00	51.40	0.00	0.00	100.00
100.00	14.70	0.00	0.00	0.00	0.00	46.50	0.00	0.00	100.00
101.00	18.20	0.00	0.00	0.00	0.00	48.20	0.00	0.00	100.00
102.00	18.60	0.00	0.00	0.00	0.00	48.70	0.00	0.00	100.00
103.00	17.20	0.00	0.00	0.00	0.00	40.80	0.00	0.00	100.00
104.00	14.30	0.00	0.00	0.00	0.00	47.90	0.00	0.00	100.00
105.00	15.50	0.00	0.00	0.00	0.00	49.00	0.00	0.00	100.00
106.00	13.70	0.00	0.00	0.00	0.00	56.00	0.00	0.00	100.00
107.00	13.30	0.00	0.00	0.00	0.00	54.00	0.00	0.00	100.00
108.00	13.50	0.00	0.00	0.00	0.00	50.20	0.00	0.00	100.00
109.00	13.30	0.00	0.00	0.00	0.00	58.20	0.00	0.00	100.00
110.00	13.50	0.00	0.00	0.00	0.00	53.60	0.00	0.00	100.00
111.00	14.40	0.00	0.00	0.00	0.00	49.60	0.00	0.00	100.00
112.00	15.80	0.00	0.00	0.00	0.00	48.90	0.00	0.00	100.00
113.00	11.00	0.00	0.00	0.00	0.00	53.20	0.00	0.00	100.00
114.00	14.70	0.00	0.00	0.00	0.00	53.50	0.00	0.00	100.00
115.00	14.70	0.00	0.00	0.00	0.00	57.00	0.00	0.00	100.00
116.00	14.30	0.00	0.00	0.00	0.00	49.00	0.00	0.00	100.00
117.00	21.99	1.93	0.00	0.00	0.00	60.27	0.00	0.00	100.00
118.00	20.11	0.91	0.00	0.00	0.00	63.82	0.00	0.00	100.00
119.00	19.18	1.01	0.00	0.00	0.00	61.13	0.00	0.00	100.00
120.00	19.85	1.11	0.00	0.00	0.00	62.93	0.00	0.00	100.00
121.00	43.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
122.00	42.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
123.00	41.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
124.00	41.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
125.00	44.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
126.00	44.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 24. Finalized data (Page 22 of 36)

	CO2	C2H4	C2H4	C2H2	NH3	N2	H2O	O2	SUM
127.00	43.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
128.00	43.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
129.00	39.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
130.00	38.54	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
131.00	46.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
132.00	39.58	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
133.00	39.38	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
134.00	45.36	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
135.00	32.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	89.00
136.00	31.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	89.00
137.00	36.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
138.00	37.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
139.00	17.33	1.22	0.00	0.00	0.00	61.29	0.00	0.00	100.00
140.00	15.96	0.89	0.00	0.00	0.00	61.22	0.00	0.00	100.00
141.00	15.90	0.92	0.00	0.00	0.00	64.02	0.00	0.00	100.00
142.00	16.00	1.23	0.00	0.00	0.00	62.53	0.00	0.00	100.00
143.00	15.55	1.11	0.00	0.00	0.00	60.93	0.00	0.00	100.00
144.00	16.35	1.08	0.00	0.00	0.00	64.17	0.00	0.00	100.00
145.00	15.27	1.18	0.00	0.00	0.00	61.46	0.00	0.00	100.00
146.00	16.32	0.43	0.00	0.00	0.00	74.75	0.00	0.00	100.00
147.00	16.26	0.63	0.00	0.00	0.00	67.49	0.00	0.00	100.00
148.00	15.49	3.24	0.00	0.00	0.00	49.28	0.00	0.00	100.00
149.00	15.70	3.50	0.00	0.00	0.00	49.26	0.00	0.00	100.00
150.00	16.16	3.29	0.00	0.00	0.00	50.20	0.00	0.00	100.00
151.00	16.48	2.75	0.00	0.00	0.00	51.38	0.00	0.00	100.00
152.00	15.83	3.53	0.00	0.00	0.00	48.26	0.00	0.00	100.00
153.00	15.97	3.25	0.00	0.00	0.00	48.84	0.00	0.00	100.00
154.00	16.39	2.74	0.00	0.00	0.00	49.97	0.00	0.00	100.00
155.00	16.54	2.77	0.00	0.00	0.00	50.60	0.00	0.00	100.00
156.00	16.36	2.87	0.00	0.00	0.00	51.48	0.00	0.00	100.00
157.00	16.81	4.07	0.00	0.00	0.00	49.71	0.00	0.00	100.00
158.00	16.80	4.08	0.00	0.00	0.00	49.71	0.00	0.00	100.00
159.00	16.80	4.06	0.00	0.00	0.00	49.72	0.00	0.00	100.00
160.00	16.80	4.07	0.00	0.00	0.00	49.71	0.00	0.00	100.00
161.00	16.81	4.05	0.00	0.00	0.00	49.73	0.00	0.00	100.00
162.00	18.10	0.00	0.00	0.00	0.00	68.00	0.00	0.00	100.00
163.00	15.22	0.00	0.00	0.00	0.00	68.65	0.00	0.00	100.00
164.00	17.10	0.00	0.00	0.00	0.00	67.10	0.00	0.00	100.00
165.00	17.10	0.00	0.00	0.00	0.00	68.40	0.00	0.00	100.00
166.00	16.52	0.00	0.00	0.00	0.00	61.86	0.00	0.00	100.00
167.00	15.70	0.00	0.00	0.00	0.00	65.20	0.00	0.00	100.00
168.00	15.00	0.00	0.00	0.00	0.00	59.50	0.00	0.00	100.00

Table 24. Finalized data (Page 23 of 36)

	CO2	C2H4	C2H4	C2H2	NH3	N2	H2O	O2	SUM
169.00	15.62	0.00	0.00	0.00	0.00	62.56	0.00	0.00	100.00
170.00	15.58	0.00	0.00	0.00	0.00	64.74	0.00	0.00	100.00
171.00	16.50	0.00	0.00	0.00	0.00	64.60	0.00	0.00	100.00
172.00	15.68	0.00	0.00	0.00	0.00	62.54	0.00	0.00	100.00
173.00	14.70	0.00	0.00	0.00	0.00	53.90	0.00	0.00	100.00
174.00	14.51	0.00	0.00	0.00	0.00	56.36	0.00	0.00	100.00
175.00	16.28	0.00	0.00	0.00	0.00	55.34	0.00	0.00	100.00
176.00	18.30	0.00	0.00	0.00	0.00	64.60	0.00	0.00	100.00
177.00	32.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	88.20
178.00	30.91	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
179.00	33.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
180.00	28.93	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
181.00	36.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
182.00	30.70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
183.00	37.48	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
184.00	35.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
185.00	35.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
186.00	29.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
187.00	34.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
188.00	29.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00
189.00	15.37	1.34	0.00	0.00	0.00	49.21	0.00	1.13	100.00
190.00	18.90	2.10	0.00	0.00	0.00	50.01	0.00	0.90	100.00
191.00	16.01	1.19	0.00	0.00	0.00	49.94	0.00	1.09	100.00
192.00	13.10	1.40	0.00	0.00	0.00	51.40	0.00	3.90	100.00
193.00	15.10	1.40	0.00	0.00	0.00	51.10	0.00	4.10	100.00
194.00	16.70	0.60	0.00	0.00	0.00	52.90	0.00	3.90	100.00
195.00	14.70	1.20	0.00	0.00	0.00	52.00	0.00	3.80	100.00
196.00	15.20	1.30	0.00	0.00	0.00	52.20	0.00	4.80	100.00
197.00	15.90	0.50	0.00	0.00	0.00	51.70	0.00	3.60	100.00
198.00	16.36	0.41	0.00	0.00	0.00	54.40	0.00	3.17	100.00
199.00	16.50	1.10	0.00	0.00	0.00	53.50	0.00	4.60	100.00
200.00	15.97	0.40	0.00	0.00	0.00	51.80	0.00	4.09	100.00
201.00	15.56	1.74	0.00	0.00	0.00	48.72	0.00	0.00	100.00
202.00	16.72	0.74	0.00	0.00	0.00	52.59	0.00	0.00	100.00
203.00	17.70	0.42	0.00	0.00	0.00	55.32	0.00	0.00	100.00
204.00	15.75	1.46	0.00	0.00	0.00	53.28	0.00	0.00	100.00
205.00	15.97	1.37	0.00	0.00	0.00	54.83	0.00	0.00	100.00
206.00	17.30	1.15	0.00	0.00	0.00	56.08	0.00	0.00	100.00
207.00	16.04	0.00	0.00	0.00	0.00	68.55	0.00	0.00	100.00
208.00	13.94	0.00	0.00	0.00	0.00	66.76	0.00	0.00	100.00
209.00	16.02	0.00	0.00	0.00	0.00	64.62	0.00	0.00	100.00
210.00	14.00	0.00	0.00	0.00	0.00	55.00	0.00	0.00	100.00

Table 24. Finalized data (Page 24 of 36)

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Table 24. Finalized data (Page 25 of 36)

	Group	H2	CH4	CO	CO2	C2H4	C2H4
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)							
1.00	1	21.99	9.27	43.50	20.28	4.95	0.00
2.00	1	28.75	8.41	39.72	19.22	3.90	0.00
3.00	1	32.66	7.68	38.01	18.26	3.38	0.00
4.00	1	36.74	7.41	34.48	18.50	2.87	0.00
5.00	1	39.84	6.29	33.52	18.99	1.36	0.00
6.00	1	32.67	8.07	37.96	17.63	3.67	0.00
7.00	1	31.52	8.28	39.62	16.91	3.66	0.00
8.00	1	31.91	7.82	40.29	16.56	3.42	0.00
9.00	1	31.50	7.57	39.79	17.74	3.41	0.00
10.00	1	32.10	6.85	38.18	19.47	3.39	0.00
11.00	1	33.50	6.35	42.91	15.65	1.59	0.00
12.00	1	30.48	8.48	40.97	16.65	3.42	0.00
13.00	1	31.39	8.36	39.68	17.17	3.40	0.00
14.00	1	32.64	7.63	38.03	18.32	3.38	0.00
15.00	1	32.47	7.50	37.67	18.97	3.38	0.00
16.00	1	30.75	7.83	41.11	16.93	3.37	0.00
17.00	1	31.73	7.66	40.32	16.92	3.37	0.00
18.00	1	31.68	7.63	37.64	19.70	3.36	0.00
19.00	1	31.58	6.68	37.17	21.22	3.34	0.00
20.00	7	22.66	6.16	39.90	31.28	0.00	0.00
21.00	7	23.47	8.37	41.84	26.33	0.00	0.00
22.00	7	20.70	2.11	32.28	44.91	0.00	0.00
23.00	7	26.29	8.27	49.63	15.81	0.00	0.00
24.00	7	28.77	8.61	39.53	23.09	0.00	0.00
25.00	7	24.68	6.96	42.19	26.16	0.00	0.00
26.00	7	15.62	3.03	35.80	45.56	0.00	0.00
27.00	7	30.02	8.28	28.57	33.13	0.00	0.00
28.00	7	31.14	8.90	29.01	30.95	0.00	0.00
29.00	7	30.39	9.76	29.65	30.20	0.00	0.00
30.00	7	31.47	8.90	24.43	35.20	0.00	0.00
31.00	2	29.04	7.35	31.74	31.87	0.00	0.00
32.00	2	27.88	10.04	33.04	29.04	0.00	0.00
33.00	3	29.42	7.03	30.68	32.87	0.00	0.00
34.00	3	27.33	9.31	32.78	30.58	0.00	0.00
35.00	3	22.08	8.25	39.64	30.03	0.00	0.00
36.00	3	20.71	7.78	46.97	24.53	0.00	0.00
37.00	3	21.28	5.76	42.39	30.57	0.00	0.00
38.00	3	20.80	7.44	43.17	28.59	0.00	0.00
39.00	3	23.89	5.87	41.61	28.63	0.00	0.00
40.00	3	24.33	7.70	42.09	25.87	0.00	0.00

Table 24. Finalized data (Page 26 of 36)

	Group	H2	CH4	CO	CO2	C2H4	C2H4
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)							
41.00	3	21.52	5.67	43.24	29.57	0.00	0.00
42.00	3	22.49	6.30	42.65	28.55	0.00	0.00
43.00	2	28.18	5.69	36.31	29.54	0.27	0.00
44.00	2	23.97	4.79	30.48	40.41	0.34	0.00
45.00	2	18.83	2.51	25.52	52.72	0.42	0.00
46.00	2	19.70	11.36	30.30	36.74	1.89	0.00
47.00	2	14.96	7.87	22.05	53.94	1.18	0.00
48.00	2	18.47	7.32	25.44	46.34	2.44	0.00
49.00	2	17.05	9.56	38.67	34.72	0.00	0.00
50.00	2	17.54	9.19	37.58	35.70	0.00	0.00
51.00	2	16.17	9.36	38.09	36.38	0.00	0.00
52.00	2	28.12	3.63	25.85	42.40	0.00	0.00
53.00	2	19.95	6.30	29.13	44.62	0.00	0.00
54.00	2	19.23	6.15	27.18	47.44	0.00	0.00
55.00	2	16.45	3.55	26.77	53.23	0.00	0.00
56.00	2	23.32	5.38	26.23	45.07	0.00	0.00
57.00	2	27.02	4.89	30.00	38.09	0.00	0.00
58.00	4	31.40	8.40	20.60	35.30	4.30	0.00
59.00	4	37.20	7.60	17.40	35.30	2.50	0.00
60.00	4	38.76	8.29	18.58	32.17	2.20	0.00
61.00	4	38.30	8.00	19.30	32.70	1.70	0.00
62.00	4	44.40	7.04	18.83	28.15	1.59	0.00
63.00	4	27.57	7.69	15.48	46.85	2.40	0.00
64.00	4	34.27	7.59	15.58	40.36	2.20	0.00
65.00	4	39.10	7.40	16.80	34.70	2.00	0.00
66.00	4	43.24	7.41	18.52	28.53	2.30	0.00
67.00	4	46.80	8.30	20.20	22.30	2.40	0.00
68.00	4	46.00	7.90	19.90	24.10	2.10	0.00
69.00	4	48.45	6.89	18.48	24.58	1.60	0.00
70.00	4	51.25	6.09	15.88	25.07	1.70	0.00
71.00	4	54.40	5.00	11.80	27.40	1.40	0.00
72.00	4	57.34	4.10	7.29	30.17	1.10	0.00
73.00	7	35.89	8.47	19.76	35.88	0.00	0.00
74.00	7	36.17	9.28	19.80	34.75	0.00	0.00
75.00	7	30.80	12.66	16.04	40.50	0.00	0.00
76.00	7	39.53	5.30	27.58	27.58	0.00	0.00
77.00	7	43.37	5.06	23.96	27.61	0.00	0.00
78.00	7	39.67	5.87	27.93	26.52	0.00	0.00
79.00	7	32.65	10.28	19.91	37.16	0.00	0.00
80.00	7	31.25	12.28	21.78	34.69	0.00	0.00

Table 24. Finalized data (Page 27 of 36)

	Group	H2	CH4	CO	CO2	C2H4	C2H4
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)							
81.00	7	35.41	7.84	20.12	36.63	0.00	0.00
82.00	7	33.61	8.50	23.28	34.61	0.00	0.00
83.00	7	34.15	8.48	22.43	34.94	0.00	0.00
84.00	7	31.95	11.41	18.47	38.17	0.00	0.00
85.00	4	14.78	8.65	34.96	41.61	0.00	0.00
86.00	4	18.21	8.66	31.79	41.34	0.00	0.00
87.00	4	19.65	8.64	32.15	39.56	0.00	0.00
88.00	4	17.73	9.09	30.84	42.35	0.00	0.00
89.00	4	14.04	8.82	31.23	45.91	0.00	0.00
90.00	5	25.58	11.63	35.27	27.52	0.00	0.00
91.00	5	25.05	10.34	34.99	29.62	0.00	0.00
92.00	5	28.06	9.42	30.06	32.46	0.00	0.00
93.00	5	28.08	9.42	32.36	30.14	0.00	0.00
94.00	5	34.38	9.62	27.60	28.39	0.00	0.00
95.00	5	37.41	9.75	28.09	24.75	0.00	0.00
96.00	6	26.10	11.03	34.56	28.31	0.00	0.00
97.00	6	26.25	11.40	35.23	27.12	0.00	0.00
98.00	6	27.83	8.45	38.20	25.53	0.00	0.00
99.00	6	27.37	8.64	33.95	30.04	0.00	0.00
100.00	6	28.41	8.60	35.51	27.48	0.00	0.00
101.00	6	33.78	8.11	22.97	35.14	0.00	0.00
102.00	6	33.14	7.99	22.61	36.26	0.00	0.00
103.00	6	35.64	8.61	26.69	29.05	0.00	0.00
104.00	6	26.49	9.40	36.66	27.45	0.00	0.00
105.00	6	26.08	9.80	33.73	30.39	0.00	0.00
106.00	6	23.41	7.73	37.73	31.14	0.00	0.00
107.00	6	25.65	8.04	37.39	28.91	0.00	0.00
108.00	6	26.31	8.63	37.95	27.11	0.00	0.00
109.00	6	22.49	7.66	38.04	31.82	0.00	0.00
110.00	6	24.14	8.41	38.36	29.09	0.00	0.00
111.00	6	25.40	10.52	35.52	28.57	0.00	0.00
112.00	6	28.38	7.44	33.27	30.92	0.00	0.00
113.00	6	27.99	7.05	41.45	23.50	0.00	0.00
114.00	6	26.45	6.88	35.05	31.61	0.00	0.00
115.00	6	23.02	7.67	35.12	34.19	0.00	0.00
116.00	6	29.02	8.43	34.51	28.04	0.00	0.00
117.00	2	13.78	4.08	21.94	55.36	4.85	0.00
118.00	2	20.39	2.23	19.27	55.59	2.51	0.00
119.00	2	24.16	2.08	21.82	49.35	2.60	0.00
120.00	2	20.65	2.45	20.38	53.53	2.99	0.00

Table 24. Finalized data (Page 28 of 36)

	Group	H2	CH4	CO	CO2	C2H4	C2H4
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)							
121.00	7	22.56	8.72	25.64	43.08	0.00	0.00
122.00	7	24.10	8.21	25.13	42.56	0.00	0.00
123.00	7	23.96	8.85	25.52	41.67	0.00	0.00
124.00	7	26.56	8.33	23.96	41.15	0.00	0.00
125.00	7	25.91	7.77	21.76	44.56	0.00	0.00
126.00	7	27.98	7.25	20.73	44.04	0.00	0.00
127.00	7	26.42	7.77	22.80	43.01	0.00	0.00
128.00	7	29.02	7.77	20.21	43.01	0.00	0.00
129.00	7	23.44	9.38	28.13	39.06	0.00	0.00
130.00	7	23.96	9.38	28.13	38.54	0.00	0.00
131.00	7	20.62	8.76	24.23	46.39	0.00	0.00
132.00	7	28.65	8.33	23.44	39.58	0.00	0.00
133.00	7	31.09	7.77	21.76	39.38	0.00	0.00
134.00	7	24.74	7.22	22.68	45.36	0.00	0.00
135.00	7	22.72	10.10	39.14	40.40	0.00	0.00
136.00	7	23.99	10.10	38.51	39.77	0.00	0.00
137.00	7	21.98	9.89	31.87	36.26	0.00	0.00
138.00	7	22.65	8.84	31.49	37.02	0.00	0.00
139.00	2	18.85	8.64	24.61	44.76	3.14	0.00
140.00	2	18.42	8.49	29.65	41.14	2.30	0.00
141.00	2	18.16	7.83	27.26	44.20	2.55	0.00
142.00	2	19.91	7.06	27.05	42.70	3.28	0.00
143.00	2	22.16	7.32	27.90	39.80	2.83	0.00
144.00	2	17.22	7.96	26.18	45.64	3.00	0.00
145.00	2	21.15	8.41	27.77	39.61	3.06	0.00
146.00	2	7.23	4.98	21.45	64.64	1.69	0.00
147.00	2	16.78	5.36	25.91	50.02	1.93	0.00
148.00	2	10.79	10.94	41.35	30.54	6.38	0.00
149.00	2	12.41	10.72	39.04	30.94	6.89	0.00
150.00	2	13.90	10.26	36.78	32.45	6.61	0.00
151.00	2	15.94	9.71	34.79	33.90	5.66	0.00
152.00	2	10.57	11.39	40.63	30.59	6.82	0.00
153.00	2	12.43	11.05	38.95	31.22	6.35	0.00
154.00	2	13.30	11.00	37.46	32.76	5.48	0.00
155.00	2	14.99	10.23	35.68	33.49	5.60	0.00
156.00	2	15.65	10.05	34.68	33.71	5.92	0.00
157.00	2	14.10	10.02	34.35	33.42	8.10	0.00
158.00	2	14.10	10.03	34.35	33.41	8.11	0.00
159.00	2	14.09	10.02	34.37	33.42	8.08	0.00
160.00	2	14.10	10.03	34.37	33.42	8.09	0.00

Table 24. Finalized data (Page 29 of 36)

	Group	H2	CH4	CO	CO2	C2H4	C2H4
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)							
161.00	2	14.10	10.03	34.37	33.44	8.07	0.00
162.00	2	17.50	4.38	21.56	56.56	0.00	0.00
163.00	2	9.97	6.11	35.37	48.55	0.00	0.00
164.00	2	9.73	5.78	32.52	51.98	0.00	0.00
165.00	2	9.49	6.01	30.38	54.11	0.00	0.00
166.00	2	10.50	7.61	38.58	43.31	0.00	0.00
167.00	2	10.92	7.76	36.21	45.11	0.00	0.00
168.00	2	13.58	8.40	40.99	37.04	0.00	0.00
169.00	2	10.43	7.49	40.37	41.71	0.00	0.00
170.00	2	9.92	7.93	37.96	44.19	0.00	0.00
171.00	2	11.58	7.06	34.75	46.61	0.00	0.00
172.00	2	11.20	8.00	38.93	41.87	0.00	0.00
173.00	2	11.71	9.98	46.42	31.89	0.00	0.00
174.00	2	11.70	9.40	45.64	33.26	0.00	0.00
175.00	2	16.33	7.16	40.04	36.47	0.00	0.00
176.00	2	16.67	3.39	28.25	51.69	0.00	0.00
177.00	1	29.31	10.41	32.39	41.26	0.00	0.00
178.00	1	27.12	6.96	35.01	30.91	0.00	0.00
179.00	1	24.02	8.50	34.22	33.26	0.00	0.00
180.00	1	25.40	8.47	37.20	28.93	0.00	0.00
181.00	1	24.48	8.82	30.60	36.10	0.00	0.00
182.00	1	25.84	7.90	35.56	30.70	0.00	0.00
183.00	1	21.67	9.34	31.51	37.48	0.00	0.00
184.00	1	22.40	8.70	33.70	35.20	0.00	0.00
185.00	1	26.83	8.61	29.43	35.14	0.00	0.00
186.00	1	28.66	8.22	33.17	29.96	0.00	0.00
187.00	1	27.10	9.40	29.30	34.20	0.00	0.00
188.00	1	27.68	8.19	35.09	29.04	0.00	0.00
189.00	2	11.43	20.02	34.90	30.95	2.70	0.00
190.00	2	10.88	17.01	29.33	38.50	4.28	0.00
191.00	2	11.76	16.96	36.15	32.70	2.44	0.00
192.00	3	18.57	8.28	40.72	29.31	3.13	0.00
193.00	3	18.97	9.15	35.04	33.71	3.13	0.00
194.00	3	23.84	7.41	28.70	38.66	1.39	0.00
195.00	3	18.55	8.14	37.33	33.26	2.71	0.00
196.00	3	17.21	9.77	34.65	35.35	3.02	0.00
197.00	3	25.73	6.26	31.32	35.57	1.12	0.00
198.00	3	19.52	6.99	33.98	38.55	0.96	0.00
199.00	3	18.38	8.83	30.79	39.38	2.63	0.00
200.00	3	24.89	7.01	31.00	36.20	0.90	0.00

Table 24. Finalized data (Page 30 of 36)

	Group	H2	CH4	CO	CO2	C2H4	C2H4
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)							
201.00	6	19.56	11.98	34.73	30.34	3.39	0.00
202.00	6	24.55	8.93	29.69	35.27	1.56	0.00
203.00	6	22.88	8.25	28.30	39.62	0.94	0.00
204.00	6	18.97	9.15	35.04	33.71	3.13	0.00
205.00	6	17.21	9.77	34.65	35.35	3.02	0.00
206.00	6	18.38	8.83	30.79	39.38	2.63	0.00
207.00	6	17.11	5.47	26.42	51.00	0.00	0.00
208.00	6	17.60	8.66	31.80	41.94	0.00	0.00
209.00	6	25.49	6.61	22.61	45.28	0.00	0.00
210.00	6	18.89	12.22	37.78	31.11	0.00	0.00
211.00	6	15.88	10.66	37.91	35.55	0.00	0.00
212.00	6	12.16	10.81	33.78	43.24	0.00	0.00
213.00	6	9.74	10.03	31.52	48.71	0.00	0.00
214.00	6	41.99	10.76	30.18	17.06	0.00	0.00
215.00	7	46.15	8.24	24.18	21.43	0.00	0.00
216.00	7	51.58	6.32	18.42	23.68	0.00	0.00
217.00	7	52.88	6.28	17.80	23.04	0.00	0.00
218.00	6	12.53	14.95	41.76	30.77	0.00	0.00
219.00	6	9.86	13.62	40.14	36.38	0.00	0.00
220.00	6	9.42	11.52	37.17	41.88	0.00	0.00
221.00	6	9.01	10.14	35.21	45.63	0.00	0.00
222.00	6	28.37	16.45	41.13	14.04	0.00	0.00
223.00	7	40.00	12.22	30.00	17.78	0.00	0.00
224.00	7	63.09	6.54	16.36	25.12	0.00	0.00
225.00	2	24.36	13.47	30.91	29.07	2.21	0.00
226.00	2	28.95	11.97	29.63	27.67	1.77	0.00
		226.00	226.00	226.00	226.00	226.00	226.00

Table 24. Finalized data (Page 31 of 36)

	C2H2	NH3	N2	H2O	O2	SUM
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)						
1.00	0.00	0.00	0.00	0.00	0.00	100.00
2.00	0.00	0.00	0.00	0.00	0.00	100.00
3.00	0.00	0.00	0.00	0.00	0.00	100.00
4.00	0.00	0.00	0.00	0.00	0.00	100.00
5.00	0.00	0.00	0.00	0.00	0.00	100.00
6.00	0.00	0.00	0.00	0.00	0.00	100.00
7.00	0.00	0.00	0.00	0.00	0.00	100.00
8.00	0.00	0.00	0.00	0.00	0.00	100.00
9.00	0.00	0.00	0.00	0.00	0.00	100.00
10.00	0.00	0.00	0.00	0.00	0.00	100.00
11.00	0.00	0.00	0.00	0.00	0.00	100.00
12.00	0.00	0.00	0.00	0.00	0.00	100.00
13.00	0.00	0.00	0.00	0.00	0.00	100.00
14.00	0.00	0.00	0.00	0.00	0.00	100.00
15.00	0.00	0.00	0.00	0.00	0.00	100.00
16.00	0.00	0.00	0.00	0.00	0.00	100.00
17.00	0.00	0.00	0.00	0.00	0.00	100.00
18.00	0.00	0.00	0.00	0.00	0.00	100.00
19.00	0.00	0.00	0.00	0.00	0.00	100.00
20.00	0.00	0.00	0.00	0.00	0.00	100.00
21.00	0.00	0.00	0.00	0.00	0.00	100.00
22.00	0.00	0.00	0.00	0.00	0.00	100.00
23.00	0.00	0.00	0.00	0.00	0.00	100.00
24.00	0.00	0.00	0.00	0.00	0.00	100.00
25.00	0.00	0.00	0.00	0.00	0.00	100.00
26.00	0.00	0.00	0.00	0.00	0.00	100.00
27.00	0.00	0.00	0.00	0.00	0.00	100.00
28.00	0.00	0.00	0.00	0.00	0.00	100.00
29.00	0.00	0.00	0.00	0.00	0.00	100.00
30.00	0.00	0.00	0.00	0.00	0.00	100.00
31.00	0.00	0.00	0.00	0.00	0.00	100.00
32.00	0.00	0.00	0.00	0.00	0.00	100.00
33.00	0.00	0.00	0.00	0.00	0.00	100.00
34.00	0.00	0.00	0.00	0.00	0.00	100.00
35.00	0.00	0.00	0.00	0.00	0.00	100.00
36.00	0.00	0.00	0.00	0.00	0.00	100.00
37.00	0.00	0.00	0.00	0.00	0.00	100.00
38.00	0.00	0.00	0.00	0.00	0.00	100.00
39.00	0.00	0.00	0.00	0.00	0.00	100.00
40.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 24. Finalized data (Page 32 of 36)

	C2H2	NH3	N2	H2O	O2	SUM
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)						
41.00	0.00	0.00	0.00	0.00	0.00	100.00
42.00	0.00	0.00	0.00	0.00	0.00	100.00
43.00	0.00	0.00	0.00	0.00	0.00	100.00
44.00	0.00	0.00	0.00	0.00	0.00	100.00
45.00	0.00	0.00	0.00	0.00	0.00	100.00
46.00	0.00	0.00	0.00	0.00	0.00	100.00
47.00	0.00	0.00	0.00	0.00	0.00	100.00
48.00	0.00	0.00	0.00	0.00	0.00	100.00
49.00	0.00	0.00	0.00	0.00	0.00	100.00
50.00	0.00	0.00	0.00	0.00	0.00	100.00
51.00	0.00	0.00	0.00	0.00	0.00	100.00
52.00	0.00	0.00	0.00	0.00	0.00	100.00
53.00	0.00	0.00	0.00	0.00	0.00	100.00
54.00	0.00	0.00	0.00	0.00	0.00	100.00
55.00	0.00	0.00	0.00	0.00	0.00	87.40
56.00	0.00	0.00	0.00	0.00	0.00	100.00
57.00	0.00	0.00	0.00	0.00	0.00	100.00
58.00	0.00	0.00	0.00	0.00	0.00	100.00
59.00	0.00	0.00	0.00	0.00	0.00	100.00
60.00	0.00	0.00	0.00	0.00	0.00	100.00
61.00	0.00	0.00	0.00	0.00	0.00	100.00
62.00	0.00	0.00	0.00	0.00	0.00	100.00
63.00	0.00	0.00	0.00	0.00	0.00	100.00
64.00	0.00	0.00	0.00	0.00	0.00	100.00
65.00	0.00	0.00	0.00	0.00	0.00	100.00
66.00	0.00	0.00	0.00	0.00	0.00	100.00
67.00	0.00	0.00	0.00	0.00	0.00	100.00
68.00	0.00	0.00	0.00	0.00	0.00	100.00
69.00	0.00	0.00	0.00	0.00	0.00	100.00
70.00	0.00	0.00	0.00	0.00	0.00	100.00
71.00	0.00	0.00	0.00	0.00	0.00	100.00
72.00	0.00	0.00	0.00	0.00	0.00	100.00
73.00	0.00	0.00	0.00	0.00	0.00	100.00
74.00	0.00	0.00	0.00	0.00	0.00	100.00
75.00	0.00	0.00	0.00	0.00	0.00	100.00
76.00	0.00	0.00	0.00	0.00	0.00	100.00
77.00	0.00	0.00	0.00	0.00	0.00	100.00
78.00	0.00	0.00	0.00	0.00	0.00	100.00
79.00	0.00	0.00	0.00	0.00	0.00	100.00
80.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 24. Finalized data (Page 33 of 36)

	C2H2	NH3	N2	H2O	O2	SUM
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)						
81.00	0.00	0.00	0.00	0.00	0.00	100.00
82.00	0.00	0.00	0.00	0.00	0.00	100.00
83.00	0.00	0.00	0.00	0.00	0.00	100.00
84.00	0.00	0.00	0.00	0.00	0.00	100.00
85.00	0.00	0.00	0.00	0.00	0.00	100.00
86.00	0.00	0.00	0.00	0.00	0.00	100.00
87.00	0.00	0.00	0.00	0.00	0.00	100.00
88.00	0.00	0.00	0.00	0.00	0.00	100.00
89.00	0.00	0.00	0.00	0.00	0.00	100.00
90.00	0.00	0.00	0.00	0.00	0.00	100.00
91.00	0.00	0.00	0.00	0.00	0.00	100.00
92.00	0.00	0.00	0.00	0.00	0.00	100.00
93.00	0.00	0.00	0.00	0.00	0.00	100.00
94.00	0.00	0.00	0.00	0.00	0.00	100.00
95.00	0.00	0.00	0.00	0.00	0.00	100.00
96.00	0.00	0.00	0.00	0.00	0.00	100.00
97.00	0.00	0.00	0.00	0.00	0.00	100.00
98.00	0.00	0.00	0.00	0.00	0.00	100.00
99.00	0.00	0.00	0.00	0.00	0.00	100.00
100.00	0.00	0.00	0.00	0.00	0.00	100.00
101.00	0.00	0.00	0.00	0.00	0.00	100.00
102.00	0.00	0.00	0.00	0.00	0.00	100.00
103.00	0.00	0.00	0.00	0.00	0.00	100.00
104.00	0.00	0.00	0.00	0.00	0.00	100.00
105.00	0.00	0.00	0.00	0.00	0.00	100.00
106.00	0.00	0.00	0.00	0.00	0.00	100.00
107.00	0.00	0.00	0.00	0.00	0.00	100.00
108.00	0.00	0.00	0.00	0.00	0.00	100.00
109.00	0.00	0.00	0.00	0.00	0.00	100.00
110.00	0.00	0.00	0.00	0.00	0.00	100.00
111.00	0.00	0.00	0.00	0.00	0.00	100.00
112.00	0.00	0.00	0.00	0.00	0.00	100.00
113.00	0.00	0.00	0.00	0.00	0.00	100.00
114.00	0.00	0.00	0.00	0.00	0.00	100.00
115.00	0.00	0.00	0.00	0.00	0.00	100.00
116.00	0.00	0.00	0.00	0.00	0.00	100.00
117.00	0.00	0.00	0.00	0.00	0.00	100.00
118.00	0.00	0.00	0.00	0.00	0.00	100.00
119.00	0.00	0.00	0.00	0.00	0.00	100.00
120.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 24. Finalized data (Page 34 of 36)

	C2H2	NH3	N2	H2O	O2	SUM
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)						
121.00	0.00	0.00	0.00	0.00	0.00	100.00
122.00	0.00	0.00	0.00	0.00	0.00	100.00
123.00	0.00	0.00	0.00	0.00	0.00	100.00
124.00	0.00	0.00	0.00	0.00	0.00	100.00
125.00	0.00	0.00	0.00	0.00	0.00	100.00
126.00	0.00	0.00	0.00	0.00	0.00	100.00
127.00	0.00	0.00	0.00	0.00	0.00	100.00
128.00	0.00	0.00	0.00	0.00	0.00	100.00
129.00	0.00	0.00	0.00	0.00	0.00	100.00
130.00	0.00	0.00	0.00	0.00	0.00	100.00
131.00	0.00	0.00	0.00	0.00	0.00	100.00
132.00	0.00	0.00	0.00	0.00	0.00	100.00
133.00	0.00	0.00	0.00	0.00	0.00	100.00
134.00	0.00	0.00	0.00	0.00	0.00	100.00
135.00	0.00	0.00	0.00	0.00	0.00	100.00
136.00	0.00	0.00	0.00	0.00	0.00	100.00
137.00	0.00	0.00	0.00	0.00	0.00	100.00
138.00	0.00	0.00	0.00	0.00	0.00	100.00
139.00	0.00	0.00	0.00	0.00	0.00	100.00
140.00	0.00	0.00	0.00	0.00	0.00	100.00
141.00	0.00	0.00	0.00	0.00	0.00	100.00
142.00	0.00	0.00	0.00	0.00	0.00	100.00
143.00	0.00	0.00	0.00	0.00	0.00	100.00
144.00	0.00	0.00	0.00	0.00	0.00	100.00
145.00	0.00	0.00	0.00	0.00	0.00	100.00
146.00	0.00	0.00	0.00	0.00	0.00	100.00
147.00	0.00	0.00	0.00	0.00	0.00	100.00
148.00	0.00	0.00	0.00	0.00	0.00	100.00
149.00	0.00	0.00	0.00	0.00	0.00	100.00
150.00	0.00	0.00	0.00	0.00	0.00	100.00
151.00	0.00	0.00	0.00	0.00	0.00	100.00
152.00	0.00	0.00	0.00	0.00	0.00	100.00
153.00	0.00	0.00	0.00	0.00	0.00	100.00
154.00	0.00	0.00	0.00	0.00	0.00	100.00
155.00	0.00	0.00	0.00	0.00	0.00	100.00
156.00	0.00	0.00	0.00	0.00	0.00	100.00
157.00	0.00	0.00	0.00	0.00	0.00	100.00
158.00	0.00	0.00	0.00	0.00	0.00	100.00
159.00	0.00	0.00	0.00	0.00	0.00	100.00
160.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 24. Finalized data (Page 35 of 36)

	C2H2	NH3	N2	H2O	O2	SUM
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)						
161.00	0.00	0.00	0.00	0.00	0.00	100.00
162.00	0.00	0.00	0.00	0.00	0.00	100.00
163.00	0.00	0.00	0.00	0.00	0.00	100.00
164.00	0.00	0.00	0.00	0.00	0.00	100.00
165.00	0.00	0.00	0.00	0.00	0.00	100.00
166.00	0.00	0.00	0.00	0.00	0.00	100.00
167.00	0.00	0.00	0.00	0.00	0.00	100.00
168.00	0.00	0.00	0.00	0.00	0.00	100.00
169.00	0.00	0.00	0.00	0.00	0.00	100.00
170.00	0.00	0.00	0.00	0.00	0.00	100.00
171.00	0.00	0.00	0.00	0.00	0.00	100.00
172.00	0.00	0.00	0.00	0.00	0.00	100.00
173.00	0.00	0.00	0.00	0.00	0.00	100.00
174.00	0.00	0.00	0.00	0.00	0.00	100.00
175.00	0.00	0.00	0.00	0.00	0.00	100.00
176.00	0.00	0.00	0.00	0.00	0.00	100.00
177.00	0.00	0.00	0.00	0.00	0.00	100.00
178.00	0.00	0.00	0.00	0.00	0.00	100.00
179.00	0.00	0.00	0.00	0.00	0.00	100.00
180.00	0.00	0.00	0.00	0.00	0.00	100.00
181.00	0.00	0.00	0.00	0.00	0.00	100.00
182.00	0.00	0.00	0.00	0.00	0.00	100.00
183.00	0.00	0.00	0.00	0.00	0.00	100.00
184.00	0.00	0.00	0.00	0.00	0.00	100.00
185.00	0.00	0.00	0.00	0.00	0.00	100.00
186.00	0.00	0.00	0.00	0.00	0.00	100.00
187.00	0.00	0.00	0.00	0.00	0.00	100.00
188.00	0.00	0.00	0.00	0.00	0.00	100.00
189.00	0.00	0.00	0.00	0.00	0.00	100.00
190.00	0.00	0.00	0.00	0.00	0.00	100.00
191.00	0.00	0.00	0.00	0.00	0.00	100.00
192.00	0.00	0.00	0.00	0.00	0.00	100.00
193.00	0.00	0.00	0.00	0.00	0.00	100.00
194.00	0.00	0.00	0.00	0.00	0.00	100.00
195.00	0.00	0.00	0.00	0.00	0.00	100.00
196.00	0.00	0.00	0.00	0.00	0.00	100.00
197.00	0.00	0.00	0.00	0.00	0.00	100.00
198.00	0.00	0.00	0.00	0.00	0.00	100.00
199.00	0.00	0.00	0.00	0.00	0.00	100.00
200.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 24. Finalized data (Page 36 of 36)

	C2H2	NH3	N2	H2O	O2	SUM
Composition Nitrogen, Oxygen and Moisture Free_5 compounds (considering C2H4)						
201.00	0.00	0.00	0.00	0.00	0.00	100.00
202.00	0.00	0.00	0.00	0.00	0.00	100.00
203.00	0.00	0.00	0.00	0.00	0.00	100.00
204.00	0.00	0.00	0.00	0.00	0.00	100.00
205.00	0.00	0.00	0.00	0.00	0.00	100.00
206.00	0.00	0.00	0.00	0.00	0.00	100.00
207.00	0.00	0.00	0.00	0.00	0.00	100.00
208.00	0.00	0.00	0.00	0.00	0.00	100.00
209.00	0.00	0.00	0.00	0.00	0.00	100.00
210.00	0.00	0.00	0.00	0.00	0.00	100.00
211.00	0.00	0.00	0.00	0.00	0.00	100.00
212.00	0.00	0.00	0.00	0.00	0.00	100.00
213.00	0.00	0.00	0.00	0.00	0.00	100.00
214.00	0.00	0.00	0.00	0.00	0.00	100.00
215.00	0.00	0.00	0.00	0.00	0.00	100.00
216.00	0.00	0.00	0.00	0.00	0.00	100.00
217.00	0.00	0.00	0.00	0.00	0.00	100.00
218.00	0.00	0.00	0.00	0.00	0.00	100.00
219.00	0.00	0.00	0.00	0.00	0.00	100.00
220.00	0.00	0.00	0.00	0.00	0.00	100.00
221.00	0.00	0.00	0.00	0.00	0.00	100.00
222.00	0.00	0.00	0.00	0.00	0.00	100.00
223.00	0.00	0.00	0.00	0.00	0.00	100.00
224.00	0.00	0.00	0.00	0.00	0.00	100.00
225.00	0.00	0.00	0.00	0.00	0.00	100.00
226.00	0.00	0.00	0.00	0.00	0.00	100.00
	226.00	226.00	226.00	226.00	226.00	226.00

APPENDIX H
TO BE PREDICTED

Table 25. To be predicted (Page 1 of 6)

Yield	CCE %	H2	CH4	CO	CO2	C2H4
0.89	78.17	21.99	9.27	43.50	20.28	4.95
1.01	80.66	28.75	8.41	39.72	19.22	3.90
1.15	85.90	32.66	7.68	38.01	18.26	3.38
1.32	92.35	36.74	7.41	34.48	18.50	2.87
1.42	92.59	39.84	6.29	33.52	18.99	1.36
1.01	76.26	32.67	8.07	37.96	17.63	3.67
1.10	84.49	31.52	8.28	39.62	16.91	3.66
1.20	90.60	31.91	7.82	40.29	16.56	3.42
1.10	84.00	31.50	7.57	39.79	17.74	3.41
0.93	70.60	32.10	6.85	38.18	19.47	3.39
0.95	68.67	33.50	6.35	42.91	15.65	1.59
1.19	92.09	30.48	8.48	40.97	16.65	3.42
1.16	88.70	31.39	8.36	39.68	17.17	3.40
1.15	85.90	32.64	7.63	38.03	18.32	3.38
1.00	75.10	32.47	7.50	37.67	18.97	3.38
1.01	77.62	30.75	7.83	41.11	16.93	3.37
1.11	84.40	31.73	7.66	40.32	16.92	3.37
1.19	90.60	31.68	7.63	37.64	19.70	3.36
1.25	95.10	31.58	6.68	37.17	21.22	3.34
1.02	100.00	22.66	6.16	39.90	31.28	0.00
0.95	100.00	23.47	8.37	41.84	26.33	0.00
0.88	100.00	20.70	2.11	32.28	44.91	0.00
0.92	100.00	26.29	8.27	49.63	15.81	0.00
0.94	100.00	28.77	8.61	39.53	23.09	0.00
1.02	100.00	24.68	6.96	42.19	26.16	0.00
0.90	100.00	15.62	3.03	35.80	45.56	0.00
0.98	100.00	30.02	8.28	28.57	33.13	0.00
0.94	100.00	31.14	8.90	29.01	30.95	0.00
1.07	100.00	30.39	9.76	29.65	30.20	0.00
0.79	100.00	31.47	8.90	24.43	35.20	0.00
0.86	88.40	29.04	7.35	31.74	31.87	0.00
2.80	92.80	27.88	10.04	33.04	29.04	0.00
3.29	100.00	29.42	7.03	30.68	32.87	0.00
2.94	100.00	27.33	9.31	32.78	30.58	0.00
1.93	100.00	22.08	8.25	39.64	30.03	0.00
1.78	100.00	20.71	7.78	46.97	24.53	0.00
2.13	100.00	21.28	5.76	42.39	30.57	0.00
1.98	100.00	20.80	7.44	43.17	28.59	0.00
2.97	100.00	23.89	5.87	41.61	28.63	0.00
3.00	100.00	24.33	7.70	42.09	25.87	0.00
3.19	100.00	21.52	5.67	43.24	29.57	0.00
2.93	100.00	22.49	6.30	42.65	28.55	0.00

Table 25. To be predicted (Page 2 of 6)

Yield	CCE %	H2	CH4	CO	CO2	C2H4
3.51	94.00	28.18	5.69	36.31	29.54	0.27
4.36	98.00	23.97	4.79	30.48	40.41	0.34
5.04	99.00	18.83	2.51	25.52	52.72	0.42
3.96	100.00	19.70	11.36	30.30	36.74	1.89
3.92	100.00	14.96	7.87	22.05	53.94	1.18
3.56	100.00	18.47	7.32	25.44	46.34	2.44
1.89	97.90	17.05	9.56	38.67	34.72	0.00
1.80	92.20	17.54	9.19	37.58	35.70	0.00
1.82	92.90	16.17	9.36	38.09	36.38	0.00
1.73	78.00	28.12	3.63	25.85	42.40	0.00
1.59	72.20	19.95	6.30	29.13	44.62	0.00
1.62	74.10	19.23	6.15	27.18	47.44	0.00
2.58	89.30	16.45	3.55	26.77	53.23	0.00
1.76	83.90	23.32	5.38	26.23	45.07	0.00
1.76	84.30	27.02	4.89	30.00	38.09	0.00
1.23	100.00	31.40	8.40	20.60	35.30	4.30
1.37	100.00	37.20	7.60	17.40	35.30	2.50
1.41	100.00	38.76	8.29	18.58	32.17	2.20
1.41	100.00	38.30	8.00	19.30	32.70	1.70
1.56	100.00	44.40	7.04	18.83	28.15	1.59
1.19	100.00	27.57	7.69	15.48	46.85	2.40
1.32	100.00	34.27	7.59	15.58	40.36	2.20
1.42	100.00	39.10	7.40	16.80	34.70	2.00
1.51	100.00	43.24	7.41	18.52	28.53	2.30
1.61	100.00	46.80	8.30	20.20	22.30	2.40
1.59	100.00	46.00	7.90	19.90	24.10	2.10
1.68	100.00	48.45	6.89	18.48	24.58	1.60
1.77	100.00	51.25	6.09	15.88	25.07	1.70
1.90	100.00	54.40	5.00	11.80	27.40	1.40
2.04	100.00	57.34	4.10	7.29	30.17	1.10
1.33	100.00	35.89	8.47	19.76	35.88	0.00
1.33	100.00	36.17	9.28	19.80	34.75	0.00
2.12	100.00	30.80	12.66	16.04	40.50	0.00
2.02	100.00	39.53	5.30	27.58	27.58	0.00
2.06	100.00	43.37	5.06	23.96	27.61	0.00
1.94	100.00	39.67	5.87	27.93	26.52	0.00
1.93	100.00	32.65	10.28	19.91	37.16	0.00
1.94	100.00	31.25	12.28	21.78	34.69	0.00
2.03	100.00	35.41	7.84	20.12	36.63	0.00
1.88	100.00	33.61	8.50	23.28	34.61	0.00
1.90	100.00	34.15	8.48	22.43	34.94	0.00
2.19	100.00	31.95	11.41	18.47	38.17	0.00

Table 25. To be predicted (Page 3 of 6)

Yield	CCE %	H2	CH4	CO	CO2	C2H4
1.01	100.00	14.78	8.65	34.96	41.61	0.00
1.05	100.00	18.21	8.66	31.79	41.34	0.00
1.07	100.00	19.65	8.64	32.15	39.56	0.00
1.04	100.00	17.73	9.09	30.84	42.35	0.00
1.00	100.00	14.04	8.82	31.23	45.91	0.00
1.10	87.00	25.58	11.63	35.27	27.52	0.00
1.10	88.00	25.05	10.34	34.99	29.62	0.00
1.17	90.00	28.06	9.42	30.06	32.46	0.00
1.23	94.00	28.08	9.42	32.36	30.14	0.00
1.36	95.00	34.38	9.62	27.60	28.39	0.00
1.44	96.00	37.41	9.75	28.09	24.75	0.00
2.34	100.00	26.10	11.03	34.56	28.31	0.00
2.21	100.00	26.25	11.40	35.23	27.12	0.00
2.50	100.00	27.83	8.45	38.20	25.53	0.00
2.67	100.00	27.37	8.64	33.95	30.04	0.00
2.46	100.00	28.41	8.60	35.51	27.48	0.00
2.75	100.00	33.78	8.11	22.97	35.14	0.00
2.75	100.00	33.14	7.99	22.61	36.26	0.00
2.47	100.00	35.64	8.61	26.69	29.05	0.00
2.46	100.00	26.49	9.40	36.66	27.45	0.00
2.50	100.00	26.08	9.80	33.73	30.39	0.00
3.46	100.00	23.41	7.73	37.73	31.14	0.00
3.41	100.00	25.65	8.04	37.39	28.91	0.00
3.18	100.00	26.31	8.63	37.95	27.11	0.00
3.60	100.00	22.49	7.66	38.04	31.82	0.00
3.32	100.00	24.14	8.41	38.36	29.09	0.00
3.10	100.00	25.40	10.52	35.52	28.57	0.00
3.19	100.00	28.38	7.44	33.27	30.92	0.00
3.46	100.00	27.99	7.05	41.45	23.50	0.00
3.21	100.00	26.45	6.88	35.05	31.61	0.00
3.31	100.00	23.02	7.67	35.12	34.19	0.00
3.03	100.00	29.02	8.43	34.51	28.04	0.00
2.02	81.00	13.78	4.08	21.94	55.36	4.85
2.64	86.90	20.39	2.23	19.27	55.59	2.51
2.47	83.50	24.16	2.08	21.82	49.35	2.60
2.51	84.70	20.65	2.45	20.38	53.53	2.99
1.28	100.00	22.56	8.72	25.64	43.08	0.00
1.31	100.00	24.10	8.21	25.13	42.56	0.00
1.31	100.00	23.96	8.85	25.52	41.67	0.00
1.35	100.00	26.56	8.33	23.96	41.15	0.00
1.30	100.00	25.91	7.77	21.76	44.56	0.00
1.34	100.00	27.98	7.25	20.73	44.04	0.00

Table 25. To be predicted (Page 4 of 6)

Yield	CCE %	H2	CH4	CO	CO2	C2H4
1.31	100.00	26.42	7.77	22.80	43.01	0.00
1.36	100.00	29.02	7.77	20.21	43.01	0.00
1.30	100.00	23.44	9.38	28.13	39.06	0.00
1.31	100.00	23.96	9.38	28.13	38.54	0.00
1.25	100.00	20.62	8.76	24.23	46.39	0.00
1.35	100.00	28.65	8.33	23.44	39.58	0.00
1.40	100.00	31.09	7.77	21.76	39.38	0.00
1.28	100.00	24.74	7.22	22.68	45.36	0.00
1.31	100.00	22.72	10.10	39.14	40.40	0.00
1.33	100.00	23.99	10.10	38.51	39.77	0.00
1.19	100.00	21.98	9.89	31.87	36.26	0.00
1.20	100.00	22.65	8.84	31.49	37.02	0.00
2.64	97.00	18.85	8.64	24.61	44.76	3.14
2.63	91.60	18.42	8.49	29.65	41.14	2.30
2.77	92.20	18.16	7.83	27.26	44.20	2.55
2.35	94.70	19.91	7.06	27.05	42.70	3.28
2.28	92.70	22.16	7.32	27.90	39.80	2.83
1.23	88.20	17.22	7.96	26.18	45.64	3.00
1.69	85.20	21.15	8.41	27.77	39.61	3.06
2.58	98.40	7.23	4.98	21.45	64.64	1.69
2.74	96.50	16.78	5.36	25.91	50.02	1.93
1.28	83.60	10.79	10.94	41.35	30.54	6.38
1.45	93.30	12.41	10.72	39.04	30.94	6.89
1.52	94.80	13.90	10.26	36.78	32.45	6.61
1.64	96.50	15.94	9.71	34.79	33.90	5.66
1.22	81.70	10.57	11.39	40.63	30.59	6.82
1.30	84.10	12.43	11.05	38.95	31.22	6.35
1.37	84.90	13.30	11.00	37.46	32.76	5.48
1.42	85.50	14.99	10.23	35.68	33.49	5.60
1.50	88.20	15.65	10.05	34.68	33.71	5.92
1.30	82.70	14.10	10.02	34.35	33.42	8.10
1.37	87.30	14.10	10.03	34.35	33.41	8.11
1.44	91.90	14.09	10.02	34.37	33.42	8.08
1.52	96.90	14.10	10.03	34.37	33.42	8.09
1.55	98.90	14.10	10.03	34.37	33.44	8.07
3.59	97.70	17.50	4.38	21.56	56.56	0.00
3.51	102.00	9.97	6.11	35.37	48.55	0.00
2.97	92.90	9.73	5.78	32.52	51.98	0.00
3.37	99.20	9.49	6.01	30.38	54.11	0.00
2.80	98.60	10.50	7.61	38.58	43.31	0.00
2.91	93.00	10.92	7.76	36.21	45.11	0.00
2.55	91.90	13.58	8.40	40.99	37.04	0.00

Table 25. To be predicted (Page 5 of 6)

Yield	CCE %	H2	CH4	CO	CO2	C2H4
2.43	85.80	10.43	7.49	40.37	41.71	0.00
2.78	91.10	9.92	7.93	37.96	44.19	0.00
2.75	87.60	11.58	7.06	34.75	46.61	0.00
2.39	81.90	11.20	8.00	38.93	41.87	0.00
1.90	81.60	11.71	9.98	46.42	31.89	0.00
2.19	89.80	11.70	9.40	45.64	33.26	0.00
2.38	94.90	16.33	7.16	40.04	36.47	0.00
3.13	98.30	16.67	3.39	28.25	51.69	0.00
1.02	75.00	29.31	10.41	32.39	41.26	0.00
0.95	78.10	27.12	6.96	35.01	30.91	0.00
0.88	75.00	24.02	8.50	34.22	33.26	0.00
0.92	77.50	25.40	8.47	37.20	28.93	0.00
0.94	78.00	24.48	8.82	30.60	36.10	0.00
1.02	83.00	25.84	7.90	35.56	30.70	0.00
0.90	78.00	21.67	9.34	31.51	37.48	0.00
0.98	84.00	22.40	8.70	33.70	35.20	0.00
0.94	80.10	26.83	8.61	29.43	35.14	0.00
1.07	88.50	28.66	8.22	33.17	29.96	0.00
0.79	67.20	27.10	9.40	29.30	34.20	0.00
0.86	72.30	27.68	8.19	35.09	29.04	0.00
1.09	57.50	11.43	20.02	34.90	30.95	2.70
1.26	67.20	10.88	17.01	29.33	38.50	4.28
1.37	70.70	11.76	16.96	36.15	32.70	2.44
2.00	100.00	18.57	8.28	40.72	29.31	3.13
2.01	100.00	18.97	9.15	35.04	33.71	3.13
2.26	100.00	23.84	7.41	28.70	38.66	1.39
2.03	100.00	18.55	8.14	37.33	33.26	2.71
2.05	100.00	17.21	9.77	34.65	35.35	3.02
2.24	100.00	25.73	6.26	31.32	35.57	1.12
2.19	100.00	19.52	6.99	33.98	38.55	0.96
2.14	100.00	18.38	8.83	30.79	39.38	2.63
2.26	100.00	24.89	7.01	31.00	36.20	0.90
1.88	100.00	19.56	11.98	34.73	30.34	3.39
2.22	100.00	24.55	8.93	29.69	35.27	1.56
2.32	100.00	22.88	8.25	28.30	39.62	0.94
1.92	100.00	18.97	9.15	35.04	33.71	3.13
1.95	100.00	17.21	9.77	34.65	35.35	3.02
2.04	100.00	18.38	8.83	30.79	39.38	2.63
3.55	100.00	17.11	5.47	26.42	51.00	0.00
3.61	100.00	17.60	8.66	31.80	41.94	0.00
3.26	100.00	25.49	6.61	22.61	45.28	0.00
2.75	100.00	18.89	12.22	37.78	31.11	0.00

Table 25. To be predicted (Page 6 of 6)

Yield	CCE %	H2	CH4	CO	CO2	C2H4
2.83	100.00	15.88	10.66	37.91	35.55	0.00
3.09	100.00	12.16	10.81	33.78	43.24	0.00
3.19	100.00	9.74	10.03	31.52	48.71	0.00
2.27	100.00	41.99	10.76	30.18	17.06	0.00
1.87	100.00	46.15	8.24	24.18	21.43	0.00
2.08	100.00	51.58	6.32	18.42	23.68	0.00
2.13	100.00	52.88	6.28	17.80	23.04	0.00
2.34	100.00	12.53	14.95	41.76	30.77	0.00
2.43	100.00	9.86	13.62	40.14	36.38	0.00
2.69	100.00	9.42	11.52	37.17	41.88	0.00
2.88	100.00	9.01	10.14	35.21	45.63	0.00
1.85	100.00	28.37	16.45	41.13	14.04	0.00
1.73	100.00	40.00	12.22	30.00	17.78	0.00
2.16	100.00	63.09	6.54	16.36	25.12	0.00
1.64	80.50	24.36	13.47	30.91	29.07	2.21
1.98	94.50	28.95	11.97	29.63	27.67	1.77

APPENDIX I

SPECIAL SUBSET OF DATA BASE (12 SETS)

Additional subset involved in the analysis from 50 runs (Figure 69.) presented in the following table 26.

Table 26. Separate results for different dynamics (BFB vs. CFB vs. Overall), (Section 1 of 2).

	C	H	O	N	S	MC	Volatile matter	Fixed carbon	Ash	ER
1.00	49.99	5.70	44.09	0.20	0.03	7.00	82.50	16.65	0.85	0.32
2.00	49.99	5.70	44.09	0.20	0.03	7.00	82.50	16.65	0.85	0.37
3.00	49.99	5.70	44.09	0.20	0.03	7.00	82.50	16.65	0.85	0.47
4.00	49.99	5.70	44.09	0.20	0.03	7.00	82.50	16.65	0.85	0.26
5.00	49.99	5.70	44.09	0.20	0.03	7.00	82.50	16.65	0.85	0.36
6.00	49.99	5.70	44.09	0.20	0.03	7.00	82.50	16.65	0.85	0.32
7.00	48.16	6.06	45.58	0.15	0.05	7.50	80.00	19.50	0.50	0.37
8.00	48.16	6.06	45.58	0.15	0.05	7.50	80.00	19.50	0.50	0.00
9.00	48.16	6.06	45.58	0.15	0.05	7.50	80.00	19.50	0.50	0.00
10.00	41.00	5.70	48.40	4.90	0.01	7.00	67.70	31.10	1.20	0.37
11.00	55.18	5.95	38.31	0.32	0.23	11.53	72.98	25.80	1.23	0.07
12.00	47.81	7.26	44.03	0.66	0.24	8.76	85.20	11.80	3.00	0.21

Table 26. Separate results for different dynamics (BFB vs. CFB vs. Overall), (Section 2 of 2).

	Steam to biomass ratio (SBR)	Temp (°C)	H2 (Vol %)	CH4 (Vol %)	CO (Vol %)	CO2 (Vol %)
1.00	0.00	800.00	18.67	8.00	37.33	36.00
2.00	0.00	800.00	23.63	6.72	32.34	37.31
3.00	0.00	810.00	24.69	7.41	30.86	37.04
4.00	0.00	800.00	23.63	6.72	32.34	37.31
5.00	0.00	790.00	23.63	6.72	32.34	37.31
6.00	0.00	800.00	20.88	9.89	39.56	29.67
7.00	0.00	850.00	28.25	7.88	15.86	48.00
8.00	2.70	800.00	55.17	5.07	11.97	27.79
9.00	4.70	800.00	57.98	4.14	7.37	30.51
10.00	0.00	850.00	25.33	7.06	31.19	36.42
11.00	0.92	800.00	50.00	8.50	12.80	28.70
12.00	0.00	800.00	22.00	16.00	32.00	30.00

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